Provenance of the calculations: programs, instructions, data, checks

PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham — the 16-component Dirac field with mass term and self-interaction on the deflating 4+4 and 7+1 spacetimes

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1. What this document is

This document is the record of how every number, table and figure of the folder `Revision_Deflation_CJT_and_Kohn-Sham` was computed. It gives

- the objects that are computed, with their defining equations (Section 2);
- the instructions for compiling and running every program (Section 3);
- the list of the commands of the program, with their source files (Section 4);
- the complete output of every check of every command, of the two independent verification scripts, of the test of those scripts with deliberate errors, of the check of the symbols and of the check of the numbers quoted in the documents (Section 5);
- every data file, with the command that wrote it, its content and its columns (Section 6);
- every figure, with the data files from which it is drawn (Section 7);
- for each table and figure of the two manuscripts, the command and the data files behind it (Section 8);
- the scripts and the source files (Sections 9 and 10);
- the complete table of the symbols and of the names used in programs and data files (Appendix A).

The statements of the checks are printed by the programs and use the spellings of the programs: `rho` for the number density n, `rho_S` for the scalar density n_S, `M` for the mass m^* of the particles in matter, `eps` for the energy density ε, `Pi` for the interaction pressure p_{int}, `tau_d` for the deflation clock τ_d, `Lambda` for the cutoff Λ, `m_ph` for the mass m_ϑ of the phase mode, `kappa` for κ_8, `a`, `b`, `H_a`, `H_b` for `a`, `b`, H_{a}, H_{b}. Appendix A lists every such name beside its symbol. In those statements "the main project" means the part of PROJECT spacetimes that treats the flat 4+4 and 7+1 spacetimes: its Rust crate (`code/spacetimes-rs` at the top of the repository) is used as a library, and its data files (`data` at the top of the repository) are what several checks compare against. "The author's document" means his document on Kohn–Sham matter and gravity on the 7+1 spacetime, whose numbers are reproduced.

2. The objects that are computed

Coordinates. $x0, \ldots, x7$: $x0$ a hidden space-like coordinate; $x1, x2, x3$ ordinary space; $x4$ the ordinary time; $x5, x6, x7$ three further coordinates, time-like on the 4+4 spacetime and space-like on the 7+1 spacetime.

The hypothesis of deflation (the author's). The coordinates $x5, x6, x7$ deflate exponentially. With three positive rates g55, g66, g77, three positive constants and the scale factors $b_t(x4) =$

$\text{const}_t\, e^{-\text{rate}_t\, x4}$ ($t = 5, 6, 7$; $\text{rate}_5 = \text{g55}$ and so on), the metrics are, with space-like components positive and time-like components negative,

$$g^{(4+4)} = \text{diag}(+1,+1,+1,+1,-1,-b_5^2,-b_6^2,-b_7^2),$$
$$g^{(7+1)} = \text{diag}(+1,+1,+1,+1,-1,+b_5^2,+b_6^2,+b_7^2).$$

The volume factor is $V_d = b_5 b_6 b_7$, the total rate $\text{gsum} = \text{g55} + \text{g66} + \text{g77}$, and the deflation clock $\tau_d = \text{gsum}\,(x4 - x4_{\text{ref}})$.

The fermion field. A column ψ of sixteen complex Grassmann components with the real 16×16 Dirac matrices γ^A of the spacetime at hand. Its Lagrangian has a kinetic term, a mass term and a self-interaction,

$$\mathcal{L} = V_d\,[\mathcal{L}_{\text{kin}} + \mathcal{L}_{\text{mass}} + \mathcal{L}_{\text{self}}], \qquad \mathcal{L}_{\text{self}} = -\tfrac{1}{2}\sum_{r=0}^{8} g_r\, I_r, \qquad I_r = \sum_{|K|=r} (\bar\psi\,\Theta_K\psi)(\bar\psi\,\Theta^K\psi).$$

Here K runs over the lists of r different indices, $\Theta^K = \zeta_r\,\gamma^{A_1}\cdots\gamma^{A_r}$ with a phase ζ_r that makes the bilinear real, Θ_K is its inverse, and $g_0, \ldots, g_8$ are nine couplings. On 7+1, $\mathcal{L}_{\text{kin}} + \mathcal{L}_{\text{mass}} = \bar\psi\,(i\gamma^\mu\nabla_\mu - m)\psi$ with $\bar\psi = \psi^\dagger\gamma^4$ and the covariant derivative ∇_μ of a spinor. On 4+4 the covariant form is $\tilde\psi\,(\gamma^\mu\nabla_\mu - m)\psi$ with $\tilde\psi = \psi^\dagger C^-_{1644}$; for fields that do not depend on $x5, x6, x7$ (the quantized sector) the theory is defined by the positive Lagrangian $\psi^\dagger(i\partial_4 - \mathsf{h})\psi$ with $\mathsf{h} = \sum_j \alpha^j(-i\partial_j) + \beta m$, and the self-interaction is taken in one of two forms: the quantized-sector form (tag 44q; bilinears with $\bar\psi = \psi^\dagger\beta$) or the covariant form (tag 44c; bilinears with $\tilde\psi$). The tag 71s denotes the self-interaction on 7+1.

Four independent couplings. Because the components of ψ anticommute, the nine invariants obey $(1 + \mathsf{F})\,I = 0$ with the Fierz matrix F; only I_0, I_1, I_7, I_8 are independent, with the effective couplings g_0^{eff} (scalar), g_1^{eff} (vector), g_7^{eff} (axial vector), g_8^{eff} (pseudoscalar).

The Vorton-Scalar. A complex scalar field Φ_{VS} with the potential $\lambda_{VS}(|\Phi_{VS}|^2 - v_{VS}^2/2)^2$ and the Yukawa coupling $y_{VS}\,\bar\psi(\Phi_{VS}P^R + \Phi_{VS}^* P^L)\psi$. In its vacuum the fermions have the mass $m = m_D + m_Y$ (bare Dirac mass plus $m_Y = y_{VS}v_{VS}/\sqrt{2}$), the radial mode has the mass $m_s = \sqrt{2\lambda_{VS}}\,v_{VS}$, and both modes couple with $g_{VS} = m_Y/v_{VS}$; $\alpha_{VS} = g_{VS}^2/4\pi$, $D_{VS} = \alpha_{VS}m/m_s$, $C = g_{VS}^2 m^2/m_s^2$.

Fields that do not depend on $x5, x6, x7$. The weighted field $\hat\psi = \sqrt{V_d}\,\psi$ obeys the flat equations with the couplings g_r/V_d, $y_{VS}/\sqrt{V_d}$, λ_{VS}/V_d and the vacuum value $v_{VS}\sqrt{V_d}$: the masses do not change, and g_r, g_{VS}^2, C, D_{VS} grow as e^{τ_d}.

The Kohn–Sham method. Uniform matter with the number density n and the scalar density n_S is described by particles of the mass $m^* = m + G_S\,n_S$ in the vector field $V_0 = G_V\,n$, with the mean-field couplings G_S, G_V (in the Hartree–Fock approximation G_S^{HF}, G_V^{HF}, linear combinations of the g_r). With $E^*(k) = \sqrt{k^2 + m^{*2}}$ and the average $\langle f\rangle = \nu\int d^Dk\, f/(2\pi)^D$ over the occupied states ($\nu = 8$ internal states, D space dimensions with momentum),

$$n_S = \left\langle \frac{m^*}{E^*} \right\rangle, \qquad \varepsilon = \langle E^*\rangle + \tfrac{1}{2}G_V n^2 - \frac{(m^*-m)^2}{2G_S},$$
$$p_j = \left\langle \frac{k_j^2}{E^*} \right\rangle + p_{\text{int}}, \quad p_{\text{int}} = \tfrac{1}{2}G_V n^2 + \frac{(m^*-m)^2}{2G_S}.$$

The CJT effective potential. For the same matter, at the chemical potential μ,

$$\Omega_{\mathrm{CJT}}(m^*, V_0; \mu) = -\nu \int \frac{d^D k}{(2\pi)^D} \, (\mu - V_0 - E^*) \, \theta(\mu - V_0 - E^*) - \frac{(m^* - m)^2}{2G_S} - \frac{V_0^2}{2G_V},$$

to which the filled states of negative energy below a cutoff Λ add $-\nu \int_{|\vec{k}|<\Lambda} d^D k \, [\sqrt{k^2 + m^{*2}} - \sqrt{k^2 + m^2}]/(2\pi)^D$. Its stationary points are the Kohn–Sham states. With the filled states a massless field acquires a mass when G_S is below the critical coupling $G_c = -(2\pi)^D (D-1)/(\nu A_D \Lambda^{D-1})$.

Gravity. For a metric with two scale factors $\mathsf{a}(x4)$ (for n_{a} coordinates) and $\mathsf{b}(x4)$ (for n_{b} coordinates), rates H_{a}, H_{b}, the Einstein–Lovelock equations are $\sum_{k=1}^{3} \alpha_k E^{(k)\mu}{}_{\nu} - \bar{\Lambda}\, \delta^{\mu}_{\nu} = \kappa_8 T^{\mu}{}_{\nu}$ with $\alpha_1 = 1$, $T^4{}_4 = \varepsilon$, $T^m{}_m = -p_m$, and $E^{(k)m}{}_m = (-1)^{k+1} k!\, 2^{k-1}$ times the sum, over the sets of $2k$ directions other than m and over their divisions into pairs, of the products of the pair curvatures $K_{4i} = H_i' + H_i^2$, $K_{ij} = H_i H_j$.

3. How to run everything

What is needed. A Rust toolchain (`cargo`); Python 3 with the packages numpy, scipy, sympy, mpmath and matplotlib; for the documents, pandoc, XeLaTeX with latexmk, and pdflatex with REVTeX 4.2 and BibTeX. No Rust crate is downloaded: the program depends only on the library crate of the same repository.

Where. All commands are run from the folder `Revision_Deflation_CJT_and_Kohn-Sham` of the repository `https://github.com/once-ere/spacetimes`. On the author's Mac that is

```
cd "/Users/nsh/Developer/github/spacetimes/Revision_Deflation_CJT_and_Kohn-Sham"
```

Step 1: compile the program and run its unit tests.

```
cargo test  --release --manifest-path code/deflation-cjt-ks-rs/Cargo.toml
cargo build --release --manifest-path code/deflation-cjt-ks-rs/Cargo.toml
```

The first command compiles the program and executes its 68 unit tests; it must end with `test result: ok`. The second builds the program `code/deflation-cjt-ks-rs/target/release/deflation_cjt_ks`.

Step 2: run the commands. Each command writes its data files and its report of checks into the folder `data` and ends with a line such as `kohnsham: 13 of 13 checks passed`.

```
B=code/deflation-cjt-ks-rs/target/release/deflation_cjt_ks
$B help
for c in selfcoupling kohnsham pairing gravity matrices modes symmetry channels \
         history quench paired cosmology matter background spectrum; do $B $c; done
$B forecast
$B superfluid
$B families
```

The first line abbreviates the path of the program; `help` lists the commands. The loop runs fifteen commands, each of which takes from a few seconds to about two minutes on the two-processor machine on which they were executed. The last three take longer: `forecast` about 13 minutes, `superfluid` about 42 minutes, `families` about 75 minutes (`$B families quick` is a short test of the same code).

Step 3: the independent verifications.

```
python3 code/scripts/verify_deflating_independent.py
python3 code/scripts/verify_cjt_ks_independent.py
python3 code/scripts/mutation_test_independent.py
python3 code/scripts/vendor_eb.py --check
```

The first two scripts repeat the central results with computer algebra and general-purpose numerical tools and write `data/deflating_checks_independent.txt` and `data/cjt_ks_checks_independent.txt`. The third introduces deliberate errors into copies of the two scripts, one at a time, and records whether the checks notice (about six minutes). The fourth verifies that the four modules of the Einstein–Boltzmann program in this folder are exactly the modules of the library crate with the listed changes.

Step 4: figures, tables and documents.

```
python3 code/scripts/make_figures.py
python3 code/scripts/write_symbol_registry.py
python3 code/scripts/write_doc_tables.py
python3 code/scripts/assemble_docs.py
python3 code/scripts/build_manuscripts.py
bash    code/scripts/build_docs.sh
python3 code/scripts/check_numbers.py
python3 code/scripts/check_symbols.py
python3 code/scripts/check_self_contained.py
```

In this order: draw every figure from the data files; write the registry of symbols; write every table and every computed number of the documents from the data files; assemble the Markdown documents; assemble and typeset the two manuscripts (REVTeX; the script fails if a reference is undefined or a box is too wide) and write their Markdown versions; typeset the other documents; verify that every number of the running text is a number that a program stated; verify the symbols; verify that no document sends its reader to another document.

4. The commands of the program

command	what it computes	source files (in `code/deflation-cjt-ks-rs/src`)	data files	checks passed (report)
`selfcoupling`	the nine channels of the self-interaction in exact arithmetic, for 7+1 and for the two forms of 4+4: reality, invariance, Fierz matrix, reduction to four couplings, Hartree-Fock couplings, pair couplings, chiral and internal symmetry	`selfint.rs`, `cmd_selfcoupling.rs`	7	34 of 34 (`selfcoupling_checks.txt`)
`kohnsham`	uniform matter by the Kohn-Sham method and by the CJT effective potential: the flat reference example; matter on the deflating spacetimes; binding thresholds along the clock; the two halves of the 4+4 field; the filled states of negative energy; bare mass, contact couplings and Vorton-Scalar together	`kohn_sham.rs`, `cjt_contact.rs`, `cmd_kohnsham.rs`	15	13 of 13 (`kohnsham_checks.txt`)
`pairing`	the gap of the pair along the hidden axis with contact couplings and exchange of the Vorton-Scalar	`pair_contact.rs`, `cmd_pairing.rs`	3	4 of 4 (`pairing_checks.txt`)
`gravity`	the Einstein-Lovelock equations with Kohn-Sham matter as the source: the reference evolution; the deflating spacetimes with matter; stability of the deflation	`gravity.rs`, `cmd_gravity.rs`	6	5 of 5 (`gravity_checks.txt`)
`matrices`	checks of the Dirac matrices of the frame, of the geometry of the deflating spacetimes, of the matrices in coordinate components, of the spin connection and of the Dirac operators	`metric.rs`, `geometry.rs`, `curved.rs`, `main.rs`	6	106 of 106 (`flat_matrix_checks.txt`); 47 of 47 (`deflating_checks.txt`)
`modes`	solutions of the Dirac equations mode by mode: creation of particles by the deflation (7+1), bounded solutions and their extinction (4+4)	`modes.rs`, `main.rs`	6	7 of 7 (`mode_checks.txt`)

command	what it computes	source files (in code/ deflation-cjt-ks-rs/src)	data files	checks passed (report)
history	the couplings and every threshold of the model along the deflation clock: gap, bound pair, matter, vacuum potential, melting, order of events	history.rs, main.rs	7	8 of 8 (history_ checks.txt)
quench	time-dependent dynamics of the paired state while the coupling grows	quench.rs, main.rs	3	5 of 5 (quench_ checks.txt)
cosmology	expansion of ordinary space on the deflating spacetime, bounds on the rate, the Hubble constant, the exponent of the electroweak scale	cosmo_defl.rs, main.rs	5	6 of 6 (cosmology_ checks.txt)
channels	exchange of the Vorton-Scalar: amplitudes in the pair channels, the field of a static source, modes of the Vorton-Scalar, creation of quanta, bounded solutions	scalar.rs, cmd_ channels.rs	7	7 of 7 (channel_ checks.txt)
matter	dark matter on the deflating spacetime: degeneracy, self-interaction, the resonance at the moment of binding, growth of structure, drift of the quantities, quanta created by the deflation	cosmo_defl.rs, cmd_ matter.rs	7	6 of 6 (matter_ checks.txt)
paired	the results of the CJT effective action along the clock: potential of the paired state, thermodynamics, Ginzburg-Landau coefficients, order-parameter space, matter	cmd_paired.rs	8	7 of 7 (paired_ checks.txt)
symmetry	isometries of the deflating spacetimes, internal symmetry of the fields that do not depend on the extra coordinates, zero modes on a vortex line, exponents of the couplings	symmetry_defl.rs, cmd_ symmetry.rs	4	19 of 19 (symmetry_ checks.txt)
background	thermal history (recombination, sound horizons, helium) on the deflating spacetime	spectra_defl.rs, eb_ cosmo.rs, cmd_spectra.rs	2	2 of 2 (background_ checks.txt)
spectrum	spectra of the microwave background and of matter on the deflating spacetime, for two closures of the perturbation equations; convergence; growth of structure	spectra_defl.rs, eb_ boltzmann.rs, eb_ spectra.rs, cmd_ spectra.rs	9	8 of 8 (spectrum_ checks.txt)
forecast	mock data of the microwave background: how well they bound the rate of deflation, and what happens to the Hubble constant	eb_fit.rs, eb_spectra.rs, cmd_spectra.rs	5	5 of 5 (forecast_ checks.txt)
families	the forecast for the seven families of dark radiation on the deflating spacetime	eb_fit.rs, eb_spectra.rs, cmd_spectra.rs	7	4 of 4 (families_ checks.txt)
superfluid	superfluid dark matter (first sound) on the deflating spacetime: suppression of structure, limits	eb_boltzmann.rs, cmd_ spectra.rs	4	2 of 2 (superfluid_ checks.txt)

The program performs 295 checks in all (each report ends with one further line, ALL, which says that every check above it passed). A check marked exact is an identity verified in exact rational or integer arithmetic; a check marked numerical states its largest deviation.

5. The complete output of the checks

5.1 Mass term and self-interaction: command selfcoupling

```
PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: exact checks of the self-interaction of
 ↪  the Dirac field (71s-..: 7+1; 44q-..: 4+4, quantized-sector form; 44c-..: 4+4, covariant form;
 ↪  self-..: all)
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks selfcoupling
34 of 34 checks passed
```

[PASS] 71s-S1 (exact) 7+1: the matrices are Gamma^A = gamma1671[A] (real); the adjoint is psibar = psi^dagger gamma1671[4]. The eight matrices anticommute in pairs and their squares are diag[-1, -1, -1, -1, 1, -1, -1, -1] times the unit matrix; the adjoint matrix is Hermitian with square 1

[PASS] 71s-S2 (exact) the 256 matrices Theta^K = phase_K Gamma^(A_1)..Gamma^(A_r) (A_1 < .. < A_r; 1, 8, 28, 56, 70, 56, 28, 8, 1 index sets for r = 0..8): with phase_K = i for the ranks r in [2, 3, 6, 7] and phase_K = 1 for the other ranks, every matrix M_K = Adj.Theta^K is Hermitian, so that every bilinear psibar Theta^K psi is real; (Theta^K)^2 = eps_K 1 with eps_K = eta_K, the product of the squares of the matrices with index in K, so that Theta_K = eta_K Theta^K (the author's definition); Theta_K := (Theta^K)^(-1) = eps_K Theta^K

[PASS] 71s-S3 (exact) Tr(Theta_K Theta^L) = 16 if K = L and 0 otherwise, for all 65536 pairs: the 256 matrices Theta^K are a basis of the complex 16x16 matrices, and Theta_K is the dual basis

[PASS] 71s-S4 (exact) invariance: for the 28 generators (1/2) Gamma^A Gamma^B (A < B), (i) Sigma^dagger.Adj + Adj.Sigma = 0, so that psibar psi is invariant, and (ii) the commutator [Theta^K, Gamma^A Gamma^B] is zero or a multiple c_K of the single matrix Theta^K' of the same rank with K' = K with A, B exchanged in or out, and eps_K c_K + eps_K' c_K' = 0: each of the nine invariants I_r = sum_K (psibar Theta_K psi)(psibar Theta^K psi) is invariant. The group is Spin(7,1)

[PASS] 71s-S5 (exact) Fierz rearrangement. Every Theta^K either commutes or anticommutes with every Theta^L; the sum of these signs over the index sets K of rank r is the same for all L of rank s and equals 16 F_rs with the closed form F_rs = (1/16) sum_l (-1)^(rs - l) C(s, l) C(8 - s, r - l); the identity sum_(K of rank r) (Theta_K)_(ab) (Theta^K)_(cd) = sum_s F_rs sum_(L of rank s) (Theta_L)_(ad) (Theta^L)_(cb) holds for all 9 x 65536 combinations of r and of the spinor indices a, b, c, d (verified with the adjoint matrix multiplied in, that is for the matrices M_K); F^2 = 1, Tr F = 16/16, rank(1 + F) = 5 and rank(1 - F) = 4: F has the eigenvalue +1 five times and -1 four times

[PASS] 71s-S6 (exact) THEOREM (anticommuting fields). Write I_r = sum T^(r)_(abcd) psi_a^* psi_b psi_c^* psi_d. Two tensors give the same polynomial in Grassmann variables exactly if their parts antisymmetric in b and d agree, A_(abcd) = T_(abcd) - T_(adcb). The nine tensors T^(r) are linearly independent (rank 9), but their antisymmetric parts satisfy sum_s (delta_rs + F_rs) A^(s) = 0 for every r and span a space of dimension 4: the nine invariants obey (1 + F) I = 0, five independent linear relations, and four invariants are independent. In terms of I_0, I_1, I_7, I_8: I_2 = -3 I_0 - I_1 - I_7 - I_8, I_3 = -4 I_0 - I_7 + 4 I_8, I_4 = -5 I_0 - 5 I_8, I_5 = -4 I_0 - I_1 + 4 I_8, I_6 = -I_0 + I_1 + I_7 - 3 I_8 (each verified on all 65536 components). Therefore sum_r g_r I_r = g_0eff I_0 + g_1eff I_1 + g_7eff I_7 + g_8eff I_8 with g_0eff = g_0 - 3 g_2 - 4 g_3 - 5 g_4 - 4 g_5 - g_6, g_1eff = g_1 - g_2 - g_5 + g_6, g_7eff = g_7 - g_2 - g_3 + g_6, g_8eff = g_8 - g_2 + 4 g_3 - 5 g_4 + 4 g_5 - 3 g_6

[PASS] 71s-S7 (exact) Hartree-Fock. With the one-body density matrix n_ab = <psi_b^dagger psi_a> of a state without pairing, <:(psi^dagger M psi)^2:> = Tr(M n)^2 - Tr(M n M n) (Wick's theorem), and the mean-field potential sum_r g_r sum_K eps_K [Tr(M_K n) M_K - M_K n M_K] is again a combination of the matrices in n; for the density matrix of uniform matter n = (rho 1 + rho_S beta)/16 (rho = <psi^dagger psi>, rho_S = <psibar psi>) the Hartree-Fock energy is E_HF = (1/2)[G_1 rho^2 + G_0 rho_S^2] with G_1 = (-g_0 + 22 g_1 - 14 g_2 + 14 g_3 - 14 g_5 + 14 g_6 - 6 g_7 + g_8)/16 and G_0 = (15 g_0 - 8 g_1 - 28 g_2 - 56 g_3 - 70 g_4 - 56 g_5 - 28 g_6 - 8 g_7 - g_8)/16, that is G = (1 - F^T) g: the exchange (Fock) terms are the Fierz transforms of the direct (Hartree) terms g_1 rho^2 and g_0 rho_S^2; the mean-field potential is U = G_1 rho 1 + G_0 rho_S beta, so that the mass of the fermions in matter is M = m + G_0 rho_S and every single-particle energy is shifted by G_1 rho. In the four effective couplings: G_1 = (-g_0eff + 22 g_1eff - 6 g_7eff + g_8eff)/16, G_0 = (15 g_0eff - 8 g_1eff - 8 g_7eff - g_8eff)/16.

[PASS] 71s-S8 (exact) the mean-field potential U(n) = sum_K eps_K [Tr(M_K n) M_K - M_K n M_K] is the derivative of the Hartree-Fock energy E(n) = (1/2) sum_K eps_K [Tr(M_K n)^2 - Tr(M_K n M_K n)]: d E(n + t d)/dt at t = 0 equals Tr(U(n) d), verified exactly for each of the nine ranks with four pairs of Hermitian integer matrices n and d. The same derivative with respect to psibar gives the nonlinear term of the Dirac equation, sum_r g_r sum_K (psibar Theta_K psi) Theta^K psi

[PASS] 71s-S9 (exact) pairs. With the pair density kappa_ab = <psi_a psi_b> the pairing energy of the interaction is (1/2) sum_r g_r Tr(kappa^dagger P_r(kappa)), P_r(kappa) = sum_K eps_K M_K kappa M_K^T. Of the 256 real matrices X_J = Cm.gamma^J, 120 are antisymmetric (pair operators psi^T X_J psi that do not vanish); each is an eigenvector of all nine maps P_r with an integer eigenvalue. Projected on two particles at rest, kappa = P_+ X_J P_+^T with P_+ = (1 + beta)/2, 56 of them are non-zero, and each is an eigenvector of the projected maps P_+ P_r(kappa) P_+^T = q_r kappa, with the pair coupling G_pair = sum_r g_r q_r (attractive if negative). X_J and X_J.gamma^4 project on the same pair, so that there are 28 different pairs, as many as there are antisymmetric states of two particles with eight internal states each (8 x 7/2 = 28): 7 pairs with G_pair = g_0 + g_1 - 9 g_2 - 9 g_3 - 5 g_4 - 5 g_5 + 5 g_6 + 5 g_7; 21 pairs with G_pair = g_0 + g_1 - g_2 - g_3 - 5 g_4 - 5 g_5 - 3 g_6 - 3 g_7. For the hidden-axis pair psi^T Cm.gamma^0 psi, G_pair = g_0 + g_1 - 9 g_2 - 9 g_3 - 5 g_4 - 5 g_5 + 5 g_6 + 5 g_7; in the four effective couplings this is g_0eff + g_1eff + 5 g_7eff.

[PASS] 71s-S10 (exact) chiral phase rotation psi -> exp(i theta chi) psi with the Hermitian matrix chi = i omega1671 (omega1671 the product of the eight matrices) (chi^2 = 1): the invariants of the ranks [1, 3, 5, 7] are invariant one by one; the interaction sum_r g_r I_r is invariant exactly if g_0 - 4 g_2 - 10 g_4 - 4 g_6 + g_8 = 0 (1 condition(s) on the nine couplings; determined from the antisymmetrized variation of the quartic tensors). In the four effective couplings the condition reads g_0eff + g_8eff = 0: three of the four independent couplings are chirally invariant (vector, axial vector, and the combination I_0 - I_8 of scalar and pseudoscalar), one is not. A mass term m psibar psi is not invariant either.

[PASS] 44q-S1 (exact) 4+4, quantized-sector form: the matrices are Gamma_eff^A = -i gamma1644[A] for A = 0..4 and Gamma_eff^t = gamma1644[t] for t = 5, 6, 7; the adjoint is psibar = psi^dagger beta1644 with beta1644 = -i gamma1644[4] = Gamma_eff^4. The eight matrices anticommute in pairs and their squares are diag[-1, -1, -1, -1, 1, -1, -1, -1] times the unit matrix; the adjoint matrix is Hermitian with square 1; W = i Cm1644.gamma1644[4] is Hermitian with W^2 = 1, commutes with beta1644, and Cm1644 = -W.beta1644

[PASS] 44q-S2 (exact) the 256 matrices Theta^K = phase_K Gamma^(A_1)..Gamma^(A_r) (A_1 < .. < A_r; 1, 8, 28, 56, 70, 56, 28, 8, 1 index sets for r = 0..8): with phase_K = i for the ranks r in [2, 3, 6, 7] and phase_K = 1 for the other ranks, every matrix M_K = Adj.Theta^K is Hermitian, so that every bilinear psibar Theta^K psi is real; (Theta^K)^2 = eps_K 1 with eps_K = eta_K, the product of the squares of the matrices with index in K, so that Theta_K = eta_K Theta^K (the author's definition); Theta_K := (Theta^K)^(-1) = eps_K Theta^K

[PASS] 44q-S3 (exact) Tr(Theta_K Theta^L) = 16 if K = L and 0 otherwise, for all 65536 pairs: the 256 matrices Theta^K are a basis of the complex 16x16 matrices, and Theta_K is the dual basis

[PASS] 44q-S4 (exact) invariance: for the 28 generators (1/2) Gamma^A Gamma^B (A < B), (i) Sigma^dagger.Adj + Adj.Sigma = 0, so that psibar psi is invariant, and (ii) the commutator [Theta^K, Gamma^A Gamma^B] is zero or a multiple c_K of the single matrix Theta^K' of the same rank with K' = K with A, B exchanged in or out, and eps_K c_K + eps_K' c_K' = 0: each of the nine invariants I_r = sum_K (psibar Theta_K psi)(psibar Theta^K psi) is invariant. The group is the group generated by (1/2) Gamma_eff^A Gamma_eff^B, of which the 13 generators that commute with W act on the coordinates (rotations and boosts of x0..x4, rotations of x5, x6, x7: Spin(4,1) x Spin(3)); the other 15 mix a coordinate x0..x4 with one of x5, x6, x7 and are not symmetries of the kinetic term

[PASS] 44q-S5 (exact) Fierz rearrangement. Every Theta^K either commutes or anticommutes with every Theta^L; the sum of these signs over the index sets K of rank r is the same for all L of rank s and equals 16 F_rs with the closed form F_rs = (1/16) sum_l (-1)^(rs - l) C(s, l) C(8 - s, r - l); the identity sum_(K of rank r) (Theta_K)_(ab) (Theta^K)_(cd) = sum_s F_rs sum_(L of rank s) (Theta_L)_(ad) (Theta^L)_(cb) holds for all 9 x 65536 combinations of r and of the spinor indices a, b, c, d (verified with the adjoint matrix multiplied in, that is for the matrices M_K); F^2 = 1, Tr F = 16/16, rank(1 + F) = 5 and rank(1 - F) = 4: F has the eigenvalue +1 five times and -1 four times

[PASS] 44q-S6 (exact) THEOREM (anticommuting fields). Write I_r = sum T^(r)_(abcd) psi_a^* psi_b psi_c^* psi_d. Two tensors give the same polynomial in Grassmann variables exactly if their parts antisymmetric in b and d agree, A_(abcd) = T_(abcd) - T_(adcb). The nine tensors T^(r) are linearly independent (rank 9), but their antisymmetric parts satisfy sum_s (delta_rs + F_rs) A^(s) = 0 for every r and span a space of dimension 4: the nine invariants obey (1 + F) I = 0, five independent linear relations, and four invariants are independent. In terms of I_0, I_1, I_7, I_8: I_2 = -3 I_0 - I_1 - I_7 - I_8, I_3 = -4 I_0 - I_7 + 4 I_8, I_4 = -5 I_0 - 5 I_8, I_5 = -4 I_0 - I_1 + 4 I_8, I_6 = -I_0 + I_1 + I_7 - 3 I_8 (each verified on all 65536 components). Therefore sum_r g_r I_r = g_0eff I_0 + g_1eff I_1 + g_7eff I_7 + g_8eff I_8 with g_0eff = g_0 - 3 g_2 - 4 g_3 - 5 g_4 - 4 g_5 - g_6, g_1eff = g_1 - g_2 - g_5 + g_6, g_7eff = g_7 - g_2 - g_3 + g_6, g_8eff = g_8 - g_2 + 4 g_3 - 5 g_4 + 4 g_5 - 3 g_6

[PASS] 44q-S7 (exact) Hartree-Fock. With the one-body density matrix n_ab = <psi_b^dagger psi_a> of a state without pairing, <:(psi^dagger M psi)^2:> = Tr(M n)^2 - Tr(M n M n) (Wick's theorem), and the mean-field potential sum_r g_r sum_K eps_K [Tr(M_K n) M_K - M_K n M_K] is again a combination of the matrices in n; for the density matrix of uniform matter n = (rho 1 + rho_S beta)/16 (rho = <psi^dagger psi>, rho_S = <psibar psi>) the Hartree-Fock energy is E_HF = (1/2)[G_1 rho^2 + G_0 rho_S^2] with G_1 = (-g_0 + 22 g_1 - 14 g_2 + 14 g_3 - 14 g_5 + 14 g_6 - 6 g_7 + g_8)/16 and G_0 = (15 g_0 - 8 g_1 - 28 g_2 - 56 g_3 - 70 g_4 - 56 g_5 - 28 g_6 - 8 g_7 - g_8)/16, that is G = (1 - F^T) g: the exchange (Fock) terms are the Fierz transforms of the direct (Hartree) terms g_1 rho^2 and g_0 rho_S^2; the mean-field potential is U = G_1 rho 1 + G_0 rho_S beta, so that the mass of the fermions in matter is M = m + G_0 rho_S and every single-particle energy is shifted by G_1 rho. In the four effective couplings: G_1 = (-g_0eff + 22 g_1eff - 6 g_7eff + g_8eff)/16, G_0 = (15 g_0eff - 8 g_1eff - 8 g_7eff - g_8eff)/16.

[PASS] 44q-S8 (exact) the mean-field potential U(n) = sum_K eps_K [Tr(M_K n) M_K - M_K n M_K] is the derivative of the Hartree-Fock energy E(n) = (1/2) sum_K eps_K [Tr(M_K n)^2 - Tr(M_K n M_K n)]: d E(n + t d)/dt at t = 0 equals Tr(U(n) d), verified exactly for each of the nine ranks with four pairs of Hermitian integer matrices n and d. The same derivative with respect to psibar gives the nonlinear term of the Dirac equation, sum_r g_r sum_K (psibar Theta_K psi) Theta^K psi

[PASS] 44q-S9 (exact) pairs. With the pair density kappa_ab = <psi_a psi_b> the pairing energy of the interaction is (1/2) sum_r g_r Tr(kappa^dagger P_r(kappa)), P_r(kappa) = sum_K eps_K M_K kappa M_K^T. Of the 256 real matrices X_J = Cm.gamma^J, 120 are antisymmetric (pair operators psi^T X_J psi that do not vanish); each is an eigenvector of all nine maps P_r with an integer eigenvalue. Projected on two particles at rest, kappa = P_+ X_J P_+^T with P_+ = (1 + beta)/2, 56 of them are non-zero, and each is an eigenvector of the projected maps P_+ P_r(kappa) P_+^T = q_r kappa, with the pair coupling G_pair = sum_r g_r q_r (attractive if negative). X_J and X_J.gamma^4 project on the same pair, so that there are 28 different pairs, as many as there are antisymmetric states of two particles with eight internal states each (8 x 7/2 = 28): 4 pairs (one particle in each of the halves W = +1 and W = -1) with G_pair = g_0 + g_1 - 9 g_2 - 9 g_3 - 5 g_4 - 5 g_5 + 5 g_6 + 5 g_7; 3 pairs (both particles in the same half W = +1 or W = -1) with G_pair = g_0 + g_1 - 9 g_2 - 9 g_3 - 5 g_4 - 5 g_5 + 5 g_6 + 5 g_7; 9 pairs (both particles in the same half W = +1 or W = -1) with G_pair = g_0 + g_1 - g_2 - g_3 - 5 g_4 - 5 g_5 - 3 g_6 - 3 g_7; 12 pairs (one particle in each of the halves W = +1 and W = -1) with G_pair = g_0 + g_1 - g_2 - g_3 - 5 g_4 - 5 g_5 - 3 g_6 - 3 g_7. For the hidden-axis pair psi^T Cm.gamma^0 psi, G_pair = g_0 + g_1 - 9 g_2 - 9 g_3 - 5 g_4 - 5 g_5 + 5 g_6 + 5 g_7; in the four effective couplings this is g_0eff + g_1eff + 5 g_7eff. In the quantized sector the hidden-axis pair satisfies W kappa W^T = - kappa: it joins a particle of the half W = +1 with a particle of the half W = -1.

[PASS] 44q-S10 (exact) chiral phase rotation psi -> exp(i theta chi) psi with the Hermitian matrix chi = omega1644 (the product of the eight matrices gamma1644, real) (chi^2 = 1): the invariants of the ranks [1, 3, 5, 7] are invariant one by one; the interaction sum_r g_r I_r is invariant exactly if g_0 - 4 g_2 - 10 g_4 - 4 g_6 + g_8 = 0 (1 condition(s) on the nine couplings; determined from the antisymmetrized variation of the quartic tensors). In the four effective couplings the condition reads g_0eff + g_8eff = 0: three of the four independent couplings are chirally invariant (vector, axial vector, and the combination I_0 - I_8 of scalar and pseudoscalar), one is not. A mass term m psibar psi is not invariant either.

[PASS] 44c-S1 (exact) 4+4, covariant form: the matrices are Gamma^A = gamma1644[A] (real); the adjoint is psitilde = psi^dagger Cm1644 with Cm1644 = gamma1644[0].gamma1644[1].gamma1644[2].gamma1644[3]. The eight matrices anticommute in pairs and their squares are diag[1, 1, 1, 1, -1, -1, -1, -1] times the unit matrix; the adjoint matrix is Hermitian with square 1; W = i Cm1644.gamma1644[4] is Hermitian with W^2 = 1, commutes with beta1644, and Cm1644 = -W.beta1644

[PASS] 44c-S2 (exact) the 256 matrices Theta^K = phase_K Gamma^(A_1)..Gamma^(A_r) (A_1 < .. < A_r; 1, 8, 28, 56, 70, 56, 28, 8, 1 index sets for r = 0..8): with phase_K = i for the ranks r in [1, 2, 5, 6] and phase_K = 1 for the other ranks, every matrix M_K = Adj.Theta^K is Hermitian, so that every bilinear psibar Theta^K psi is real; (Theta^K)^2 = eps_K 1 with eps_K = (-1)^r eta_K with eta_K the product of the squares of the matrices with index in K (the inverse matrix differs from eta_K Theta^K by the sign (-1)^r, because the adjoint is Cm1644 and not a single Dirac matrix); Theta_K := (Theta^K)^(-1) = eps_K Theta^K

[PASS] 44c-S3 (exact) Tr(Theta_K Theta^L) = 16 if K = L and 0 otherwise, for all 65536 pairs: the 256 matrices Theta^K are a basis of the complex 16x16 matrices, and Theta_K is the dual basis

[PASS] 44c-S4 (exact) invariance: for the 28 generators (1/2) Gamma^A Gamma^B (A < B), (i) Sigma^dagger.Adj + Adj.Sigma = 0, so that psibar psi is invariant, and (ii) the commutator [Theta^K, Gamma^A Gamma^B] is zero or a multiple c_K of the single matrix Theta^K' of the same rank with K' = K with A, B exchanged in or out, and eps_K c_K + eps_K' c_K' = 0: each of the nine invariants I_r = sum_K (psibar Theta_K psi)(psibar Theta^K psi) is invariant. The group is Spin(4,4); 13 of the 28 generators commute with W (the rotations and boosts of x0..x4 and the rotations of x5, x6, x7), and only these are symmetries of the positive Lagrangian of the quantized sector

[PASS] 44c-S5 (exact) Fierz rearrangement. Every Theta^K either commutes or anticommutes with every Theta^L; the sum of these signs over the index sets K of rank r is the same for all L of rank s and equals 16 F_rs with the closed form F_rs = (1/16) sum_l (-1)^(rs - l) C(s, l) C(8 - s, r - l); the identity sum_(K of rank r) (Theta_K)_(ab) (Theta^K)_(cd) = sum_s F_rs sum_(L of rank s) (Theta_L)_(ad) (Theta^L)_(cb) holds for all 9 x 65536 combinations of r and of the spinor indices a, b, c, d (verified with the adjoint matrix multiplied in, that is for the matrices M_K); F^2 = 1, Tr F = 16/16, rank(1 + F) = 5 and rank(1 - F) = 4: F has the eigenvalue +1 five times and -1 four times

[PASS] 44c-S6 (exact) THEOREM (anticommuting fields). Write I_r = sum T^(r)_(abcd) psi_a^* psi_b psi_c^* psi_d. Two tensors give the same polynomial in Grassmann variables exactly if their parts antisymmetric in b and d agree, A_(abcd) = T_(abcd) - T_(adcb). The nine tensors T^(r) are linearly independent (rank 9), but their antisymmetric parts satisfy sum_s (delta_rs + F_rs) A^(s) = 0 for every r and span a space of dimension 4: the nine invariants obey (1 + F) I = 0, five independent linear relations, and four invariants are independent. In terms of I_0, I_1, I_7, I_8: I_2 = -3 I_0 - I_1 - I_7 - I_8, I_3 = -4 I_0 - I_7 + 4 I_8, I_4 = -5 I_0 - 5 I_8, I_5 = -4 I_0 - I_1 + 4 I_8, I_6 = -I_0 + I_1 + I_7 - 3 I_8 (each verified on all 65536 components). Therefore sum_r g_r I_r = g_0eff I_0 + g_1eff I_1 + g_7eff I_7 + g_8eff I_8 with g_0eff = g_0 - 3 g_2 - 4 g_3 - 5 g_4 - 4 g_5 - g_6, g_1eff = g_1 - g_2 - g_5 + g_6, g_7eff = g_7 - g_2 - g_3 + g_6, g_8eff = g_8 - g_2 + 4 g_3 - 5 g_4 + 4 g_5 - 3 g_6

[PASS] 44c-S7 (exact) Hartree-Fock. With the one-body density matrix n_ab = <psi_b^dagger psi_a> of a state without pairing, <:(psi^dagger M psi)^2:> = Tr(M n)^2 - Tr(M n M n) (Wick's theorem), and the mean-field potential sum_r g_r sum_K eps_K [Tr(M_K n) M_K - M_K n M_K] is again a combination of the matrices in n; for the density matrix n = (rho 1 + rho_S beta + rho_W W + rho_SW W.beta)/16 (rho = <psi^dagger psi>, rho_S = <psi^dagger beta psi>, rho_W = <psi^dagger W psi> the covariant charge density, rho_SW = <psi^dagger W.beta psi> = -<psitilde psi>) the Hartree-Fock energy is diagonal, E_HF = (1/2)[G_V0 rho^2 + G_S0 rho_S^2 + G_1 rho_W^2 + G_0 rho_SW^2], with G_V0 = (-g_0 + 4 g_2 + 10 g_4 + 4 g_6 - g_8)/16, G_S0 = (-g_0 - 2 g_1 + 2 g_2 + 6 g_3 + 10 g_5 - 2 g_6 + 2 g_7 + g_8)/16, G_1 = (-g_0 + 22 g_1 - 14 g_2 + 14 g_3 - 14 g_5 + 14 g_6 - 6 g_7 + g_8)/16, G_0 = (15 g_0 - 8 g_1 - 28 g_2 - 56 g_3 - 70 g_4 - 56 g_5 - 28 g_6 - 8 g_7 - g_8)/16; the mean-field potential is U = G_V0 rho 1 + G_S0 rho_S beta + G_1 rho_W W + G_0 rho_SW W.beta: the two halves W = +1 and W = -1 of the field have the masses m + G_S0 rho_S +- G_0 rho_SW and the energy shifts G_V0 rho +- G_1 rho_W. In the four effective couplings: G_V0 = (-g_0eff - g_8eff)/16, G_S0 = (-g_0eff - 2 g_1eff + 2 g_7eff + g_8eff)/16, G_1 = (-g_0eff + 22 g_1eff - 6 g_7eff + g_8eff)/16, G_0 = (15 g_0eff - 8 g_1eff - 8 g_7eff - g_8eff)/16.

[PASS] 44c-S8 (exact) the mean-field potential U(n) = sum_K eps_K [Tr(M_K n) M_K - M_K n M_K] is the derivative of the Hartree-Fock energy E(n) = (1/2) sum_K eps_K [Tr(M_K n)^2 - Tr(M_K n M_K n)]: d E(n + t d)/dt at t = 0 equals Tr(U(n) d), verified exactly for each of the nine ranks with four pairs of Hermitian integer matrices n and d. The same derivative with respect to psibar gives the nonlinear term of the Dirac equation, sum_r g_r sum_K (psibar Theta_K psi) Theta^K psi

[PASS] 44c-S9 (exact) pairs. With the pair density kappa_ab = <psi_a psi_b> the pairing energy of the interaction is (1/2) sum_r g_r Tr(kappa^dagger P_r(kappa)), P_r(kappa) = sum_K eps_K M_K kappa M_K^T. Of the 256 real matrices X_J = Cm.gamma^J, 120 are antisymmetric (pair operators psi^T X_J psi that do not vanish); each is an eigenvector of all nine maps P_r with an integer eigenvalue. Projected on two particles at rest, kappa = P_+ X_J P_+^T with P_+ = (1 + beta)/2, 56 of them are non-zero, and each is an eigenvector of the projected maps P_+ P_r(kappa) P_+^T = q_r kappa, with the pair coupling G_pair = sum_r g_r q_r (attractive if negative). X_J and X_J.gamma^4 project on the same pair, so that there are 28 different pairs, as many as there are antisymmetric states of two particles with eight internal states each (8 x 7/2 = 28): 4 pairs (one particle in each of the halves W = +1 and W = -1) with G_pair = -g_0 - g_1 + 9 g_2 + 9 g_3 + 5 g_4 + 5 g_5 - 5 g_6 - 5 g_7; 3 pairs (both particles in the same half W = +1 or W = -1) with G_pair = g_0 + g_1 - 9 g_2 - 9 g_3 - 5 g_4 - 5 g_5 + 5 g_6 + 5 g_7; 9 pairs (both particles in the same half W = +1 or W = -1) with G_pair = g_0 + g_1 - g_2 - g_3 - 5 g_4 - 5 g_5 - 3 g_6 - 3 g_7; 12 pairs (one particle in each of the halves W = +1 and W = -1) with G_pair = -g_0 - g_1 + g_2 + g_3 + 5 g_4 + 5 g_5 + 3 g_6 + 3 g_7. For the hidden-axis pair psi^T Cm.gamma^0 psi, G_pair = -g_0 - g_1 + 9 g_2 + 9 g_3 + 5 g_4 + 5 g_5 - 5 g_6 - 5 g_7; in the four effective couplings this is -g_0eff - g_1eff - 5 g_7eff. In the quantized sector the hidden-axis pair satisfies W kappa W^T = - kappa: it joins a particle of the half W = +1 with a particle of the half W = -1.

[PASS] 44c-S10 (exact) chiral phase rotation psi -> exp(i theta chi) psi with the Hermitian matrix chi = omega1644 (the product of the eight matrices gamma1644, real) (chi^2 = 1): the invariants of the ranks [0, 2, 4, 6, 8] are invariant one by one; the interaction sum_r g_r I_r is invariant exactly if g_1 + g_3 - g_5 - g_7 = 0 (1 condition(s) on the nine couplings; determined from the antisymmetrized variation of the quartic tensors). In the four effective couplings the condition reads g_1eff - g_7eff = 0. (For the covariant form this rotation is the symmetry of the kinetic term of the quantized sector; the covariant kinetic term itself is not invariant under it, and the covariant mass term is. The roles of the even and of the odd ranks are exchanged with respect to the forms with a Dirac adjoint, because the matrix Cm1644 commutes with omega1644.)

[PASS] 44c-S11 (exact) relation between the two forms of the 4+4 interaction: for every index set K the covariant matrix is M_K(covariant) = z_K W.M_K(quantized sector) with z_K = +-1 if K contains an even number of the indices 5, 6, 7 (128 index sets; W commutes with M_K) and z_K = +-i if it contains an odd number (128 index sets; W anticommutes with M_K, and the bilinear connects the two halves W = +1 and W = -1 of the field). With eps_K and M_K of the quantized-sector form, I_r(covariant) = sum_K eps_K (psi^dagger W.M_K psi)^2 for all 256 index sets: the covariant form is the quantized-sector form with the matrix W inserted in every bilinear, and nothing else. Consequence: between two particles of the same half the two forms give the same interaction; between a particle of the half W = +1 and a particle of the half W = -1 they give interactions of opposite sign

[PASS] 71s-S12 (exact) internal symmetry of the fields that do not depend on the extra coordinates x0, x5, x6, x7 (their kinetic term contains alpha^1, alpha^2, alpha^3 and beta only). The eight Hermitian generators of the symmetry of the model with the Vorton-Scalar are the six rotations of the extra axes, (phase) gamma^a gamma^b with a < b in {0, 5, 6, 7}, the unit matrix (fermion number), and P_E = (phase) gamma^0 gamma^5 gamma^6 gamma^7 with P_E^2 = 1 and trace 0; all commute with the four matrices of the kinetic term, and P_E commutes with the chirality matrix. The self-interaction is invariant under the six rotations and the phase for all values of the nine couplings (0 conditions found), because the rotations are among the 28 generators of check S4. Under psi -> exp(i theta P_E) psi the invariants of the ranks [0, 4, 8] are invariant one by one, and the interaction sum_r g_r I_r is invariant exactly if g_1 + g_3 - g_5 - g_7 (that is g_1eff - g_7eff) = 0 and g_2 + g_3 - g_6 - g_7 (that is -g_7eff) = 0 (2 condition(s))

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[PASS] 44q-S12  (exact) internal symmetry of the fields that do not depend on the extra
 ↪  coordinates x0, x5, x6, x7 (their kinetic term contains alpha^1, alpha^2, alpha^3 and beta
 ↪  only). The eight Hermitian generators of the symmetry of the model with the Vorton-Scalar are
 ↪  the six rotations of the extra axes, (phase) gamma^a gamma^b with a < b in {0, 5, 6, 7}, the
 ↪  unit matrix (fermion number), and P_E = (phase) gamma^0 gamma^5 gamma^6 gamma^7 with P_E^2 = 1
 ↪  and trace 0; all commute with the four matrices of the kinetic term, and P_E commutes with the
 ↪  chirality matrix. The self-interaction is invariant under the six rotations and the phase for
 ↪  all values of the nine couplings (0 conditions found), because the rotations are among the 28
 ↪  generators of check S4. Under psi -> exp(i theta P_E) psi the invariants of the ranks [0, 4,
 ↪  8] are invariant one by one, and the interaction sum_r g_r I_r is invariant exactly if g_1 +
 ↪  g_3 - g_5 - g_7 (that is g_1eff - g_7eff) = 0 and g_2 + g_3 - g_6 - g_7 (that is -g_7eff) = 0
 ↪  (2 condition(s))
[PASS] self-X1  (numerical) the example of the author's document, g_0 = g_S = -1e5 and g_1 = g_V =
 ↪  +1e4 with all other couplings zero, gives the Hartree-Fock couplings G_0 = -98750 and G_1 =
 ↪  +20000 (7+1 and the quantized-sector form of 4+4), the values stated in that document
```

5.2 Kohn–Sham and CJT calculations: command `kohnsham`

```
PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the Kohn-Sham calculations
 ↪  (ks-..), of the CJT effective potential with contact interaction (cjt-..) and of the
 ↪  functional with the Vorton-Scalar (uni-..)
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks kohnsham
13 of 13 checks passed

[PASS] ks-K1  (numerical) flat 7+1 spacetime, the example of the author's document (m = 1, G_S =
 ↪  -1e5, G_V = +1e4, k_F = 0.756636150701493 in seven space dimensions): this program gives M =
 ↪  0.348097340146, rho = 1.388100213e-5, eps = 1.348787233e-5, E/A - m = -0.02832142796, and
 ↪  finds the saturation point (the minimum of the energy per particle, where the pressure
 ↪  vanishes: p = -4.2e-22) at k_F = 0.756636150701; the document prints 0.348097340146,
 ↪  1.388100213e-5, 1.348787233e-5, -0.02832142796 and 0.756636150701493. Largest relative
 ↪  difference 1.7e-10
[PASS] ks-K2  (numerical) thermodynamic consistency of the Kohn-Sham state (flat, seven space
 ↪  dimensions): the pressure computed from the momentum integral, <k^2/(7 w)> + Pi, equals mu rho
 ↪  - eps with mu = sqrt(k_F^2 + M^2) + G_V rho along the whole equation of state (141 Fermi
 ↪  momenta for each of the two sets; largest deviation 7.7e-16 of eps); at saturation the
 ↪  chemical potential equals the energy per particle (theorem of Hugenholtz and Van Hove): mu -
 ↪  E/A = 3.3e-16. A remark on the author's example: his document uses G_S = -1e5, G_V = +1e4 in
 ↪  its Kohn-Sham section and calls them the Hartree-Fock couplings G_0, G_1, while the
 ↪  Hartree-Fock couplings of g_0 = -1e5, g_1 = +1e4 are G_0 = -98750, G_1 = 20000 (his section 4
 ↪  and check 71s-S7 here). For the second pair the lowest energy per particle is E/A - m =
 ↪  0.000156 at k_F = 0.0200: the doubled vector repulsion removes the binding, and uniform matter
 ↪  is not self-bound. The numbers of the author's Kohn-Sham section therefore belong to G_S =
 ↪  -1e5, G_V = +1e4 taken as the mean-field couplings themselves (as couplings of the Lagrangian
 ↪  with g_2 = .. = g_8 = 0 that would be g_0 = -105341.6, g_1 = +2484.5)
[PASS] ks-K3  (numerical) the momentum integral over the ellipsoid of occupied states (four flat
 ↪  and three deflating directions; 115200 nodes): at the reference time (all stretch factors 1)
 ↪  it reproduces the integral over the ball in seven dimensions, M, rho and eps to 2.2e-14 and
 ↪  the pressures along flat and deflating directions to 3.3e-15 of the kinetic energy density;
 ↪  with 401408 nodes instead, M, eps and the pressures change by at most 9.2e-10 (relative) at
 ↪  the clock readings 1.5, 3 and 4 for both sets of rates
[PASS] ks-K4  (numerical) conservation of energy on the deflating 7+1 spacetime: the Kohn-Sham
 ↪  energy density obeys d eps/dx4 = sum_t rate_t (eps + p_t) with the pressures p_t = <s_t^2
 ↪  u_t^2/w> + Pi along the deflating coordinates (the covariant divergence of the energy-momentum
 ↪  tensor vanishes); numerical derivative against the right-hand side at the clock readings -1,
 ↪  0, 0.7, 2 and 3.5, for equal rates and for rates in the ratio 1 : 2 : 3: largest relative
 ↪  deviation 5.6e-9
```

[PASS] ks-K5 (numerical) history of saturated matter while x5, x6, x7 deflate (7+1, all momenta, collisionless; equal rates / rates 1 : 2 : 3): the density grows as exp(tau_d); the mass of the particles falls to half its value at the reference time at tau_d = 0.983 / 0.990; the matter is self-bound (energy per particle below m) only for -0.675 < tau_d < 0.537 / -0.658 < tau_d < 0.523; at tau_d = 4 the vector repulsion (1/2) G_V rho^2 is 69.4 / 65.0 per cent of the energy density, which then grows as exp(2 tau_d); the collisionless state lies above the state of lowest energy of the same density by at most 8.652e-1 m per particle on -2 <= tau_d <= 4 (energy that collisions would turn into heat)
[PASS] ks-K6 (numerical) fields that do not depend on x5, x6, x7: the reduced couplings grow as exp(tau_d), and uniform matter becomes self-bound when the scalar attraction reaches a threshold (verified: bound 0.1 per cent above it, not bound 0.1 per cent below it). Thresholds in units m = 1: D = 4, scalar attraction alone (G_V = 0): |G_S| m^3 = 62.0767; D = 4, G_V = 0.1 |G_S| (the ratio of the values G_S = -1e5, G_V = +1e4 of the author's document): |G_S| m^3 = 85.6504; D = 4, G_V = 0.2025 |G_S| (the ratio of the Hartree-Fock couplings of g_0 = -1e5, g_1 = +1e4): |G_S| m^3 = 119.5723; D = 3, scalar attraction alone (G_V = 0): |G_S| m^2 = 8.0671; D = 3, G_V = 0.1 |G_S| (the ratio of the values G_S = -1e5, G_V = +1e4 of the author's document): |G_S| m^2 = 9.8364; D = 3, G_V = 0.2025 |G_S| (the ratio of the Hartree-Fock couplings of g_0 = -1e5, g_1 = +1e4): |G_S| m^2 = 12.2272. The deflation therefore turns a gas into self-bound matter at the clock reading tau_bind = ln(|G_S|(threshold)/|G_S|(reference))
[PASS] ks-K8 (numerical) covariant form of the interaction on 4+4 (the two halves W = +1 and W = -1 carry opposite scalar and vector charge): for the attractive scalar coupling of the author's example (g_1 = -0.1 g_0) the state with equal halves is never the state of lowest energy; matter polarizes spontaneously. Polarization P = (rho_+ - rho_-)/rho of the state of lowest energy, for couplings 1.5, 3 and 6 times the binding threshold of the quantized-sector form and total densities of Fermi momentum 0.4, 0.6, 0.8, 1.0 (in this order): 1.00, 1.00, 1.00, 0.35, 1.00, 1.00, 0.65, 0.20, 1.00, 1.00, 0.40, 0.20 (grid of 0.05). The fully polarized state is the Kohn-Sham state of one kind of particle with four states and the couplings G_S0 + G_0, G_V0 + G_1 (verified to 1e-9). Example: for couplings three times the binding threshold of the quantized-sector form and total density of Fermi momentum 0.8: E/A - m = +0.3799 for equal halves, +0.3731 for all particles in one half, and +0.1323 in the quantized-sector form
[PASS] cjt-J1 (numerical) the Kohn-Sham method and the CJT effective potential agree (no-sea approximation): for 30 states (five sets of parameters in D = 7, 4, 3; Fermi momenta 0.3 to 1.4) the stationary point of Omega(M, V_0; mu) at the chemical potential of the Kohn-Sham state has the same mass, density and energy density, and -Omega equals the Kohn-Sham pressure (sum of the four relative deviations at most 1.6e-12; residuals of the two stationarity conditions at most 8.2e-15). Of the 23 states with positive pressure, 20 are the global minimum of Omega; in the other 3 a denser phase has the higher pressure at the same chemical potential (the Kohn-Sham state of that density is then metastable: a dilute gas above saturated matter)
[PASS] cjt-J2 (numerical) with the filled states of negative energy (cutoff Lambda) a massless field acquires a mass in vacuum when the scalar attraction exceeds the critical coupling G_c = -(2 pi)^D (D - 1)/(NU A_D Lambda^(D-1)) (A_D: area of the unit sphere): G_c Lambda^6 = -90 pi^4 for D = 7 (the value of the author's document DIRAC16COMPLEX), G_c Lambda^3 = -3 pi^2 for D = 4, G_c Lambda^2 = -pi^2/2 for D = 3; the minimization of the effective potential gives no mass 2 per cent below G_c and a mass 2 per cent above it, for all three D. On the deflating spacetime the reduced coupling crosses G_c at the clock reading tau_c = ln(G_c/G_S(reference))
[PASS] cjt-J3 (numerical) the Dirac sea and the author's example. The coupling G_S = -1e5 m^-6 of the example exceeds the critical coupling 90 pi^4/Lambda^6 of the vacuum as soon as the cutoff is larger than Lambda = (90 pi^4/1e5)^(1/6) m = 0.6665 m: the no-sea approximation presupposes a cutoff below this value. With the sea included (bare mass adjusted so that the vacuum mass stays 1): without sea: saturation at k_F = 0.7566, M = 0.3481, E/A - M_vac = -0.02832; Lambda = 0.3: bare mass 0.9979, saturation at k_F = 0.7566, M = 0.3479, E/A - M_vac = -0.02837; Lambda = 0.45: bare mass 0.9660, saturation at k_F = 0.7568, M = 0.3438, E/A - M_vac = -0.02984; Lambda = 0.6: bare mass 0.7578, saturation at k_F = 0.7565, M = 0.3072, E/A - M_vac = -0.04483
[PASS] uni-U1 (numerical) contact interaction, bare mass and Vorton-Scalar together: for f_D = 0 and G_S = G_V = 0 the functional is the one of the main project (masses and energy densities for C = 4, 6, 10 and k_F = 0.3, 0.6, 1.0 agree to 2.1e-8), and the smallest C at which matter is self-bound is 3.701101 here against 3.701102 there

```
[PASS] uni-U2  (numerical) at low density the Vorton-Scalar acts on the mass of the fermions
 ↪  exactly as a scalar contact coupling of strength -C = -g_VS^2/m_s^2 (units m = 1), whatever
 ↪  fraction of the mass is a bare mass: (M - 1)/rho_S tends to G_S - C (five sets of parameters;
 ↪  at k_F <= 0.05 the deviation is at most 1.8e-4). The exchange of the radial mode of the scalar
 ↪  and the scalar channel g_0 of the self-interaction are the same interaction at momenta far
 ↪  below m_s
[PASS] uni-U3  (numerical) a bare Dirac mass gives the phase mode of the Vorton-Scalar a mass. The
 ↪  chiral symmetry U(1)_VS is then broken explicitly; for the phase theta of the scalar the
 ↪  fermions have the mass m(theta) = sqrt(m_D^2 + m_Y^2 + 2 m_D m_Y cos(theta)) (m_Y = g_VS
 ↪  v_VS), and the energy of the filled states gives m_ph^2 v_VS^2 = (|<psibar psi>| - rho_S) m_D
 ↪  m_Y/m, the relation of Gell-Mann, Oakes and Renner with the scalar density rho_S of the matter
 ↪  subtracted (numerical second derivative against this formula, 24 cases, and the closed form of
 ↪  the condensate for D = 3, (nu/(2 pi^2)) [Lambda sqrt(Lambda^2 + m^2) - m^2 asinh(Lambda/m)]
 ↪  m/2: largest deviation 3.2e-7 of the vacuum value). In units of the couplings of the model,
 ↪  m_ph^2 = C m_s^2 f_D (|<psibar psi>| - rho_S)/((1 - f_D) m^3): on the deflating spacetime m_ph
 ↪  grows as exp(tau_d/2), and its value depends on the cutoff Lambda through the condensate. In
 ↪  matter the mass falls, m_ph^2(k_F)/m_ph^2(0) = 1 - rho_S(k_F)/|<psibar psi>|: Lambda = 1 m:
 ↪  |<psibar psi>| = 0.10798 m^3, ratio 0.8540 at k_F = Lambda/2; Lambda = 2 m: |<psibar psi>| =
 ↪  0.61370 m^3, ratio 0.8241 at k_F = Lambda/2; Lambda = 4 m: |<psibar psi>| = 2.91759 m^3, ratio
 ↪  0.7897 at k_F = Lambda/2. The ratio vanishes only at k_F = Lambda (6.7e-16), where the filled
 ↪  states of positive energy cancel those of negative energy below the cutoff: that is the limit
 ↪  of the effective theory and not a prediction
```

5.3 Pairing with contact couplings: command `pairing`

```
PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the pairing with contact
 ↪  self-interaction and exchange of the Vorton-Scalar (pc-..)
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks pairing
4 of 4 checks passed

[PASS] pc-P1  (numerical) pairing amplitudes of the four independent contact channels in the
 ↪  hidden-axis channel, for two particles of mass M with momenta p, -p in ordinary space that
 ↪  scatter into p', -p' (E = sqrt(M^2 + p^2), E' likewise): F_0 = (E E' + M^2 - p.p')/(2 E E'),
 ↪  F_1 = (3 E E' - 2 M^2 - p.p')/(E E'), F_7 = (3 E E' + 2 M^2 - p.p')/(E E'), F_8 = -(E E' - M^2
 ↪  - p.p')/(2 E E'). Direct evaluation with the 256 matrices M_K and the projectors on the states
 ↪  of positive energy agrees with these closed forms for 40 random sets of mass and momenta on
 ↪  each of the two spacetimes (7+1; 4+4 in the quantized-sector form): largest deviation 1.8e-15;
 ↪  the two spacetimes give the same amplitudes (largest difference 7.1e-15). The pairing
 ↪  potential of the contact interaction is V = g_0eff F_0 + g_1eff F_1 + g_7eff F_7 + g_8eff F_8
[PASS] pc-P2  (numerical) the reduction to four invariants and the pairing amplitudes. For two
 ↪  particles at rest the nine amplitudes are the exact pair couplings of the hidden-axis pair, q
 ↪  = (1, 1, -9, -9, -5, -5, 5, 5, 0) (largest deviation 0.0e0). At general momenta the relations
 ↪  (1 + F) I = 0 hold for the part of the amplitudes that is even in the angle between p and p'
 ↪  (the sum of the direct and of the exchange scattering of two identical particles): the
 ↪  amplitudes of the ranks 2 to 6, evaluated directly, equal the prescribed combinations of F_0,
 ↪  F_1, F_7, F_8 (largest deviation 1.2e-14); the parts odd in the angle do not (deviation up to
 ↪  13.79), and they do not contribute to a contact interaction in a pair state of zero orbital
 ↪  angular momentum. Consequence for the model: with a strictly local interaction four couplings
 ↪  suffice; when the contact interaction is given a range, the model is DEFINED by giving this
 ↪  range to the four channels S, V, A, P. The closed forms are derived for momenta in ordinary
 ↪  space; if the momentum has a component along the hidden axis x0 itself they do not hold
 ↪  (deviation up to 0.92), because the pair operator points along that axis
[PASS] pc-P3  (numerical) the exchange of the Vorton-Scalar is a self-interaction of finite range:
 ↪  (i) without contact couplings the pairing kernel of this program equals the kernel of the main
 ↪  project (four sets of parameters, five pairs of momenta each: largest relative difference
 ↪  1.1e-15); (ii) the scalar contact coupling g_0eff = -g_VS^2/m_s^2 with the range parameter
 ↪  m_ct = m_s (the mass of the heavy boson whose exchange the contact coupling stands for) gives
 ↪  exactly the kernel of the exchange of the radial mode (largest relative difference 3.3e-16):
 ↪  g_0eff(q) = -g_VS^2/(q^2 + m_s^2), and likewise g_8eff(q) = +g_VS^2/(q^2 + m_ph^2) for the
 ↪  phase mode
```

```
[PASS] pc-P4  (numerical) the gap of the hidden-axis pair along the deflation clock with contact
 ↪  couplings (72 clock readings in four sets of (m_s, k_F), seven variants each): without contact
 ↪  couplings the gaps of the earlier calculation (exchange of the two modes of the Vorton-Scalar)
 ↪  are reproduced (largest relative difference 6.6e-16); a scalar contact attraction raises the
 ↪  gap, a vector contact repulsion and a mass of the phase mode lower it (verified at every clock
 ↪  reading with a gap above 1e-6 E_F); for small gaps T_c/gap is within 0.001 of the
 ↪  weak-coupling value 0.567. Example: for m_s = m, k_F = 0.5 m at tau_d = 2.04 (g_VS = 2.771):
 ↪  gap/E_F = 0.0246 (exchange alone), 0.2968 (with scalar contact), 0.1961 (author's ratio),
 ↪  0.0006 (with vector repulsion 0.3), 0.1287 (with axial coupling), 0.0006 (contact alone),
 ↪  0.0197 (phase mode of mass 0.3 m)
```

5.4 Gravity with Kohn–Sham matter: command `gravity`

```
PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the Einstein-Lovelock
 ↪  equations with Kohn-Sham matter (gr-..)
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks gravity
5 of 5 checks passed

[PASS] gr-G1  (numerical) the Einstein-Lovelock tensors of a spacetime with two scale factors,
 ↪  computed from the pair curvatures K_(4i) = H_i' + H_i^2, K_(ij) = H_i H_j by E^(k)m_m =
 ↪  (-1)^(k+1) k! 2^(k-1) times the sum over the sets of 2k directions other than m of the sum
 ↪  over their divisions into pairs of the product of the pair curvatures: for 3 + 4 directions
 ↪  (the cosmological spacetime of the author's document KOHN_SHAM_GRAVITY) the nine components
 ↪  E^(1), E^(2), E^(3) (time, ordinary space, auxiliary space), each with its coefficients of
 ↪  H_a' and H_b', equal the 21 polynomials printed in that document at four pairs of rates
 ↪  (largest deviation 2.6e-15)
[PASS] gr-G2  (numerical) the self-consistent evolution of the author's document is reproduced:
 ↪  with his data (kappa = 1000, LambdaBar = 0, H_a(0) = 0.05, a(0) = b(0) = 1, Kohn-Sham matter
 ↪  of the saturation density) the constraint gives H_b(0), and the integration to x4 = 20 gives
 ↪  a, b, H_a, H_b; for Einstein gravity and for Einstein-Lovelock gravity with alpha_2 = 20,
 ↪  alpha_3 = 500 the ten numbers agree with the printed ones to 4.8e-8 (relative); the constraint
 ↪  is preserved to 1.3e-14. The conservation law eps' + 3 H_a (eps + p_a) + 4 H_b (eps + p_b) = 0
 ↪  holds at the test point of the document (a = 1 + 0.3 x + 0.05 x^2, b = 1/(1 + 0.2 x), x = 0.7)
 ↪  with the relative residual -9.2e-10
[PASS] gr-G3  (numerical) the deflating spacetime in the author's Einstein-Lovelock gravity (four
 ↪  flat coordinates x0..x3 with H_a = 0, three deflating coordinates with equal rates, H_b =
 ↪  -g55): (i) it solves the three field equations without matter for LambdaBar = 3 g55^2 and
 ↪  alpha_2 = 1/(4 g55^2) (residuals 5.4e-16 g55^2; alpha_3 is arbitrary because E^(3) vanishes),
 ↪  in agreement with the exact checks 44d-G10 and 71d-G10; (ii) at this solution the two
 ↪  equations for the accelerations are -6 H_a' + .. and +2 H_b' + .. (deviation of the matrix of
 ↪  coefficients from diag(-6, 2): 2.2e-15): the evolution is well defined; (iii) in Einstein
 ↪  gravity without the Gauss-Bonnet term and without LambdaBar the same spacetime needs a source
 ↪  with kappa eps = 3.0 g55^2, kappa p_a = -6.0 g55^2, kappa p_b = -3.0 g55^2, that is p_a = -2
 ↪  eps along the flat coordinates and p_b = -eps along the deflating ones. Kohn-Sham matter
 ↪  cannot be this source: its pressure along a coordinate on which the fields do not depend is Pi
 ↪  = (1/2) G_V rho^2 + (M - m)^2/(2 G_S), and Pi = -eps would require e_kin + G_V rho^2 = 0
```

```
[PASS] gr-G4  (numerical) the deflating spacetime with Kohn-Sham matter as a source (rate g55 =
 ↪  0.01 m; initial data a = b = 1, H_a = 0, H_b from the constraint; loading = kappa eps(0)/(3
 ↪  g55^2)): matter makes the three coordinates deflate FASTER than exponentially and makes the
 ↪  four flat coordinates EXPAND; in 4 of the 8 evolutions the rates exceed 50 g55 before the
 ↪  clock of the undisturbed deflation reads 6 (they grow without bound in a finite time). The
 ↪  constraint is preserved along every evolution to 6.4e-10 (relative). Results: [7+1, all
 ↪  momenta; tuned Einstein-Lovelock; loading 1e-3] H_b(0) = -1.00050 g55; at tau_ref = 6.00: a =
 ↪  1.2508, b = 4.908e-2, H_a = +1.4052 g55, H_b = -7.9908 g55 | [7+1, all momenta; tuned
 ↪  Einstein-Lovelock; loading 1e-1] H_b(0) = -1.04881 g55; the rates run away; at tau_ref = 3.58:
 ↪  a = 1.5647, b = 2.320e-2, H_a = +8.9453 g55, H_b = -50.0355 g55 | [7+1, all momenta; Einstein;
 ↪  loading 1e-3] H_b(0) = -0.03162 g55; at tau_ref = 6.00: a = 1.0011, b = 9.335e-1, H_a =
 ↪  +0.0012 g55, H_b = -0.0375 g55 | [7+1, all momenta; Einstein; loading 1e-1] H_b(0) = -0.31623
 ↪  g55; the rates run away; at tau_ref = 4.17: a = 1.5278, b = 9.199e-2, H_a = +11.2603 g55, H_b
 ↪  = -50.0924 g55 | [sector without momenta along x5, x6, x7; tuned Einstein-Lovelock; loading
 ↪  1e-3] H_b(0) = -1.00050 g55; at tau_ref = 6.00: a = 1.2333, b = 5.683e-2, H_a = +1.1823 g55,
 ↪  H_b = -6.6163 g55 | [sector without momenta along x5, x6, x7; tuned Einstein-Lovelock; loading
 ↪  1e-1] H_b(0) = -1.04881 g55; the rates run away; at tau_ref = 3.74: a = 1.6348, b = 2.202e-2,
 ↪  H_a = +9.1382 g55, H_b = -50.0730 g55 | [sector without momenta along x5, x6, x7; Einstein;
 ↪  loading 1e-3] H_b(0) = -0.03162 g55; at tau_ref = 6.00: a = 1.0016, b = 9.331e-1, H_a =
 ↪  +0.0017 g55, H_b = -0.0380 g55 | [sector without momenta along x5, x6, x7; Einstein; loading
 ↪  1e-1] H_b(0) = -0.31623 g55; the rates run away; at tau_ref = 4.05: a = 1.7118, b = 8.801e-2,
 ↪  H_a = +13.1968 g55, H_b = -50.2782 g55
[PASS] gr-G5  (numerical) the deflating vacuum solution of the tuned Einstein-Lovelock gravity,
 ↪  perturbed without matter. Near the solution the constraint reads -6 g55 dH_b - 48 H_a^2 =
 ↪  kappa eps (to lowest order; dH_b = H_b + g55): (a) for a deflation that is SLOWER than g55
 ↪  there are no initial data without matter near the solution (no root H_a between -g55 and +g55
 ↪  for H_b = -0.999 g55; the constraint is a polynomial of degree four in H_a whose roots for H_b
 ↪  = -g55 are 0, 0 and (6 +- sqrt(20)) g55, the two last ones far from the deflating solution);
 ↪  (b) for H_b = -1.001 g55 the constraint has the two roots H_a = +-g55 sqrt(0.001/8) =
 ↪  +-0.01118 g55 to lowest order (found to 4 per cent), and in both cases the deflation does not
 ↪  return to the rate g55 but becomes faster and faster: H_a(0) = +0.01147 g55: at tau_ref =
 ↪  6.00, H_a = +1.73339 g55, H_b = -6.42224 g55; H_a(0) = -0.01088 g55: the rates run away; at
 ↪  tau_ref = 5.20, H_a = -0.00498 g55, H_b = -50.15518 g55. The exactly exponential deflation is
 ↪  therefore an UNSTABLE solution of the gravity in which it is a vacuum solution: it lies on the
 ↪  edge of the allowed initial data, and every allowed perturbation, with or without matter,
 ↪  grows. Constraint preserved to 8.3e-9
```

5.5 Geometry, matrices, spin connection, Dirac operators: command `matrices`

The Dirac matrices of the frame:

```
PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: the checks of the real 16x16 Dirac
 ↪  matrices of the main project, executed again (these matrices are the matrices of the
 ↪  orthonormal frame of the deflating spacetimes); identifiers 44-.. refer to gamma1644, 71-.. to
 ↪  gamma1671
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks matrices
106 of 106 checks passed

[PASS] 44-R1  (exact) every entry of gamma1644[A], A = 0..7, is one of the integers -1, 0, +1;
 ↪  hence the eight matrices are real
[PASS] 44-R2  (exact) every gamma1644[A] is a signed permutation matrix (one entry +1 or -1 in
 ↪  each row and in each column)
[PASS] 44-C1  (exact) gamma1644[A].gamma1644[B] + gamma1644[B].gamma1644[A] = 2 eta1644^(AB) 1
 ↪  for all 64 pairs (A, B), with eta1644 = diag[1, 1, 1, 1, -1, -1, -1, -1]
[PASS] 44-C2  (exact) transpose(gamma1644[A]) = eta1644^(AA) gamma1644[A]: matrices with square
 ↪  +1 are symmetric, matrices with square -1 are antisymmetric; hence transpose(gamma1644[A]) =
 ↪  inverse(gamma1644[A])
```

```
[PASS] 44-C3  (exact) the 256 ordered products gamma1644[A1]...gamma1644[Ak] (A1 < ... < Ak, k =
↪  0..8) are linearly independent: they span all real 16x16 matrices, so the representation is
↪  irreducible over the real numbers
[PASS] 44-C4  (exact) every one of the 255 non-empty products has trace 0, and the square of every
↪  product is +1 or -1
[PASS] 44-L1  (exact) Sigma1644^(AB) := (1/4)[gamma1644[A], gamma1644[B]] equals (1/2)
↪  gamma1644[A].gamma1644[B] for A != B, and Sigma1644^(AB) = -Sigma1644^(BA)
[PASS] 44-L2  (exact) the 28 matrices Sigma1644^(AB), A < B, are linearly independent
[PASS] 44-L3  (exact) [Sigma1644^(AB), gamma1644[C]] = eta1644^(BC) gamma1644[A] - eta1644^(AC)
↪  gamma1644[B] for all A, B, C: the Sigma1644^(AB) rotate the eight matrices as the components
↪  of a vector
[PASS] 44-L4  (exact) [Sigma1644^(AB), Sigma1644^(CD)] = eta1644^(BC) Sigma1644^(AD) -
↪  eta1644^(AC) Sigma1644^(BD) - eta1644^(BD) Sigma1644^(AC) + eta1644^(AD) Sigma1644^(BC) for
↪  all 4096 index combinations: the commutation relations of the Lie algebra so(p,q) of the
↪  metric eta1644
[PASS] 44-L5  (exact) the structure constants are integers and the Killing form is diagonal in the
↪  basis Sigma1644^(AB): B(Sigma1644^(AB), Sigma1644^(AB)) = -12 eta1644^(AA) eta1644^(BB); it
↪  has 16 positive and 12 negative eigenvalues (no zero eigenvalue: the algebra is semisimple)
[PASS] 44-L6  (exact) 12 generators are antisymmetric matrices (compact: they generate rotations)
↪  and 16 are symmetric (non-compact: they generate boosts); these are the numbers of the real
↪  form so(4,4), which has 12 compact and 16 non-compact generators
[PASS] 44-G1  (exact) each gamma1644[A] is invertible with inverse eta1644^(AA) gamma1644[A], and
↪  -gamma1644[A].gamma1644[B].inverse(gamma1644[A]) = gamma1644[B] for B != A, = -gamma1644[A]
↪  for B = A: gamma1644[A] acts on the vector index as the reflection that reverses the axis xA
[PASS] 44-G2  (numerical) exp(theta Sigma1644^(AB)) = cos(theta/2) 1 + sin(theta/2)
↪  gamma1644[A].gamma1644[B] when (gamma1644[A].gamma1644[B])^2 = -1 (rotation), and =
↪  cosh(theta/2) 1 + sinh(theta/2) gamma1644[A].gamma1644[B] when (gamma1644[A].gamma1644[B])^2
↪  = +1 (boost), for the 28 planes and random theta (largest deviation 2.3e-14)
[PASS] 44-G3  (numerical) for 20 random elements S = exp(sum_(A<B) theta_AB Sigma1644^(AB))
↪  (theta_AB random real numbers):  S.gamma1644[C].inverse(S) = sum_D Lambda^C_D gamma1644[D]
↪  with a real 8x8 matrix Lambda that preserves eta1644 (Lambda.eta1644.transpose(Lambda) =
↪  eta1644) and has determinant +1 (largest deviations 2.0e-13, 2.2e-15, 2.7e-15)
[PASS] 44-N1  (exact) the eight matrices rebuilt from the definitions equal, entry by entry, the
↪  eight matrices T16[0..7] printed by Mathematica in 4+4.nb (cell 314, Out[211])
[PASS] 44-N2  (exact) the rebuilt volume element equals T16[8] printed in 4+4.nb (cell 136,
↪  Out[99])
[PASS] 44-N3  (exact) the metric printed in 4+4.nb (cell 13, Out[9]) is eta4488 =
↪  diag(1,1,1,1,-1,-1,-1,-1) (PROJECT name: eta1644)
[PASS] 44-B1  (exact) the 8x8 matrix sigma (= sigma88) of 4+4.nb, PROJECT name sigma88: sigma88 =
↪  tau[1].tau[2].tau[3] = tau[4].tau[5].tau[6].tau[7], sigma88.sigma88 = 1,
↪  tau[1].tau[2]...tau[7] = 1, and tau[7] = diag(-1_4, +1_4) (4+4.nb Out[72], Out[89], Out[87],
↪  Out[63])
[PASS] 44-B2  (exact) tau[A].taubar[B] + tau[B].taubar[A] = 2 eta1644^(AB) 1 and taubar[A].tau[B]
↪  + taubar[B].tau[A] = 2 eta1644^(AB) 1 (8x8), A, B = 0..7 (4+4.nb Out[77])
[PASS] 44-B3  (exact) sigma88.taubar[A] = transpose(sigma88.tau[A]) for A = 0..7 (4+4.nb Out[76])
[PASS] 44-B4  (exact) T16[A] = [[0, taubar[A]], [tau[A], 0]] in 8x8 blocks, with tau[0] =
↪  taubar[0] = 1.  PROJECT names of these objects of 4+4.nb: gamma1644[A] := T16[A], tau1644[A]
↪  := tau[A], taubar1644[A] := taubar[A]
[PASS] 44-V1  (exact) omega1644 := gamma1644[0]...gamma1644[7] (= T16[8]) = diag(-1_8, +1_8):
↪  real, diagonal, symmetric, with square +1
[PASS] 44-V2  (exact) omega1644 anticommutes with every gamma1644[A] and commutes with every
↪  Sigma1644^(AB)
[PASS] 44-P1  (exact) PL1644 := (1 - omega1644)/2 = diag(1_8, 0_8) and PR1644 := (1 + omega1644)/2
↪  = diag(0_8, 1_8) are real matrices with integer entries
[PASS] 44-P2  (exact) PL1644 + PR1644 = 1, PL1644.PL1644 = PL1644, PR1644.PR1644 = PR1644,
↪  PL1644.PR1644 = PR1644.PL1644 = 0, trace PL1644 = trace PR1644 = 8 (4+4.nb Out[114])
[PASS] 44-P3  (exact) the rebuilt projectors PL1644 and PR1644 equal the matrices P_L and P_R
↪  printed in 4+4.nb (cells 163 and 165, Out[112] and Out[113])
[PASS] 44-P4  (exact) gamma1644[A].PL1644 = PR1644.gamma1644[A] and gamma1644[A].PR1644 =
↪  PL1644.gamma1644[A]: every Dirac matrix maps left-handed spinors to right-handed ones and
↪  conversely
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[PASS] 44-P5  (exact) PL1644 and PR1644 commute with every Sigma1644^(AB): chirality is invariant
↪  under Spin(4,4)
[PASS] 44-S1  (exact) Cm1644 := gamma1644[0].gamma1644[1].gamma1644[2].gamma1644[3] (the matrix
↪  sigma16 of 4+4.nb) = diag(-sigma88, sigma88), symmetric, Cm1644.Cm1644 = 1 (4+4.nb Out[106])
[PASS] 44-S2  (exact) Cp1644 := gamma1644[4].gamma1644[5].gamma1644[6].gamma1644[7] (the matrix
↪  S44 of 7+1.nb) = diag(sigma88, sigma88), symmetric, Cp1644.Cp1644 = 1, and omega1644 =
↪  Cm1644.Cp1644
[PASS] 44-S3  (exact) Cm1644.gamma1644[A].Cm1644 = -transpose(gamma1644[A]) and
↪  Cp1644.gamma1644[A].Cp1644 = +transpose(gamma1644[A]) for A = 0..7
[PASS] 44-S4  (exact) Cm1644.gamma1644[A] is antisymmetric for A = 0..7 (4+4.nb Out[105]) and
↪  Cp1644.gamma1644[A] is symmetric
[PASS] 44-S5  (exact) gamma1644[4] (the matrix JT4 = T16[4] of 4+4.nb) is a complex structure:
↪  gamma1644[4].gamma1644[4] = -1, and transpose(gamma1644[4]).Cm1644.gamma1644[4] = Cm1644
[PASS] 71-R1  (exact) every entry of gamma1671[A], A = 0..7, is one of the integers -1, 0, +1;
↪  hence the eight matrices are real
[PASS] 71-R2  (exact) every gamma1671[A] is a signed permutation matrix (one entry +1 or -1 in
↪  each row and in each column)
[PASS] 71-C1  (exact) gamma1671[A].gamma1671[B] + gamma1671[B].gamma1671[A] = 2 eta1671^(AB) 1
↪  for all 64 pairs (A, B), with eta1671 = diag[-1, -1, -1, -1, 1, -1, -1, -1]
[PASS] 71-C2  (exact) transpose(gamma1671[A]) = eta1671^(AA) gamma1671[A]: matrices with square
↪  +1 are symmetric, matrices with square -1 are antisymmetric; hence transpose(gamma1671[A]) =
↪  inverse(gamma1671[A])
[PASS] 71-C3  (exact) the 256 ordered products gamma1671[A1]...gamma1671[Ak] (A1 < ... < Ak, k =
↪  0..8) are linearly independent: they span all real 16x16 matrices, so the representation is
↪  irreducible over the real numbers
[PASS] 71-C4  (exact) every one of the 255 non-empty products has trace 0, and the square of every
↪  product is +1 or -1
[PASS] 71-L1  (exact) Sigma1671^(AB) := (1/4)[gamma1671[A], gamma1671[B]] equals (1/2)
↪  gamma1671[A].gamma1671[B] for A != B, and Sigma1671^(AB) = -Sigma1671^(BA)
[PASS] 71-L2  (exact) the 28 matrices Sigma1671^(AB), A < B, are linearly independent
[PASS] 71-L3  (exact) [Sigma1671^(AB), gamma1671[C]] = eta1671^(BC) gamma1671[A] - eta1671^(AC)
↪  gamma1671[B] for all A, B, C: the Sigma1671^(AB) rotate the eight matrices as the components
↪  of a vector
[PASS] 71-L4  (exact) [Sigma1671^(AB), Sigma1671^(CD)] = eta1671^(BC) Sigma1671^(AD) -
↪  eta1671^(AC) Sigma1671^(BD) - eta1671^(BD) Sigma1671^(AC) + eta1671^(AD) Sigma1671^(BC) for
↪  all 4096 index combinations: the commutation relations of the Lie algebra so(p,q) of the
↪  metric eta1671
[PASS] 71-L5  (exact) the structure constants are integers and the Killing form is diagonal in the
↪  basis Sigma1671^(AB): B(Sigma1671^(AB), Sigma1671^(AB)) = -12 eta1671^(AA) eta1671^(BB); it
↪  has 7 positive and 21 negative eigenvalues (no zero eigenvalue: the algebra is semisimple)
[PASS] 71-L6  (exact) 21 generators are antisymmetric matrices (compact: they generate rotations)
↪  and 7 are symmetric (non-compact: they generate boosts); these are the numbers of the real
↪  form so(7,1), which has 21 compact and 7 non-compact generators
[PASS] 71-G1  (exact) each gamma1671[A] is invertible with inverse eta1671^(AA) gamma1671[A], and
↪  -gamma1671[A].gamma1671[B].inverse(gamma1671[A]) = gamma1671[B] for B != A, = -gamma1671[A]
↪  for B = A: gamma1671[A] acts on the vector index as the reflection that reverses the axis xA
[PASS] 71-G2  (numerical) exp(theta Sigma1671^(AB)) = cos(theta/2) 1 + sin(theta/2)
↪  gamma1671[A].gamma1671[B] when (gamma1671[A].gamma1671[B])^2 = -1 (rotation), and =
↪  cosh(theta/2) 1 + sinh(theta/2) gamma1671[A].gamma1671[B] when (gamma1671[A].gamma1671[B])^2
↪  = +1 (boost), for the 28 planes and random theta (largest deviation 7.1e-15)
[PASS] 71-G3  (numerical) for 20 random elements S = exp(sum_(A<B) theta_AB Sigma1671^(AB))
↪  (theta_AB random real numbers):  S.gamma1671[C].inverse(S) = sum_D Lambda^C_D gamma1671[D]
↪  with a real 8x8 matrix Lambda that preserves eta1671 (Lambda.eta1671.transpose(Lambda) =
↪  eta1671) and has determinant +1 (largest deviations 5.0e-14, 2.0e-15, 2.0e-15)
[PASS] 71-N1  (exact) the eight matrices rebuilt from the definitions equal, entry by entry, the
↪  eight matrices Gamma17[0..7] printed by Mathematica in 7+1.nb (cell 204, Out[178])
[PASS] 71-N2  (exact) PROJECT matrices: gamma1671[A] = Gamma17up[swap04(A)] with swap04 = (0 4)
↪  and Gamma17up[A] = eta17^(AA) Gamma17[A] (7+1.nb section 5.3); explicitly gamma1671[4] =
↪  Gamma17[0] (time-like), gamma1671[0] = -Gamma17[4], gamma1671[j] = -Gamma17[j] for j = 1, 2,
↪  3, 5, 6, 7.  (gamma1671 is NOT the complex set Gamma71 of 7+1.nb section 4.3, and eta1671 is
↪  NOT the matrix eta71 of 7+1.nb)
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[PASS] 71-N3  (exact) in the labelling of 7+1.nb: Gamma17[A].Gamma17[B] + Gamma17[B].Gamma17[A] =
 ↪  2 eta17^(AB) 1 with eta17 = diag(+1,-1,-1,-1,-1,-1,-1,-1) (7+1.nb Out[173])
[PASS] 71-B1  (exact) tauO[A] = tau[A] (A = 1, 2, 3), tauO[A] = tau[A].sigma (A = 4..7): real,
 ↪  antisymmetric, tauO[A].tauO[A] = -1, mutually anticommuting, tauO[1]...tauO[7] = 1 (7+1.nb
 ↪  section 3.1)
[PASS] 71-B2  (exact) block form: Gamma17[0] = [[0, 1], [1, 0]] and Gamma17[i] = [[tauO[i], 0],
 ↪  [0, tauObar[i]]] with tauObar[i] = transpose(tauO[i]) = -tauO[i], i = 1..7 (7+1.nb Out[175])
[PASS] 71-B3  (exact) built from the 4+4 matrices only: Gamma17[0] = T16[0], Gamma17[i] =
 ↪  T16[0].T16[i] (i = 1, 2, 3), Gamma17[k] = T16[0].T16[k].S44 (k = 4..7) (7+1.nb Out[176])
[PASS] 71-B4  (exact) in the labels of this project: gamma1671[4] = gamma1644[0]; gamma1671[j] =
 ↪  -gamma1644[0].gamma1644[j] (j = 1, 2, 3); gamma1671[0] =
 ↪  +gamma1644[0].gamma1644[5].gamma1644[6].gamma1644[7]; gamma1671[5] =
 ↪  -gamma1644[0].gamma1644[4].gamma1644[6].gamma1644[7]; gamma1671[6] =
 ↪  +gamma1644[0].gamma1644[4].gamma1644[5].gamma1644[7]; gamma1671[7] =
 ↪  -gamma1644[0].gamma1644[4].gamma1644[5].gamma1644[6].  The time-like matrix of 7+1 IS the
 ↪  matrix of the hidden axis of 4+4; the other seven are products of two or of four of the 4+4
 ↪  matrices and therefore not linear combinations of them
[PASS] 71-V1  (exact) omega1671 := gamma1671[0]...gamma1671[7] = Gamma17[8] = [[0, -1], [1, 0]] in
 ↪  8x8 blocks (the omega17 of 7+1.nb: the seven sign changes of Gamma17up and the odd permutation
 ↪  swap04 compensate): real, antisymmetric, with square -1
[PASS] 71-V2  (exact) omega1671 anticommutes with every gamma1671[A] and commutes with every
 ↪  Sigma1671^(AB)
[PASS] 71-V3  (exact) the linear system X.gamma1671[A] + gamma1671[A].X = 0 (A = 0..7) for a real
 ↪  16x16 matrix X has rank 255, and omega1671 solves it (71-V2): the real matrices that
 ↪  anticommute with all eight gamma1671[A] are exactly the multiples of omega1671; since
 ↪  omega1671.omega1671 = -1, no real matrix with square +1 anticommutes with all of them: there
 ↪  is NO real chirality matrix in 7+1
[PASS] 71-P1  (exact) chi1671 := i omega1671 = i Gamma17[8] (= chi17 of 7+1.nb) is Hermitian,
 ↪  chi1671.chi1671 = 1, anticommutes with every gamma1671[A]
[PASS] 71-P2  (exact) PL1671 := (1 - chi1671)/2 and PR1671 := (1 + chi1671)/2: PL1671 + PR1671 =
 ↪  1, PL1671.PL1671 = PL1671, PR1671.PR1671 = PR1671, PL1671.PR1671 = 0, Hermitian, trace 8 each;
 ↪  they are complex (not real) matrices
[PASS] 71-P3  (exact) gamma1671[A].PL1671 = PR1671.gamma1671[A] for A = 0..7, and PL1671, PR1671
 ↪  commute with every Sigma1671^(AB)
[PASS] 71-P4  (exact) in 8x8 blocks chi1671 = [[0, -i], [i, 0]], 2 PL1671 = [[1, i], [-i, 1]] and
 ↪  2 PR1671 = [[1, -i], [i, 1]]; these are chi17, PL17 and PR17 of 7+1.nb (sections 4.6 and 7.10)
[PASS] 71-P5  (exact) with s := 1 - i Gamma17[0] (s.s^dagger = 2, so Uc1671 := s/sqrt(2) is
 ↪  unitary): s.chi1671 = -T16[8].s, i.e. Uc1671.chi1671.inverse(Uc1671) = -omega1644.  Hence
 ↪  Uc1671.PL1671.inverse(Uc1671) = PR1644 and Uc1671.PR1671.inverse(Uc1671) = PL1644: the
 ↪  complex chiral projectors of 7+1 are unitarily equivalent to the real projectors of 4+4.nb,
 ↪  with the names L and R exchanged.  (7+1.nb defines chi17 = +i Gamma17[8] for its real matrices
 ↪  and chi71 = -i Gamma71[8] for its complex matrices Gamma71; the PROJECT matrix chi1671 equals
 ↪  chi17, NOT chi71)
[PASS] 71-S1  (exact) Dirac adjoint: transpose(gamma1671[A]) =
 ↪  gamma1671[4].gamma1671[A].gamma1671[4] for A = 0..7 (gamma1671[4] is the time-like matrix; it
 ↪  is symmetric and squares to +1)
[PASS] 44-K1  (exact) 4+4: the matrices Cm1644.gamma1644[A1]...gamma1644[Ar] (A1 < ... < Ar) of
 ↪  rank r = 0..8 are all symmetric (+) or all antisymmetric (-): signs [1, -1, -1, 1, 1, -1, -1,
 ↪  1, 1]; the matrices Cp1644.gamma1644[A1]...gamma1644[Ar]: signs [1, 1, -1, -1, 1, 1, -1, -1,
 ↪  1]; in both cases exactly 120 of the 256 matrices are antisymmetric
[PASS] 44-K2  (exact) 4+4: the two Lorentz-scalar pair matrices Cm1644 (rank 0) and
 ↪  Cm1644.omega1644 (rank 8; equal to Cp1644 up to sign) are SYMMETRIC: the Lorentz-scalar
 ↪  bilinears dirac1644^T Cm1644 dirac1644 and dirac1644^T Cp1644 dirac1644 of one anticommuting
 ↪  16-component field vanish identically
[PASS] 71-K0  (exact) 7+1: Cp1671 := gamma1671[4] satisfies Cp1671.gamma1671[A] =
 ↪  +transpose(gamma1671[A]).Cp1671 and Cm1671 := gamma1671[4].omega1671 satisfies
 ↪  Cm1671.gamma1671[A] = -transpose(gamma1671[A]).Cm1671; both are symmetric matrices
[PASS] 71-K1  (exact) 7+1: signs (+ symmetric, - antisymmetric) of the matrices
 ↪  Cm1671.gamma1671[A1]...gamma1671[Ar] (A1 < ... < Ar) by rank r = 0..8: [1, -1, -1, 1, 1, -1,
 ↪  -1, 1, 1]; of Cp1671.gamma1671[A1]...gamma1671[Ar]: [1, 1, -1, -1, 1, 1, -1, -1, 1]; exactly
 ↪  120 of the 256 matrices are antisymmetric in both cases
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[PASS] 71-K2  (exact) 7+1: the two Lorentz-scalar pair matrices Cm1671 (rank 0) and
 ↪  Cm1671.omega1671 = -Cp1671 (rank 8) are SYMMETRIC: the Lorentz-scalar bilinears dirac1671^T
 ↪  Cm1671 dirac1671 and dirac1671^T Cp1671 dirac1671 of one anticommuting 16-component field
 ↪  vanish identically
[PASS] 44-K3  (exact) 4+4: for all A != B in 1..7, transpose(Sigma1644^(AB)).Cm1644.gamma1644[0] +
 ↪  Cm1644.gamma1644[0].Sigma1644^(AB) = 0, while the same expression with A = 0 is not zero: the
 ↪  bilinear DeltaV1644^0 = dirac1644^T Cm1644 gamma1644^0 dirac1644 is INVARIANT under the 21
 ↪  generators of the pseudo-rotations of the seven coordinates x1..x7 (the group Spin(3,4), which
 ↪  contains the Lorentz group of x1 x2 x3 x4), and only the 7 generators that involve the hidden
 ↪  axis x0 change it
[PASS] 71-K3  (exact) 7+1: for all A != B in 1..7, transpose(Sigma1671^(AB)).Cm1671.gamma1671[0] +
 ↪  Cm1671.gamma1671[0].Sigma1671^(AB) = 0, while the same expression with A = 0 is not zero: the
 ↪  bilinear DeltaV1671^0 = dirac1671^T Cm1671 gamma1671^0 dirac1671 is INVARIANT under the 21
 ↪  generators of the Lorentz group of the seven coordinates x1..x7 (the group Spin(6,1), which
 ↪  contains the Lorentz group of x1 x2 x3 x4), and only the 7 generators that involve the hidden
 ↪  axis x0 change it
[PASS] 44-D1  (exact) (gamma1644^A k_A)^2 = (eta1644^(AB) k_A k_B) 1 for 40 random integer vectors
 ↪  k: the Dirac operator gamma1644^A d_A squares to the wave operator eta1644^(AB) d_A d_B = d0^2
 ↪  + d1^2 + d2^2 + d3^2 - d4^2 - d5^2 - d6^2 - d7^2
[PASS] 44-D2  (exact) (i gamma1644.k + m1644)(i gamma1644.k - m1644) = -(k.k + m1644^2) 1 for 40
 ↪  random integer (k, m1644): every component of a solution of (gamma1644^A d_A - m1644)
 ↪  dirac1644 = 0 obeys the Klein-Gordon equation (eta1644^(AB) d_A d_B - m1644^2) dirac1644 = 0
[PASS] 44-D3  (exact) worked example k = (0,2,2,4; 5,0,0,0), m1644 = 1 (on the mass shell:
 ↪  k4^2+k5^2+k6^2+k7^2 = k0^2+k1^2+k2^2+k3^2 + m1644^2): Pi1644 := (i gamma1644.k + m1644)/(2
 ↪  m1644) satisfies Pi1644.Pi1644 = Pi1644, trace Pi1644 = 8 and (i gamma1644.k - m1644).Pi1644 =
 ↪  0, so the plane-wave solutions u exp(i k.x) form an 8-dimensional space, u = Pi1644 w
[PASS] 44-D4  (exact) PL1644.gamma1644[A].PR1644 = [[0, taubar1644[A]], [0, 0]],
 ↪  PR1644.gamma1644[A].PL1644 = [[0, 0], [tau1644[A], 0]], PL1644.gamma1644[A].PL1644 =
 ↪  PR1644.gamma1644[A].PR1644 = 0: with dirac1644 = (psiL1644, psiR1644) the 16-component
 ↪  equation is the pair  taubar1644^A d_A psiR1644 = m1644 psiL1644,  tau1644^A d_A psiL1644 =
 ↪  m1644 psiR1644  of 8-component equations
[PASS] 44-D5  (exact) (tau1644^A k_A)(taubar1644^B k_B) = (taubar1644^A k_A)(tau1644^B k_B) =
 ↪  (eta1644^(AB) k_A k_B) 1_8 for 40 random integer k: each 8-component field obeys the
 ↪  Klein-Gordon equation
[PASS] 44-D6  (exact) gamma1644[4].gamma1644[a] (a = 0..3) are real symmetric with square +1 (and
 ↪  so are alpha1644^a := -gamma1644[4].gamma1644[a], the matrices of the Hamiltonian h1644 =
 ↪  alpha1644^a k_a + beta1644 m1644, beta1644 := -i gamma1644[4]); gamma1644[4].gamma1644[t] (t
 ↪  = 5, 6, 7) are real antisymmetric with square -1; gamma1644[4] is real antisymmetric with
 ↪  square -1
[PASS] 44-D7  (exact) x4-evolution d_4 dirac1644 = K1644 dirac1644 with K1644 = -m1644 gamma1644^4
 ↪  + i k_a gamma1644^4 gamma1644^a + i kappa_t gamma1644^4 gamma1644^t (a = 0..3 space, t = 5, 6,
 ↪  7 extra times): K1644.K1644 = -(m1644^2 + |k|^2 - |kappa|^2) 1 for 40 random integer (k,
 ↪  kappa, m1644): the frequencies in x4 are +-sqrt(m1644^2 + |k|^2 - |kappa|^2)
[PASS] 44-D8  (exact) K1644 + K1644^dagger = 2 i kappa_t gamma1644^4 gamma1644^t: K1644 is
 ↪  anti-Hermitian (so that h1644 := i K1644 is Hermitian and the x4-evolution preserves
 ↪  dirac1644^dagger dirac1644) if and only if kappa = 0, i.e. if the field does not depend on the
 ↪  extra times x5, x6, x7
[PASS] 44-D9  (exact) example of the instability: m1644 = 3, k = 0, kappa = (5, 0, 0) gives
 ↪  K1644.K1644 = +16, (K1644 - 4)(K1644 + 4) = 0 with trace K1644 = 0: K1644 has the real
 ↪  eigenvalues +4 and -4, eight times each, so eight solutions grow like exp(+4 x4)
[PASS] 44-D10  (exact) W1644 := i Cm1644.gamma1644[4] is Hermitian with W1644.W1644 = 1 and trace
 ↪  0 (eigenvalues +1 and -1, eight times each); W1644 commutes with gamma1644[4] and with
 ↪  gamma1644[4].gamma1644[a] (a = 0..3), hence with K1644 at kappa = 0, and anticommutes with
 ↪  gamma1644[4].gamma1644[t] (t = 5, 6, 7)
[PASS] 44-D11  (exact) the density of the conserved current of the Spin(4,4)-invariant Lagrangian,
 ↪  j1644^4 = dirac1644^dagger (i Cm1644.gamma1644[4]) dirac1644 = dirac1644^dagger W1644
 ↪  dirac1644, is NOT positive: W1644 has eight eigenvalues -1; the positive density
 ↪  dirac1644^dagger dirac1644 is conserved only for kappa = 0 (44-D8)
[PASS] 71-D1  (exact) (gamma1671^A k_A)^2 = (eta1671^(AB) k_A k_B) 1 = (k4^2 - k0^2 - k1^2 - k2^2
 ↪  - k3^2 - k5^2 - k6^2 - k7^2) 1 for 40 random integer vectors k
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[PASS] 71-D2 (exact) (gamma1671.k + m1671)(gamma1671.k - m1671) = (k.k - m1671^2) 1 for 40 random integer (k, m1671): with dirac1671 = u exp(-i k_A x^A) the operator i gamma1671^A d_A - m1671 becomes gamma1671.k - m1671, and every component of a solution obeys the Klein-Gordon equation (d4^2 - d0^2 - d1^2 - d2^2 - d3^2 - d5^2 - d6^2 - d7^2 + m1671^2) dirac1671 = 0
[PASS] 71-D3 (exact) worked example E = 5, momentum p = (p1, p2, p3) = (2, 2, 4), m1671 = 1 (on the mass shell, 25 = 24 + 1): Pi1671 := (gamma1671.k + m1671)/(2 m1671) satisfies Pi1671.Pi1671 = Pi1671, trace Pi1671 = 8, (gamma1671.k - m1671).Pi1671 = 0: eight independent positive-energy plane waves
[PASS] 71-D4 (exact) beta1671 := gamma1671[4] and alpha1671^j := gamma1671[4].gamma1671[j] (j = 0, 1, 2, 3, 5, 6, 7) are real symmetric; alpha1671^i.alpha1671^j + alpha1671^j.alpha1671^i = 2 delta^(ij), alpha1671^j.beta1671 + beta1671.alpha1671^j = 0, beta1671.beta1671 = 1: the Hamiltonian h1671 = -i alpha1671^j d_j + beta1671 m1671 is Hermitian
[PASS] 71-D5 (exact) h1671(p) = p^j alpha1671^j + m1671 beta1671 is a real symmetric matrix with h1671(p).h1671(p) = (|p|^2 + m1671^2) 1 for 40 random integer (p, m1671): the energies are +E and -E, E = sqrt(|p|^2 + m1671^2), eight times each
[PASS] 71-D6 (exact) tau1671^j := tauO[swap04(j)] (swap04 = (0 4)): tau1671^4 = 1_8; for the seven space indices tau1671^j is real, antisymmetric, tau1671^i.tau1671^j + tau1671^j.tau1671^i = -2 delta^(ij); in 8x8 blocks beta1671 = [[0, 1], [1, 0]] and alpha1671^j = [[0, tau1671^j], [-tau1671^j, 0]] (beta1671 = betaD and alpha1671^j = alphaD[swap04(j)] of 7+1.nb section 7.1)
[PASS] 71-D7 (exact) chiral form: with EL1671 = (1; -i)/sqrt(2), ER1671 = (-i; 1)/sqrt(2) (16x8, in 8x8 blocks; chi1671.EL1671 = -EL1671, chi1671.ER1671 = +ER1671, orthonormal): EL1671^dagger.h1671(p).EL1671 = -i p^j tau1671^j, ER1671^dagger.h1671(p).ER1671 = +i p^j tau1671^j, EL1671^dagger.h1671(p).ER1671 = ER1671^dagger.h1671(p).EL1671 = m1671 (20 random integer (p, m1671)); hence (i d_4 + tau1671^j d_j) phiL1671 = m1671 phiR1671 and (i d_4 - tau1671^j d_j) phiR1671 = m1671 phiL1671 for phiL1671 = EL1671^dagger dirac1671, phiR1671 = ER1671^dagger dirac1671 (EL1671, ER1671, phiL1671, phiR1671 are the objects EL, ER, phiL, phiR of 7+1.nb sections 7.10 and 7.11, in the PROJECT labelling of the coordinates)
[PASS] 71-D8 (exact) worked example E = 5, p = (2, 2, 4), m1671 = 1: with M1671 = m1671 + p^j tau1671^j (M1671.transpose(M1671) = E^2 1_8), the columns of Upos1671 := (E 1; transpose(M1671)) satisfy h1671(p).Upos1671 = +E Upos1671 and those of Vneg1671 := (E 1; -M1671) satisfy h1671(-p).Vneg1671 = -E Vneg1671; transpose(Upos1671).Upos1671 = transpose(Vneg1671).Vneg1671 = 2 E^2 1_8: the spinors u_s = Upos1671 e_s/sqrt(E), v_s = Vneg1671 e_s/sqrt(E) are REAL and normalized to 2E
[PASS] 71-D9 (exact) charge conjugation dirac1671_c := omega1671 conj(dirac1671): omega1671 is real and omega1671.gamma1671[A] = -gamma1671[A].omega1671, so (i gamma1671^A d_A - m1671) dirac1671_c = omega1671 conj((i gamma1671^A d_A - m1671) dirac1671); omega1671.omega1671 = -1 gives (dirac1671_c)_c = -dirac1671: a 16-component Majorana condition dirac1671_c = dirac1671 has only the solution dirac1671 = 0
[PASS] 71-Q1 (numerical) 7+1, one particle at rest: the 8 spin states form the spinor of SO(7) (Casimir 21/4) and split under SO(3) x SU(2)_A x SU(2)_B (ordinary rotations x rotations of the extra space x0 x5 x6 x7) as (2; 2, 1) + (2; 1, 2); [A_a, B_b] = 0 to 0e0, residual 6e-16
[PASS] 71-Q2 (numerical) 7+1, two identical particles in an s wave (28 states allowed by the Pauli principle): under SO(7) they form 7 + 21 (Casimir 6 and 10); there is NO SO(7) singlet. Under SO(3) x SU(2)_A x SU(2)_B: 7 = (1; 2, 2) + (3; 1, 1), 21 = (1; 3, 1) + (1; 1, 3) + (3; 1, 1) + (3; 2, 2). The pairs of ordinary spin 0 are the 10 states (1; 2, 2) + (1; 3, 1) + (1; 1, 3) (residual 3e-13)
[PASS] 71-Q3 (exact) 7+1: for space indices i != j, gamma1671[i].gamma1671[j] = diag(tau1671^i tau1671^j, tau1671^i tau1671^j) in 8x8 blocks, so on the particles at rest, u = (e; e)/sqrt(2), the rotation generator Sigma1671^(ij) acts as the real antisymmetric 8x8 matrix (1/2) tau1671^i tau1671^j
[PASS] 71-Q4 (exact) 7+1: the 7 matrices tau1671^j and the 21 matrices tau1671^i tau1671^j (i < j; space indices) are antisymmetric and linearly independent: together they span all 28 antisymmetric 8x8 matrices, i.e. all s-wave pair amplitudes
[PASS] 71-Q5 (exact) 7+1: the quadratic Casimir operator of SO(7), Cas7(Phi) := -sum_(i<j) [Sigma1671^(ij), [Sigma1671^(ij), Phi]], gives Cas7(Phi) = 6 Phi for every Phi = tau1671^j (the vector, 7) and Cas7(Phi) = 10 Phi for every Phi = tau1671^i tau1671^j (the adjoint, 21)
[PASS] 71-Q6 (exact) 7+1: the linear system [tau1671^i tau1671^j, X] = 0 (all 21 planes) for an 8x8 matrix X has rank 63 and is solved by X = 1: the only rotation-invariant pair matrix is the unit matrix, which is SYMMETRIC and therefore excluded by the Pauli principle. Two identical particles of the field cannot form an s-wave pair that is a singlet under all rotations of the seven space dimensions

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[PASS] 71-Q7  (exact) 7+1: the ten antisymmetric matrices tau1671^0, tau1671^5, tau1671^6,
 ↪  tau1671^7 and tau1671^a tau1671^b (a < b in {0, 5, 6, 7}) are linearly independent and commute
 ↪  with the generators tau1671^2 tau1671^3, tau1671^3 tau1671^1, tau1671^1 tau1671^2 of ordinary
 ↪  rotations; the system [tau1671^i tau1671^j, Phi] = 0 (ordinary rotations) on antisymmetric Phi
 ↪  has rank 18 = 28 - 10: these ten matrices are ALL the s-wave pairs of ordinary spin 0
[PASS] 71-Q8  (exact) 7+1: each of tau1671^j (any space index j) and tau1671^i tau1671^j (i != j)
 ↪  is an orthogonal matrix: Phi.transpose(Phi) = 1.  A condensate whose pair matrix is
 ↪  proportional to tau1671^j or to tau1671^i tau1671^j therefore gives the SAME gap to all eight
 ↪  spin states (a 'unitary' state)
[PASS] 71-Q9  (exact) 7+1: two-particle content of the field bilinears, u_s^T X u_r for particles
 ↪  at rest: X = Cm1671.gamma1671[j] gives -tau1671^j (space index j) and X = Cm1671.gamma1671[4]
 ↪  gives 0; X = Cp1671.gamma1671[i].gamma1671[j] gives tau1671^i tau1671^j.  Hence the vector
 ↪  pair field is DeltaV1671^j := dirac1671^T Cm1671 gamma1671^j dirac1671 (its time component
 ↪  contains no pair of two particles) and the adjoint pair field is DeltaT1671^(ij) :=
 ↪  dirac1671^T Cp1671 gamma1671^i gamma1671^j dirac1671
[PASS] 71-Q10  (exact) 7+1: the system [tau1671^i tau1671^j, Phi] = 0 for the six rotation planes
 ↪  (1,2), (1,3), (2,3) of ordinary space and (5,6), (5,7), (6,7) of the extra space, on
 ↪  antisymmetric Phi, has rank 27 = 28 - 1, and Phi = tau1671^0 solves it: EXACTLY ONE s-wave
 ↪  pair is a singlet both of the ordinary rotations and of the rotations of x5 x6 x7.  Its pair
 ↪  matrix is tau1671^0, the two-particle content of DeltaV1671^0 = dirac1671^T Cm1671
 ↪  gamma1671^0 dirac1671, the component of the vector pair field along the hidden space dimension
 ↪  x0 (the same result as in 4+4, check 44-Q4)
[PASS] 44-Q1  (numerical) 4+4, fields independent of x5 x6 x7, one particle at rest (gamma1644[4]
 ↪  u = i u, 8 states; deviation 0e0): under SU(2)_X x SU(2)_Y (the two factors of the rotation
 ↪  group SO(4) of the four space coordinates x0 x1 x2 x3; they are NOT the factors SU(2)_A x
 ↪  SU(2)_B of the internal symmetry of the zero-mode sector, which do not act on ordinary space)
 ↪  x SU(2)_T (rotations of the extra times) the states are (2, 1; 2) with one sign of W1644 and
 ↪  (1, 2; 2) with the other; every state has ordinary spin 1/2 (residual 2e-16)
[PASS] 44-Q2  (numerical) 4+4, fields independent of x5 x6 x7, two identical particles in an s
 ↪  wave (28 states): (1, 1; 3) twice, (3, 1; 1), (1, 3; 1), (2, 2; 1), (2, 2; 3) under SU(2)_X x
 ↪  SU(2)_Y x SU(2)_T (X, Y as in 44-Q1).  The pairs that are singlets under ALL rotations of the
 ↪  four space dimensions are the 6 states (1, 1; 3) + (1, 1; 3): two triplets of the extra-time
 ↪  rotations, none of them a singlet of SU(2)_T.  10 states have ordinary spin 0 (residual 1e-14)
[PASS] 44-Q3  (numerical) 4+4: among the 120 Pauli-allowed bilinears dirac1644^T Cm1644
 ↪  gamma1644[A1]...gamma1644[Ar] dirac1644 (A1 < ... < Ar), those whose two-particle content
 ↪  (particles at rest) lies entirely in the SO(4)-singlet pairs have the index sets {A1,...,Ar} =
 ↪  {5} {0,1,2,3,5} {4,5} {0,1,2,3,4,5} {6} {0,1,2,3,6} {4,6} {0,1,2,3,4,6} {7} {0,1,2,3,7} {4,7}
 ↪  {0,1,2,3,4,7}; they include the components A = 5, 6, 7 (the extra-time components) of the
 ↪  vector pair field DeltaV1644^A := dirac1644^T Cm1644 gamma1644^A dirac1644
[PASS] 44-Q4  (numerical) 4+4: exactly 1 of the 28 pair states is a singlet both of the ordinary
 ↪  rotations (x1 x2 x3) and of the extra-time rotations (x5 x6 x7); it belongs to the space
 ↪  vector (2, 2; 1), is made of one particle with W1644 = +1 and one with W1644 = -1, and is the
 ↪  two-particle content of the bilinears with the index sets {A1,...,Ar} = {0} {0,4}: in
 ↪  particular of DeltaV1644^0 = dirac1644^T Cm1644 gamma1644^0 dirac1644, the component of the
 ↪  vector pair field along the hidden space dimension x0
[PASS] ALL  (exact) every check above passed
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The deflating spacetimes:

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PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the geometry, of the Dirac
 ↪  matrices, of the spin connection and of the Dirac operators of the deflating spacetimes
 ↪  (44d-..: 4+4 with deflating x5, x6, x7; 71d-..: 7+1 with deflating x5, x6, x7)
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks matrices
47 of 47 checks passed
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[PASS] 44d-G1 (exact) metric of the deflating 4+4 spacetime in the author's sign convention (space-like +, time-like -): g = diag(gs_00, .., gs_44, gs_55 b_5^2, gs_66 b_6^2, gs_77 b_7^2) with the metric signs gs = diag[1, 1, 1, 1, -1, -1, -1, -1] and the scale factors b_5 = const5 exp(-g55 x4), b_6 = const6 exp(-g66 x4), b_7 = const7 exp(-g77 x4), that is g_55 = -const5^2 exp(-2 g55 x4) and likewise g_66, g_77 (the author's formulas for this spacetime, on which x5, x6, x7 are time-like; g55, g66, g77 > 0 are the three rates, and in statements with a general index t = 5, 6, 7 the rate of xt is written rate_t, gsum = g55 + g66 + g77). Verified: g_tt = -b_t^2 and d g_tt/dx4 = -2 rate_t g_tt; the frame e^A = (dx0, .., dx4, b_5 dx5, b_6 dx6, b_7 dx7) is orthonormal. All exact geometry checks use five rational parameter sets: (g55, g66, g77) = (1, 1, 1) at a point where (b_5, b_6, b_7) = (1, 1, 1); (g55, g66, g77) = (1, 2, 3) at a point where (b_5, b_6, b_7) = (1/2, 1/3, 1/5); (g55, g66, g77) = (1/2, 3/4, 5/3) at a point where (b_5, b_6, b_7) = (2, 1/3, 5/7); (g55, g66, g77) = (2, 2, 5) at a point where (b_5, b_6, b_7) = (3/2, 3/2, 1/4); (g55, g66, g77) = (7/3, 7/3, 7/3) at a point where (b_5, b_6, b_7) = (1/2, 3, 5/4)

[PASS] 44d-G2 (exact) the Christoffel symbols computed from their definition are Gamma^t_(4t) = Gamma^t_(t4) = -rate_t and Gamma^4_(tt) = -gs_tt rate_t b_t^2 = + rate_t b_t^2 (t = 5, 6, 7); all others vanish

[PASS] 44d-G3 (exact) Cartan's structure equations hold with the spin-connection coefficients of the tetrad postulate: the torsion d e^A + omega^A_B ^ e^B vanishes, and d omega^A_B + omega^A_C ^ omega^C_B equals the Riemann tensor computed from the Christoffel symbols (two independent routes to the curvature)

[PASS] 44d-G4 (exact) Riemann tensor in the orthonormal frame: R^t_(4t4) = -rate_t^2, R^4_(t4t) = gs_tt rate_t^2 = -rate_t^2, R^t_(sts) = gs_ss rate_t rate_s = -rate_t rate_s (t != s in {5, 6, 7}), and the components that follow from antisymmetry; all other components vanish. No component involves x0, x1, x2 or x3: the four space coordinates x0, x1, x2, x3 stay flat

[PASS] 44d-G5 (exact) with gsq = g55^2 + g66^2 + g77^2 and gsum = g55 + g66 + g77, in the author's sign convention of the metric: Ricci tensor R^4_4 = gsq, R^t_t = rate_t gsum, R^a_a = 0 (a = 0..3); curvature scalar R = gsq + gsum^2; Einstein tensor G^4_4 = -(g55 g66 + g55 g77 + g66 g77), G^t_t = rate_t gsum - (gsq + gsum^2)/2, G^a_a = -(gsq + gsum^2)/2; all are constants (they do not depend on x4), the mixed components are the same on the 4+4 and on the 7+1 spacetime, and the covariant divergence of the Einstein tensor vanishes (contracted Bianchi identity)

[PASS] 44d-G6 (exact) Killing vectors (the Lie derivative of the metric along each vanishes identically): 7 translations (x0, x1, x2, x3, x5, x6, x7), 6 rotations of (x0, x1, x2, x3), and the scaling d/dx4 + sum_t rate_t xt d/dxt, 14 in all; one more for every pair of equal rates (a rotation of the two coordinates), and three more if all three rates are equal (20 in all). Numbers found for the five parameter sets: [20, 14, 14, 15, 20]

[PASS] 44d-G7 (exact) if the three rates are equal, g55 = g66 = g77, the part (x4, x5, x6, x7) of the metric has constant curvature, R_(rho sigma mu nu) = +g55^2 (g_(rho mu) g_(sigma nu) - g_(rho nu) g_(sigma mu)): the four time-like coordinates carry minus the metric of hyperbolic space H^4 of radius 1/g55, written in horospherical coordinates (the upper-half-space form)

[PASS] 44d-G10 (exact) Lovelock gravity, in the author's sign convention of the metric (space-like components positive), with the action R - 2 Lambda_8 + alpha_GB (R^2 - 4 R_(mu nu) R^(mu nu) + R_(mu nu rho sigma) R^(mu nu rho sigma)) and the field equations G^mu_nu + Lambda_8 delta^mu_nu + alpha_GB Lanczos^mu_nu = 0: the curvature acts diagonally on pairs of frame directions; the Lanczos tensor of the Gauss-Bonnet term is Lanczos^a_a = -4 g55 g66 g77 gsum (a = 0..3), Lanczos^4_4 = Lanczos^t_t = 0; the Lovelock tensor of third order vanishes identically. For three equal rates (2 of the five parameter sets) the deflating spacetime solves the vacuum field equations with Lambda_8 = 3 g55^2 and alpha_GB = -1/(4 g55^2), that is 4 alpha_GB Lambda_8 = -3: the deflation needs no matter as its source in Einstein-Gauss-Bonnet gravity. For rates that are not all equal no choice of Lambda_8 and alpha_GB satisfies the components 4, 5, 6, 7

[PASS] 44d-G9 (numerical) geodesic with m = 1, momentum P_1 = 0.5 along x1 and P_5 = 0.3 along the deflating time x5 (g55 = 1, const5 = 0.8), started at x4 = -2 with increasing x4: P_1, P_5 and the norm are conserved to 1.7e-14; the path reaches the largest value x4 = 1.09240103 (predicted turning value ln(E_perp const5/|P_5|)/g55 with E_perp = sqrt(m^2 + P_1^2) = 1.09240103) after the affine parameter 3.38543615 (closed form 3.38543615) and then returns to smaller x4: a particle with momentum along a deflating time cannot pass the turning value

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[PASS] 44d-C1  (exact) the coordinate matrices gamma^mu = e_A^mu gamma1644[A] of the deflating 4+4
↪  spacetime (gamma^t = gamma1644[t]/b_t for t = 5, 6, 7; the other five unchanged) are real and
↪  satisfy gamma^mu.gamma^nu + gamma^nu.gamma^mu = 2 c g^(mu nu) 1 for all 64 pairs, with c = +1
↪  and the inverse metric g^(mu nu) of the author's sign convention (space-like +, time-like -;
↪  g^(tt) = -1/b_t^2). The squares of the matrices are therefore equal to the signs of the
↪  metric. Verified for five rational parameter sets: (g55, g66, g77) = (1, 1, 1) at a point
↪  where (b_5, b_6, b_7) = (1, 1, 1); (g55, g66, g77) = (1, 2, 3) at a point where (b_5, b_6,
↪  b_7) = (1/2, 1/3, 1/5); (g55, g66, g77) = (1/2, 3/4, 5/3) at a point where (b_5, b_6, b_7) =
↪  (2, 1/3, 5/7); (g55, g66, g77) = (2, 2, 5) at a point where (b_5, b_6, b_7) = (3/2, 3/2, 1/4);
↪  (g55, g66, g77) = (7/3, 7/3, 7/3) at a point where (b_5, b_6, b_7) = (1/2, 3, 5/4)
[PASS] 44d-C2  (exact) the spin-connection coefficients computed from the tetrad postulate
↪  omega_mu^A_B = e^A_nu (d_mu e_B^nu + Gamma^nu_(mu lam) e_B^lam), with the Christoffel symbols
↪  obtained from the metric, and with the first frame index lowered by the metric signs gs,
↪  omega_(mu A B) = gs_AA omega_mu^A_B, are antisymmetric, omega_(mu A B) = -omega_(mu B A), and
↪  the only non-zero ones are omega_(t,t4) = -omega_(t,4t) = -gs_tt rate_t b_t = + rate_t b_t (t
↪  = 5, 6, 7); same five parameter sets
[PASS] 44d-C3  (exact) Omega_mu := (c/4) omega_(mu A B) gamma1644[A].gamma1644[B] with c = +1
↪  equals +(1/2) rate_t b_t gamma1644[t].gamma1644[4] for mu = t in {5, 6, 7} and vanishes for mu
↪  = 0..4 (the same formula on both spacetimes, because c gs_tt = -1 on both): the spin
↪  connection is an element of the Lie algebra spanned by the scaled commutators Sigma^(t4)
[PASS] 44d-C4  (exact) d_mu gamma^nu + Gamma^nu_(mu lam) gamma^lam + [Omega_mu, gamma^nu] = 0 for
↪  all 64 pairs (mu, nu), with d_4 gamma^t = +rate_t gamma^t: the coordinate matrices are
↪  covariantly constant
[PASS] 44d-C5  (exact) sum_mu gamma^mu.Omega_mu = -(1/2) (g55 + g66 + g77) gamma1644[4]: in the
↪  Dirac operator the spin connection amounts to the replacement d_4 -> d_4 - gsum/2, which the
↪  weighted field psihat = sqrt(b_5 b_6 b_7) psi removes
[PASS] 44d-C6  (exact) d_mu Omega_nu - d_nu Omega_mu + [Omega_mu, Omega_nu] = (c/4) R_(A B mu nu)
↪  gamma1644[A].gamma1644[B] for all 64 pairs (mu, nu), with c = +1 and with the Riemann tensor
↪  that metric.rs computes from the metric, its first index lowered by the metric (d_4 Omega_t =
↪  -rate_t Omega_t): the curvature of the spinor derivative is the Riemann tensor
[PASS] 44d-C7  (exact) (1/2) sum_(mu nu) gamma^mu.gamma^nu.(d_mu Omega_nu - d_nu Omega_mu +
↪  [Omega_mu, Omega_nu]) = -(c R/4) 1 = - (R/4) 1 with the curvature scalar R = g55^2 + g66^2 +
↪  g77^2 + gsum^2 of the author's sign convention: the square of the Dirac operator differs from
↪  c times the wave operator of the spinor derivative by this term
[PASS] 44d-C8  (exact) transpose(Omega_mu).Cm1644 + Cm1644.Omega_mu = 0 (Cm1644 =
↪  gamma1644[0].gamma1644[1].gamma1644[2].gamma1644[3]): the adjoint field psi^dagger Cm1644 is
↪  differentiated with -Omega_mu on its right, as the conserved current requires
[PASS] 44d-D1  (exact) in momentum space (d_mu -> i k_mu) the Dirac operator of the weighted field
↪  at fixed x4, D = i gamma^mu k_mu with the coordinate matrices, satisfies (D + m)(D - m) =
↪  -(g^(mu nu) k_mu k_nu + m^2) 1 with g^(mu nu) k_mu k_nu = k_0^2 + k_1^2 + k_2^2 + k_3^2 -
↪  k_4^2 - sum_t k_t^2/b_t^2: the instantaneous mass shell is k_4^2 + sum_t k_t^2/b_t(x4)^2 =
↪  |k|^2 + m^2. Verified for five rational parameter sets and four rational momenta each
[PASS] 44d-D2  (exact) the generator of the evolution of the weighted field in x4, K(x4) = -m
↪  gamma1644[4] + i sum_a k_a gamma1644[4].gamma1644[a] + i sum_t (k_t/b_t(x4))
↪  gamma1644[4].gamma1644[t], satisfies K^2 = (sum_t k_t^2/b_t^2 - m^2 - |k|^2) 1 and K +
↪  K^dagger = 2 i sum_t (k_t/b_t) gamma1644[4].gamma1644[t]
[PASS] 44d-D3  (exact) dK/dx4 = i sum_t rate_t (k_t/b_t) gamma1644[4].gamma1644[t] is a Hermitian
↪  matrix whose square is sum_t (rate_t k_t/b_t)^2 times the unit matrix; the weighted field
↪  obeys the second-order equation d_4^2 psihat = (K^2 + dK/dx4) psihat, a Klein-Gordon equation
↪  with a term that depends on the spin
[PASS] 44d-D4  (exact) W = i Cm1644.gamma1644[4] is Hermitian with W^2 = 1 and trace 0, and W.K +
↪  K^dagger.W = 0 for every momentum and all k_5, k_6, k_7: the covariant charge psihat^dagger W
↪  psihat of a mode is conserved on the deflating spacetime, and it is not positive
[PASS] 44d-D5  (exact) K is anti-Hermitian (the evolution in x4 is unitary and psihat^dagger
↪  psihat is conserved) if and only if k_5 = k_6 = k_7 = 0, that is, if the field does not depend
↪  on the deflating times
[PASS] 44d-D6  (exact) with gamma1644[A] = [[0, taubar1644[A]], [tau1644[A], 0]] the operator
↪  gamma^mu d_mu of the weighted field has the blocks [[0, sum_A taubar1644[A] e_A^mu d_mu],
↪  [sum_A tau1644[A] e_A^mu d_mu, 0]], e_A^mu d_mu = (d_0, .., d_4, d_5/b_5, d_6/b_6, d_7/b_7):
↪  the equation is the pair taubar^A e_A^mu d_mu psihat^R = m psihat^L, tau^A e_A^mu d_mu
↪  psihat^L = m psihat^R of real 8-component equations
```

[PASS] 44d-D7 (exact) worked example: m = 3, k = 0, (k_5, k_6, k_7) = (1, 0, 0) at the value of x4 where b_5 = 1/5 (so k_5/b_5 = 5 > m): K^2 = +16 1, and (4 + K)/8 and (4 - K)/8 are complementary projectors of trace 8: at this x4 eight independent solutions grow as exp(+4 x4) and eight decay as exp(-4 x4) to first order. A mode that depends on a deflating time is oscillatory only while k_t/b_t(x4) < sqrt(m^2 + |k|^2)

[PASS] 71d-G1 (exact) metric of the deflating 7+1 spacetime in the author's sign convention (space-like +, time-like -): g = diag(gs_00, .., gs_44, gs_55 b_5^2, gs_66 b_6^2, gs_77 b_7^2) with the metric signs gs = diag[1, 1, 1, 1, -1, 1, 1, 1] and the scale factors b_5 = const5 exp(-g55 x4), b_6 = const6 exp(-g66 x4), b_7 = const7 exp(-g77 x4), that is g_55 = +const5^2 exp(-2 g55 x4) and likewise g_66, g_77 (the author's formulas for this spacetime, on which x5, x6, x7 are space-like; g55, g66, g77 > 0 are the three rates, and in statements with a general index t = 5, 6, 7 the rate of xt is written rate_t, gsum = g55 + g66 + g77). Verified: g_tt = +b_t^2 and d g_tt/dx4 = -2 rate_t g_tt; the frame e^A = (dx0, .., dx4, b_5 dx5, b_6 dx6, b_7 dx7) is orthonormal. All exact geometry checks use five rational parameter sets: (g55, g66, g77) = (1, 1, 1) at a point where (b_5, b_6, b_7) = (1, 1, 1); (g55, g66, g77) = (1, 2, 3) at a point where (b_5, b_6, b_7) = (1/2, 1/3, 1/5); (g55, g66, g77) = (1/2, 3/4, 5/3) at a point where (b_5, b_6, b_7) = (2, 1/3, 5/7); (g55, g66, g77) = (2, 2, 5) at a point where (b_5, b_6, b_7) = (3/2, 3/2, 1/4); (g55, g66, g77) = (7/3, 7/3, 7/3) at a point where (b_5, b_6, b_7) = (1/2, 3, 5/4)

[PASS] 71d-G2 (exact) the Christoffel symbols computed from their definition are Gamma^t_(4t) = Gamma^t_(t4) = -rate_t and Gamma^4_(tt) = -gs_tt rate_t b_t^2 = - rate_t b_t^2 (t = 5, 6, 7); all others vanish

[PASS] 71d-G3 (exact) Cartan's structure equations hold with the spin-connection coefficients of the tetrad postulate: the torsion d e^A + omega^A_B ^ e^B vanishes, and d omega^A_B + omega^A_C ^ omega^C_B equals the Riemann tensor computed from the Christoffel symbols (two independent routes to the curvature)

[PASS] 71d-G4 (exact) Riemann tensor in the orthonormal frame: R^t_(4t4) = -rate_t^2, R^4_(t4t) = gs_tt rate_t^2 = +rate_t^2, R^t_(sts) = gs_ss rate_t rate_s = +rate_t rate_s (t != s in {5, 6, 7}), and the components that follow from antisymmetry; all other components vanish. No component involves x0, x1, x2 or x3: the four space coordinates x0, x1, x2, x3 stay flat

[PASS] 71d-G5 (exact) with gsq = g55^2 + g66^2 + g77^2 and gsum = g55 + g66 + g77, in the author's sign convention of the metric: Ricci tensor R^4_4 = gsq, R^t_t = rate_t gsum, R^a_a = 0 (a = 0..3); curvature scalar R = gsq + gsum^2; Einstein tensor G^4_4 = -(g55 g66 + g55 g77 + g66 g77), G^t_t = rate_t gsum - (gsq + gsum^2)/2, G^a_a = -(gsq + gsum^2)/2; all are constants (they do not depend on x4), the mixed components are the same on the 4+4 and on the 7+1 spacetime, and the covariant divergence of the Einstein tensor vanishes (contracted Bianchi identity)

[PASS] 71d-G6 (exact) Killing vectors (the Lie derivative of the metric along each vanishes identically): 7 translations (x0, x1, x2, x3, x5, x6, x7), 6 rotations of (x0, x1, x2, x3), and the scaling d/dx4 + sum_t rate_t xt d/dxt, 14 in all; one more for every pair of equal rates (a rotation of the two coordinates), and three more if all three rates are equal (20 in all). Numbers found for the five parameter sets: [20, 14, 14, 15, 20]

[PASS] 71d-G7 (exact) if the three rates are equal, g55 = g66 = g77, the part (x4, x5, x6, x7) of the metric has constant curvature, R_(rho sigma mu nu) = +g55^2 (g_(rho mu) g_(sigma nu) - g_(rho nu) g_(sigma mu)) in the author's sign convention: it is de Sitter space of radius 1/g55 in the contracting half of its flat slicing

[PASS] 71d-G10 (exact) Lovelock gravity, in the author's sign convention of the metric (space-like components positive), with the action R - 2 Lambda_8 + alpha_GB (R^2 - 4 R_(mu nu) R^(mu nu) + R_(mu nu rho sigma) R^(mu nu rho sigma)) and the field equations G^mu_nu + Lambda_8 delta^mu_nu + alpha_GB Lanczos^mu_nu = 0: the curvature acts diagonally on pairs of frame directions; the Lanczos tensor of the Gauss-Bonnet term is Lanczos^a_a = -4 g55 g66 g77 gsum (a = 0..3), Lanczos^4_4 = Lanczos^t_t = 0; the Lovelock tensor of third order vanishes identically. For three equal rates (2 of the five parameter sets) the deflating spacetime solves the vacuum field equations with Lambda_8 = 3 g55^2 and alpha_GB = -1/(4 g55^2), that is 4 alpha_GB Lambda_8 = -3: the deflation needs no matter as its source in Einstein-Gauss-Bonnet gravity. For rates that are not all equal no choice of Lambda_8 and alpha_GB satisfies the components 4, 5, 6, 7

[PASS] 71d-G8 (numerical) geodesic of a particle of mass m = 1 with momentum P_1 = 0.5 along x1 and P_5 = 0.3 along the deflating coordinate x5 (g55 = 1, const5 = 0.8), integrated from x4 = -2 to x4 = 4 by the Runge-Kutta method: P_1, P_5 and the norm are conserved to 2.0e-10; the affine parameter (the proper time, since m = 1) that elapses is 3.3375401839, the closed form gives 3.3375401839; the proper time to x4 = infinity is finite, 3.386358: a particle that moves along a deflating coordinate leaves the coordinate patch after a finite proper time, with unbounded energy
[PASS] 71d-C1 (exact) the coordinate matrices gamma^mu = e_A^mu gamma1671[A] of the deflating 7+1 spacetime (gamma^t = gamma1671[t]/b_t for t = 5, 6, 7; the other five unchanged) are real and satisfy gamma^mu.gamma^nu + gamma^nu.gamma^mu = 2 c g^(mu nu) 1 for all 64 pairs, with c = -1 and the inverse metric g^(mu nu) of the author's sign convention (space-like +, time-like -; g^(tt) = +1/b_t^2). The squares of the matrices are therefore opposite to the signs of the metric. Verified for five rational parameter sets: (g55, g66, g77) = (1, 1, 1) at a point where (b_5, b_6, b_7) = (1, 1, 1); (g55, g66, g77) = (1, 2, 3) at a point where (b_5, b_6, b_7) = (1/2, 1/3, 1/5); (g55, g66, g77) = (1/2, 3/4, 5/3) at a point where (b_5, b_6, b_7) = (2, 1/3, 5/7); (g55, g66, g77) = (2, 2, 5) at a point where (b_5, b_6, b_7) = (3/2, 3/2, 1/4); (g55, g66, g77) = (7/3, 7/3, 7/3) at a point where (b_5, b_6, b_7) = (1/2, 3, 5/4)
[PASS] 71d-C2 (exact) the spin-connection coefficients computed from the tetrad postulate omega_mu^A_B = e^A_nu (d_mu e_B^nu + Gamma^nu_(mu lam) e_B^lam), with the Christoffel symbols obtained from the metric, and with the first frame index lowered by the metric signs gs, omega_(mu A B) = gs_AA omega_mu^A_B, are antisymmetric, omega_(mu A B) = -omega_(mu B A), and the only non-zero ones are omega_(t,t4) = -omega_(t,4t) = -gs_tt rate_t b_t = - rate_t b_t (t = 5, 6, 7); same five parameter sets
[PASS] 71d-C3 (exact) Omega_mu := (c/4) omega_(mu A B) gamma1671[A].gamma1671[B] with c = -1 equals +(1/2) rate_t b_t gamma1671[t].gamma1671[4] for mu = t in {5, 6, 7} and vanishes for mu = 0..4 (the same formula on both spacetimes, because c gs_tt = -1 on both): the spin connection is an element of the Lie algebra spanned by the scaled commutators Sigma^(t4)
[PASS] 71d-C4 (exact) d_mu gamma^nu + Gamma^nu_(mu lam) gamma^lam + [Omega_mu, gamma^nu] = 0 for all 64 pairs (mu, nu), with d_4 gamma^t = +rate_t gamma^t: the coordinate matrices are covariantly constant
[PASS] 71d-C5 (exact) sum_mu gamma^mu.Omega_mu = -(1/2) (g55 + g66 + g77) gamma1671[4]: in the Dirac operator the spin connection amounts to the replacement d_4 -> d_4 - gsum/2, which the weighted field psihat = sqrt(b_5 b_6 b_7) psi removes
[PASS] 71d-C6 (exact) d_mu Omega_nu - d_nu Omega_mu + [Omega_mu, Omega_nu] = (c/4) R_(A B mu nu) gamma1671[A].gamma1671[B] for all 64 pairs (mu, nu), with c = -1 and with the Riemann tensor that metric.rs computes from the metric, its first index lowered by the metric (d_4 Omega_t = -rate_t Omega_t): the curvature of the spinor derivative is the Riemann tensor
[PASS] 71d-C7 (exact) (1/2) sum_(mu nu) gamma^mu.gamma^nu.(d_mu Omega_nu - d_nu Omega_mu + [Omega_mu, Omega_nu]) = -(c R/4) 1 = + (R/4) 1 with the curvature scalar R = g55^2 + g66^2 + g77^2 + gsum^2 of the author's sign convention: the square of the Dirac operator differs from c times the wave operator of the spinor derivative by this term
[PASS] 71d-C8 (exact) transpose(Omega_mu).gamma1671[4] + gamma1671[4].Omega_mu = 0: the adjoint field psi^dagger gamma1671[4] is differentiated with -Omega_mu on its right, as the conserved current requires
[PASS] 71d-D1 (exact) in momentum space (d_mu -> -i k_mu) the Dirac operator of the weighted field at fixed x4 is gamma^mu k_mu - m with the coordinate matrices, and (gamma^mu k_mu + m)(gamma^nu k_nu - m) = -(g^(mu nu) k_mu k_nu + m^2) 1 with g^(mu nu) k_mu k_nu = -k_4^2 + k_0^2 + k_1^2 + k_2^2 + k_3^2 + sum_t k_t^2/b_t^2 (author's sign convention of the metric): the instantaneous mass shell is k_4^2 = |p|^2 + sum_t k_t^2/b_t(x4)^2 + m^2. Verified for five rational parameter sets and four rational momenta each
[PASS] 71d-D2 (exact) the Hamiltonian of the weighted field, h(x4) = m beta1671 + sum_j p_j alpha1671^j + sum_t (k_t/b_t(x4)) alpha1671^t (beta1671 = gamma1671[4], alpha1671^A = gamma1671[4].gamma1671[A]), is real and symmetric for every momentum and every x4, and h^2 = (m^2 + |p|^2 + sum_t k_t^2/b_t^2) 1: the evolution in x4 is unitary and psihat^dagger psihat is conserved, but the energy of a mode with k_t != 0 grows as exp(rate_t x4)
[PASS] 71d-D3 (exact) dh/dx4 = sum_t rate_t (k_t/b_t) alpha1671^t is real symmetric and anticommutes with beta1671: the Hamiltonian changes with x4 only through the momenta along the deflating coordinates, which is what creates pairs
[PASS] 71d-D5 (exact) omega1671 commutes with every alpha1671^A and anticommutes with beta1671; hence the charge-conjugate field omega1671 psi^* solves the Dirac equation on the deflating spacetime whenever psi does (the scale factors are real)

```
[PASS] 71d-D6  (exact) with the chiral 16x8 matrices E^L = (1_8, -i 1_8)^T/sqrt(2), E^R = (-i 1_8,
 ↪  1_8)^T/sqrt(2) (columns of 8 + 8 components) and the real 8x8 matrices tau1671^j of the
 ↪  author's notebook, (E^L)^dagger h E^L = -i sum_j (p_j/e_j) tau1671^j, (E^R)^dagger h E^R = +i
 ↪  sum_j (p_j/e_j) tau1671^j and (E^L)^dagger h E^R = (E^R)^dagger h E^L = m, where p_j/e_j =
 ↪  (p_0, p_1, p_2, p_3, k_5/b_5, k_6/b_6, k_7/b_7): the equation of the weighted field is the
 ↪  pair (i d_4 + tau^j e_j^mu d_mu) phihat^L = m phihat^R, (i d_4 - tau^j e_j^mu d_mu) phihat^R =
 ↪  m phihat^L of 8-component equations
[PASS] 71d-D7  (exact) worked example: m = 3, p = 0, (k_5, k_6, k_7) = (2, 0, 0) at the value of
 ↪  x4 where b_5 = 1/2 (so k_5/b_5 = 4): h^2 = 25 1, and (5 + h)/10, (5 - h)/10 are complementary
 ↪  projectors of trace 8: eight states of energy +5 and eight of energy -5 at this x4; one
 ↪  e-folding of the deflation later the energy is sqrt(9 + 16 e^2) = 11.28
[PASS] ALL  (exact) every check above passed
```

5.6 Solutions mode by mode: command `modes`

```
PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the solutions of the Dirac
 ↪  equations on the deflating spacetimes (71d-M..: pair creation on 7+1; 44d-M..: bounded
 ↪  solutions on 4+4)
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks modes
7 of 7 checks passed

[PASS] 71d-M1  (numerical) pair creation by the deflation, one direction: for E_perp/g55 = 0.25,
 ↪  0.5, .., 3.0 the number of particles created per spin state in a mode that starts empty agrees
 ↪  with 1/(exp(2 pi E_perp/g55) + 1) to a relative deviation of at most 3.2e-6 (the numbers range
 ↪  from 0.17 to 6.5e-9); the eight spin states are created equally; the evolved states stay
 ↪  orthonormal to 2.0e-13
[PASS] 71d-M2  (numerical) equal rates g55 = g66 = g77 = 1.3 (constants 1, 0.7, 2), mode with m =
 ↪  1, p = (0, 0.1, 0.2, 0.3), (k_5, k_6, k_7) = (0.5, -0.8, 0.3): the created number per spin
 ↪  state is 5.70627486e-3 for all eight states, the law 1/(exp(2 pi E_perp/g55) + 1) gives
 ↪  5.70627482e-3 (relative deviation 6.7e-9): with equal rates the result depends neither on the
 ↪  direction of k_t nor on the constants
[PASS] 71d-M3  (numerical) unequal rates (g55 = 1 with g66 = 2, 1.2 or 0.5), mode with wavenumbers
 ↪  along x5 and x6: whenever the second physical wavenumber is still below exp(-3) E_perp at the
 ↪  moment at which the first reaches E_perp, the mean created number is the Fermi-Dirac law with
 ↪  the rate of the coordinate that becomes energetic first, to within 1.2 per cent; when the two
 ↪  become energetic together the eight spin states are created unequally (largest ratio of the
 ↪  largest to the smallest eigenvalue in the scan: 110) and no closed form is known
 ↪  (data/creation_anisotropic.csv)
[PASS] 44d-M1  (numerical) modes of the 4+4 field that depend on the deflating times (five cases:
 ↪  one, two or three non-zero k_t; equal and unequal rates and constants): beyond the turning
 ↪  value of x4 exactly eight of the sixteen solutions stay bounded; integrated backwards to early
 ↪  times they contain positive- and negative-energy parts of equal weight (relative difference of
 ↪  the two Gram matrices at most 2.0e-14): they are standing waves in x4, not travelling particles
[PASS] 44d-M2  (numerical) the covariant charge psihat^dagger W psihat of every bounded solution
 ↪  vanishes (ratio to psihat^dagger psihat at most 1.0e-14), as it must for a conserved quantity
 ↪  of a solution that dies out
[PASS] 44d-M3  (numerical) between the turning value and the late x4 every bounded solution falls
 ↪  by exp(-S) times a factor of order one, S = Int sqrt(k_d^2 - E_perp^2) dx4 = 30 being the WKB
 ↪  exponent (largest difference between the logarithm of the factor and S in the five cases:
 ↪  0.53). Where one rate governs (cases 1 to 4) the eight solutions fall by the same factor
 ↪  (spread of the logarithms at most 1.1e-14) and its logarithm is S + (1/2) ln 2 - (1/2) ln(1 +
 ↪  exp(-2 pi E_perp/rate_t)), as the exact solution requires (largest deviation 3.6e-4); with
 ↪  three unequal rates (case 5) the factor depends on the spin state (spread 0.40). The
 ↪  dependence on a deflating time dies out as exp(-S), and S grows as exp(rate_t x4)
[PASS] 44d-M4  (numerical) where one rate governs (one non-zero k_t, or equal rates) the computed
 ↪  |psihat|^2 equals the exact form 2 u_d |K_(1/2 + i E_perp/rate_t)(u_d)|^2 (modified Bessel
 ↪  function of the order 1/2 + i E_perp/rate_t, u_d = k_d/rate_t) over the range 0.02 < u_d < 12
 ↪  with a largest relative deviation of 3.6e-12
[PASS] ALL  (numerical) every check above passed
```

5.7 Symmetries and exponents of the couplings: command `symmetry`

```
PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the symmetries on the
 ↪  deflating spacetimes (isometries, internal symmetry of the fields that do not depend on the
 ↪  extra coordinates, zero modes on a vortex line, reduction of the couplings)
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks symmetry
19 of 19 checks passed

[PASS] 44d-S1  (exact) of the 28 rotations and boosts of the flat 4+4 spacetime (one for each pair
 ↪  of coordinates) exactly 6 remain isometries of the deflating spacetime for three different
 ↪  rates (the rotations of x0, x1, x2, x3), 7 if two rates are equal (in addition the rotation of
 ↪  the two coordinates that deflate at the same rate) and 9 if the three rates are equal (in
 ↪  addition the three rotations of x5, x6, x7); no transformation that involves x4 survives, and
 ↪  none that mixes x0 with a deflating coordinate. Together with the 7 translations, the scaling
 ↪  and, for equal rates, the three special vectors of metric.rs this gives 14, 15 and 20 Killing
 ↪  vectors (Lie derivative of the metric zero at two points, exact arithmetic)
[PASS] 44d-S2  (exact) the products gamma^A gamma^B that commute with the five matrices alpha^1,
 ↪  alpha^2, alpha^3, beta, i beta chi (and so act as internal symmetries on fields that do not
 ↪  depend on the extra coordinates) are exactly those of the six pairs taken from x0, x5, x6, x7.
 ↪  Of these six, the deflating spacetime keeps as isometries 0 (three different rates), 1 (two
 ↪  equal rates) and 3 (three equal rates): the remaining 6, 5 and 3 are symmetries of the sector
 ↪  of such fields that have no counterpart among the isometries of the spacetime (hidden
 ↪  symmetries)
[PASS] 71d-S1  (exact) of the 28 rotations and boosts of the flat 7+1 spacetime (one for each pair
 ↪  of coordinates) exactly 6 remain isometries of the deflating spacetime for three different
 ↪  rates (the rotations of x0, x1, x2, x3), 7 if two rates are equal (in addition the rotation of
 ↪  the two coordinates that deflate at the same rate) and 9 if the three rates are equal (in
 ↪  addition the three rotations of x5, x6, x7); no transformation that involves x4 survives, and
 ↪  none that mixes x0 with a deflating coordinate. Together with the 7 translations, the scaling
 ↪  and, for equal rates, the three special vectors of metric.rs this gives 14, 15 and 20 Killing
 ↪  vectors (Lie derivative of the metric zero at two points, exact arithmetic)
[PASS] 71d-S2  (exact) the products gamma^A gamma^B that commute with the five matrices alpha^1,
 ↪  alpha^2, alpha^3, beta, i beta chi (and so act as internal symmetries on fields that do not
 ↪  depend on the extra coordinates) are exactly those of the six pairs taken from x0, x5, x6, x7.
 ↪  Of these six, the deflating spacetime keeps as isometries 0 (three different rates), 1 (two
 ↪  equal rates) and 3 (three equal rates): the remaining 6, 5 and 3 are symmetries of the sector
 ↪  of such fields that have no counterpart among the isometries of the spacetime (hidden
 ↪  symmetries)
[PASS] def-S3  (numerical) the internal symmetry of the fields that do not depend on the extra
 ↪  coordinates, computed again for both spacetimes (24 items of data/reduced_symmetry.csv, all
 ↪  as expected): the commutant of the five matrices has dimension 8 and is u(2) + u(2) = su(2)_A
 ↪  + su(2)_B + u(1)_F + u(1)_E; the eight one-particle states are (2,1) + (1,2); the four vector
 ↪  pair operators are the (2,2) with fermion number 2, and their invariant quadratic form is the
 ↪  frame metric of the axes x0, x5, x6, x7. The deflation does not change this symmetry, because
 ↪  the weighted field of this sector obeys the flat equations (checks 44d-D.., 71d-D.. of the
 ↪  command matrices)
[PASS] 44d-V1  (exact) alpha^1, alpha^2, alpha^3, beta and i beta chi are Hermitian, have square
 ↪  one and anticommute with one another
[PASS] 44d-V2  (exact) i beta alpha^1 and alpha^2 beta chi are commuting Hermitian matrices with
 ↪  square one
[PASS] 44d-V3  (exact) each of the four joint eigenspaces of the two matrices has dimension 4: a
 ↪  vortex line of unit winding carries four zero modes of the sixteen-component field
[PASS] 44d-V4  (exact) Pc = i alpha^1 alpha^2 alpha^3 chi equals the product of the five matrices,
 ↪  commutes with each of them, is Hermitian and traceless and has square one: it is, up to sign,
 ↪  the generator P of U(1)_E
[PASS] 44d-V5  (exact) winding +1: every w_z with (i beta alpha^1) w_z = -w_z, (alpha^2 beta chi)
 ↪  w_z = -w_z solves the zero-energy equations (i alpha^1 + beta) w_z = 0, (i alpha^2 + i beta
 ↪  chi) w_z = 0; on these four spinors alpha^3 = Pc, and its trace vanishes: two zero modes have
 ↪  alpha^3 = +1 (energy +k_3) and two alpha^3 = -1 (energy -k_3)
```

```
[PASS] 44d-V6  (exact) winding -1: the spinors w_z with (i beta alpha^1) w_z = -w_z, (alpha^2 beta
 ↪  chi) w_z = +w_z solve (i alpha^1 + beta) w_z = 0, (i alpha^2 - i beta chi) w_z = 0, and on
 ↪  them alpha^3 = -Pc: the directions of motion are exchanged
[PASS] 71d-V1  (exact) alpha^1, alpha^2, alpha^3, beta and i beta chi are Hermitian, have square
 ↪  one and anticommute with one another
[PASS] 71d-V2  (exact) i beta alpha^1 and alpha^2 beta chi are commuting Hermitian matrices with
 ↪  square one
[PASS] 71d-V3  (exact) each of the four joint eigenspaces of the two matrices has dimension 4: a
 ↪  vortex line of unit winding carries four zero modes of the sixteen-component field
[PASS] 71d-V4  (exact) Pc = i alpha^1 alpha^2 alpha^3 chi equals the product of the five matrices,
 ↪  commutes with each of them, is Hermitian and traceless and has square one: it is, up to sign,
 ↪  the generator P of U(1)_E
[PASS] 71d-V5  (exact) winding +1: every w_z with (i beta alpha^1) w_z = -w_z, (alpha^2 beta chi)
 ↪  w_z = -w_z solves the zero-energy equations (i alpha^1 + beta) w_z = 0, (i alpha^2 + i beta
 ↪  chi) w_z = 0; on these four spinors alpha^3 = Pc, and its trace vanishes: two zero modes have
 ↪  alpha^3 = +1 (energy +k_3) and two alpha^3 = -1 (energy -k_3)
[PASS] 71d-V6  (exact) winding -1: the spinors w_z with (i beta alpha^1) w_z = -w_z, (alpha^2 beta
 ↪  chi) w_z = +w_z solve (i alpha^1 + beta) w_z = 0, (i alpha^2 - i beta chi) w_z = 0, and on
 ↪  them alpha^3 = -Pc: the directions of motion are exchanged
[PASS] def-S4  (exact) reduction to fields that do not depend on x5, x6, x7: sqrt|g| times the
 ↪  Yukawa term and the potential of the Vorton-Scalar, written with the weighted fields psihat =
 ↪  sqrt(Vd) psi and Phihat = sqrt(Vd) Phi_VS, are the same expressions with the couplings
 ↪  y_VS/sqrt(Vd), lambda_VS/Vd and the vacuum value v_VS sqrt(Vd); the products y_VS v_VS and
 ↪  lambda_VS v_VS^2, hence the fermion mass and the mass of the radial mode, do not change
 ↪  (rational sample values, exact). The exponents of data/coupling_exponents.csv follow, with Vd
 ↪  proportional to exp(-gsum x4)
[PASS] def-S5  (exact) reduction to fields that do not depend on x5, x6, x7, mass term and
 ↪  self-interaction of the fermion field: sqrt|g| times [-m_D psibar psi - (1/2) g_r (psibar
 ↪  Theta_K psi)(psibar Theta^K psi)], written with the weighted field psihat = sqrt(Vd) psi, is
 ↪  the same expression with the unchanged bare Dirac mass m_D and with the coupling g_r/Vd; every
 ↪  coupling of the self-interaction therefore grows as exp(tau_d), as g_VS^2 does, and the ratio
 ↪  of the contact couplings to g_VS^2 is constant (rational sample values, three sets)
[PASS] ALL  (exact) every check above passed
```

5.8 Exchange of the Vorton-Scalar: command `channels`

```
PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the exchange of the
 ↪  Vorton-Scalar on the deflating spacetimes (def-K..: amplitudes in the pair channels; def-B..:
 ↪  the exchanged field)
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks channels
7 of 7 checks passed

[PASS] def-K1  (numerical) the exchange amplitudes of the Vorton-Scalar in the 30 representative
 ↪  pair channels of the main project (15 on each spacetime; three momenta and 13 scattering
 ↪  angles each: 1170 rows) computed again with the frame matrices agree with every number of the
 ↪  file data/pair_channel_kernel.csv of the main project to the digits printed there (1170 rows
 ↪  compared). For fields that do not depend on x5, x6, x7 the amplitude of a channel on the
 ↪  deflating spacetime is the flat amplitude times the square of the coupling of the moment
[PASS] def-K2  (numerical) the amplitudes of the hidden-axis channel reproduce the closed forms of
 ↪  the main project (largest deviation 1.1e-15)
[PASS] def-K3  (numerical) the quartic invariant of the gap matrix of the four vector pairs,
 ↪  Tr[(M_gap M_gap^dagger)^2]/8 = 2 (sum |Delta^a|^2)^2 - |eta_ab Delta^a Delta^b|^2 with the
 ↪  frame metric of x0, x5, x6, x7, holds on both deflating spacetimes (largest deviation
 ↪  2.1e-14); in coordinate components Delta_t = b_t(x4) Delta^t the same form reads with the
 ↪  metric g_tt of the moment
[PASS] def-B1  (numerical) the field that a static source builds up from zero approaches exp(gsum
 ↪  x4)/(k^2 + m_bos^2): after tau_d = gsum x4 = 20 the relative difference is 1.9e-5 4.3e-6
 ↪  2.4e-5 3.5e-5 for (k^2 + m_bos^2, gsum) = (1, 0.2), (0.25, 0.2), (1, 1), (4, 0.5) (transient
 ↪  proportional to exp(-tau_d/2)). The potential between two fermions at rest is therefore the
 ↪  Yukawa potential of the unshifted mass m_s with the strength alpha_VS of the moment
```

```
[PASS] def-B2  (exact) a mode of the Vorton-Scalar that does not depend on x5, x6, x7 goes as
 ↪  exp(exponent x4) with exponent = gsum/2 +- sqrt(gsum^2/4 - k^2 - m_bos^2): it oscillates if
 ↪  and only if k^2 + m_bos^2 > gsum^2/4, and its amplitude then grows as exp(gsum x4/2) (the
 ↪  weighted field has constant amplitude: the number of quanta is conserved). The phase mode
 ↪  (m_bos = 0) does not oscillate for k < gsum/2; the radial mode oscillates at every k if m_s >
 ↪  gsum/2 (table data/scalar_zero_modes.csv)
[PASS] def-B3  (numerical) 7+1: a mode of the Vorton-Scalar that depends on one deflating
 ↪  coordinate and starts empty ends with 1/(exp(2 pi varpi_b/rate_t) - 1) quanta, a Bose-Einstein
 ↪  law with the temperature rate_t/(2 pi): 36 cases (varpi_b/rate_t = 0.25 to 3, each of the
 ↪  three coordinates; the number is read off where the physical wavenumber is 3000 varpi_b),
 ↪  largest relative deviation 1.9e-5; the Bogoliubov coefficients satisfy |alpha|^2 - |beta|^2 =
 ↪  1 to 1.3e-13
[PASS] def-B4  (numerical) 4+4: a mode of the Vorton-Scalar that depends on a deflating time has
 ↪  one bounded solution; at early x4 it is a standing wave (the two travelling waves have equal
 ↪  amplitudes) and its profile is the modified Bessel function K of the imaginary order i
 ↪  varpi_b/rate_t (ratio to the exact form constant to 1e-4 away from the zeros): varpi_b/rate_t
 ↪  = 0.5: |c_-/c_+| - 1 = 0.0e0, 23 of 23 samples; varpi_b/rate_t = 1.5: |c_-/c_+| - 1 = 0.0e0,
 ↪  24 of 24 samples; varpi_b/rate_t = 3: |c_-/c_+| - 1 = 0.0e0, 26 of 26 samples;
[PASS] ALL  (numerical) every check above passed
```

5.9 The model along the clock: commands `history`, `quench`, `paired`

```
PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the history of the model
 ↪  along the deflation clock (couplings that grow as exp(gsum x4))
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks history
8 of 8 checks passed

[PASS] def-H1  (numerical) thresholds of the model, computed again: a pair of fermions is bound if
 ↪  D_VS > 1.67981; matter is self-bound if C > 3.7011 (3 pi^2/8 = 3.7011) or, with the
 ↪  negative-energy states, C > 5.9218 (3 pi^2/5 = 5.9218). On a deflating spacetime D_VS and C
 ↪  grow as exp(gsum x4), so each threshold is passed at a definite x4
[PASS] def-H2  (numerical) along the clock, wherever the gap is below 0.05 E_F the critical
 ↪  temperature is 0.567 times the gap (largest deviation 0.0020); the gap of the vector channel
 ↪  is at every moment at least as large as the gap with the radial mode alone, and that at least
 ↪  as large as the gap of the two-form channel; the gap never decreases as the deflation proceeds
 ↪  (four sets of m_s/m and k_F/m, 34 moments each)
[PASS] def-H3  (numerical) growth of the gap: for a weakly coupled state the gap is proportional
 ↪  to exp(-1/lambda), where lambda = lambda_S + lambda_P is the sum of the pairing strengths of
 ↪  the radial mode and of the phase mode in the vector channel, and lambda grows as exp(tau_d),
 ↪  which gives d ln(gap)/d tau_d = 1/lambda; in the computed histories lambda d ln(gap)/d tau_d
 ↪  lies between 1.00 and 1.03 wherever 1e-6 < gap/E_F < 0.02: the gap grows much faster than the
 ↪  couplings, by one e-folding in the time 1/(gsum/lambda)
[PASS] def-H4  (numerical) the moment at which a pair becomes bound: just below D_c the scattering
 ↪  length is large and negative (-134.0/m_s at 0.99 D_c), just above large and positive
 ↪  (135.9/m_s at 1.01 D_c); the binding energy starts as a power of the distance from the
 ↪  threshold with the exponents 2.01 (between 2 and 4 per cent above D_c) and 2.01 (between 4 and
 ↪  8 per cent), that is quadratically: E_b ~ (exp(tau_pair) - 1)^2 m_s^2/m. Because E_b vanishes
 ↪  at the threshold, the pair cannot follow the deflation adiabatically at first, however slow
 ↪  the deflation is
[PASS] def-H5  (numerical) matter of fixed density loses the fermion mass (the Vorton-Scalar loses
 ↪  its expectation value in the matter) when C passes a critical value that falls with the
 ↪  density: k_F = 0.2 m: C = 151.892 (2.46 times pi^2 m^2/(4 k_F^2)); k_F = 0.5 m: C = 12.971
 ↪  (1.31 times pi^2 m^2/(4 k_F^2)); k_F = 1 m: C = 2.556 (1.04 times pi^2 m^2/(4 k_F^2)); k_F = 2
 ↪  m: C = 0.617 (1.00 times pi^2 m^2/(4 k_F^2)). The estimate C = pi^2 m^2/(4 k_F^2) of the main
 ↪  project is reached at high density, where the transition happens at weak coupling; at low
 ↪  density the critical C is larger. On a deflating spacetime every piece of matter becomes
 ↪  abnormal at a definite x4, the denser the earlier
```

```
[PASS] def-H6  (numerical) the field beyond which the one-loop potential of the Vorton-Scalar lies
 ↪  below its vacuum value decreases monotonically as C grows: 24.7 v_VS at C = 1, 1.77 v_VS at C
 ↪  = 10. As the deflation proceeds the barrier that protects the vacuum moves in towards the
 ↪  vacuum value
[PASS] def-H7  (numerical) the temperature T_half at which the fermions lose half of their mass,
 ↪  in units of the (constant) fermion mass, falls monotonically as g_VS grows, for m_s/m = 0.3, 1
 ↪  and 3 (26 of the 30 moments examined; it is of the order of the four-dimensional vacuum value
 ↪  m/g_VS, which falls as exp(-gsum x4/2)): matter of a fixed temperature melts at a definite x4.
 ↪  No melting temperature can be given for m_s/m = 0.3 from g_VS = 1.69 (C = 32) on; there the
 ↪  Hartree potential ceases to exist at 1.05 v_VS before the expectation value has fallen,
 ↪  because the one-loop potential turns over too close to the vacuum value
[PASS] def-H8  (numerical) order of events: the pair of fermions becomes bound before the one-loop
 ↪  potential of the Vorton-Scalar has fallen below its vacuum value at a field of 3 v_VS only for
 ↪  the mass ratios m_s/m = 10, 30 of the six ratios 0.1, 0.3, 1, 3, 10, 30 examined (the
 ↪  condition is 4 pi D_c m/m_s < 4.27, i.e. m_s/m > 4.9): for a light radial mode the deflation
 ↪  destabilizes the vacuum first
[PASS] ALL  (numerical) every check above passed

PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the time-dependent dynamics
 ↪  of the paired state under a coupling that grows as exp(gsum x4)
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks quench
5 of 5 checks passed

[PASS] def-Q1  (numerical) the equations of motion conserve |2F|^2 + (1 - 2n)^2 = 1 for every
 ↪  pair: largest deviation at the end of the seven runs 4.2e-11
[PASS] def-Q2  (numerical) convergence: with twice as many energies (4000) and half the time step
 ↪  the run with gsum = 1e-3 E_sh gives the smallest ratio 0.73710 (against 0.73710) and the final
 ↪  density of broken pairs 4.4255e-4 (against 4.4255e-4)
[PASS] def-Q3  (numerical) the faster the deflation, the further the order parameter falls behind:
 ↪  the smallest value of |Delta|/Delta_eq is 0.958, 0.890, 0.737, 0.539, 0.328, 0.189, 0.098 for
 ↪  gsum/E_sh = 1e-4, 3e-4, 1e-3, 3e-3, 1e-2, 3e-2, 1e-1 (Xi_d at the start: 0.26 to 262)
[PASS] def-Q4  (numerical) the order parameter catches up with the equilibrium gap of the moment
 ↪  (to within 10 per cent) when Xi_d = gsum/(lambda Delta_eq) has fallen to a value between 0.23
 ↪  and 0.52 (six runs; in the fastest run, gsum = 0.1 E_sh, it does not catch up before lambda =
 ↪  0.5): the paired state follows the deflation if and only if gsum is small compared with lambda
 ↪  times the gap
[PASS] def-Q5  (numerical) the lag leaves broken pairs behind: their density at lambda = 0.5 is
 ↪  6.0e-6, 5.1e-5, 4.4e-4, 2.4e-3, 1.2e-2, 4.1e-2 (units of E_sh times the density of states) for
 ↪  gsum/E_sh = 1e-4 to 3e-2, growing with the rate; for the equilibrium state it is zero
[PASS] ALL  (numerical) every check above passed

PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: the results of the CJT effective action
 ↪  along the deflation clock
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks paired
7 of 7 checks passed

[PASS] def-P1  (numerical) the paired state along the clock (13 moments, two particles): at every
 ↪  moment the solution of the gap equation is a minimum of the CJT potential (Newton's method on
 ↪  the potential reproduces the iterated gap function to 7.7e-12; the minimum along the ray lies
 ↪  at s = 1 to 4.6e-7; the smallest eigenvalue of the second variation is positive), the critical
 ↪  temperature from the potential equals that of the linearized gap equation to 2.6e-12, and the
 ↪  condensation energy is within 15 per cent of -nu N(0) gap^2/4 wherever the gap is below 0.1
 ↪  E_F. The condensation energy deepens monotonically, in weak coupling with twice the
 ↪  logarithmic slope of the gap (to within 0.1 per cent), and the coherence length of the pair
 ↪  shrinks: for the first particle from 25 to 1.0 in units of 1/k_F. At the last 5 moments of
 ↪  that particle the gap exceeds 0.1 E_F, and at the last one the coherence length has fallen to
 ↪  the distance 1/k_F between the particles: the deflation carries the paired state from
 ↪  overlapping pairs towards bound pairs
```

```
[PASS] def-P2  (numerical) thermodynamics along the clock: while the critical temperature rises
 ↪  from 6.34e-3 to 1.04e-1 E_F (g_VS = 2.6 to 3.4), the ratio gap(0)/T_c stays within 0.019 of
 ↪  the weak-coupling value 1.7639 and the jump of the heat capacity within 0.048 of 1.4261: both
 ↪  ratios are universal while the coupling is weak, and they move only slightly as the gap
 ↪  approaches 0.2 E_F at the last moment
[PASS] def-P3  (numerical) Ginzburg-Landau energy along the clock (three moments, T_c = 6.3e-3,
 ↪  3.3e-2, 1.0e-1 E_F): while T_c is below 0.05 E_F the coefficients from the CJT potential agree
 ↪  with the weak-coupling formulas to 1.0 per cent; at the third moment, T_c = 0.10 E_F, they
 ↪  differ by up to 7.9 per cent, the state leaving the weak-coupling regime. The quadratic
 ↪  coefficient keeps its slope 4 N(0), the quartic and the gradient coefficients fall as 1/T_c^2,
 ↪  and the coherence length, sqrt(7 zeta(3)/(48 pi^2)) v_F/T_c = 0.1333 v_F/T_c in weak coupling
 ↪  (computed: 0.1333, 0.1340, 0.1390), shrinks from 80 to 5.1 (units 1/m). Vortex cores shrink as
 ↪  the deflation proceeds
[PASS] def-P4  (numerical) the space of the order parameter on the deflating spacetimes (frame
 ↪  components along x0, x5, x6, x7; both spacetimes): every unitary state, whichever its
 ↪  direction, has all eight gaps equal and the potential -2 (units of N(0) gap^2), with four flat
 ↪  directions (the Goldstone modes) and four directions of positive curvature; the half-paired
 ↪  state has four gaps closed and the potential -1. The potential does not distinguish the
 ↪  deflating axes from x0: in this sector the deflation selects no direction of the order
 ↪  parameter
[PASS] def-P5  (numerical) the test of the CJT method on the zero-dimensional model, executed
 ↪  again: the two-loop truncation, the one used for the models, has a solution for every coupling
 ↪  from 0.01 to 30 and stays within 10.7 per cent of the exact propagator
[PASS] def-P6  (numerical) the moment of binding from the gap equation: the linearized gap
 ↪  equation in vacuum reaches the eigenvalue 1 at D_VS = 1.7347, 1.6902, 1.6810 for m_s/m = 0.3,
 ↪  0.1, 0.03, approaching the value 1.67981 of the Schroedinger equation as the binding becomes
 ↪  non-relativistic (relative differences 3.3e-2, 6.2e-3, 6.9e-4)
[PASS] def-P7  (numerical) self-bound matter along the clock: once C has passed 3.7011 the lowest
 ↪  energy per particle lies below m, and it falls monotonically as the deflation proceeds: from
 ↪  -0.0192 m at C = 4 to -0.405 m at C = 29.6. At every one of these moments the matter of lowest
 ↪  energy is in the abnormal state (fermion mass zero inside the matter, largest value of m*/m in
 ↪  the table 0.0e0), at a Fermi momentum that falls from 0.98 m to 0.59 m. The exchange energies
 ↪  of the radial and of the phase mode at that density are 0.44 and 0.51 per cent of the
 ↪  mean-field energy at C = 4, and 0.90 and 1.22 per cent at C = 29.6 (m_s/m = 0.3): the mean
 ↪  field remains a good approximation along the whole history
[PASS] ALL  (numerical) every check above passed
```

5.10 Cosmology: commands cosmology, matter, background, spectrum, forecast, families, superfluid

```
PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the cosmological calculations
 ↪  on the deflating spacetime
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks cosmology
6 of 6 checks passed

[PASS] def-C1  (numerical) without deflation the background integrals of this module agree with
 ↪  the main project (its recombination calculation gives z_* = 1089.92): 100 theta_* = 1.03965
 ↪  here against 1.03965, sound horizon 144.449 Mpc against 144.449 Mpc; age 13.814 Gyr
[PASS] def-C2  (numerical) with deflation at a constant rate, gravity was weaker in the past, the
 ↪  early universe expanded more slowly at a given density, and the sound horizon is larger;
 ↪  holding 100 theta_*, omega_b and omega_c fixed, the inferred Hubble constant FALLS
 ↪  monotonically with the rate: d ln H_0/d(gsum/H_0) = -0.969 at small rates; H_0 = 66.70, 65.36,
 ↪  60.37 km/s/Mpc for gsum = 0.01, 0.03, 0.1 H_0 (67.36 without deflation). Deflation moves the
 ↪  Hubble constant away from the locally measured value. These numbers come from the full
 ↪  calculation of recombination on the deflating background (last scattering moves from z_* =
 ↪  1089.92 to 1088.04 at gsum = 0.1 H_0, and the helium fraction falls). If z_* and the helium
 ↪  fraction are held at their values without deflation, the slope is -0.911 and the values are
 ↪  66.74, 65.48, 60.77 km/s/Mpc: that approximation underestimates the fall by 5.8 to 6.0 per
 ↪  cent
```

```
[PASS] def-C3  (numerical) bounds on the present rate of deflation if gravity is eight-dimensional
 ↪  (G_N grows at the rate gsum): lunar laser ranging, (dG_N/dt)/G_N = (7.1 +- 7.6) 10^-14 per
 ↪  year, gives gsum < 2.23e-13 per year = 3.24e-3 H_0 at two sigma; nucleosynthesis,
 ↪  G_N(nucleosynthesis)/G_N(today) > 0.94 at two sigma, gives gsum < 4.43e-12 per year = 0.064
 ↪  H_0 for a rate that was constant since then. At these rates the inferred Hubble constant
 ↪  changes by -0.211 and -4.38 km/s/Mpc, in the direction away from the local measurement, 73.50
 ↪  +- 0.81 km/s/Mpc. Over the age t_0 of the model without deflation (H_0 t_0 = 0.9516) the
 ↪  deflation clock has advanced by gsum t_0 = 3.1e-3 at the first of these rates
[PASS] def-C5  (numerical) bounds that do not rest on gravity: (i) under the assumption HV (the
 ↪  vacuum value of the Vorton-Scalar sets the electroweak scale) the Fermi constant grows at the
 ↪  rate gsum while every mass stays constant; nucleosynthesis with fixed masses allows G_Fermi at
 ↪  t = 1 s to differ by 10 to 20 per cent from today's value (Scherrer and Spergel 1993), which
 ↪  gives gsum < 0.111 to 0.234 H_0, weaker than the bound from Newton's constant; (ii) in the
 ↪  model as it stands (without H2 and without HV) no mass of ordinary matter depends on the
 ↪  deflation, and the measured constancy of the ratio of the proton mass to the electron mass
 ↪  does not bound the rate; (iii) the fine-structure constant would bound it, gsum < 3.2e-18 per
 ↪  year = 4.6e-8 H_0, only if the electromagnetic field propagated along the deflating
 ↪  coordinates, which is not part of the model
[PASS] def-C6  (numerical) the author's hypothesis H2 (the superfluid state is the Higgs field) on
 ↪  a deflating spacetime: the electroweak scale is then the scale of the pair condensate, which
 ↪  depends on alpha_VS, and alpha_VS grows as exp(gsum x4). With n_EW = d ln v_EW/d ln alpha_VS
 ↪  the electron mass changes at the rate n_EW gsum and the proton mass does not; atomic clocks
 ↪  (ratio of the two masses constant to 8.0e-17 per year at two sigma) then give gsum < 8.0e-17
 ↪  per year / n_EW = 1.16e-6 H_0 / n_EW, a bound 2787 times stronger than that of lunar laser
 ↪  ranging for n_EW = 1, and one that needs no assumption about gravity; over the age of the
 ↪  universe the deflation clock can then have advanced by at most 1.1e-6/n_EW. In the
 ↪  scale-invariant ladder equation of the main project n_EW = (1 + 4 rho_c^2) atan(1/(2
 ↪  rho_c))/(8 rho_c^3) + 1/(4 rho_c^2) with rho_c^2 = (alpha_VS/alpha_c - 1)/4 (agreement with
 ↪  the numerical derivative: 2.8e-8): n_EW = 15.1, 2.57, 1.15, 0.54 for alpha_VS/alpha_c = 1.25,
 ↪  2, 3, 5; it diverges at the critical coupling and exceeds 1 for alpha_VS < 3.27 alpha_c. For
 ↪  the model itself (massive fermions, massive radial mode) n_EW has not been computed
[PASS] def-C4  (numerical) dark matter of bound pairs loses rest mass while the coupling grows,
 ↪  because the binding energy increases (four examples with alpha_VS = 0.4 to 1 today; binding
 ↪  energy at most 8.2 per cent of the rest mass). If gravity feels the deflation, the inferred
 ↪  Hubble constant changes by between -4.11 and -0.19 km/s/Mpc at the rates that lunar laser
 ↪  ranging and nucleosynthesis allow: it falls, the weaker gravity of the past outweighing the
 ↪  loss of mass. (The runs with a varying rest mass hold the redshift of last scattering and the
 ↪  helium fraction at their values without deflation, which by def-C2 underestimates a fall by
 ↪  about six per cent of itself.) If Newton's constant is held fixed and the deflation acts on
 ↪  the dark sector alone, rates between 0.03 and 1 H_0 change the inferred Hubble constant by at
 ↪  most +0.48 km/s/Mpc, for a loss of up to 7.7 per cent of the rest mass
 ↪  (data/hubble_mass_loss.csv). The loss of rest mass is the only effect of the deflation that
 ↪  raises the inferred Hubble constant; in these examples it supplies less than a tenth of the
 ↪  6.1 km/s/Mpc that separate 67.36 from the local value 73.50
[PASS] ALL  (numerical) every check above passed

PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the dark-matter estimates on
 ↪  the deflating spacetime
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks matter
6 of 6 checks passed

[PASS] def-D1  (numerical) the estimates of degeneracy and of condensation in halos contain no
 ↪  coupling: fermions with eight internal states are degenerate in the core of a dwarf galaxy
 ↪  only below 0.276 keV, bound pairs are condensed there only below 262 eV. Computed again; every
 ↪  number agrees with the file of the main project (the masses do not change along the deflation)
[PASS] def-D2  (numerical) self-interaction of bound pairs: with the coupling of today the table
 ↪  of the main project is reproduced (smallest fermion mass 918 MeV for alpha_VS = 0.1). The
 ↪  smallest mass falls as alpha_VS^(-2/3), that is, as exp(-2 tau_d/3) along the deflation. For a
 ↪  given particle the masses m and m_s are constant, hence so is the cross-section 8 pi/m_s^2 of
 ↪  the pairs; the deflation changes whether the pair exists: it is bound only after D_VS =
 ↪  alpha_VS m/m_s has grown to 1.67981
```

```
[PASS] def-D3  (numerical) the moment at which the pair becomes bound is a zero-energy resonance
 ↪  of the scattering of two fermions: the scattering length diverges at D_VS = D_c. For a
 ↪  particle with m = 918 MeV and m_s = 54.7 MeV the cross-section per mass at the relative
 ↪  velocities 10, 156 and 1000 km/s rises from 8.4e-1, 8.4e-1, 8.4e-1 cm^2/g one unit of the
 ↪  deflation clock before the binding to the peak values 1.3e7, 5.2e4, 1.3e3 cm^2/g (limit of
 ↪  unitarity at 10 km/s: 1.3e7); it exceeds 100 cm^2/g during an interval of the clock of 0.268,
 ↪  0.268, 0.258 around the moment of binding. A deflating universe in which the dark matter is
 ↪  made of these fermions passes through an epoch of strong, velocity-dependent self-interaction
[PASS] def-D4  (numerical) growth of structure on the deflating spacetime (A1, A2, A3;
 ↪  data/growth_deflation.csv). The growth depends on the closure of the perturbation equations
 ↪  at the first order in the rate. (i) At a fixed Hubble constant the amplitude of the density
 ↪  contrast today (sigma_8 for a given primordial amplitude) falls with the rate in both
 ↪  closures: d ln sigma_8/d(gsum/H_0) = -0.443 (closure "Momentum") and -0.207 (closure
 ↪  "Pressure"). (ii) The Hubble constant that keeps the angular scale of the sound horizon is
 ↪  lower, the cosmological term takes over later, and matter clusters for longer. With that
 ↪  Hubble constant the net change is d ln sigma_8/d(gsum/H_0) = +0.003 ("Momentum") and +0.239
 ↪  ("Pressure"): +0.001 and +0.077 per cent at the rate that lunar laser ranging allows, +0.01
 ↪  and +1.60 per cent at the rate that nucleosynthesis allows. The growth rate f_growth = d ln
 ↪  delta_m/d ln a today rises in both closures, by the factors 1.0021 and 1.0035 (lunar laser
 ↪  ranging), 1.0463 and 1.0773 (nucleosynthesis). Common to both closures is that, at a fixed
 ↪  angle, the amplitude does not fall and the growth rate rises; how large these changes are, and
 ↪  whether the amplitude changes at all, depends on which perturbations the agent carries that
 ↪  holds the extra scale factors, and is not a prediction of the hypotheses. (iii) The
 ↪  four-dimensional equation delta_m'' + 2 H delta_m' = 4 pi G_N(x4) rho_m delta_m, which an
 ↪  earlier version of this command integrated, is neither closure: it gives d ln
 ↪  sigma_8/d(gsum/H_0) = -0.317 at a fixed Hubble constant and +0.128 at a fixed angle
[PASS] def-D5  (numerical) drift of the quantities (table data/dm_drift.csv): the couplings change
 ↪  by 3.1e-3 of themselves over the age of the universe at the rate that lunar laser ranging
 ↪  allows and by 0.06 at the rate that nucleosynthesis allows; quantities near the threshold of
 ↪  binding move much faster (scattering length at D_VS = 1.5: n = 9.1; binding energy at D_VS =
 ↪  1.8: n = 30.1), and the gap of a weakly paired state moves as 1/lambda (n = 5 for lambda =
 ↪  0.2). The masses do not move
[PASS] def-D6  (numerical) the quanta that the deflation creates (7+1; modes that depend on a
 ↪  deflating coordinate): at the present rate that lunar laser ranging allows the temperature of
 ↪  the spectrum is 2.9e-33 K; at the largest rate that any of the three bounds of the
 ↪  cosmological part allows (gsum = 0.234 H_0) it is 2.1e-31 K, and hbar gsum is then 3.4e-37 of
 ↪  a mass of 1 keV. The creation of massive particles is suppressed by exp(-2 pi m/rate_t) with
 ↪  an exponent above 1e37 for every mass above 1 keV: no particles are created today. Creation
 ↪  matters only for an epoch in which the rate was comparable with the masses
[PASS] ALL  (numerical) every check above passed

PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the thermal history on the
 ↪  deflating spacetime
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks background
2 of 2 checks passed

[PASS] def-B1  (exact) thermal history without deflation: for the three models of the command
 ↪  `background` of the main project (fiducial; fluid dark radiation; stepped fluid) the copy of
 ↪  the background and recombination modules gives the numbers of the main project to the last
 ↪  bit: conformal age, redshifts of last scattering, of the maximum of the visibility and of the
 ↪  drag epoch, the two sound horizons, the two angles, the redshift of reionization and the
 ↪  helium fraction
```

```
[PASS] def-B2  (numerical) thermal history on the deflating spacetime (assumptions A1, A2;
 ↪  data/background_deflation.csv). Gravity was weaker in the past. At the rate that
 ↪  nucleosynthesis allows (gsum = 0.06436 H_0) and the Hubble constant of the main project,
 ↪  Newton's constant at last scattering is 0.9400 of today's, the sound horizon at last
 ↪  scattering grows from 144.449 to 149.092 Mpc, 100 theta_* from 1.03965 to 1.05252, last
 ↪  scattering moves from z_* = 1089.92 to 1088.76, the helium fraction falls from 0.24540 to
 ↪  0.24023, and the age grows from 13.814 to 13.955 Gyr; the same directions hold for the two
 ↪  models with dark radiation and at the rate of lunar laser ranging. The Hubble constant that
 ↪  restores the angle is 62.976 km/s/Mpc for the fiducial model, 62.970 with fluid dark radiation
 ↪  (N = 0.5) and 62.972 with the stepped fluid of the model (against 67.360, 67.360, 67.360
 ↪  without deflation at the same angles)
[PASS] ALL  (numerical) every check above passed

PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the spectra of the microwave
 ↪  background on the deflating spacetime
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks spectrum
8 of 8 checks passed

[PASS] def-E1  (exact) the Einstein-Boltzmann modules of the revision (copies of four modules of
 ↪  the main project with a listed set of changes, written by code/scripts/vendor_eb.py) reproduce
 ↪  the program of the main project when the rates of deflation are zero, for both closures: the
 ↪  spectra TT, TE and EE for 2 <= l <= 2500, the power spectrum of matter at 164 wavenumbers,
 ↪  sigma_8, the angle theta_* and the sound horizon agree bit for bit; and the spectra agree with
 ↪  the file data/cmb_fiducial_cl.csv of the main project to the digits written there (largest
 ↪  relative deviation 4.9e-11)
[PASS] def-E2  (numerical) spectra at a fixed angle of the sound horizon (assumptions A1, A2, A3;
 ↪  table data/cmb_deflation_summary.csv). With the full calculation of recombination on the
 ↪  deflating background the Hubble constant that keeps 100 theta_* = 1.03965 (kept to 5e-15) is
 ↪  67.149, 66.704, 65.365, 62.976, 60.368 km/s/Mpc for gsum = 0.00324, 0.01, 0.03, 0.06436, 0.1
 ↪  H_0 (H_0 = 67.36 km/s/Mpc): it falls as the rate grows. Last scattering moves from z_* =
 ↪  1089.92 to 1088.04 at the largest rate, and the helium fraction falls from 0.24540 to 0.23696.
 ↪  If z_* and the helium fraction are held at their values without deflation, the fall of the
 ↪  Hubble constant is underestimated by 5.8 to 6.0 per cent of itself. The acoustic peaks stay
 ↪  where they are (the first five move by at most 0.32 in l); their heights change by at most
 ↪  0.12 per cent at the rate of lunar laser ranging and 2.5 per cent at the rate of
 ↪  nucleosynthesis (data/cmb_deflation_peaks.csv)
[PASS] def-E3  (numerical) what remains in the spectra when the angle is kept. At the rate that
 ↪  lunar laser ranging allows (gsum = 0.00324 H_0) the temperature spectrum differs from that of
 ↪  the model without deflation by at most 2.7e-3 of itself for 30 <= l <= 2500 (0.134 of the
 ↪  cosmic variance of a single multipole); chi^2 against Planck-like mock data with all other
 ↪  parameters fixed is 1.12 (closure "Momentum") and 1.12 ("Pressure"), less than the 3.84 of a
 ↪  deviation of two standard deviations: such data cannot tell this rate from no deflation. At
 ↪  the rate that nucleosynthesis allows (gsum = 0.06436 H_0, Newton's constant at last scattering
 ↪  0.9383 of today's) the difference reaches 0.056 of the spectrum and chi^2 = 468: the microwave
 ↪  background is then the stronger test (command `forecast`)
[PASS] def-E4  (numerical) dependence on the closure of the perturbation equations ("Momentum":
 ↪  the agent that holds the extra scale factors carries no momentum density; "Pressure": it
 ↪  carries no pressure perturbation; file data/cmb_deflation_closure.csv). For 30 <= l <= 2500,
 ↪  the multipoles that the forecast uses, the temperature spectra of the two closures differ by
 ↪  at most 1.4e-3 and 2.3e-2 of the cosmic variance of a multipole at gsum = 0.00324 and 0.06436
 ↪  H_0, which is 1.4e-3 and 1.2e-3 of the effect of the deflation itself (root of the sum of
 ↪  squares); chi^2 against Planck-like mock data is 1.122 against 1.122, and 468.30 against
 ↪  468.30. The spectra at these multipoles are a prediction of assumptions A1 and A2 alone. For l
 ↪  < 30, where the potentials of today act (late integrated Sachs-Wolfe effect), the closures
 ↪  differ by up to 0.011 and 0.330 of the cosmic variance, 2.14 and 2.13 of the effect; and the
 ↪  amplitude sigma_8 of matter is 0.9762 of its value without deflation in the closure "Momentum"
 ↪  against 0.9918 in the closure "Pressure" at gsum = 0.06436 H_0 (0.99892 against 0.99969 at
 ↪  0.00324 H_0). There the result depends on the closure and is not a prediction of the hypotheses
```

```
[PASS] def-E5  (numerical) what the agent supplies (file data/cmb_deflation_gaps.csv; gsum =
 ↪  0.06436 H_0, three wavenumbers, five times). Each closure is a consistent theory: the quantity
 ↪  eta' + alpha'' that the source of the temperature uses equals the derivative of the two
 ↪  potentials along the solution (largest relative difference 1.7e-4), and the two identities
 ↪  from which the program computes the gaps hold along the solution to 8.6e-4, the accuracy of a
 ↪  numerical derivative. After recombination the gaps are of the first order in the rate: in
 ↪  units of (a gsum/Hc) times the terms of their equations, the closure "Momentum" lacks 0.31 to
 ↪  0.49 in the equation of the shear and 0.54 to 0.98 in the trace equation; the closure
 ↪  "Pressure" lacks 0.62 to 0.94 in the momentum constraint, and in the equation of the shear
 ↪  2.94 to 4.84 times (Hc/k)^2 (far inside the horizon, k > 20 Hc: 2.5e-5 to 7.0e-3): there the
 ↪  closure "Pressure" satisfies the equation of the shear as well, and is the closure "Shear".
 ↪  The ratio a gsum/Hc = gsum/H is 2.7e-6 at the first of the five times (recombination), 5.5e-2
 ↪  at the last (conformal time 13000 Mpc) and 6.9e-2 today: the closure matters only for what
 ↪  happens late
[PASS] def-E6  (numerical) numerical convergence on the deflating spacetime (gsum = 0.03 H_0, both
 ↪  closures; file data/cmb_deflation_convergence.csv): halving all tolerances and wavenumber
 ↪  steps changes the spectra by at most 1.5e-4 (TT), 2.0e-4 (TE, in units of sqrt(TT EE)), 3.6e-4
 ↪  (EE) for 100 <= l <= 2500, and by 1.2e-3, 3.3e-3, 3.5e-3 for 30 <= l <= 99; sigma_8 changes by
 ↪  2.5e-7. The program of the main project has 1.4e-4, 1.8e-4, 3.4e-4 and 1.2e-3, 4.1e-3, 5.9e-3
 ↪  for its fiducial model (its file data/cmb_convergence.csv); the requirements are those of the
 ↪  main project (5e-4 for l >= 100; 2e-3 for TT and 1e-2 for TE and EE below)
[PASS] def-E7  (numerical) the helium fraction. Newton's constant at nucleosynthesis was 0.9383 of
 ↪  today's at gsum = 0.06436 H_0; the slower expansion is equivalent to -0.379 neutrino species,
 ↪  and the helium fraction of standard nucleosynthesis falls from 0.24540 to 0.24008. If this is
 ↪  ignored (helium fraction 0.24540), the temperature spectrum changes by at most 8.6e-3 of
 ↪  itself for 30 <= l <= 2500, the Hubble constant at fixed angle by 0.045 km/s/Mpc, and chi^2
 ↪  against Planck-like mock data (other parameters fixed) from 468.3 to 335.3
[PASS] def-E8  (numerical) the growth of structure (file data/cmb_deflation_growth.csv). From a =
 ↪  0.01 to today, at k = 0.1/Mpc, the density contrast of matter in the Einstein-Boltzmann
 ↪  calculation grows by the factor that the growth equation of the same closure for pressureless
 ↪  matter gives: the ratios to the model without deflation agree to 1.3e-4 for the five rates and
 ↪  both closures, at the Hubble constant of each model and at a fixed Hubble constant; with the
 ↪  equation of the other closure they would disagree by at least 0.24 gsum/H_0. At a fixed Hubble
 ↪  constant the growth is 0.99856 (closure "Momentum") and 0.99933 ("Pressure") of its value
 ↪  without deflation at the rate of lunar laser ranging, 0.9712 and 0.9862 at the rate of
 ↪  nucleosynthesis; at a fixed angle, 1.00001 and 1.00077, and 1.0000 and 1.0160. The amplitude
 ↪  sigma_8 of the full calculation for the same primordial amplitude is 0.99892 and 0.99969, and
 ↪  0.9762 and 0.9918, of its value without deflation (0.82275): it contains, besides this growth,
 ↪  the change of the scale 8 Mpc/h with the Hubble constant and the evolution before a = 0.01
 ↪  with a weaker gravity
[PASS] ALL  (numerical) every check above passed

PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the forecast for the rate of
 ↪  deflation from mock data of the microwave background
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks forecast
5 of 5 checks passed

[PASS] def-F1  (numerical) the mock data and the Fisher matrices of the main project, computed
 ↪  with the copy of its program: the fiducial model has chi^2 = 0, and the Fisher errors of the
 ↪  six LambdaCDM parameters agree with the file data/dr_fisher.csv of the main project to 3.1e-11
 ↪  for both data sets
[PASS] def-F2  (numerical) forecast for the rate of deflation (Fisher matrix;
 ↪  data/deflation_fisher.csv). With the six LambdaCDM parameters free, Planck-like data
 ↪  determine gsum/H_0 to 0.01963 (one standard deviation; closure "Momentum") and 0.01974
 ↪  ("Pressure"); the upper bound at 95 per cent is gsum < 0.0385 H_0 = 2.65e-12 per year. Adding
 ↪  the deep survey gives 0.01056 and a bound of 0.0207 H_0. With the six parameters held fixed
 ↪  the error would be 0.00108: most of the effect of the deflation on the spectra can be absorbed
 ↪  by the other parameters. The bound lies between that of lunar laser ranging (0.00324 H_0) and
 ↪  that of nucleosynthesis (0.0644 H_0): 11.9 times weaker than the first and 1.7 times stronger
 ↪  than the second. Halving the step of the derivative changes the error by 8.2e-4
```

```
[PASS] def-F3  (numerical) the Hubble constant along the degeneracy: per unit gsum/H_0 the
 ↪  best-fit Hubble constant changes by -84.35 km/s/Mpc (Planck-like) and -82.11 km/s/Mpc (with
 ↪  the deep survey), -3.245 and -1.699 km/s/Mpc at the bound of 95 per cent: it FALLS. (At a
 ↪  fixed angle theta_* and fixed densities the command `cosmology` finds -65.24 km/s/Mpc per unit
 ↪  gsum/H_0 at small rates; in the fit the densities and the other parameters move as well.) The
 ↪  error of the Hubble constant grows from 0.561 to 1.748 km/s/Mpc when the rate is free.
 ↪  Deflation cannot raise the Hubble constant inferred from the microwave background
[PASS] def-F4  (numerical) fits at fixed rate (profile; data/deflation_profile.csv). For gsum up
 ↪  to 0.01 H_0 the minimum chi^2 is 1.00 to 1.01 times (gsum/sigma)^2 with the Fisher error
 ↪  sigma, and the shift of the best-fit Hubble constant 1.00 to 1.01 times the Fisher slope: the
 ↪  Fisher matrix describes the fits. The profile reaches chi^2 = 3.84 at gsum = 0.0376 H_0
 ↪  (Planck-like; Fisher bound 0.0385) and 0.0205 H_0 (with the deep survey; 0.0207). At the rate
 ↪  of lunar laser ranging the best fit has chi^2 = 0.027 and H_0 = 67.086 km/s/Mpc; at the rate
 ↪  of nucleosynthesis chi^2 = 11.6 (3.4 standard deviations) and H_0 = 61.70 km/s/Mpc
 ↪  (Planck-like), chi^2 = 39.8 (6.3 standard deviations) with the deep survey: the largest rate
 ↪  that nucleosynthesis allows would be excluded by such data. At every rate the best-fit Hubble
 ↪  constant is lower than without deflation
[PASS] def-F5  (numerical) accuracy of the fits. (i) Recomputing the derivatives of the spectra at
 ↪  the best fit for gsum = 0.02 H_0 and continuing the fit changes chi^2 by at most 0.000 and the
 ↪  Hubble constant by at most 0.001 km/s/Mpc (data/deflation_chord_check.csv). (ii) The closure:
 ↪  refitting with the closure "Pressure" at gsum = 0.01 H_0 and at the rate of nucleosynthesis
 ↪  changes chi^2 by at most 0.9 per cent of itself (or of 1, where it is smaller than 1) and the
 ↪  best-fit Hubble constant by at most 0.0005 km/s/Mpc: the forecast, which uses the multipoles l
 ↪  >= 30, does not depend on the closure
[PASS] ALL  (numerical) every check above passed

PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the forecast for the
 ↪  dark-radiation families of the main project on the deflating spacetime
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks families
4 of 4 checks passed

[PASS] def-N1  (numerical) the forecast of the main project for its dark-radiation families,
 ↪  repeated on the deflating spacetime at the rate that lunar laser ranging allows (gsum =
 ↪  0.00324 H_0). The data are the spectra of the fiducial model without deflation (chi^2 = 0 for
 ↪  that model). The deflating model without extra radiation fits them with chi^2 = 0.027 and H_0
 ↪  = 67.086 km/s/Mpc (Planck-like), chi^2 = 0.094 and H_0 = 67.094 km/s/Mpc with the deep survey
[PASS] def-N2  (numerical) the families on the deflating background against the main project (its
 ↪  files data/dr_fisher.csv and data/dr_profiles.csv): the Fisher errors of N agree to 0.2 per
 ↪  cent (Planck-like) and 0.3 per cent (with the deep survey) for the 7 families. In the fits at
 ↪  fixed N = 0.1 to 1 the best-fit Hubble constant is lower than in the main project by 0.274 to
 ↪  0.279 km/s/Mpc (Planck-like) and 0.266 to 0.270 km/s/Mpc (deep survey); the model without
 ↪  extra radiation is lower by 0.274 and 0.266 km/s/Mpc. The minimum chi^2 is LOWER than in the
 ↪  main project, by up to 3.11 and 8.06 at N = 1 (check def-N4 explains why). Deflation at this
 ↪  rate lowers every Hubble constant of the main project by about a quarter of a km/s/Mpc
[PASS] def-N3  (numerical) accuracy of the fits on the deflating background: recomputing the
 ↪  derivatives of the spectra at the N = 0.5 best fit of each family and continuing the fit
 ↪  changes chi^2 by at most 0.054 and the Hubble constant by at most 0.050 km/s/Mpc
 ↪  (data/dr_deflation_chord_check.csv). The scan of the transition redshift of the stepped fluid
 ↪  of the model at N = 0.4 (data/dr_deflation_zt_scan.csv) has its best fit at z_t = 10^3.00
 ↪  (chi^2 = 0.944); its Hubble constants are lower than those of the main project by 0.272 to
 ↪  0.277 km/s/Mpc
```

```
[PASS] def-N4  (numerical) the rate of deflation and the density N of extra radiation as free
 ↪  parameters together (Fisher matrices with eight parameters at the fiducial model;
 ↪  data/dr_deflation_joint.csv). (i) With the rate held at zero the errors of N are those of the
 ↪  main project (to 2.6e-11), and the marginal Fisher matrix of the pair reproduces the errors of
 ↪  each parameter alone (to 4.4e-16). (ii) N and the rate are positively correlated for every
 ↪  family: correlation coefficient 0.37 to 0.89 (Planck-like) and 0.28 to 0.93 (with the deep
 ↪  survey). Extra radiation makes the early universe expand faster, the weaker gravity of the
 ↪  deflating spacetime makes it expand more slowly, and the two compensate in part. With both
 ↪  free the error of gsum/H_0 grows from 0.0196 to between 0.0211 and 0.0429 (Planck-like), and
 ↪  the error of N by the same factor. (iii) At the largest rate that lunar laser ranging allows
 ↪  the best-fitting density is N = 0.005 to 0.023 instead of zero (chi^2 = 0.006 to 0.023 instead
 ↪  of 0.027). The difference between the minimum chi^2 of the fits on the deflating background
 ↪  and those of the main project at N = 0.1 and 0.2 is 0.99 to 1.01 times the cross term of the
 ↪  Fisher matrix. (iv) For every family and both data sets the largest Hubble constant with chi^2
 ↪  <= 3.84 (N >= 0, gsum >= 0) is reached at ZERO rate (a rise of 0.78 to 2.95 km/s/Mpc,
 ↪  Planck-like): the deflation lowers the Hubble constant faster than it makes room for extra
 ↪  radiation, and it cannot help the dark radiation to raise the Hubble constant
[PASS] ALL  (numerical) every check above passed
```

```
PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of superfluid dark matter on the
 ↪  deflating spacetime
generated by: deflation_cjt_ks (Rust crate spacetimes-deflation-cjt-ks, version 0.1.0; PROJECT
 ↪  spacetimes, Revision_Deflation_CJT_and_Kohn-Sham); command: deflation_cjt_ks superfluid
2 of 2 checks passed

[PASS] def-U1  (numerical) superfluid dark matter without deflation: the copy of the
 ↪  Einstein-Boltzmann program reproduces the files data/sfdm_summary.csv and
 ↪  data/sfdm_bounds.csv of the main project (half-mode wavenumbers of 15 models; limits on the
 ↪  fermion mass and on the sound speed of a condensate) to the digits written there (largest
 ↪  relative deviations 3.4e-11 and 3.1e-11)
[PASS] def-U2  (numerical) superfluid dark matter on the deflating spacetime
 ↪  (data/sfdm_deflation_summary.csv, sfdm_deflation_bounds.csv). The wavenumber below which the
 ↪  first sound of the fluid suppresses structure (half-mode) changes by -0.164 to -0.149 per cent
 ↪  at the rate that lunar laser ranging allows and by -3.28 to -3.04 per cent at the rate that
 ↪  nucleosynthesis allows (15 models; each background at the Hubble constant that keeps the angle
 ↪  theta_*). The limits that follow from the Lyman-alpha bound on thermal warm dark matter of 5.3
 ↪  keV move from m > 1.514 keV (fermions, weak pairing) and c_s0 < 1.645e-11 c (condensate;
 ↪  Thomas-Fermi radius < 0.365 pc) without deflation to m > 1.516 keV, c_s0 < 1.641e-11 c (0.364
 ↪  pc) at the rate of lunar laser ranging and m > 1.555 keV, c_s0 < 1.565e-11 c (0.347 pc) at the
 ↪  rate of nucleosynthesis. The suppression is set while the fluid still has pressure, long
 ↪  before today, when gravity was weaker: the Jeans wavenumber is proportional to the square root
 ↪  of Newton's constant, which in the early universe was 0.9969 and 0.9383 of today's value at
 ↪  the two rates (square root minus one: -0.154 and -3.13 per cent). The two closures give the
 ↪  same half-mode wavenumbers to 1.5e-5
[PASS] ALL  (numerical) every check above passed
```

5.11 Independent verification of the geometry, the Dirac field, the modes and the cosmology

```
PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: independent verification of the
 ↪  deflating spacetimes and of the model without contact terms (computer algebra and
 ↪  general-purpose numerical tools)
generated by: code/scripts/verify_deflating_independent.py (sympy 1.14.0, scipy, numpy 2.5.3,
 ↪  mpmath 1.3.0)
43 of 43 checks passed

[PASS] I44d-G1  4+4, arbitrary rates and constants (computer algebra), metric in the author's sign
 ↪  convention with g_tt = -b_t^2: the Christoffel symbols obtained from the metric by
 ↪  differentiation are Gamma^t_(4t) = -rate_t and Gamma^4_(tt) = -gs_tt rate_t b_t^2 = + rate_t
 ↪  b_t^2; all others vanish
```

```
[PASS] I44d-G2  4+4: Ricci tensor R^4_4 = gsq, R^t_t = rate_t gsum, zero along x0..x3; curvature
 ↪  scalar gsq + gsum^2; Einstein tensor -gprod (component 44), rate_t gsum - (gsq + gsum^2)/2
 ↪  (components tt), -(gsq + gsum^2)/2 (x0..x3), with gsq = g55^2 + g66^2 + g77^2, gsum = g55 +
 ↪  g66 + g77, gprod = g55 g66 + g55 g77 + g66 g77: identities in the six symbols g55, g66, g77,
 ↪  const5, const6, const7 (mixed components; the same on both spacetimes in the author's sign
 ↪  convention)
[PASS] I44d-G3  4+4: Riemann tensor in the orthonormal frame, R^t_(4t4) = -rate_t^2 and R^t_(sts)
 ↪  = gs_ss rate_t rate_s = -rate_t rate_s; no component carries an index 0, 1, 2 or 3
[PASS] I44d-G4  4+4: the Lie derivative of the metric vanishes identically along 14 vector fields
 ↪  (7 translations, 6 rotations of x0..x3, the scaling d_4 + sum rate_t xt d_t); it does not
 ↪  vanish for the rotation of x0 and x5 nor for the boost of x4 and x1; for g55 = g66 = g77 it
 ↪  vanishes in addition along the three rotations of x5, x6, x7 and three special vector fields
 ↪  (20 in all)
[PASS] I44d-G5  4+4: the list of Killing vectors is complete. The matrices of the Lorentz algebra
 ↪  of the tangent space that leave the Riemann tensor unchanged form a space whose dimension,
 ↪  plus 8, bounds the number of Killing vectors from above; exact ranks for (g55, g66, g77) = (1,
 ↪  2, 3): 6 + 8 = 14; (1, 2, 4): 6 + 8 = 14; (3, 5, 7): 6 + 8 = 14; (2, 2, 5): 7 + 8 = 15; (5, 2,
 ↪  2): 7 + 8 = 15; (1, 1, 1): 12 + 8 = 20; (7, 7, 7): 12 + 8 = 20: the bounds 14 (three different
 ↪  rates), 15 (two equal) and 20 (three equal) equal the numbers of Killing vectors exhibited
[PASS] I44d-L1  4+4, Einstein-Gauss-Bonnet gravity (metric in the author's sign convention;
 ↪  standard formula of the Lanczos tensor, computer algebra): Lanczos^a_a = -4 g55 g66 g77 gsum
 ↪  for a = 0..3 and Lanczos^4_4 = Lanczos^t_t = 0, identically in the rates and constants; for
 ↪  g55 = g66 = g77 the deflating spacetime solves G^mu_nu + Lambda_8 delta^mu_nu + alpha_GB
 ↪  Lanczos^mu_nu = 0 with Lambda_8 = 3 g55^2 and alpha_GB = -1/(4 g55^2); for unequal rates it
 ↪  does not, because G^5_5 - G^6_6 = +-(g55 - g66) gsum while the Lanczos tensor vanishes there
[PASS] I44d-C1  4+4: the coordinate matrices gamma^mu = e_A^mu gamma^A satisfy gamma^mu gamma^nu +
 ↪  gamma^nu gamma^mu = 2 c g^(mu nu) with c = +1 and the inverse metric of the author's sign
 ↪  convention, for all 64 pairs, identically in b_5, b_6, b_7
[PASS] I44d-C2  4+4: the spin connection Omega_mu = (c/4) omega_(mu A B) gamma^A gamma^B computed
 ↪  from the tetrad postulate (indices lowered with the metric) is antisymmetric in its frame
 ↪  indices and equals Omega_t = +(1/2) rate_t b_t gamma^t gamma^4 (frame matrices) for t = 5, 6,
 ↪  7, zero otherwise
[PASS] I44d-C3  4+4: d_mu gamma^nu + Gamma^nu_(mu lam) gamma^lam + [Omega_mu, gamma^nu] = 0 for
 ↪  all 64 pairs
[PASS] I44d-C4  4+4: sum_mu gamma^mu Omega_mu = -(gsum/2) gamma^4: the Dirac operator is gamma^4
 ↪  (d_4 - gsum/2) + sum_a gamma^a d_a + sum_t gamma^t d_t / b_t
[PASS] I44d-C5  4+4: (1/2) gamma^mu gamma^nu (d_mu Omega_nu - d_nu Omega_mu + [Omega_mu,
 ↪  Omega_nu]) = -(c R/4) 1 with c = +1 and R = gsq + gsum^2
[PASS] I44d-D1  4+4: the generator K = -m gamma^4 + i k_a gamma^4 gamma^a + i (k_t/b_t) gamma^4
 ↪  gamma^t of the evolution of the weighted field satisfies K^2 = (k_d^2 - E_perp^2) 1
 ↪  identically in the eight symbols m, k_0..k_3, k_5, k_6, k_7 and in b_5, b_6, b_7
[PASS] I71d-G1  7+1, arbitrary rates and constants (computer algebra), metric in the author's sign
 ↪  convention with g_tt = +b_t^2: the Christoffel symbols obtained from the metric by
 ↪  differentiation are Gamma^t_(4t) = -rate_t and Gamma^4_(tt) = -gs_tt rate_t b_t^2 = - rate_t
 ↪  b_t^2; all others vanish
[PASS] I71d-G2  7+1: Ricci tensor R^4_4 = gsq, R^t_t = rate_t gsum, zero along x0..x3; curvature
 ↪  scalar gsq + gsum^2; Einstein tensor -gprod (component 44), rate_t gsum - (gsq + gsum^2)/2
 ↪  (components tt), -(gsq + gsum^2)/2 (x0..x3), with gsq = g55^2 + g66^2 + g77^2, gsum = g55 +
 ↪  g66 + g77, gprod = g55 g66 + g55 g77 + g66 g77: identities in the six symbols g55, g66, g77,
 ↪  const5, const6, const7 (mixed components; the same on both spacetimes in the author's sign
 ↪  convention)
[PASS] I71d-G3  7+1: Riemann tensor in the orthonormal frame, R^t_(4t4) = -rate_t^2 and R^t_(sts)
 ↪  = gs_ss rate_t rate_s = +rate_t rate_s; no component carries an index 0, 1, 2 or 3
[PASS] I71d-G4  7+1: the Lie derivative of the metric vanishes identically along 14 vector fields
 ↪  (7 translations, 6 rotations of x0..x3, the scaling d_4 + sum rate_t xt d_t); it does not
 ↪  vanish for the rotation of x0 and x5 nor for the boost of x4 and x1; for g55 = g66 = g77 it
 ↪  vanishes in addition along the three rotations of x5, x6, x7 and three special vector fields
 ↪  (20 in all)
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[PASS] I71d-G5  7+1: the list of Killing vectors is complete. The matrices of the Lorentz algebra
↪  of the tangent space that leave the Riemann tensor unchanged form a space whose dimension,
↪  plus 8, bounds the number of Killing vectors from above; exact ranks for (g55, g66, g77) = (1,
↪  2, 3): 6 + 8 = 14; (1, 2, 4): 6 + 8 = 14; (3, 5, 7): 6 + 8 = 14; (2, 2, 5): 7 + 8 = 15; (5, 2,
↪  2): 7 + 8 = 15; (1, 1, 1): 12 + 8 = 20; (7, 7, 7): 12 + 8 = 20: the bounds 14 (three different
↪  rates), 15 (two equal) and 20 (three equal) equal the numbers of Killing vectors exhibited
[PASS] I71d-L1  7+1, Einstein-Gauss-Bonnet gravity (metric in the author's sign convention;
↪  standard formula of the Lanczos tensor, computer algebra): Lanczos^a_a = -4 g55 g66 g77 gsum
↪  for a = 0..3 and Lanczos^4_4 = Lanczos^t_t = 0, identically in the rates and constants; for
↪  g55 = g66 = g77 the deflating spacetime solves G^mu_nu + Lambda_8 delta^mu_nu + alpha_GB
↪  Lanczos^mu_nu = 0 with Lambda_8 = 3 g55^2 and alpha_GB = -1/(4 g55^2); for unequal rates it
↪  does not, because G^5_5 - G^6_6 = +-(g55 - g66) gsum while the Lanczos tensor vanishes there
[PASS] I71d-C1  7+1: the coordinate matrices gamma^mu = e_A^mu gamma^A satisfy gamma^mu gamma^nu +
↪  gamma^nu gamma^mu = 2 c g^(mu nu) with c = -1 and the inverse metric of the author's sign
↪  convention, for all 64 pairs, identically in b_5, b_6, b_7
[PASS] I71d-C2  7+1: the spin connection Omega_mu = (c/4) omega_(mu A B) gamma^A gamma^B computed
↪  from the tetrad postulate (indices lowered with the metric) is antisymmetric in its frame
↪  indices and equals Omega_t = +(1/2) rate_t b_t gamma^t gamma^4 (frame matrices) for t = 5, 6,
↪  7, zero otherwise
[PASS] I71d-C3  7+1: d_mu gamma^nu + Gamma^nu_(mu lam) gamma^lam + [Omega_mu, gamma^nu] = 0 for
↪  all 64 pairs
[PASS] I71d-C4  7+1: sum_mu gamma^mu Omega_mu = -(gsum/2) gamma^4: the Dirac operator is gamma^4
↪  (d_4 - gsum/2) + sum_a gamma^a d_a + sum_t gamma^t d_t / b_t
[PASS] I71d-C5  7+1: (1/2) gamma^mu gamma^nu (d_mu Omega_nu - d_nu Omega_mu + [Omega_mu,
↪  Omega_nu]) = -(c R/4) 1 with c = -1 and R = gsq + gsum^2
[PASS] I71d-D1  7+1: the Hamiltonian h = m beta + p_j alpha^j + (k_t/b_t) alpha^t of the weighted
↪  field is symmetric and satisfies h^2 = (E_perp^2 + k_d^2) 1 identically in the eight symbols
↪  m, p_0..p_3, k_5, k_6, k_7 and in b_5, b_6, b_7
[PASS] I71d-M1  7+1: psihat = sqrt(u_d) [J_(1/2 + i E_perp/rate_t)(u_d) w + i J_(-1/2 + i
↪  E_perp/rate_t)(u_d) alpha^d w], with h_perp w = E_perp w and u_d = k_d/rate_t, satisfies i d
↪  psihat/dx4 = h psihat (Bessel functions of mpmath; relative residual 1.1e-15 at four values of
↪  x4; mode m = 1, p = (0, 0.3, -0.4, 0.5), k_5 = 0.7, g55 = 2)
[PASS] I71d-M2  7+1: the number of particles created in a mode that starts empty, integrated with
↪  the Runge-Kutta method DOP853 of scipy and projected on the negative-energy states at k_d = 80
↪  E_perp, equals 1/(exp(2 pi E_perp/rate_t) + 1) for all eight internal states in three modes
↪  (largest relative deviation 2.2e-05)
[PASS] I71d-M3  7+1, equal rates 1.3 and a mode that depends on all three deflating coordinates
↪  (constants 1, 0.7, 2): the same law holds (relative deviation 1.4e-05)
[PASS] I44d-M1  4+4: psihat = sqrt(u_d) [K_(1/2 - i E_perp/rate_t)(u_d) w - K_(1/2 + i
↪  E_perp/rate_t)(u_d) i gamma^4 gamma^d w], with K_perp w = i E_perp w and u_d = k_d/rate_t,
↪  satisfies d psihat/dx4 = K psihat (modified Bessel functions of mpmath; relative residual
↪  2.1e-15); the eigenspace of K_perp with the eigenvalue i E_perp has dimension 8, so that there
↪  are eight such solutions; they fall as exp(-u_d)
[PASS] I44d-M2  4+4: between early x4 and the value at which the WKB exponent is S_wkb = 7.63 the
↪  bounded solution falls by exp(-S_wkb) times a constant: ln(amplitude early/late) - S_wkb =
↪  0.33534, the exact value (1/2) ln 2 - (1/2) ln(1 + exp(-2 pi E_perp/rate_t)) = 0.33654 (the
↪  difference, 1.2e-03, is the correction of relative order 1/u_d to the WKB form)
[PASS] Idef-B1  7+1: a scalar mode that depends on one deflating coordinate and starts empty ends
↪  with 1/(exp(2 pi varpi_b/rate_t) - 1) quanta (DOP853 of scipy, three modes, largest relative
↪  deviation 4.6e-05)
[PASS] I44d-V1  4+4: the two zero-energy equations of a vortex line of unit winding have a
↪  4-dimensional space of constant spinor solutions (singular values, numpy); on it alpha^3 has
↪  the eigenvalues -1 -1 +1 +1 and equals the central matrix P_c = i alpha^1 alpha^2 alpha^3 chi:
↪  four zero modes, two for each direction along the line
[PASS] I71d-V1  7+1: the two zero-energy equations of a vortex line of unit winding have a
↪  4-dimensional space of constant spinor solutions (singular values, numpy); on it alpha^3 has
↪  the eigenvalues -1 -1 +1 +1 and equals the central matrix P_c = i alpha^1 alpha^2 alpha^3 chi:
↪  four zero modes, two for each direction along the line
[PASS] Idef-C1  cosmology: an independent integration (scipy) of the expansion with three equal
↪  rates gives, at fixed 100 theta_* = 1.03958 and with last scattering held at z_* = 1089.92,
↪  the Hubble constant for gsum = 0.01 H_0: 66.743 km/s/Mpc (Rust program: 66.743); for gsum =
↪  0.1 H_0: 60.773 km/s/Mpc (Rust program: 60.773)
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[PASS] Idef-C2  recombination on the deflating background, independently: Peebles' three-level
↪  atom (scipy; not the recombination program of the main project) gives z_* = 1090.84 without
↪  deflation (Rust program: 1089.92) and 100 theta_* = 1.03905 (1.03965). For gsum = 0.03 H_0:
↪  last scattering moves by -0.551 (Rust: -0.550), the helium fraction by -0.00243 (-0.00243);
↪  Hubble constant 65.365 km/s/Mpc (Rust: 65.365), and 65.483 (65.483) if last scattering and
↪  helium are held fixed; for gsum = 0.1 H_0: last scattering moves by -1.882 (Rust: -1.877), the
↪  helium fraction by -0.00844 (-0.00844); Hubble constant 60.367 km/s/Mpc (Rust: 60.368), and
↪  60.773 (60.773) if last scattering and helium are held fixed. The shifts of last scattering
↪  agree to 0.3 per cent, the changes of the Hubble constant at the fixed angle to 0.01 per cent
↪  (full) and 0.00 per cent (last scattering and helium held fixed): the larger fall of the
↪  Hubble constant in the full calculation is confirmed by a calculation that shares no
↪  recombination code with it
[PASS] I44d-P1  4+4: Einstein tensor of ordinary space to first order in a perturbation (plane
↪  wave along x3, synchronous gauge, conformal time, scale factors of x5, x6, x7 unperturbed;
↪  arbitrary rates; sympy). With Hc = a'/a, He = Hc - a gsum/2 and the signs chosen so that the
↪  time-time component is -(8 pi G) rho: background 3 Hc^2 - 3 a gsum Hc + a^2 gprod (time-time)
↪  and -(2 Hc' + Hc^2 - 2 a gsum Hc + a^2 (gsq + gprod)) (space-space); G^4_4: (2 k^2 eta - He
↪  h') cos(k x3); G^4_3: 2 k eta' sin(k x3); trace of the space-space part: (2 k^2 eta - 2 He h'
↪  - h'') cos(k x3); G^3_3 - G^1_1: (1/2)[(h'' + 6 eta'') + 2 He (h' + 6 eta') - 2 k^2 eta] cos(k
↪  x3); all divided by a^2. These are the equations (E), (M), (T), (S) of assumption A3: those of
↪  Ma and Bertschinger with Hc replaced by He in three of them. The components G^0_0 and G^5_5
↪  along the extra coordinates are perturbed as well: an agent must hold those coordinates
[PASS] I71d-P1  7+1: Einstein tensor of ordinary space to first order in a perturbation (plane
↪  wave along x3, synchronous gauge, conformal time, scale factors of x5, x6, x7 unperturbed;
↪  arbitrary rates; sympy). With Hc = a'/a, He = Hc - a gsum/2 and the signs chosen so that the
↪  time-time component is -(8 pi G) rho: background 3 Hc^2 - 3 a gsum Hc + a^2 gprod (time-time)
↪  and -(2 Hc' + Hc^2 - 2 a gsum Hc + a^2 (gsq + gprod)) (space-space); G^4_4: (2 k^2 eta - He
↪  h') cos(k x3); G^4_3: 2 k eta' sin(k x3); trace of the space-space part: (2 k^2 eta - 2 He h'
↪  - h'') cos(k x3); G^3_3 - G^1_1: (1/2)[(h'' + 6 eta'') + 2 He (h' + 6 eta') - 2 k^2 eta] cos(k
↪  x3); all divided by a^2. These are the equations (E), (M), (T), (S) of assumption A3: those of
↪  Ma and Bertschinger with Hc replaced by He in three of them. The components G^0_0 and G^5_5
↪  along the extra coordinates are perturbed as well: an agent must hold those coordinates
[PASS] Idef-P2  background on the deflating spacetime (sympy; four species: pressureless matter,
↪  radiation, a fluid with a variable equation of state, cosmological term): the expression of
↪  the program for the derivative of the conformal Hubble rate is the derivative of the equation
↪  of the expansion Hc^2 - a gsum Hc + a^2 gprod/3 = (sum of 8 pi G_N a^2 rho)/3 with (8 pi G_N
↪  a^2 rho)' = (a gsum - (1 + 3 w) Hc) 8 pi G_N a^2 rho; the quantity B = He' - He^2 + 4 pi G_N
↪  a^2 (rho + P) equals a gsum [ (1/6) sum of 8 pi G_N a^2 rho (1 - 3 w) + (a gsum)^2/4 - a^2
↪  gprod/3 ] / (2 He), which vanishes without deflation; and the expression of the program for B'
↪  is its derivative
[PASS] Idef-P3  the two identities between the gaps (sympy; pressureless matter, radiation with
↪  anisotropic stress, a fluid with variable equation of state and its own sound speed,
↪  cosmological term; arbitrary rates). With D, T, DP, Pi the sums of 8 pi G_N a^2 times the
↪  density, momentum, pressure and shear perturbations, eta eliminated by the energy constraint
↪  (E), and the equations of motion of matter of Ma and Bertschinger: momentum gap = 2 k^2 eta' -
↪  T, trace gap = h'' + 2 He h' - 2 k^2 eta + 3 DP, shear gap = alpha' + 2 He alpha - eta + (3/2)
↪  Pi/k^2 with alpha = (h' + 6 eta')/(2 k^2) obey  momentum gap = He (trace gap) + h' B - (a
↪  gsum/2)(D - 3 DP)  and  2 k^2 (shear gap) = (trace gap) + (3/k^2)[(momentum gap)' + 2 He
↪  (momentum gap)].  Without deflation B = 0 and a gsum = 0: the three gaps then vanish together,
↪  as in general relativity; with deflation they cannot all vanish
[PASS] Idef-P4  growth of pressureless matter (sympy). From (E) with the momentum constraint:
↪  delta'' + [(2 He' + g)/(2 He)] delta' = [(Hc - a gsum)/(2 He)] g delta (closure "Momentum");
↪  from (E) with the trace equation: delta'' + He delta' = g delta/2, that is, in the time x4,
↪  delta'' + (2 H - gsum/2) delta' = 4 pi G_N rho_m delta (closure "Pressure"); g = 8 pi G_N a^2
↪  rho_m, primes in conformal time. Both hold for every wavenumber. For matter alone and three
↪  equal rates, to the first order in r = gsum/H the growth rate is d ln delta/d ln a = 1 + c r
↪  with c = -9/16 ("Pressure"), -13/16 ("Momentum"), and -11/16 for the four-dimensional equation
↪  delta'' + 2 H delta' = 4 pi G_N rho_m delta, which is neither
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[PASS] Idef-P5  growth of pressureless matter from a = 0.01 to today at a fixed Hubble constant,
 ↪  relative to no deflation: an independent integration (scipy) of the three growth equations
 ↪  agrees with the Rust program (data/growth_deflation.csv) to 6.3e-07 for the rates up to 0.1
 ↪  H_0; at gsum = 0.1 H_0: closure "Momentum" 0.95525 (Rust 0.95525); closure "Pressure" 0.97830
 ↪  (Rust 0.97830); four-dimensional equation 0.96740 (Rust 0.96740)
[PASS] Idef-P6  the three closures as systems of equations for the density contrast of
 ↪  pressureless matter and the metric (matter and cosmological term; implicit integrator of
 ↪  scipy; gsum = 0.1 H_0, Hubble constant fixed; growth from a = 0.01 to today relative to no
 ↪  deflation). Closure "Momentum": 0.955197; closure "Pressure": 0.978268; neither depends on
 ↪  the wavenumber (to 7e-14 for k = 0.1, 0.03, 0.01, 0.003/Mpc). Closure "Shear" (energy
 ↪  constraint and equation of the shear; its equations are stiff, because the momentum density of
 ↪  the agent relaxes at the rate k^2/(3 Hc)): 0.978266 at k = 0.1/Mpc, 0.978251 at k = 0.03/Mpc,
 ↪  0.978119 at k = 0.01/Mpc, 0.976788 at k = 0.003/Mpc; it differs from the closure "Pressure" by
 ↪  1.5e-06, 1.7e-05, 1.5e-04, 1.5e-03, that is by 0.31, 0.30, 0.30, 0.27 times (H_0/k)^2: far
 ↪  inside the horizon the closure "Shear" is the closure "Pressure"
[PASS] Idef-P7  the Einstein-Boltzmann program of the revision against the growth equations: for
 ↪  the five rates of data/cmb_deflation_growth.csv and both closures, the growth of the density
 ↪  contrast of matter at k = 0.1/Mpc from a = 0.01 to today in the full calculation (photons,
 ↪  neutrinos, baryons, cold dark matter), relative to no deflation and at a fixed Hubble
 ↪  constant, agrees with the independent integration (scipy) of the growth equation of the
 ↪  closure to 1.3e-04
[PASS] Idef-P8  the agent in the unperturbed universe (sympy; arbitrary rates): if the time-time
 ↪  Einstein equation holds with matter alone (assumption A2: the agent that keeps the rates
 ↪  constant has no energy density), the space-space equation and the conservation of matter
 ↪  require the agent to have the pressure p_agent in ordinary space with 8 pi G_N p_agent =
 ↪  -(gsum/H) dH/dx4 - 2 gsum H + 2 gprod - gsum gprod/(3 H). To the first order in the rates this
 ↪  is -gsum H/2 in an era of matter: a tension, of the relative size gsum/H
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5.12 Independent verification of the self-interaction, of the Kohn–Sham and CJT calculations, of the pairing and of the gravity

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PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: independent verification of the
 ↪  self-interaction, of the Kohn-Sham and CJT calculations, of the pairing and of the gravity
 ↪  with Kohn-Sham matter (Grassmann algebra, computer algebra and general-purpose numerical
 ↪  tools)
generated by: code/scripts/verify_cjt_ks_independent.py (python3 with numpy, scipy, sympy)
37 of 37 checks passed

[PASS] Icks-S1  the Fierz matrix F_rs = (1/16) sum_l (-1)^(rs-l) C(s,l) C(8-s,r-l) (exact rational
 ↪  arithmetic of sympy): F^2 = 1, eigenvalue +1 five times and -1 four times, so that (1 + F) I =
 ↪  0 are five independent relations among the nine invariants and four invariants remain; solved
 ↪  for I_2 .. I_6 the relations are I_2 = -3 I_0 - I_1 - I_7 - I_8, I_3 = -4 I_0 - I_7 + 4 I_8,
 ↪  I_4 = -5 I_0 - 5 I_8, I_5 = -4 I_0 - I_1 + 4 I_8, I_6 = -I_0 + I_1 + I_7 - 3 I_8, the table
 ↪  selfcoupling_reduction.csv of the Rust program
[PASS] Icks-S2-71s  7+1: each of the 256 matrices Theta^K (products of the Dirac matrices with the
 ↪  phases of the model) squares to eps_K = +-1, with eps_K = eta_K (eta_K: product of the squares
 ↪  of the matrices in K) as in the table of the Rust program; every vertex matrix M_K = A Theta^K
 ↪  is Hermitian, so that every bilinear psibar Theta^K psi is real (largest deviation 0.0e+00);
 ↪  Tr(Theta_K Theta^L) = 16 delta_KL (0.0e+00): the 256 matrices are a basis of all 16 x 16
 ↪  matrices
[PASS] Icks-S3-71s  7+1: the Fierz matrix computed from the matrices, F_rs = (1/256) sum_(|K| = r)
 ↪  Tr(Theta_K Theta_L Theta^K Theta^L) for an L of rank s (the completeness relation of the
 ↪  basis; the same for every L of the rank, checked for six of each rank: 0.0e+00), equals the
 ↪  closed form (largest deviation 0.0e+00)
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[PASS] Icks-S4-71s  7+1: the nine invariants evaluated on Grassmann-valued fields (psi and psi*
 ↪  independent random complex combinations of 6 anticommuting generators; each I_r is then an
 ↪  element of degree four with 15 coefficients; three random fields) obey (1 + F) I = 0 (largest
 ↪  residual 6.3e-16 of the largest coefficient) and the five relations of Icks-S1 (4.9e-16); the
 ↪  rank of the nine invariants over all samples is 4 (singular values 4.28e-01 and 2.68e-16 on
 ↪  either side of the gap): exactly four invariants are independent. (The products of the
 ↪  Grassmann-valued components, done one by one with the Grassmann class, give the same
 ↪  bilinears: 7.2e-15.)
[PASS] Icks-S5-71s  7+1: Hartree-Fock couplings by Wick's theorem, <:(psi^dagger M psi)^2:> = Tr(M
 ↪  n)^2 - Tr(M n M n). The density matrix n of a Fermi sea, summed from the eigenvectors of
 ↪  positive energy of h(k) = alpha.k + beta M for 40 random pairs k, -k, has the form (rho +
 ↪  rho_S beta)/16 (deviation 1.1e-17) with rho_S = sum M/w. The energy of each of the nine
 ↪  channels, E_HF = (1/2)(G_1 rho^2 + G_0 rho_S^2), reproduces the coefficients of the table
 ↪  selfcoupling_hartree_fock.csv of the Rust program (largest deviation 2.1e-13)
[PASS] Icks-S6-71s  7+1: the contact interaction among the 28 pairs of two particles at rest (the
 ↪  eight states with beta = +1), diagonalized numerically for nine random couplings: seven pairs
 ↪  (among them the pair along the hidden axis) have the coupling g_0eff + g_1eff + 5 g_7eff and
 ↪  twenty-one have g_0eff + g_1eff - 3 g_7eff (largest deviation 6.4e-14)
[PASS] Icks-S7-71s  7+1: under the finite chiral rotation psi -> exp(0.37 i chi) psi (chi
 ↪  Hermitian, chi^2 = 1) of Grassmann-valued fields the interaction sum_r g_r I_r is unchanged
 ↪  when g_0eff + g_8eff = g_0 - 4 g_2 - 10 g_4 - 4 g_6 + g_8 = 0 (relative change 5.8e-16) and
 ↪  changes for generic couplings (1.43)
[PASS] Icks-S2-44q  4+4, quantized-sector form: each of the 256 matrices Theta^K (products of the
 ↪  Dirac matrices with the phases of the model) squares to eps_K = +-1, with eps_K = eta_K
 ↪  (eta_K: product of the squares of the matrices in K) as in the table of the Rust program;
 ↪  every vertex matrix M_K = A Theta^K is Hermitian, so that every bilinear psibar Theta^K psi is
 ↪  real (largest deviation 0.0e+00); Tr(Theta_K Theta^L) = 16 delta_KL (0.0e+00): the 256
 ↪  matrices are a basis of all 16 x 16 matrices
[PASS] Icks-S3-44q  4+4, quantized-sector form: the Fierz matrix computed from the matrices, F_rs
 ↪  = (1/256) sum_(|K| = r) Tr(Theta_K Theta_L Theta^K Theta^L) for an L of rank s (the
 ↪  completeness relation of the basis; the same for every L of the rank, checked for six of each
 ↪  rank: 0.0e+00), equals the closed form (largest deviation 0.0e+00)
[PASS] Icks-S4-44q  4+4, quantized-sector form: the nine invariants evaluated on Grassmann-valued
 ↪  fields (psi and psi* independent random complex combinations of 6 anticommuting generators;
 ↪  each I_r is then an element of degree four with 15 coefficients; three random fields) obey (1
 ↪  + F) I = 0 (largest residual 9.4e-16 of the largest coefficient) and the five relations of
 ↪  Icks-S1 (7.6e-16); the rank of the nine invariants over all samples is 4 (singular values
 ↪  5.67e-01 and 2.73e-16 on either side of the gap): exactly four invariants are independent.
 ↪  (The products of the Grassmann-valued components, done one by one with the Grassmann class,
 ↪  give the same bilinears: 7.9e-15.)
[PASS] Icks-S5-44q  4+4, quantized-sector form: Hartree-Fock couplings by Wick's theorem,
 ↪  <:(psi^dagger M psi)^2:> = Tr(M n)^2 - Tr(M n M n). The density matrix n of a Fermi sea, summed
 ↪  from the eigenvectors of positive energy of h(k) = alpha.k + beta M for 40 random pairs k, -k,
 ↪  has the form (rho + rho_S beta)/16 (deviation 2.2e-17) with rho_S = sum M/w. The energy of
 ↪  each of the nine channels, E_HF = (1/2)(G_1 rho^2 + G_0 rho_S^2), reproduces the coefficients
 ↪  of the table selfcoupling_hartree_fock.csv of the Rust program (largest deviation 8.0e-15)
[PASS] Icks-S6-44q  4+4, quantized-sector form: the contact interaction among the 28 pairs of two
 ↪  particles at rest (the eight states with beta = +1), diagonalized numerically for nine random
 ↪  couplings: seven pairs (among them the pair along the hidden axis) have the coupling g_0eff +
 ↪  g_1eff + 5 g_7eff and twenty-one have g_0eff + g_1eff - 3 g_7eff (largest deviation 7.9e-14)
[PASS] Icks-S7-44q  4+4, quantized-sector form: under the finite chiral rotation psi -> exp(0.37 i
 ↪  chi) psi (chi Hermitian, chi^2 = 1) of Grassmann-valued fields the interaction sum_r g_r I_r
 ↪  is unchanged when g_0eff + g_8eff = g_0 - 4 g_2 - 10 g_4 - 4 g_6 + g_8 = 0 (relative change
 ↪  8.6e-16) and changes for generic couplings (0.59)
[PASS] Icks-S2-44c  4+4, covariant form: each of the 256 matrices Theta^K (products of the Dirac
 ↪  matrices with the phases of the model) squares to eps_K = +-1, with eps_K = (-1)^r eta_K
 ↪  (eta_K: product of the squares of the matrices in K) as in the table of the Rust program;
 ↪  every vertex matrix M_K = A Theta^K is Hermitian, so that every bilinear psibar Theta^K psi is
 ↪  real (largest deviation 0.0e+00); Tr(Theta_K Theta^L) = 16 delta_KL (0.0e+00): the 256
 ↪  matrices are a basis of all 16 x 16 matrices
```

```
[PASS] Icks-S3-44c  4+4, covariant form: the Fierz matrix computed from the matrices, F_rs =
↪  (1/256) sum_(|K| = r) Tr(Theta_K Theta_L Theta^K Theta^L) for an L of rank s (the completeness
↪  relation of the basis; the same for every L of the rank, checked for six of each rank:
↪  0.0e+00), equals the closed form (largest deviation 0.0e+00)
[PASS] Icks-S4-44c  4+4, covariant form: the nine invariants evaluated on Grassmann-valued fields
↪  (psi and psi* independent random complex combinations of 6 anticommuting generators; each I_r
↪  is then an element of degree four with 15 coefficients; three random fields) obey (1 + F) I =
↪  0 (largest residual 6.9e-16 of the largest coefficient) and the five relations of Icks-S1
↪  (4.3e-16); the rank of the nine invariants over all samples is 4 (singular values 4.26e-01 and
↪  2.47e-16 on either side of the gap): exactly four invariants are independent. (The products of
↪  the Grassmann-valued components, done one by one with the Grassmann class, give the same
↪  bilinears: 7.2e-15.)
[PASS] Icks-S5-44c  4+4, covariant form: Hartree-Fock couplings by Wick's theorem, <:(psi^dagger M
↪  psi)^2:> = Tr(M n)^2 - Tr(M n M n). The density matrix n of a Fermi sea, summed from the
↪  eigenvectors of positive energy of h(k) = alpha.k + beta M for 40 random pairs k, -k, has the
↪  form (rho + rho_S beta)/16 (deviation 2.2e-17) with rho_S = sum M/w. The energy of each of the
↪  nine channels, E_HF = (1/2)(G_V0 rho^2 + G_S0 rho_S^2 + G_1 rho_W^2 + G_0 rho_SW^2), W = i C^-
↪  gamma^4, without cross terms, reproduces the coefficients of the table
↪  selfcoupling_hartree_fock.csv of the Rust program (largest deviation 2.2e-16)
[PASS] Icks-S6-44c  4+4, covariant form: the contact interaction among the 28 pairs of two
↪  particles at rest (the eight states with beta = +1), diagonalized numerically for nine random
↪  couplings: the 12 pairs of two particles in the same half W = +-1 have the couplings of the
↪  quantized-sector form (3 with g_0eff + g_1eff + 5 g_7eff, 9 with g_0eff + g_1eff - 3 g_7eff),
↪  and the 16 pairs of particles in different halves have the OPPOSITE sign (4 with -(g_0eff +
↪  g_1eff + 5 g_7eff), 12 with -(g_0eff + g_1eff - 3 g_7eff)) (largest deviation 1.5e-14)
[PASS] Icks-S7-44c  4+4, covariant form: under the finite chiral rotation psi -> exp(0.37 i chi)
↪  psi (chi Hermitian, chi^2 = 1) of Grassmann-valued fields the interaction sum_r g_r I_r is
↪  unchanged when g_1eff - g_7eff = g_1 + g_3 - g_5 - g_7 = 0 (relative change 6.0e-16) and
↪  changes for generic couplings (0.27)
[PASS] Icks-S8  the two forms of the self-interaction on 4+4: with W = i C^- gamma^4 (Hermitian,
↪  W^2 = 1, commuting with beta) the invariants of the covariant form are those of the
↪  quantized-sector form with W inserted in every bilinear, I_r(covariant) = sum_K eps_K
↪  (psi^dagger W M_K psi)^2, on Grassmann-valued fields (largest relative deviation 3.0e-16):
↪  the two forms agree for particles in the same half W = +-1 and differ in sign for particles in
↪  different halves
[PASS] Icks-K1  the Fermi integrals rho_S = <M/w>, e_kin = <w> and <k^2/(D w)> over a ball of
↪  momenta in D = 7, 4, 3 space dimensions, three ways: Gauss-Legendre nodes of numpy (used
↪  below), the adaptive quadrature of scipy, and the closed forms from the recurrence I_n =
↪  [k_F^(n-1) E_F - (n-1) M^2 I_(n-2)]/n with I_0 = asinh(k_F/M), I_1 = E_F - M (40 digits):
↪  largest relative difference 1.2e-14
[PASS] Icks-K2  flat 7+1 spacetime, the example of the author's document KOHN_SHAM_GRAVITY (m = 1,
↪  G_S = -1e5, G_V = +1e4, k_F = 0.756636150701493): M = 0.348097340146, rho = 1.388100213e-05,
↪  eps = 1.348787233e-05, E/A - m = -0.02832142796 here, against the printed 0.348097340146,
↪  1.388100213e-5, 1.348787233e-5, -0.02832142796 (largest relative difference 1.7e-10); the
↪  minimum of the energy per particle, located by the vanishing of the pressure, is at k_F =
↪  0.756636150701 (printed: 0.756636150701493; pressure there 1.3e-21) and the chemical potential
↪  equals the energy per particle (difference 8.9e-15); p = mu rho - eps along the equation of
↪  state to 9.2e-15. With G_S = -98750, G_V = 20000, the Hartree-Fock couplings that g_0 = -1e5,
↪  g_1 = +1e4 give (Icks-S5; the same numbers are printed in section 4 of the author's document),
↪  the energy per particle stays above m for all Fermi momenta from 0.02 to 1.5 (lowest value
↪  +0.000156): uniform matter is not self-bound for that pair of couplings. The Kohn-Sham numbers
↪  of the document belong to G_S = -1e5, G_V = +1e4 taken as the mean-field couplings themselves
[PASS] Icks-K3  Kohn-Sham matter of 7+1 with momenta along all seven space coordinates while x5,
↪  x6, x7 deflate (collisionless): the integral over the ellipsoid of occupied states with the
↪  four flat momenta integrated in closed form reproduces the table ks_deflating_71.csv of the
↪  Rust program at the clock readings -2, -1, 0, 1, 2, 4 for equal rates and for rates 1 : 2 : 3
↪  (12 states; M, rho, rho_S, eps and the five pressures: largest deviation 1.4e-08); at the
↪  reference time it reproduces the ball (4.4e-15); the energy density obeys d eps/dx4 = sum_t
↪  rate_t (eps + p_t) (numerical derivative, four cases: 1.9e-08)
```

```
[PASS] Icks-K4  fields that do not depend on x5, x6, x7 (D = 4) nor on x0 (D = 3): the smallest
↪  scalar attraction |G_S| m^(D-1) at which uniform matter is self-bound, found with scipy
↪  (lowest energy per particle over the Fermi momentum equal to m): D = 4, G_V/|G_S| = 0.0000:
↪  62.0767; D = 4, G_V/|G_S| = 0.1000: 85.6504; D = 4, G_V/|G_S| = 0.2025: 119.5723; D = 3,
↪  G_V/|G_S| = 0.0000: 8.0671; D = 3, G_V/|G_S| = 0.1000: 9.8364; D = 3, G_V/|G_S| = 0.2025:
↪  12.2272; the table ks_reduced_thresholds.csv of the Rust program agrees to 1.0e-09
↪  (relative). The third ratio, 0.202532, is the ratio G_V/|G_S| of the Hartree-Fock couplings of
↪  g_1 = -0.1 g_0 (Icks-S5)
[PASS] Icks-K5  covariant form on 4+4, the two halves W = +-1 as two kinds of particles with four
↪  states each (g_0 = -363.26, g_1 = -0.1 g_0, total density of Fermi momentum 0.8; Hartree-Fock
↪  couplings of Icks-S5): the two gap equations M_+- = m + G_S0 rho_S +- G_0 rho_SW, solved with
↪  the hybrid Powell method of scipy from 81 start values that include negative masses, have
↪  exactly one solution at each of the polarizations 0, 0.5 and 1, with positive masses, and it
↪  is the state of the table ks_44c_polarization.csv of the Rust program (masses and energy:
↪  largest difference 1.5e-09): E/A - m = +0.3799 for equal halves, +0.3348 for the polarization
↪  0.5 and +0.3731 with all particles in one half; the same density in the quantized-sector form
↪  has E/A - m = +0.1323 (difference 1.1e-09). The state with equal halves is not the state of
↪  lowest energy. (A remark on the no-sea approximation: the Hartree-Fock energy as a function of
↪  unrestricted trial masses has no lowest value, because trial masses of opposite sign and large
↪  magnitude put particles into states of negative rest energy; for equal halves it falls to E/A
↪  - m = -2.74. The Kohn-Sham state is the solution of the gap equations, not the lowest value of
↪  that function.)
[PASS] Icks-J1  the CJT effective potential Omega(M, V_0; mu) = Omega_1(M, mu - V_0) - (M -
↪  m)^2/(2 G_S) - V_0^2/(2 G_V) evaluated in Python: for the 30 states of the table cjt_vs_ks.csv
↪  (D = 7, 4, 3) its stationary point at the chemical potential of the Kohn-Sham state has the
↪  mass, the density and the energy density Omega + mu rho of the Kohn-Sham state, and -Omega is
↪  the Kohn-Sham pressure, both of the Rust program and of the Kohn-Sham routine of this script
↪  (largest relative deviation 2.0e-09)
[PASS] Icks-J2  with the filled states of negative energy below a cutoff Lambda, a massless field
↪  acquires a mass in vacuum when the scalar attraction exceeds G_c = -(2 pi)^D (D - 1)/(nu A_D
↪  Lambda^(D-1)); computer algebra gives D = 7: G_c = -90*pi**4/Lambda**6; D = 4: G_c =
↪  -3*pi**2/Lambda**3; D = 3: G_c = -pi**2/(2*Lambda**2); the numerical minimum of the vacuum
↪  potential is at M = 0 for G_S = 0.98 G_c and at M > 0 for G_S = 1.02 G_c in all three cases.
↪  The coupling G_S = -1e5 of the author's example equals G_c for Lambda = (90 pi^4/1e5)^(1/6) =
↪  0.6665 m
[PASS] Icks-J3  the author's example with the filled states of negative energy below the cutoff
↪  included (vacuum mass kept at 1 by the choice of the bare mass m = M_vac + G_S nu Int^Lambda
↪  M_vac/w; energy of matter relative to the vacuum, minimized over the mass and the Fermi
↪  momentum with scipy): Lambda = 0.30: bare mass 0.9979, saturation at k_F = 0.7566 with E/A -
↪  M_vac = -0.02837; Lambda = 0.45: bare mass 0.9660, saturation at k_F = 0.7568 with E/A - M_vac
↪  = -0.02984; Lambda = 0.60: bare mass 0.7578, saturation at k_F = 0.7565 with E/A - M_vac =
↪  -0.04483; the values quoted by the Rust program (check cjt-J3) agree to 5e-05
[PASS] Icks-U1  the functional with bare Dirac mass, contact couplings and Vorton-Scalar in
↪  ordinary space (D = 3), minimized directly with scipy: 12 states of the table
↪  unified_states.csv of the Rust program are reproduced (eps, M and x: largest deviation
↪  1.7e-08); without bare mass and contact couplings matter is first self-bound at C = g_VS^2
↪  m^2/m_s^2 = 3.701102, the exact value 3 pi^2/8 = 3.701102 of a gas of massless fermions with
↪  the energy density m^2 m_s^2/(8 g_VS^2) of the scalar at phi = 0. Phase mode with a bare mass:
↪  the second derivative of the energy of the filled states with respect to the phase of the
↪  scalar equals (|<psibar psi>| - rho_S) m_D m_Y/m (cutoff 2 m, f_D = 0.3, k_F = m; adaptive
↪  quadrature; agreement with the formula and with the Rust table to 2.7e-07)
[PASS] Icks-P1  pairing amplitudes in the hidden-axis channel (pair matrix X = C^- gamma^0,
↪  antisymmetric; momenta p, -p -> p', -p' in ordinary space) from projectors that are summed
↪  from numerical eigenvectors of the single-particle Hamiltonian, twelve random cases on each of
↪  7+1 and 4+4 (quantized-sector form): the amplitudes of the ranks 0, 1, 7, 8 equal the closed
↪  forms F_0 = (E E' + M^2 - p.p')/(2 E E'), F_1 = (3 E E' - 2 M^2 - p.p')/(E E'), F_7 = (3 E E'
↪  + 2 M^2 - p.p')/(E E'), F_8 = -(E E' - M^2 - p.p')/(2 E E') (largest deviation 1.8e-15); the
↪  two spacetimes give the same amplitudes (3.7e-15); at rest the nine amplitudes are (1, 1, -9,
↪  -9, -5, -5, 5, 5, 0) (0.0e+00); the relations of Icks-S1 hold for the part of the amplitudes
↪  that is even under p' -> -p' (4.4e-15) and not for the odd part (deviation up to 6.55)
```

```
[PASS] Icks-P2  the gap equation of the hidden-axis pair solved again in Python (another grid of
↪  momenta: geometric panels around the Fermi momentum, ten Gauss-Legendre nodes each, up to 80
↪  m; plain iteration to a relative change below 1e-12; the gap at the Fermi momentum from the
↪  equation itself). The closed angular average of the kernel agrees with the adaptive quadrature
↪  of scipy (1.2e-14). Gap in units of the Fermi kinetic energy at the clock reading 2.04, k_F =
↪  0.5 m, for the exchange of the Vorton-Scalar alone (variant 0), with the scalar contact
↪  attraction g_0eff = -g_VS^2/m^2 (variant 1) and with the contact couplings in the ratio of the
↪  author's example (variant 2): m_s = 1.0, variant 0: 0.0247 (Rust: 0.0246); m_s = 1.0, variant
↪  1: 0.2971 (Rust: 0.2968); m_s = 1.0, variant 2: 0.1962 (Rust: 0.1961); m_s = 0.3, variant 0:
↪  0.4117 (Rust: 0.4117); m_s = 0.3, variant 1: 0.9029 (Rust: 0.9023); m_s = 0.3, variant 2:
↪  0.7405 (Rust: 0.7403); largest relative difference 1.1e-03
[PASS] Icks-G1  spacetimes ds^2 = -dt^2 + a(t)^2 (dx_a)^2 + b(t)^2 (dy_b)^2 with 3 + 4 and with 4
↪  + 3 space coordinates, arbitrary functions a(t), b(t) (computer algebra, from the metric:
↪  Christoffel symbols, Riemann tensor): the Riemann tensor is diagonal in the pairs of
↪  directions, R^(ij)_(ij) = K_ij with K_(t i) = a_i''/a_i and K_(ij) = H_i H_j, and has no other
↪  components; the Einstein tensor is G^m_m = -sum of K over the pairs that do not contain m, and
↪  the Lanczos tensor 2 (R R^m_n - 2 R^m_a R^a_n - 2 R^(m a)_(n b) R^b_a + R^(m a)_(b c) R^(b
↪  c)_(n a)) - (1/2) delta^m_n (R^2 - 4 Ric^2 + Riem^2) is -4 times the sum over the sets of four
↪  directions without m of the sum over their three divisions into pairs of the products K K: the
↪  formula E^(k)m_m = (-1)^(k+1) k! 2^(k-1) [pair sum] of the Rust program for k = 1 (E^(1) = -G)
↪  and k = 2 (E^(2) = Lanczos tensor), time component and both space components
[PASS] Icks-G2  the Lovelock tensors of orders k = 1, 2, 3 derived from the reduced action S_k =
↪  Int dt N a^(n_a) b^(n_b) L_k, with the lapse N, the Lovelock density L_k = k! 2^k [sum over
↪  the sets of 2k directions and their divisions into pairs of the products of K] (L_1 =
↪  curvature scalar, L_2 = Gauss-Bonnet density, L_3 = cubic Lovelock density) and K_(t i) = (N
↪  a_i)^(-1) d/dt (a_i'/N): the Euler-Lagrange expressions for N, a and b at N = 1 (computer
↪  algebra, with second derivatives in the Lagrangian) reproduce E^(k)m_m = (-1)^(k+1) k! 2^(k-1)
↪  [pair sum without m] exactly, for the time component and both space components, for 3 + 4 and
↪  for 4 + 3 space coordinates (18 identities). All terms with third and fourth derivatives
↪  cancel: the equations are of second order
[PASS] Icks-G3  the self-consistent evolution of the author's document KOHN_SHAM_GRAVITY (three
↪  ordinary and four auxiliary space coordinates, kappa = 1000, LambdaBar = 0, H_a(0) = 0.05,
↪  a(0) = b(0) = 1, Kohn-Sham matter of the saturation density with G_S = -1e5, G_V = +1e4)
↪  integrated again with the Runge-Kutta method of order 8 of scipy, the field equations built
↪  from the pair sums with sympy and the matter from an independent quadrature: H_b(0), a(20),
↪  b(20), H_a(20), H_b(20) agree with the ten numbers printed in the document, for Einstein
↪  gravity and for Einstein-Lovelock gravity with alpha_2 = 20, alpha_3 = 500, to 4.8e-08
↪  (relative); the constraint is preserved to 9.7e-14
[PASS] Icks-G4  the deflating spacetime (four flat coordinates, H_a = 0; three coordinates
↪  deflating at the same rate, H_b = -g55) in the author's Einstein-Lovelock gravity, by computer
↪  algebra in the symbol g55: it solves the three field equations without matter for LambdaBar =
↪  3 g55^2 and alpha_2 = 1/(4 g55^2); E^(3) vanishes identically there (alpha_3 is arbitrary);
↪  the matrix of the coefficients of (H_a', H_b') in the two space equations is diag(-6, 2); in
↪  Einstein gravity alone the same spacetime needs kappa eps = 3 g55^2, kappa p_a = -6 g55^2,
↪  kappa p_b = -3 g55^2; near the solution the constraint is -6 g55 dH_b - 48 H_a^2 + (higher
↪  orders) = kappa eps, with no term linear in H_a; for H_b = -g55 its four roots in H_a are 0, 0
↪  and (6 +- sqrt(20)) g55
[PASS] Icks-G6  the Einstein-Lovelock polynomials that the documents print for four directions
↪  with the rate H_a and three with the rate H_b (E^(1) and E^(2): time component and both space
↪  components; E^(3): time component) are the sums over pair curvatures, term by term (computer
↪  algebra, seven identities); every term of the three components of E^(3) contains at least two
↪  factors of H_a or of its derivative, so that E^(3) and its first variation vanish at H_a = 0
[PASS] Icks-G5  the deflating spacetime with Kohn-Sham matter of 7+1 (momenta along all seven
↪  space coordinates) as a source, g55 = 0.01 m, loading kappa eps(0)/(3 g55^2) = 1e-3,
↪  integrated with scipy to the clock reading 6 of the undisturbed deflation: tuned
↪  Einstein-Lovelock: H_b(0) = -1.00050 g55; at tau_ref = 6: a = 1.2508, b = 4.9085e-02, H_a =
↪  +1.4052 g55, H_b = -7.9908 g55; Einstein: H_b(0) = -0.03162 g55; at tau_ref = 6: a = 1.0011, b
↪  = 9.3353e-01, H_a = +0.0012 g55, H_b = -0.0375 g55; the summary of the Rust program (six
↪  decimals) agrees to 6.4e-06 (relative to |value| + 0.05; constraint preserved to 1.3e-11).
↪  Without matter, from H_b(0) = -1.001 g55 and the positive root H_a(0) = 0.01147 g55 of the
↪  constraint: H_a = +1.73339 g55, H_b = -6.42224 g55 at the same clock reading (Rust: +1.73339,
↪  -6.42224; relative difference 6.7e-10): the deflation does not return to the rate g55
```

5.13 The test of the independent scripts with deliberate errors

```
PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: mutation test of the independent
 ↪  verifications
generated by: code/scripts/mutation_test_independent.py

A. code/scripts/verify_cjt_ks_independent.py

the unchanged script: 37 checks, all pass

A01. Fierz matrix with the sign (-1)^(rs) instead of (-1)^(rs-l): detected by the script stops
 ↪  (AssertionError)
A02. covariant form of 4+4 with the phases of the 7+1 model (i for the ranks 2, 3, 6, 7): detected
 ↪  by Icks-S2-44c
A03. Wick's theorem without the exchange term: detected by Icks-S5-71s, Icks-S5-44q, Icks-S5-44c
A04. interaction of two particles without the transposition of the second vertex: detected by
 ↪  Icks-S6-71s, Icks-S6-44q, Icks-S6-44c
A05. chiral condition of 7+1 with +g_8 instead of -g_8: detected by Icks-S7-71s, Icks-S7-44q
A06. gap equation M = m - G_S rho_S instead of M = m + G_S rho_S: detected by the script stops
 ↪  (ValueError)
A07. Kohn-Sham energy density with the scalar term +(M - m)^2/(2 G_S) instead of -(M - m)^2/(2
 ↪  G_S): detected by the script stops (ValueError)
A08. closed form of the integral over the flat momenta with c^3/3 instead of 2 c^3/3: detected by
 ↪  the script stops (ValueError)
A09. density of the deflating matter without the factor s5 s6 s7: detected by Icks-K3
A10. second gap equation of the two halves with -G_0 rho_SW replaced by +G_0 rho_SW: detected by
 ↪  Icks-K5
A11. CJT potential with +V_0^2/(2 G_V) instead of -V_0^2/(2 G_V): detected by Icks-J1
A12. critical coupling from the integral of k^(D-1) instead of k^(D-2): detected by Icks-J2
A13. potential of the Vorton-Scalar with 1/(4 C) instead of 1/(8 C): detected by the script stops
 ↪  (ValueError)
A14. pair along x1 instead of the hidden axis x0 (7+1): detected by Icks-P1
A15. vector amplitude F_1 with +2 M^2 instead of -2 M^2 in the gap equation: detected by Icks-P2
A16. Lovelock tensors with the factor 2^k instead of 2^(k-1): detected by the script stops
 ↪  (ValueError)
A17. Lanczos tensor with the last term doubled: detected by Icks-G1
A18. Lovelock density of the reduced action with k! 2^(k-1) instead of k! 2^k: detected by Icks-G2
A19. pressure of the matter along the first group of coordinates without the interaction part Pi:
 ↪  detected by Icks-G3, Icks-G5
A20. field equation of the first group of coordinates with +kappa p_a instead of -kappa p_a:
 ↪  detected by the script stops (ValueError)

B. code/scripts/verify_deflating_independent.py, cosmological perturbations

the unchanged script: 11 checks (Idef-C1, Idef-C2, I44d-P1, I71d-P1, Idef-P2, Idef-P8, Idef-P3,
 ↪  Idef-P4, Idef-P5, Idef-P6, Idef-P7), all pass

B01. energy constraint with He = Hc - a gsum instead of Hc - a gsum/2: detected by I44d-P1, I71d-P1
B02. trace equation with He h' instead of 2 He h': detected by I44d-P1, I71d-P1
B03. closed form of B with (a gsum)^2/2 instead of (a gsum)^2/4: detected by Idef-P2
B04. pressure of the agent with gprod instead of 2 gprod: detected by Idef-P8
B05. first identity between the gaps with (D - DP) instead of (D - 3 DP): detected by Idef-P3
B06. second identity between the gaps with He instead of 2 He: detected by Idef-P3
B07. growth equation of the closure "Momentum" with the source not reduced: detected by Idef-P4
B08. growth equation of the closure "Pressure" with the friction Hc instead of He: detected by
 ↪  Idef-P4
B09. numerical growth, closure "Pressure": friction reduced by gsum instead of gsum/2: detected by
 ↪  Idef-P5, Idef-P7
B10. numerical growth, closure "Momentum": source not reduced: detected by Idef-P5, Idef-P7
B11. system of the closure "Pressure" with a gsum instead of a gsum/2: detected by Idef-P6
B12. system of the closure "Shear" with Hc instead of He: detected by Idef-P6
B13. recombination on a background in which Newton's constant does not change: detected by Idef-C2
```

```
B14. response of the helium fraction to Newton's constant doubled: detected by Idef-C2
B15. Peebles' factor with 4 pi instead of 8 pi (an error of atomic physics): detected by Idef-C2
B16. equation of the expansion with the term -gsum H doubled (integration at fixed last
 ↪  scattering): detected by Idef-C1

36 of 36 deliberate errors detected
```

5.14 The symbols

```
PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the symbols (MASTER registry
 ↪  provenance/symbols_master.csv)
generated by: code/scripts/check_symbols.py
7 of 7 checks passed

[PASS] MS1  the 229 symbol texts of the MASTER registry are pairwise different, and so are its 339
 ↪  names in programs and data files
[PASS] MS2  no definition of the MASTER registry sends the reader to another document (searched
 ↪  for the names of the documents of the project and for the words "main registry", "supplement",
 ↪  "see the")
[PASS] MS3  none of the 41 names in programs that this sub-project introduces itself (entries of
 ↪  the groups 10 to 15 with the origin "new"; names of more than two characters) is a symbol of
 ↪  one of the author's notebooks (1921 symbols compared). Names that are the author's own (G_S,
 ↪  G_V, V_0, p_a, p_b and others of his Kohn-Sham sections) are used for the objects for which he
 ↪  uses them
[PASS] MS4  in the 27 documents of this folder the rates g55, g66, g77 are written upright 243
 ↪  times and never as a letter with a subscript; each of the 24 components of the metric g_{55},
 ↪  g_{66}, g_{77}, g_{tt} carries the superscript (4+4) or (7+1) or stands in a paragraph that
 ↪  names it a component of the metric
[PASS] MS5  the retired symbols do not occur in the documents: \varkappa nowhere; g_{1644d},
 ↪  g_{1671d} only where a paragraph says that they belong to the earlier sign convention
[PASS] MS6  the 14 retired names of programs (tau, kappa_t, Q_over_eps, Omega0, S2, mu and nu as
 ↪  orders of Bessel functions, A, Xi without index, m_b, w0, omega_c for the shell, the metric
 ↪  names g_1644d and g_1671d of the earlier sign convention, and Lambda as the range of the
 ↪  contact couplings) do not occur in the 132 data files that the routines of this folder write
[PASS] MS7  each of the 1093 columns of the 111 data files of this folder is a registered name, a
 ↪  column of the flat-space part of the project, one of the 249 descriptive words listed in the
 ↪  script, or (213 columns) a systematic name
```

5.15 The numbers quoted in the running text of the documents

```
PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: are the numbers of the running text
 ↪  numbers that the programs stated?
generated by: code/scripts/check_numbers.py
429 numbers in the running text of 21 documents: (a) 404 equal a number that a check, a generated
 ↪  table or the description of a data file states; (r) 18 are the rounded value of such a number;
 ↪  (b) 6 are listed in the script as values from the literature, definitions, counts, parameters
 ↪  or superseded values quoted by the error log; (c) 1 are bounds, within a decade, of a stated
 ↪  number; 0 could not be traced

[PASS] every number of the running text is traced, and each of the 15 different numbers of kinds
 ↪  (r) and (c) has been judged

docs_src/manuscript/00_preamble.tex                              0 stated,   0 rounded,  0 listed,
 ↪  0 bounds,  0 NOT TRACED
docs_src/manuscript/01_introduction.tex                          4 stated,   0 rounded,  0 listed,
 ↪  0 bounds,  0 NOT TRACED
docs_src/manuscript/02_geometry.tex                              8 stated,   2 rounded,  0 listed,
 ↪  0 bounds,  0 NOT TRACED
docs_src/manuscript/03_dirac.tex                                13 stated,   0 rounded,  0 listed,
 ↪  0 bounds,  0 NOT TRACED
```

```
docs_src/manuscript/03b_selfint.tex                              5 stated,   0 rounded,  0 listed,
 ↪  0 bounds,  0 NOT TRACED
docs_src/manuscript/04_modes.tex                                18 stated,   2 rounded,  0 listed,
 ↪  0 bounds,  0 NOT TRACED
docs_src/manuscript/05_clock.tex                                 4 stated,   0 rounded,  0 listed,
 ↪  0 bounds,  0 NOT TRACED
docs_src/manuscript/05b_ks.tex                                  25 stated,   3 rounded,  0 listed,
 ↪  0 bounds,  0 NOT TRACED
docs_src/manuscript/06_pairing.tex                              11 stated,   0 rounded,  0 listed,
 ↪  0 bounds,  0 NOT TRACED
docs_src/manuscript/06b_gapcontact.tex                          14 stated,   1 rounded,  0 listed,
 ↪  0 bounds,  0 NOT TRACED
docs_src/manuscript/07_pair.tex                                 18 stated,   0 rounded,  0 listed,
 ↪  0 bounds,  0 NOT TRACED
docs_src/manuscript/08_quench.tex                               13 stated,   0 rounded,  1 listed,
 ↪  0 bounds,  0 NOT TRACED
docs_src/manuscript/09_cjt.tex                                  55 stated,   1 rounded,  0 listed,
 ↪  1 bounds,  0 NOT TRACED
docs_src/manuscript/09b_cjtcontact.tex                          14 stated,   0 rounded,  0 listed,
 ↪  0 bounds,  0 NOT TRACED
docs_src/manuscript/10_order.tex                                 1 stated,   0 rounded,  0 listed,
 ↪  0 bounds,  0 NOT TRACED
docs_src/manuscript/10b_gravity.tex                             19 stated,   2 rounded,  0 listed,
 ↪  0 bounds,  0 NOT TRACED
docs_src/manuscript/11_cosmology.tex                           162 stated,   2 rounded,  2 listed,
 ↪  0 bounds,  0 NOT TRACED
docs_src/manuscript/12_higgs.tex                                 7 stated,   5 rounded,  2 listed,
 ↪  0 bounds,  0 NOT TRACED
docs_src/manuscript/13_status_conclusions.tex                    9 stated,   0 rounded,  0 listed,
 ↪  0 bounds,  0 NOT TRACED
docs_src/manuscript/14_appendices.tex                            4 stated,   0 rounded,  1 listed,
 ↪  0 bounds,  0 NOT TRACED
docs_src/manuscript/99_end.tex                                   0 stated,   0 rounded,  0 listed,
 ↪  0 bounds,  0 NOT TRACED

kind (r), numbers of the text that are rounded values of stated numbers (the number of the text
 ↪  and the documents that quote it; the stated numbers that match; the stated number that the
 ↪  text means, as judged):
  60                    in: 12_higgs
                        matches: 59.89 in tables_cjt_ks.md; 60.367 in
                         ↪  deflating_checks_independent.txt; 60.368 in
                         ↪  deflating_checks_independent.txt; 60.37 in cosmology_checks.txt
                       judged:  half of the mass of the scalar particle of 125 GeV, the estimate of
                         ↪  condition C5 (list LITERATURE of this script); that it is also the
                         ↪  rounded value of a stated number is a coincidence
  84                    in: 11_cosmology
                        matches: -84.35 in forecast_checks.txt; -84.4 in tables_cjt_ks.md
                      judged:  check def-F3: the best-fit Hubble constant changes by -84.35 km/s/Mpc
                         ↪  per unit of gsum/H_0 (Planck-like mock data), quoted without decimals
  125                   in: 12_higgs
                        matches: 125.047 in tables_cjt_ks.md; 124.857 in tables_cjt_ks.md
                        judged:  the mass of the Higgs boson in GeV, a value from the literature
                         ↪  (list LITERATURE of this script); that it is also the rounded value of
                         ↪  two entries of a generated table is a coincidence
  2.1                   in: 11_cosmology
                        matches: 2.115 in tables_cjt_ks.md; 2.13 in spectrum_checks.txt; 2.0643 in
                         ↪  tables_cjt_ks.md; 2.14 in spectrum_checks.txt
                       judged:  check def-E4: below multipole 30 the two closures differ by 2.14 and
                         ↪  2.13 times the effect of the deflation at the two bounds, quoted as 2.1
  2.33                  in: 09_cjt
                        matches: 2.325 in tables_cjt_ks.md
```

```
                       judged:  table of melting temperatures (data/thermal_history.csv): T_half =
                        ↪  2.325 m at g_VS = 0.3, m_s = m (2.32536 unrounded)
  3.386                in: 02_geometry
                       matches: 3.38636 in deflating_checks.txt
                       judged:  check 71d-G8: proper time 3.38636 of the geodesic on the 7+1
                        ↪  spacetime
  1.0924               in: 02_geometry
                       matches: 1.0924 in deflating_checks.txt
                       judged:  check 44d-G9: turning value 1.09240103 of the geodesic on the 4+4
                        ↪  spacetime
  -256.95              in: 10b_gravity
                       matches: -256.951 in tables_cjt_ks.md
                      judged:  description of data/gravity_deflating.csv: the matter of the sector
                        ↪  has G_S = -256.951 (D = 4), quoted with two decimals
  10^{-15}             in: 04_modes, 05b_ks
                       matches: 9.4e-16 in cjt_ks_checks_independent.txt; 1.1e-15 in
                        ↪  channel_checks.txt; 8.6e-16 in cjt_ks_checks_independent.txt; 7.7e-16
                        ↪  in kohnsham_checks.txt
                      judged:  check I71d-M1: relative residual 1.1e-15 of the exact solution with
                        ↪  the Bessel functions of mpmath
  10^{-30}             in: 04_modes
                       matches: 8.85e-31 in tables_cjt_ks.md
                       judged:  check def-D6: the temperature at the largest allowed rate is
                        ↪  2.1e-31 K, below the bound quoted; in RS5, Sec. 5.1, and in RP9 (R-E39)
                        ↪  the same power of ten is the rate, in units of 100 km/s/Mpc, of the
                        ↪  unit test that runs the program at a negligible rate (a parameter of
                        ↪  the test)
  2\times10^{-8}       in: 05b_ks
                       matches: 1.9e-08 in cjt_ks_checks_independent.txt; 2.1e-08 in
                        ↪  kohnsham_checks.txt; 2.2e-08 in tables_cjt_ks.md; 1.7e-08 in
                        ↪  cjt_ks_checks_independent.txt
                       judged:  check Icks-K3 of the independent script: the conservation law d
                        ↪  eps/dx4 = sum_t rate_t (eps + p_t) holds to 1.9e-08
  3\times10^{-8}       in: 12_higgs
                       matches: 2.8e-08 in cosmology_checks.txt
                       judged:  check def-C6: the formula for n_EW agrees with the numerical
                        ↪  derivative to 2.8e-08
  6\times10^{-9}       in: 05b_ks
                       matches: 5.6e-09 in kohnsham_checks.txt; 6.5e-09 in mode_checks.txt
                      judged:  check ks-K4: the same conservation law in the Rust program, largest
                        ↪  relative deviation 5.6e-9
  7\times10^{-16}      in: 06b_gapcontact
                       matches: 6.9e-16 in cjt_ks_checks_independent.txt; 6.7e-16 in
                        ↪  kohnsham_checks.txt; 6.6e-16 in pairing_checks.txt
                       judged:  check pc-P4: without contact couplings the gaps of the earlier
                        ↪  calculation are reproduced to 6.6e-16

kind (c), bounds within a decade of a stated number (the number of the text and the documents that
 ↪  quote it; the stated numbers that match; the stated number that the text means, as judged):
  10^{-36}             in: 09_cjt
                       matches: 3.4e-37 in matter_checks.txt
                      judged:  check def-D6: hbar gsum is 3.4e-37 of 1 keV at the largest allowed
                        ↪  rate, below the bound quoted

kind (b), numbers that the programs do not state, as listed in the script (those that occur in the
 ↪  documents):
  0.02237            density of baryons of the model of the main project
  10^{18}            order of magnitude of the clock comparisons, (1.0 +- 1.1) x 10^-18 per year
   ↪  (Ref. Lange2021)
  2000               number of energies of the shell model (parameter of the run)
  246                electroweak scale in GeV (Standard Model)
  420                4 * 3 * 35 (test of the formula for the Lanczos tensor)
```

5.16 No document sends its reader to another document

```
PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: does a document send its reader to
 ↪  another document?
generated by: code/scripts/check_self_contained.py
23 sources of documents and 4 built documents searched for 9 wordings of referral

[PASS] no document of this folder sends its reader to another manuscript, document, provenance or
 ↪  symbol file

wordings searched for:
  - a supplement or provenance document of the first revision named as a place to read
  - a supplement or provenance document of the flat-spacetime part named as a place to read
  - the word supplement
  - a companion paper
  - an earlier paper
  - the registry of another part of the project
  - see another document
  - defined in another document
  - a citation of an unpublished manuscript of the project as the source of a definition

hits per file:
docs_src/P1_master_symbols_spacetimes.src.md
 ↪  0
docs_src/P2_calculations_code_and_checks.src.md
 ↪  0
docs_src/manuscript/00_preamble.tex
 ↪  0
docs_src/manuscript/01_introduction.tex
 ↪  0
docs_src/manuscript/02_geometry.tex
 ↪  0
docs_src/manuscript/03_dirac.tex
 ↪  0
docs_src/manuscript/03b_selfint.tex
 ↪  0
docs_src/manuscript/04_modes.tex
 ↪  0
docs_src/manuscript/05_clock.tex
 ↪  0
docs_src/manuscript/05b_ks.tex
 ↪  0
docs_src/manuscript/06_pairing.tex
 ↪  0
docs_src/manuscript/06b_gapcontact.tex
 ↪  0
docs_src/manuscript/07_pair.tex
 ↪  0
docs_src/manuscript/08_quench.tex
 ↪  0
docs_src/manuscript/09_cjt.tex
 ↪  0
docs_src/manuscript/09b_cjtcontact.tex
 ↪  0
docs_src/manuscript/10_order.tex
 ↪  0
docs_src/manuscript/10b_gravity.tex
 ↪  0
docs_src/manuscript/11_cosmology.tex
 ↪  0
docs_src/manuscript/12_higgs.tex
 ↪  0
```

```
docs_src/manuscript/13_status_conclusions.tex
 ↪  0
docs_src/manuscript/14_appendices.tex
 ↪  0
docs_src/manuscript/99_end.tex
 ↪  0
manuscript/16 Component Dirac Fermion Condensate Singlet Superfluid on 4+4 Deflating Spacetime.md
 ↪  0
manuscript/16 Component Dirac Fermion Condensate Singlet Superfluid on 7+1 Deflating Spacetime.md
 ↪  0
manuscript/16 Component Dirac Fermion Condensate Singlet Superfluid on 4+4 Deflating
 ↪  Spacetime.tex             0
manuscript/16 Component Dirac Fermion Condensate Singlet Superfluid on 7+1 Deflating
 ↪  Spacetime.tex             0
```

6. The data files

The folder `data` holds 111 tables of numbers (files `.csv`) besides the reports of Section 5. Each begins with lines that say which program and command wrote it, what it contains, and what its columns are; the table repeats these lines.

data file	command	content	columns
`background_checks.csv`	`background`	PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the thermal history on the deflating spacetime	
`background_deflation.csv`	`background`	thermal history on the deflating spacetime (assumptions A1, A2: Newton's constant G_N(x4) = G_N(today) exp(gsum (x4 - x4_today)), three equal rates) for the three models of the command 'background' of the main project	gsum_over_H0: total rate of deflation in units of 67.36 km/s/Mpc (0, the rate that lunar laser ranging allows, the rate that nucleosynthesis allows); fixed: 0 = the Hubble constant of the main project, 67.36 km/s/Mpc; 1 = the Hubble constant for which the angle theta_* has its value without deflation; H0 in km/s/Mpc; tau0_Mpc: conformal age; z_star, z_rec, z_drag: redshifts of last scattering (optical depth 1), of the maximum of the visibility function, and of the end of the baryon drag; rs_star, rs_drag: sound horizons in Mpc; theta_s_100, theta_star_100: 100 times the angles of the sound horizon at z_rec and at z_star; z_reio; Y_He: helium mass fraction; GN_star, GN_early: Newton's constant at last scattering and in the early universe over its value today; age_Gyr
`boson_bounded.csv`	`channels`	4+4: the bounded solution of a scalar mode that depends on a deflating time (weighted field squared, normalized to its maximum) and the exact form K_(i varpi_b/rate_t)(u_d)^2 (not normalized), u_d = k_d/rate_t; equal rates 1	order: varpi_b/rate_t; x4; u_d = k_d/rate_t; Shat2: computed, normalized to its maximum; K2: exact form
`boson_creation.csv`	`channels`	7+1: number of quanta of a scalar mode that the deflation creates in a mode that starts empty; the mode depends on one deflating coordinate; rates (g55, g66, g77) = (1, 2, 0.5)	coordinate: 5, 6 or 7; rate of that coordinate; varpi_b = sqrt(k^2 + m_bos^2 - gsum^2/4); order = varpi_b/rate_t; n_cr: computed number; n_exact = 1/(exp(2 pi order) - 1); relative deviation; deviation of the Bogoliubov coefficients from |alpha|^2 - |beta|^2 = 1

data file	command	content	columns
channel_checks.csv	channels	PROJECT spacetimes, Revision_ Deflation_CJT_and_Kohn-Sham: checks of the exchange of the Vorton-Scalar on the deflating spacetimes (def-K..: amplitudes in the pair channels; def-B..: the exchanged field)	
cjt_gl_history.csv	paired	Ginzburg-Landau coefficients of the paired state at three moments of the deflation, particle (m_s, k_F) = (1.0, 0.5) m, from the CJT potential, with their ratios to the weak-coupling formulas	tau_d; g; Tc; dalpha_dlnT (formula 4 N(0)) and its ratio to the formula; beta at T_c (formula 7 zeta(3) N(0)/(2 pi^2 T_c^2)) and ratio; K: gradient coefficient (formula 7 zeta(3) N(0) v_ F^2/(96 pi^2 T_c^2)) and ratio; xi_ GL = sqrt(K/|a|) for |ln(T/T_c)| = 1, a = alpha/8: coherence length (units 1/m); xi_GL T_c/v_F (formula sqrt(7 zeta(3)/(48 pi^2)) = 0.1333)
cjt_history.csv	paired	the paired state at zero temperature along the deflation clock, from the CJT potential of the main project (vector channel): two particles, the coupling g_ VS growing as exp(tau_d/2)	set: 1 = (m_s, k_F) = (0.3, 0.5), 2 = (1.0, 0.5) in units of m; tau_d = 2 ln(g/g_first); g; lambda: pairing strength; E_F; gap_fixed_point, gap_ newton: gap at the Fermi momentum from the iteration of the gap equation and from Newton's method on the potential; max_rel_diff of the two gap functions; V: potential per unit volume at the solution (units m^4), the condensation energy; V_BCS = -nu N(0) gap^2/4; V_over_V_BCS; s_ min_minus_1: position of the minimum of V along the ray s x solution, minus 1; curvature_min: smallest eigenvalue of the second variation (positive: a minimum); Tc_ linear, Tc_second_variation: critical temperature from the linearized gap equation and from the potential; Tc_ over_gap; xi_kF: coherence length v_ F/(pi gap) in units of 1/k_F
cjt_sea_ saturation.csv	kohnsham	the author's example (seven space dimensions, G_S = -1e5, G_V = +1e4) with the filled states of negative energy below the cutoff Lambda included (CJT effective potential, one-loop vacuum term); the bare mass m is chosen so that the mass of the fermions in vacuum is 1. Lambda = 0: no-sea approximation	Lambda: cutoff in units of the vacuum mass; m_bare; kF; M: mass of the particles in matter; e_bind: energy per particle relative to the vacuum, minus the vacuum mass
cjt_sea_vacuum.csv	kohnsham	the vacuum with the filled states of negative energy (cutoff Lambda = 1: all quantities in units of the cutoff): the mass of the fermions in vacuum, the minimum of Omega_vac(M) = -NU Int^Lambda d^Dk/(2 pi)^D [w(M) - w(m)] - (M - m)^2/(2 G_S), as a function of the scalar attraction. In the sector of fields that do not depend on x5, x6, x7 the attraction grows as exp(tau_d); tau_ d is counted from the moment at which it reaches the critical coupling G_c	D: space dimensions; m_bare: bare mass m in units of Lambda; tau_d = ln(G_S/G_c); G_S Lambda^(D-1); M_ vac: mass in vacuum in units of Lambda; condensate: scalar density of the sea, <psibar psi> = -NU Int^Lambda M/w, in units of Lambda^D
cjt_thermo_ history.csv	paired	thermodynamic ratios of the paired state at three moments of the deflation, particle (m_s, k_F) = (1.0, 0.5) m, vector channel: the critical temperature and the gap grow, their ratio and the jump of the heat capacity at T_c do not change	tau_d = 2 ln(g/2.6); g; lambda; Tc; Tc_ over_EF; gap0: gap at T = 0; gap0_ over_Tc (weak coupling: 1.7639); heat_capacity_jump: (C_paired - C_ unpaired)/C_unpaired at T_c (weak coupling: 1.4261)

data file	command	content	columns
cjt_threshold.csv	paired	critical value of D_VS = alpha_VS m/m_ s at which the linearized gap equation in vacuum has the eigenvalue 1 (relativistic kernel of the radial mode), compared with the critical coupling of the Yukawa potential from the Schroedinger equation; on the deflating spacetime D_VS reaches this value at a definite x4	
cjt_toy.csv	paired	zero-dimensional phi^4 model S = phi^2/2 + lambda phi^4/24: exact propagator G = <phi^2> and solutions of the CJT gap equation with Gamma_2 truncated at 2 to 5 loops (NaN: the truncated equation has no real solution). The test of the method does not involve the spacetime; it is executed again here	
cjt_vs_ks.csv	kohnsham	the two methods compared (no-sea approximation): for each Kohn-Sham state (fixed Fermi momentum; gap equation; Gauss-Legendre integrals) the chemical potential is computed and passed to the CJT effective potential Omega(M, V_0; mu) (Simpson integrals), whose stationary point is searched independently	case; D; m; G_S; G_V; kF; mu; M_ks; M_cjt: stationary point of Omega nearest to M_ks; M_cjt_global: global minimum of Omega (computed where the pressure is positive; empty otherwise); rho_ks; rho_cjt; eps_ks; eps_cjt = Omega + mu rho; p_ks; minus_Omega
cmb_deflation_ cl.csv	spectrum	angular power spectra of the microwave background on the deflating spacetime (without gravitational lensing), D_l = l(l+1) C_l/(2 pi) in microkelvin^2, for the models of data/cmb_deflation_summary.csv and the two closures (suffix _M: “Momentum”, _P: “Pressure”)	
cmb_deflation_ closure.csv	spectrum	dependence of the temperature spectrum on the closure of the perturbation equations (assumption A3): closure_i = D_l^TT with the closure “Momentum” minus D_l^TT with the closure “Pressure”, and for comparison effect_i = the effect of the deflation itself (closure “Momentum” minus no deflation, both at the same angle theta_ *), in microkelvin^2, for the models i of data/cmb_deflation_summary.csv	
cmb_deflation_ convergence.csv	spectrum	numerical convergence of the spectra on the deflating spacetime, gsum = 0.03 H_0, for the two closures: accuracy setting 1 (used everywhere) relative to setting 2 (all tolerances and wavenumber steps halved)	rel_TT = TT_1/TT_2 - 1; rel_EE likewise; dTE = (TE_1 - TE_ 2)/sqrt(TT_2 EE_2); suffix _M: closure “Momentum”, _P: “Pressure”

data file	command	content	columns
cmb_deflation_gaps.csv	spectrum	what the agent supplies (assumption A3), at gsum = 0.06436 H_0: with a Newton's constant that depends on the time the momentum constraint, the trace equation and the equation of the shear of ordinary space cannot all hold with matter and radiation alone. The gaps are what each equation lacks: momentum gap = 2 k^2 eta' - 8 pi G_N a^2 (rho + P) theta, trace gap = h'' + 2 He h' - 2 k^2 eta + 24 pi G_N a^2 delta P, shear gap = alpha' + 2 He alpha - eta + 12 pi G_N a^2 (rho + P) sigma/k^2, with He = Hc - a gsum/2. The table gives them in units of (a gsum/Hc) times the size of the terms of the equation	k: wavenumber in 1/Mpc; closure: 0 = "Momentum", 1 = "Pressure"; tau_cf: conformal time in Mpc; a_gsum_over_Hc; potential: eta + alpha', the sum of the two gravitational potentials (unit curvature perturbation); difference: its derivative by a central difference; expression: eta' + alpha'', the quantity that the source of the temperature uses; shear_gap: in units of (a gsum/Hc) |potential|; trace_gap: in units of (a gsum/Hc) (k^2 + Hc^2) |potential|; momentum_gap: in units of (a gsum/Hc) k^2 Hc |potential|; shear_check, trace_check: numerical checks of the two identities from which the program computes the gaps (alpha' and h'' by differences along the solution), relative to |potential| and to (k^2 + Hc^2) |potential|
cmb_deflation_growth.csv	spectrum	growth of the density contrast of matter (cold dark matter and baryons) from a = 0.01 to today at the wavenumber k = 0.1/Mpc, relative to the model without deflation, for the two closures (suffix _M: "Momentum", _P: "Pressure"): from the Einstein-Boltzmann equations of the command 'spectrum' (growth_EB) and from the growth equation of the closure for pressureless matter that the command 'matter' integrates (growth_ODE); and the amplitude sigma_8 of the full calculation for the same primordial amplitude	gsum_over_H0; H0: Hubble constant of the model in km/s/Mpc (fixed angle theta_*); growth_EB, growth_ODE: at that Hubble constant; growth_EB_fixed_H0, growth_ODE_fixed_H0: with the Hubble constant held at 67.36 km/s/Mpc; sigma_8; sigma_8_ratio: sigma_8 over its value without deflation
cmb_deflation_peaks.csv	spectrum	positions and heights (microkelvin^2) of the first five acoustic peaks of the temperature spectrum, for the models of data/cmb_deflation_summary.csv (closure "Momentum"; without gravitational lensing)	
cmb_deflation_pk.csv	spectrum	linear power spectrum of matter today, P(k) in Mpc^3 at the wavenumbers k in 1/Mpc, for the models of data/cmb_deflation_summary.csv and the two closures (same numbering and suffixes as in data/cmb_deflation_cl.csv)	

data file	command	content	columns
cmb_deflation_summary.csv	spectrum	the models whose spectra are computed: three equal rates of deflation with the total rate gsum, Newton's constant G_N(x4) = G_N(today) exp(gsum (x4 - x4_today)) (assumptions A1, A2), perturbations under assumption A3 with the closure "Momentum" (suffix _M) or "Pressure" (suffix _P); the Hubble constant of each model is the one for which the angle theta_* of the sound horizon at last scattering has its value without deflation; the densities of baryons and of cold dark matter, the primordial amplitude and tilt and the optical depth are those of the fiducial model of the main project	gsum_over_H0: gsum in units of 67.36 km/s/Mpc; gsum_per_yr; H0: Hubble constant in km/s/Mpc; dH0: its change; H0_fixed_zstar: the Hubble constant that the same condition gives if the redshift of last scattering and the helium fraction are held at their values without deflation; theta_star_100; z_star; rs_star (Mpc); z_drag; rs_drag (Mpc); GN_star, GN_early: Newton's constant at last scattering and in the early universe over its value today; Y_He: helium fraction (it responds to Newton's constant at nucleosynthesis); age_Gyr; tau0_Mpc: conformal age; sigma_8_M, sigma_8_P; max_rel_TT: largest |TT/TT_0 - 1| for 30 <= l <= 2500 against the model without deflation; cv_max, cv_rss: largest difference of TT in units of the cosmic variance of one multipole, and the root of the sum of squares over 30 <= l <= 2500 (the last three for the closure "Momentum"); chi2_planck_M, chi2_planck_P: chi^2 of the model against mock data of the fiducial model for the Planck-like data set of the main project, all other parameters fixed; closure_cv_low, closure_cv_high: largest difference between the temperature spectra of the two closures in units of the cosmic variance of one multipole, for 2 <= l <= 29 and for 30 <= l <= 2500; closure_over_effect_low, closure_over_effect_high: that difference relative to the effect of the deflation itself (roots of the sums of squares over the same ranges)
cosmology_checks.csv	cosmology	PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the cosmological calculations on the deflating spacetime	
coupling_exponents.csv	symmetry	how the quantities of the model follow the deflation in the sector of fields that do not depend on x5, x6, x7: quantity proportional to exp(n tau_d), tau_d = gsum (x4 - x4_ref), gsum = g55 + g66 + g77	quantity; meaning; n; note
creation_anisotropic.csv	modes	7+1 deflating spacetime, unequal rates: particles created in a mode with wavenumbers along x5 and x6 (g55 = 1; E_perp = m = 1, p = 0, k_7 = 0, constants 1)	g66; delay: x4 at which q_6 = E_perp minus x4 at which q_5 = E_perp, in units of 1/g55 (delay > 0: x5 becomes energetic first); ratio: the other physical wavenumber divided by E_perp at the moment when the first one reaches E_perp; n_min, n_mean, n_max: smallest, mean and largest eigenvalue of the 8x8 matrix of created particles; fd5 = 1/(exp(2 pi E_perp/g55) + 1); fd6 = 1/(exp(2 pi E_perp/g66) + 1)

data file	command	content	columns
creation_history.csv	modes	7+1 deflating spacetime: mean number of particles (per spin state, first superadiabatic basis) in a mode that starts empty, as a function of x4	case: 1 = E_perp/g55 = 1, one direction; 2 = E_perp/g55 = 0.5, one direction; 3 = E_perp = 1, wavenumbers along x5 (g55 = 1) and x6 (g66 = 2); x4 (the physical wavenumber along x5 equals 1 at x4 = 0); k_d_over_E_perp; n_cr: mean adiabatic particle number
creation_single.csv	modes	7+1 deflating spacetime: number of particles created by the deflation in a mode with one non-zero wavenumber along a deflating coordinate (k_5 = 1, g55 = 1, const5 = 1, p = 0, m = E_perp), per spin state, numerical and exact	E_perp_over_g55: E_perp/g55; n_min, n_max: smallest and largest eigenvalue of the 8x8 matrix of created particles; n_exact = 1/(exp(2 pi E_perp/g55) + 1); rel_dev = mean/exact - 1; unitarity: deviation of the evolved states from orthonormality; steps: number of integration steps
curvature.csv	matrices	curvature of the deflating spacetimes for several sets of rates, from the closed forms of check G5 (mixed components; they do not depend on x4)	spacetime: 1644 = 4+4, 1671 = 7+1; g55, g66, g77: rates; Ric44, Ric55, Ric66, Ric77: mixed Ricci tensor, components (mu, mu); Rscalar: curvature scalar; Ein00 (= Ein11 = Ein22 = Ein33), Ein44, Ein55, Ein66, Ein77: mixed Einstein tensor, components (mu, mu)
deflating_checks.csv	matrices	PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the geometry, of the Dirac matrices, of the spin connection and of the Dirac operators of the deflating spacetimes (44d-..: 4+4 with deflating x5, x6, x7; 71d-..: 7+1 with deflating x5, x6, x7)	
deflation_bounds.csv	cosmology	upper bounds (two sigma) on the present total rate of deflation gsum = g55 + g66 + g77, each under the stated assumption	observation; measured value; relation to gsum; bound on gsum per year; bound in units of H_0 = 67.36 km/s/Mpc; assumption
deflation_chord_check.csv	forecast	effect of recomputing the derivatives of the spectra at the end point of a fit (gsum = 0.02 H_0, closure “Momentum”) and continuing the fit: chi2 and parameters before (a) and after (b)	
deflation_fisher.csv	forecast	Fisher-matrix forecasts for the mock data sets of the main project with the total rate of deflation gsum as a seventh parameter (three equal rates; assumptions A1, A2, A3): error of the rate; errors of the six LambdaCDM parameters without (sigma6_) and with (sigma7_) the rate as a free parameter; shift of the best-fit parameters per unit gsum/H_0 at fixed rate (slope_)	sigma_rate: standard error of gsum/H_0 (H_0 = 67.36 km/s/Mpc); bound95_rate: 1.96 sigma_rate, the upper bound at 95 per cent for a parameter that cannot be negative and whose best value is zero; sigma_rate_fixed: the error if the six parameters are held fixed; dH0_drate: change of the best-fit Hubble constant in km/s/Mpc per unit gsum/H_0
deflation_profile.csv	forecast	best fits of the six LambdaCDM parameters to mock microwave-background data (fiducial LambdaCDM spectra without deflation, no lensing) for a fixed total rate of deflation gsum (three equal rates; assumptions A1, A2, A3)	gsum_over_H0: the rate in units of 67.36 km/s/Mpc; chi2: minimum chi^2 (0 without deflation by construction); tau_reio: optical depth of reionization; H0 in km/s/Mpc; rs_drag in Mpc; S8 = sigma8 sqrt(Omega_m/0.3); dN_bbn: extra radiation during nucleosynthesis (none here); GN_star: Newton’s constant at last scattering over today’s; jacobian_refreshed = 1 if the derivatives of the spectra were recomputed at the best fit for gsum = 0.02 H_0

data file	command	content	columns
deflation_residuals.csv	forecast	spectra on the deflating spacetime at gsum = 0.02 H_0 relative to the fiducial LambdaCDM spectra (Planck-like mock, closure “Momentum”): fit_ = the best fit of the six parameters; angle_ = the model with the fiducial densities, amplitude, tilt and optical depth and the Hubble constant that keeps the angle theta_*	_TT = TT_model/TT_fid - 1; _EE likewise; _TE = (TE_model - TE_fid)/sqrt(TT_fid EE_fid)
deflation_temperature.csv	matter	7+1: the creation of quanta by the deflation has the spectrum of a temperature k T = hbar rate_t/(2 pi) (modes that depend on the deflating coordinate x_t). The table gives that temperature for three equal rates at the largest total rates that the bounds allow, and the exponent 2 pi m/rate_t that suppresses the creation of particles of mass m	bound; gsum/H_0; hbar rate_t in eV; T in kelvin; exponent for m = 1 keV; exponent for m = 1 GeV (the number of created particles per mode is exp(-exponent))
dimer_history.csv	history	two fermions in the Yukawa potential of the radial mode along the deflation clock: D_VS = alpha_VS m/m_s grows as exp(gsum x4)	D; tau_pair = ln(D/D_c) = gsum (x4 - x4 at which the pair becomes bound); a_ms: scattering length times m_s; bound_states; Eb: binding energy times m/m_s^2 (0 = unbound)
dm_drift.csv	matter	how fast the quantities of the model drift on a deflating spacetime: n = d ln(quantity)/d tau_d with tau_d = gsum x4, and the relative change n gsum t_0 over the age of the universe at the largest rates that the three bounds allow (gsum = 3.24e-3, 0.0644 and 0.234 H_0), with the age t_0 of the model without deflation (H_0 t_0 = 0.9516)	quantity; where evaluated; n; change over the age at the bound from lunar laser ranging (assumption A2); at the bound from nucleosynthesis through G_N (A2); at the bound from nucleosynthesis through the Fermi constant (assumption HV, 20 per cent)
dm_halo_estimates.csv	matter	could the particles of the model be degenerate (fermions) or Bose-condensed (bound pairs) in observed dark-matter halos? These estimates contain the masses, the densities and the velocities only; the masses do not change on a deflating spacetime, so the numbers are those of the main project	m_deg_keV: fermions with 8 internal states are degenerate (v_F >= sigma) only if lighter than this; mB_max_eV: bosons are condensed (n lambda^3 >= 2.612 at k T = m sigma^2) only if lighter than this; rho_deg_1p5keV: density (Msun/pc^3) at which fermions of 1.5 keV would be degenerate at this velocity dispersion, and its ratio to the actual density; vF_1p5keV: their Fermi velocity at the actual density (km/s) and its ratio to sigma
dm_resonance.csv	matter	scattering of two unbound fermions of the model at low energy while the deflation carries D_VS = alpha_VS m/m_s through the value D_c at which the pair becomes bound: s-wave cross-section per mass, sigma/m = 4 pi a^2/[(1 + k^2 a^2) m], with the scattering length a(D_VS) of the Yukawa potential (routine dimer::scattering_length of the main project) and k = m v/2	tau_pair = gsum (x4 - x4_binding) = ln(D_VS/D_c); D = D_VS = D_c exp(tau_pair); a_ms: scattering length in units of 1/m_s; sigma/m in cm^2/g at the relative velocities 10, 156 and 1000 km/s (effective-range corrections neglected)

data file	command	content	columns
dm_selfinteraction.csv	matter	smallest fermion mass for which a gas of bound pairs satisfies sigma/m <= 1 cm^2/g, if the pair-pair scattering length is the range 1/m_s of the binding potential and the potential binds (alpha m/m_s >= D_c), evaluated with the coupling of the moment alpha = alpha_today exp(tau_d), tau_d = gsum (x4 - x4_today)	
dr_deflation_chord_check.csv	families	the dark-radiation families on the deflating spacetime: effect of recomputing the derivatives of the spectra at the end point of a fit (N = 0.5) and continuing the fit: chi2 and parameters before (a) and after (b)	
dr_deflation_fisher.csv	families	the dark-radiation families of the main project on the deflating spacetime: Fisher-matrix forecasts for the mock data sets: errors of the six LambdaCDM parameters without (sigma6_) and with (sigma7_) the density N of extra radiation as a free parameter; error of N; shift of the best-fit parameters per unit N at fixed N (slope_)	
dr_deflation_joint.csv	families	the density N of extra radiation of each family and the total rate of deflation gsum as free parameters together: Fisher matrices with eight parameters (the six of LambdaCDM, N, gsum) at the fiducial model without deflation, for the mock data sets of the main project	sigma_N, sigma_N_joint: error of N with the rate held at zero and with the rate free; sigma_rate, sigma_rate_joint: error of gsum/H_0 (H_0 = 67.36 km/s/Mpc) with N held at zero and with N free; correlation: correlation coefficient of N and gsum; dH0_dN, dH0_drate: change of the best-fit Hubble constant in km/s/Mpc per unit N and per unit gsum/H_0; N_best_llr, chi2_best_llr: the best-fitting N >= 0 and its chi^2 at the largest rate that lunar laser ranging allows; H0_gain_zero_rate: the largest rise of the Hubble constant with chi^2 <= 3.84 at zero rate; H0_gain, N_at_gain, rate_at_gain: the largest rise with chi^2 <= 3.84 for N >= 0 and gsum >= 0, and where it is reached; cross_N0.1, cross_N0.2: (minimum chi^2 of the fit on the deflating background at the rate of lunar laser ranging minus that of the main project at the same N) divided by the prediction of the Fisher matrix
dr_deflation_profiles.csv	families	the dark-radiation families of the main project on the deflating spacetime: best fits of the six LambdaCDM parameters to mock microwave-background data (fiducial LambdaCDM spectra without deflation, no lensing) for fixed density N of extra radiation	chi2 = minimum chi^2; tau_reio: optical depth of reionization; H0 in km/s/Mpc; rs_drag in Mpc; S8 = sigma8 sqrt(Omega_m/0.3); dN_bbn = extra radiation during nucleosynthesis used for the helium abundance (the response of the helium abundance to Newton's constant at nucleosynthesis is included in Y_He); GN_star: Newton's constant at last scattering over today's; jacobian_refreshed = 1 if the derivatives of the spectra were recomputed at the N = 0.5 best fit of the family

data file	command	content	columns
dr_deflation_residuals.csv	families	the dark-radiation families on the deflating spacetime: best-fit spectra at N = 0.5 (Planck-like mock) relative to the fiducial LambdaCDM spectra without deflation	famK_TT = TT_model/TT_fid - 1; famK_EE likewise; famK_TE = (TE_model - TE_fid)/sqrt(TT_fid EE_fid)
dr_deflation_zt_scan.csv	families	the stepped fluid of the model (r_step = 15^(1/3)) with N = 0.4 after the step, on the deflating spacetime: best fit to the Planck-like mock as a function of the transition redshift z_t	
events.csv	history	the order of events on a deflating spacetime: values of alpha_VS = g_VS^2/(4 pi) at which the thresholds of the model are passed, for several mass ratios m_s/m; matter of Fermi momentum 0.5 m; cutoff 100 m for the vacuum condensate	m_s_over_m; event; quantity that passes a critical value; that value; alpha_VS at the event; tau_pair = ln(alpha/alpha at which a pair becomes bound) = gsum (x4 - x4 of that event): negative = earlier
extinction_checks.csv	modes	4+4 deflating spacetime: the eight bounded solutions of modes that depend on the deflating times	case; E_perp; x4_turn: x4 at which k_d = E_perp; x4_late: x4 at which the backward integration starts (WKB exponent 30 beyond the turning value); S_wkb: integral of sqrt(k_d^2 - E_perp^2) dx4 between them; log_amp_min, log_amp_max: logarithm of the amplification of the eight solutions between x4_late and the early time (k_d = 1e-9 E_perp); standing_wave_dev: relative difference of the Gram matrices of the positive- and negative-energy parts at the early time; charge_ratio: |psihat^dagger W psihat|/|psihat^dagger psihat|; steps
extinction_profile.csv	modes	4+4 deflating spacetime: |psihat|^2 of one bounded solution as a function of x4, normalized to its maximum, for the five cases of data/extinction_checks.csv	case; x4; k_d_over_E_perp; norm2: numerical; exact: 2 u_d |K_(1/2 + i E_perp/rate_t)(u_d)|^2 with u_d = k_d/rate_t, scaled to the numerical curve by least squares (cases 1 to 4, where one rate governs; nan where not evaluated)
families_checks.csv	families	PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the forecast for the dark-radiation families of the main project on the deflating spacetime	
flat_matrix_checks.csv	matrices	PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: the checks of the real 16x16 Dirac matrices of the main project, executed again (these matrices are the matrices of the orthonormal frame of the deflating spacetimes); identifiers 44-.. refer to gamma1644, 71-.. to gamma1671	
forecast_checks.csv	forecast	PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the forecast for the rate of deflation from mock data of the microwave background	

data file	command	content	columns
gap_history.csv	history	the gap of the paired state along the deflation clock tau_d = gsum (x4 - x4_ref): equilibrium gap at the Fermi surface and critical temperature for the couplings of each moment (adiabatic history)	set: 1 = (m_s, k_F) = (0.1, 0.5), 2 = (0.3, 0.5), 3 = (1.0, 0.5), 4 = (0.1, 0.2), in units of m; tau_d; g; alpha = g^2/(4 pi); D_VS = alpha m/m_s; C = g^2 m^2/m_s^2; lambda_S, lambda_P: dimensionless pairing strengths of the two modes; gap_V, gap_T, gap_S: gap of the vector channel, of the two-form channel and with the radial mode only, in units of the Fermi kinetic energy E_F (0 = below 1e-7 E_F); Tc_over_gap_V; dln_gap_dtau = d ln(gap_V)/d tau_d; follow = gap_V/(d ln gap_V/d tau_d) in units of E_F: the gap can follow the deflation only if gsum is much smaller than this
geodesic_1644d.csv	matrices	a geodesic of the deflating 4+4 spacetime with momentum along the deflating coordinate x5 (check 44d-G9): g55 = 1, const5 = 0.8, m = 1, P_1 = 0.5, P_5 = 0.3, start at x4 = -2	affine: affine parameter (proper time for m = 1); x4; u4 = dx4/dlam
geodesic_1671d.csv	matrices	a geodesic of the deflating 7+1 spacetime with momentum along the deflating coordinate x5 (check 71d-G8): g55 = 1, const5 = 0.8, m = 1, P_1 = 0.5, P_5 = 0.3, start at x4 = -2	affine: affine parameter (proper time for m = 1); x4; u4 = dx4/dlam
gl_quartic_check.csv	channels	check of the identities Tr[M_gap M_gap^dagger]/8 = sum_a |Delta^a|^2 and Tr[(M_gap M_gap^dagger)^2]/8 = 2 (sum_a |Delta^a|^2)^2 - |eta_ab Delta^a Delta^b|^2 for the gap matrix M_gap of the four spin-0 vector pairs (frame components a = 0, 5, 6, 7), with eta_ab the frame metric restricted to these axes: (-1,-1,-1,-1) in 7+1 and (+1,-1,-1,-1) in 4+4; five complex test vectors, at rest and at |p| = m	
gravity_checks.csv	gravity	PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the Einstein-Lovelock equations with Kohn-Sham matter (gr-..)	
gravity_deflating.csv	gravity	the deflating spacetime with Kohn-Sham matter as a source of gravity: evolution of the scale factor a of the four flat coordinates x0..x3 and of the scale factor b of the three deflating coordinates (equal rates), from the initial data a = b = 1, H_a = 0 and H_b given by the constraint; rate of the undisturbed deflation g55 = 0.01 m. matter 0: 7+1, momenta along all seven space coordinates, saturated matter of the author's example (m = 1, G_S = -1e5, G_V = +1e4); matter 1: fields that do not depend on x5, x6, x7 (D = 4; G_S = -256.951, G_V = 25.695, Fermi momentum 0.66682 of saturation). gravity 0: Einstein-Lovelock with LambdaBar = 3 g55^2 and alpha_2 = 1/(4 g55^2), for which the deflating spacetime without matter is an exact solution; gravity 1: Einstein gravity with the matter as the only source. loading = kappa eps(0)/(3 g55^2)	matter; gravity; loading; x4 (units 1/m); tau_ref = 3 g55 x4: clock of the undisturbed deflation; a; b; H_a/g55; H_b/g55; tau_d = -3 ln b: the deflation clock actually reached; eps/eps(0); p_a/eps(0); p_b/eps(0); M: mass of the particles in matter; kappa eps/(3 g55^2)

data file	command	content	columns
gravity_deflating_summary.csv	gravity	summary of the evolutions of gravity_deflating.csv: initial rate from the constraint and the state at the end of the integration (six e-foldings of the clock of the undisturbed deflation, or the point at which the integration stopped)	matter; gravity; loading = kappa eps(0)/(3 g55^2); Hb0_over_g: initial H_b/g55; stop: 'end' if the integration reached tau_ref = 6, 'runaway' if a rate exceeded 50 g55 before; tau_ref_end; a_end; b_end; Ha_end_over_g; Hb_end_over_g; constraint: largest relative violation of the constraint
gravity_reference.csv	gravity	the self-consistent evolution of the author's document KOHN_SHAM_GRAVITY (section 16.3), computed again by this program: gravity with three ordinary and four auxiliary space coordinates driven by Kohn-Sham matter (m = 1, G_S = -1e5, G_V = +1e4, comoving Fermi momentum of saturation), kappa = 1000, LambdaBar = 0, H_a(0) = 0.05, a(0) = b(0) = 1, x4 from 0 to 20 (the time is called x0 in that document); classical Runge-Kutta method with 8000 steps (check gr-G2)	gravity; quantity; here: this program; document: the value printed in the author's document; rel_diff
gravity_reference_path.csv	gravity	the two evolutions of gravity_reference.csv as functions of the time	case: 0 = Einstein, 1 = Einstein-Lovelock (alpha_2 = 20, alpha_3 = 500); x4: time; a, b: scale factors of ordinary and auxiliary space; H_a, H_b: their rates; eps: energy density of the matter; p_a, p_b: its pressures; M: mass of the particles in matter
gravity_vacuum_perturbation.csv	gravity	the deflating vacuum solution of the tuned Einstein-Lovelock gravity (LambdaBar = 3 g55^2, alpha_2 = 1/(4 g55^2), g55 = 0.01), perturbed without matter: initial rate H_b = -1.001 g55 and H_a from the constraint (two roots of opposite sign)	case: 0 = positive root H_a(0), 1 = negative root; x4; tau_ref = 3 g55 x4; a; b; H_a/g55; H_b/g55
growth_deflation.csv	matter	linear growth of the density contrast delta_m of pressureless matter in ordinary space on the deflating spacetime (assumptions A1, A2, A3), from a = 0.01 to today. With a Newton's constant that depends on the time the growth depends on the closure of the perturbation equations (A3). Closure "Pressure" (suffix _P; also the closure "Shear" far inside the horizon): delta_m'' + (2 H - gsum/2) delta_m' = 4 pi G_N(x4) rho_m delta_m, primes for derivatives with respect to x4. Closure "Momentum" (suffix _M): in conformal time, delta_m'' + [(2 He' + R)/(2 He)] delta_m' = [(Hc - a gsum)/(2 He)] g delta_m, with Hc = a H, He = Hc - a gsum/2, g = 8 pi G_N(x4) a^2 rho_m and R = g + (4/3) 8 pi G_N(x4) a^2 rho_r. Suffix _N4: the four-dimensional equation delta_m'' + 2 H delta_m' = 4 pi G_N(x4) rho_m delta_m, which is not a consequence of the eight-dimensional equations and is listed only because an earlier version of this file contained it	gsum_over_H0; h: the Hubble constant that keeps the angular scale of the sound horizon (as in data/hubble_deflation.csv); then for each of the three equations: growth = delta_m at a = 1 for delta_m = a + 2 a_eq/3 at a = 0.01; f_growth = d ln delta_m/d ln a at a = 1; growth_ratio, f_growth_ratio, f_sigma8_ratio: ratios to the model without deflation (f_sigma8_ratio is the ratio of the products growth times f_growth); growth_ratio_fixed_h, f_growth_ratio_fixed_h: the same ratios if the Hubble constant is held at 67.36 km/s/Mpc instead

data file	command	content	columns
higgs_scale_exponent.csv	cosmology	the author's hypothesis H2 on a deflating spacetime: if the Higgs field is the pair condensate, the electroweak scale v_EW is the scale of that condensate and depends on alpha_VS, which grows as exp(gsum x4). Table: n_EW = d ln v_EW/d ln alpha_VS in the scale-invariant ladder equation (massless fermions and bosons; the test case of the supplement S4 of the main project, Sec. 7.2), where rho_c^2 = (alpha_VS/alpha_c - 1)/4 and rho_c ln(cutoff/scale) = atan(1/(2 rho_c))	alpha_over_alpha_c; ln_cutoff_over_scale; n_EW; gsum_max_per_yr and gsum_max_over_H0: the bound on the rate of deflation from the constancy of the ratio mu_pe of the proton mass to the electron mass, 8.0e-17 per year (two sigma), divided by n_EW
history_checks.csv	history	PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the history of the model along the deflation clock (couplings that grow as exp(gsum x4))	
hubble_deflation.csv	cosmology	the Hubble constant inferred from the angular scale of the sound horizon when x5, x6, x7 deflate at a constant total rate gsum throughout cosmic history (assumptions A1, A2 of cosmo_defl.rs; three equal rates); densities omega_b, omega_c and 100 theta_* held at the values of the model without deflation; recombination and the helium fraction computed on the deflating background (the thermal history of the main project with Newton's constant of the moment)	gsum_over_H0: gsum divided by the Hubble constant of the model without deflation (67.36 km/s/Mpc); gsum_per_yr; h; dH0: change of the inferred Hubble constant in km/s/Mpc; GN_early: Newton's constant in the early universe over its value today, exp(-gsum t_0); GN_star: the same at last scattering; rs_Mpc: sound horizon at last scattering; age_Gyr; z_star: redshift of last scattering; Y_He: helium mass fraction; h_fixed_zstar, dH0_fixed_zstar: the Hubble constant and its change if the redshift of last scattering and the helium fraction are held at their values without deflation (the approximation of the module cosmo_defl)
hubble_mass_loss.csv	cosmology	dark matter made of bound pairs on a deflating spacetime: the pair loses rest mass as its binding energy grows with the coupling; inferred Hubble constant with the dark-matter density at early times, omega_b and 100 theta_* held fixed	D_today = alpha_VS m/m_s today; m_s_over_m; alpha_today; gsum_over_H0; gravity: 1 = gravity feels the deflation (assumptions A1, A2), 0 = constant G_N, the deflation acts on the dark sector only; mass_ratio: rest mass of a pair today over its mass in the early universe; h; dH0 (km/s/Mpc)
isometries.csv	symmetry	the 28 rotations and boosts of the flat eight-dimensional spacetimes, one for each pair (A, B) of coordinates: which of them are isometries of the deflating spacetimes, and which act as internal symmetries on fields that do not depend on the extra coordinates x0, x5, x6, x7	spacetime: 1644d = 4+4, 1671d = 7+1; axis_A, axis_B: the pair of coordinates (A, B); kind; flat: 1 = isometry of the flat spacetime (all are); rates_1_2_3, rates_2_2_5, rates_1_1_1: 1 = isometry of the deflating spacetime with (g55, g66, g77) = (1, 2, 3), (2, 2, 5), (1, 1, 1), verified by exact arithmetic (symmetry_defl.rs); internal: 1 = gamma^A gamma^B commutes with alpha^1, alpha^2, alpha^3, beta and i beta chi
kohnsham_checks.csv	kohnsham	PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the Kohn-Sham calculations (ks-..), of the CJT effective potential with contact interaction (cjt-..) and of the functional with the Vorton-Scalar (uni-..)	

data file	command	content	columns
ks_44c_polarization.csv	kohnsham	the covariant form of the self-interaction on 4+4 in the quantized sector (D = 4): the two halves W = +1 and W = -1 of the field as two kinds of particles with four states each. Couplings g_0 = -|g_0|, g_1 = 0.1 |g_0| with |g_0| = mult times 121.08583 (the value at which the quantized-sector form with the same couplings first binds matter); Hartree-Fock couplings of module selfint (check 44c-S7)	mult: strength of the couplings; kF_total: Fermi momentum that the total density would have with all eight states filled equally; rho: total density; pol_W = (rho_+ - rho_-)/rho: polarization; M_plus, M_minus: masses of the particles of the two halves; eps: energy density; e_bind: energy per particle minus m; e_bind_q: the same density in the quantized-sector form
ks_deflating_71.csv	kohnsham	Kohn-Sham matter on the deflating 7+1 spacetime, momenta along all seven space coordinates, collisionless (every particle keeps its wave numbers): the matter is the saturated matter of the author's example at the reference time (m = 1, G_S = -1e5, G_V = +1e4, k_F = 0.756636150701493); set 0 = equal rates, set 1 = rates in the ratio 1 : 2 : 3 (g55 : g66 : g77)	set; tau_d = gsum (x4 - x4_ref): deflation clock; s5, s6, s7: stretch factors exp(rate_t (x4 - x4_ref)) of the proper momenta along x5, x6, x7; M: mass of the particles in matter; rho: density (= rho_ref exp(tau_d)); rho_S: scalar density; eps: energy density; p_flat: pressure along each of x0..x3; p5, p6, p7: pressures along x5, x6, x7; Pi: interaction part of the pressure; e_bind: energy per particle minus m; e_bind_relaxed: the same for the state of lowest energy at the same density (a ball of occupied states); M_relaxed: its mass
ks_flat_eos.csv	kohnsham	equation of state of uniform matter on flat 7+1 spacetime (seven space dimensions, no-sea approximation) for two sets of the mean-field couplings: set 0 = G_S = -1e5, G_V = +1e4, the values that the Kohn-Sham section of the author's document KOHN_SHAM_GRAVITY uses; set 1 = G_S = -98750, G_V = 20000, the Hartree-Fock couplings G = (1 - F^T) g that the couplings g_0 = -1e5, g_1 = +1e4 of the Lagrangian give (module selfint; the same two numbers are printed in section 4 of the author's document)	set; kF: Fermi momentum; rho: density; M: mass of the particles in matter; eps: energy density; p: pressure; e_bind: energy per particle minus m; mu: chemical potential; cs2: square of the speed of sound dp/d eps (units m = 1)
ks_flat_reference.csv	kohnsham	the Kohn-Sham state of uniform matter on flat 7+1 spacetime for the example of the author's document KOHN_SHAM_GRAVITY (m = 1, G_S = -1e5, G_V = +1e4, Fermi momentum 0.756636150701493 in all seven space dimensions), computed by this program and compared with the numbers printed in that document (check ks-K1)	quantity; here: value of this program; document: value printed in the author's document; rel_diff: relative difference
ks_reduced_fixed_density.csv	kohnsham	a gas of fixed density (fixed Fermi momentum) in the sector of fields that do not depend on x5, x6, x7, while the couplings grow with the deflation clock; tau_d is counted from the binding threshold of the same ratio set	D; ratio_set; kF; tau_d; GS_abs; M; eps; p: pressure along the flat directions; Pi: pressure along the deflating coordinates (interaction part alone); e_bind: energy per particle minus m
ks_reduced_gas.csv	kohnsham	fields that do not depend on x5, x6, x7: a gas of fixed Fermi momentum while the couplings grow as exp(tau_d): the coupling and the clock reading (counted from the binding threshold of the same ratio set) at which the pressure of the gas becomes negative	D; ratio_set: 0 = G_V = 0, 1 = G_V = 0.1 |G_S|, 2 = ratio 0.2025 of the Hartree-Fock couplings of g_1 = -0.1 g_0; ratio G_V/|G_S|; kF; GS_abs: |G_S| m^(D-1) at which p = 0; tau_d: clock reading relative to the binding threshold

data file	command	content	columns
ks_reduced_saturation.csv	kohnsham	self-bound matter in the sector of fields that do not depend on x5, x6, x7, along the deflation clock: at each clock reading the state of lowest energy per particle (the saturation point, pressure zero) for the couplings of that moment, G_S = -|G_S(threshold)| exp(tau_d), G_V = ratio |G_S|; tau_d is counted from the moment at which matter first becomes self-bound	D: number of space dimensions with momentum (4 or 3); ratio_set: 0 = G_V = 0, 1 = G_V = 0.1 |G_S|, 2 = ratio 0.2025 of the Hartree-Fock couplings of g_1 = -0.1 g_0; tau_d; GS_abs = |G_S| m^(D-1); kF: Fermi momentum at saturation; rho; M: mass of the particles in matter; e_bind: energy per particle minus m; p: pressure (zero at an interior minimum); cs2: square of the speed of sound
ks_reduced_thresholds.csv	kohnsham	fields that do not depend on x5, x6, x7 (D = 4 space dimensions: the quantized sector of 4+4 with the quantized-sector form of the interaction, and the low-energy sector of 7+1; D = 3: fields that do not depend on the hidden coordinate x0 either). The couplings of the reduced Lagrangian grow as exp(tau_d). The smallest scalar attraction at which uniform matter is self-bound (energy per particle below m at some density), in units m = 1, the Fermi momentum of the bound state just above the threshold, and for comparison the critical coupling of the vacuum for a cutoff equal to m	D; ratio_set: 0 = G_V = 0, 1 = G_V = 0.1 |G_S|, 2 = ratio 0.2025 of the Hartree-Fock couplings of g_1 = -0.1 g_0; description; ratio G_V/|G_S|; GS_abs: |G_S| m^(D-1) at the threshold; kF: Fermi momentum of the bound state; Gc: critical coupling G_c m^(D-1) of the vacuum for Lambda = m
matter_checks.csv	matter	PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the dark-matter estimates on the deflating spacetime	
matter_saturation.csv	paired	self-bound normal matter along the deflation clock (mean field without the negative-energy states): C = g_VS^2 m^2/m_s^2 = 4 exp(tau_d); energy per particle at the density of lowest energy, and the exchange (Fock) energies of the two modes there for m_s/m = 0.3	tau_d; C; e_bind: lowest energy per particle minus m (units of m; negative: bound); kF_sat: Fermi momentum there; x_sat: m*/m there; g2 = g_VS^2; hartree: mean-field energy density minus that of free fermions at the same density; fock_radial, fock_phase: exchange energy densities; their ratios to |hartree|
medium_history.csv	history	normal matter of fixed density along the deflation clock: C = g_VS^2 m^2/m_s^2 grows as exp(gsum x4)	k_F (units of m); sea: 1 if the filled negative-energy states are included; C; x = m*/m: fermion mass in the matter; e_per_n: energy per fermion (units of m; below 1 = bound); pressure (units of m^4); cs2: square of the sound speed
mode_checks.csv	modes	PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the solutions of the Dirac equations on the deflating spacetimes (71d-M..: pair creation on 7+1; 44d-M..: bounded solutions on 4+4)	
order_parameter_states.csv	paired	the weak-coupling CJT potential at zero temperature as a function of the order parameter Delta^a (frame components a = 0, 5, 6, 7), with the gap matrix built from the Dirac matrices of each deflating spacetime; unit: gap of the unitary state = 1, N(0) = 1, shell half-width 200	spacetime; state; gap2_small, gap2_large: the two fourfold eigenvalues of M M^dagger; V: potential (unitary: -2; half-paired: -1); grad: norm of the gradient; flat, positive: numbers of vanishing and of positive eigenvalues of the matrix of second derivatives in the eight real coordinates (unitary states)

data file	command	content	columns
pair_amplitudes.csv	pairing	pairing amplitudes of the contact channels in the hidden-axis pair channel X = Cm.gamma^0, for momenta p, p' in ordinary space: direct evaluation with the 16x16 matrices, F_r = sum_(K of rank r) eps_K Tr[kappa(p')^dagger M_K kappa(p) M_K^T]/sqrt(N N'), against the closed forms F_0 = (E E' + M^2 - p.p')/(2 E E'), F_1 = (3 E E' - 2 M^2 - p.p')/(E E'), F_7 = (3 E E' + 2 M^2 - p.p')/(E E'), F_8 = -(E E' - M^2 - p.p')/(2 E E') (twelve of the forty random momenta of each model; checks pc-P1, pc-P2)	model: 0 = 7+1, 1 = 4+4 (quantized-sector form); M: mass; k = |p|; kp = |p'|; cos: cosine of the angle between p and p'; F0, F1, F7, F8: direct evaluation; F0c, F1c, F7c, F8c: closed forms
pair_channel_kernel.csv	channels	on-shell exchange amplitude F_X(p', p) of the Vorton-Scalar in representative pair channels on the deflating spacetimes, for fields that do not depend on x5, x6, x7 (frame matrices, weighted fields); F > 0 is attraction. The amplitude of a channel is F_X times the square of the coupling of the moment, which grows as exp(gsum x4)	
pair_classification.csv	matrices	s-wave pair states of two identical particles of the 16-component field at rest, by multiplet	
pair_gap_history.csv	pairing	the gap of the hidden-axis pair at the Fermi surface along the deflation clock, with contact couplings and the exchange of the Vorton-Scalar; g_VS = exp(tau_d/2), fermion mass m = 1; contact couplings proportional to g_VS^2 (all couplings of the reduced Lagrangian grow as exp(tau_d)), with the range parameter m_ct = m (each contact coupling g is replaced by g m_ct^2/(q^2 + m_ct^2)). Variants: 0 = exchange of the Vorton-Scalar alone (as before); 1 = exchange and scalar contact attraction, g_0eff = -g^2/m^2; 2 = exchange and contact couplings in the ratio of the author's example, g_0eff = -g^2/m^2, g_1eff = +0.1 g^2/m^2; 3 = exchange and vector contact repulsion alone, g_1eff = +0.3 g^2/m^2; 4 = exchange and axial contact coupling, g_7eff = -0.1 g^2/m^2; 5 = contact couplings alone (no Vorton-Scalar), g_0eff = -g^2/m^2, g_1eff = +0.1 g^2/m^2; 6 = exchange alone with a phase mode of mass 0.3 m (explicitly broken symmetry)	set: 1 = (m_s, k_F) = (0.1, 0.5), 2 = (0.3, 0.5), 3 = (1.0, 0.5), 4 = (0.1, 0.2), in units of m; variant; m_s; k_F; tau_d; g = g_VS; lambda = N(0) W(k_F, k_F): dimensionless pairing strength; gap: gap at the Fermi surface in units of the Fermi kinetic energy E_F (0: below 1e-7 E_F or no solution); Tc_over_gap (computed for the variants 0, 2 and 5 where the kernel has no sign change; NaN otherwise)
paired_checks.csv	paired	PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: the results of the CJT effective action along the deflation clock	
pairing_checks.csv	pairing	PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the pairing with contact self-interaction and exchange of the Vorton-Scalar (pc-..)	
quench_checks.csv	quench	PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the time-dependent dynamics of the paired state under a coupling that grows as exp(gsum x4)	

data file	command	content	columns
quench_history.csv	quench	the order parameter of the paired state while the pairing strength grows as lambda = 0.15 exp(tau_d), tau_d = gsum (x4 - x4_i), up to lambda = 0.5; time-dependent mean-field dynamics of the shell model (quench.rs), started in equilibrium	gsum; tau_d; lambda; gap_abs: |Delta| of the evolving state; gap_eq: equilibrium gap for the coupling of the moment, 1/sinh(1/lambda); ratio = gap_abs/gap_eq; n_ex: density of broken pairs; Xi_d = gsum/(lambda gap_eq): the parameter that decides whether the state can follow
quench_summary.csv	quench	summary of data/quench_history.csv, one row per rate of deflation	gsum (units of E_sh); Xi_d_start, Xi_d_end: Xi_d = gsum/(lambda gap_eq) at the start and at the end; ratio_min: smallest gap_abs/gap_eq during the run; tau_d_min: clock reading there; tau_d_catch, Xi_d_catch: clock reading and Xi_d at which gap_abs/gap_eq first exceeds 0.9 after the minimum (nan: never); ratio_final: mean of gap_abs/gap_eq over the last tenth of the run; n_ex_final: density of broken pairs at the end
reduced_symmetry.csv	symmetry	internal symmetries of the two models on the deflating spacetimes in the sector of fields that do not depend on the extra coordinates x0 x5 x6 x7: commutant of alpha^1 alpha^2 alpha^3 beta and i beta chi (frame matrices; the weighted field obeys the flat equations in this sector), computed again with the routine symmetry::analyse of the crate spacetimes	item; value for 7+1 (matrices gamma1671); value for 4+4 (matrices gamma1644, quantized sector); expected value; ok = 1 if the computed values agree with the expectation
scalar_zero_modes.csv	channels	a scalar field of mass m_bos that does not depend on x5, x6, x7 on the deflating spacetimes: phi_bos ~ exp(i k.x + exponent x4), exponent = gsum/2 +- sqrt(gsum^2/4 - k^2 - m_bos^2); units gsum = 1	m_bos (0: the phase mode; m_s for the radial mode); k; real and imaginary parts of the two exponents
selfcoupling_channels.csv	selfcoupling	the 256 channels of the self-interaction for each model: Theta^K = phase_K Gamma^(A_1)..Gamma^(A_r), Theta_K = eps_K Theta^K = (Theta^K)^(-1) (checks ..-S2, ..-S3)	model; mask: bit mask of K; K: its indices; rank r; phase: 1 or i; eta_K: product of the squares of the matrices in K; eps_K
selfcoupling_checks.csv	selfcoupling	PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: exact checks of the self-interaction of the Dirac field (71s-..: 7+1; 44q-..: 4+4, quantized-sector form; 44c-..: 4+4, covariant form; self-..: all)	
selfcoupling_examples.csv	selfcoupling	the effective couplings, the Hartree-Fock couplings and the pair coupling of the hidden-axis pair for six sets of couplings: author = the example of the author's document: g_0 = g_S = -1e5, g_1 = g_V = +1e4 (units m = 1); scalar = attractive scalar coupling alone, g_0 = -1; vector = repulsive vector coupling alone, g_1 = +1; axial = g_7 = +1 alone (axial vector); pseudoscalar = g_8 = +1 alone; chiral = g_0 = -1, g_8 = +1: the chirally invariant combination of scalar and pseudoscalar	set; model; g0eff, g1eff, g7eff, g8eff: the four independent couplings; hf_rho, hf_rhoS: the Hartree-Fock couplings of rho^2 and rho_S^2 (7+1 and quantized-sector form: G_1 and G_0; covariant form: G_V0 and G_S0); hf_rhoW, hf_rhoSW: covariant form only (G_1 and G_0); G_pair: contact coupling of the hidden-axis pair of two particles at rest (attractive if negative)

data file	command	content	columns
selfcoupling_fierz.csv	selfcoupling	the Fierz matrix of the nine channels of the self-interaction, multiplied by 16: sum_(K of rank r) (Theta_K)_(ab) (Theta^K)_(cd) = sum_s F_rs sum_(L of rank s) (Theta_L)_(ad) (Theta^L)_(cb); 16 F_rs = sum_l (-1)^(rs - l) C(s, l) C(8 - s, r - l); F^2 = 1 (checks 71s-S5, 44q-S5, 44c-S5: the same matrix for all three models)	r: rank of the left-hand side; F_r0 .. F_r8: 16 times the element of the Fierz matrix in the row r and the column 0 .. 8 (integers)
selfcoupling_hartree_fock.csv	selfcoupling	Hartree-Fock couplings of uniform matter: E_HF = (1/2) sum_i G_i x_i^2 for the densities x_i of the column 'density' (rho = <psi^dagger psi>, rho_S = <psi^dagger beta psi>, and for the covariant form of 4+4 also rho_W = <psi^dagger W psi>, rho_SW = <psi^dagger W beta psi>); each G_i is a linear combination of the nine couplings (checks ..-S7)	model: 71s = 7+1, 44q = 4+4 quantized-sector form, 44c = 4+4 covariant form; density; coupling: name of G_i; c0 .. c8: 16 times the coefficient of g_0 .. g_8 in G_i; formula: G_i written out
selfcoupling_pairs.csv	selfcoupling	the 120 pair operators psi^T X_J psi with antisymmetric X_J = Cm.gamma^J (Cm = gamma1671[4] . omega1671 on 7+1, Cm = gamma1644[0] . gamma1644[1] . gamma1644[2] . gamma1644[3] on 4+4; gamma^J the product of the real Dirac matrices with index in J) and their contact couplings in the nine channels (checks ..-S9). lam_r: eigenvalue of the pairing map P_r(kappa) = sum_K eps_K M_K kappa M_K^T on kappa = X_J. q_r: the same for the pair of two particles at rest, kappa = P_+ X_J P_+^T; the pair coupling is G_pair = sum_r g_r q_r, attractive if negative. The hidden-axis pair is J = {0}	model; mask: bit mask of J; J: its indices; rank: number of indices; eta_J: product of the squares of the gamma matrices in J; at_rest: whether the pair of two particles at rest exists (kappa not zero); w_parity (4+4): +1 if both particles are in the same half W = +-1, -1 if in different halves; lam0 .. lam8; q0 .. q8 (rational numbers; empty if the pair does not exist at rest)
selfcoupling_reduction.csv	selfcoupling	for anticommuting fields the nine invariants obey (1 + F) I = 0; this table expresses each I_r by the four independent ones: I_r = c0 I_0 + c1 I_1 + c7 I_7 + c8 I_8 (checks 71s-S6, 44q-S6, 44c-S6). The column of I_j, read downwards and multiplied by the couplings g_r, is the effective coupling g_jeff	r: rank; c0, c1, c7, c8: the coefficients of I_0, I_1, I_7, I_8
sfdm_deflation_bounds.csv	superfluid	limits on superfluid dark matter on the deflating spacetime from the requirement that its half-mode wavenumber is not smaller than that of thermal warm dark matter at the Lyman-alpha mass limits (benchmark wavenumbers of the standard model)	m_wdm_keV: thermal-relic mass limit used as benchmark; m_fermion_min_keV: smallest mass of the fermions (8 internal states) in the weak-pairing regime; vF0: their Fermi velocity today at the mean density; cs0_max: largest sound speed of a Bose condensate today at the mean density; R_TF_max_kpc: the corresponding largest Thomas-Fermi core radius
sfdm_deflation_summary.csv	superfluid	half-mode wavenumber (where P/P_cdm first falls to 1/4) of superfluid dark matter on the deflating spacetime	cs0: sound speed today at the mean density (units of c); m_thermal_keV: mass of the thermal warm-dark-matter relic with the same half-mode in the standard model (fit of Viel et al. 2005); R_TF_kpc: radius of the Thomas-Fermi core, pi c_s/sqrt(4 pi G rho) with Newton's constant of today (Bose only); dN_bbn: density of the fluid at a = 1e-9 in units of one neutrino species; GN_early: Newton's constant in the early universe over its value today

data file	command	content	columns
sfdm_deflation_transfer.csv	superfluid	linear matter power spectrum today if the dark matter is a superfluid, relative to cold dark matter on the same background: T2 = P/P_cdm, on the deflating spacetime	
spectrum_checks.csv	spectrum	PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the spectra of the microwave background on the deflating spacetime	
static_source.csv	channels	the field of a static source on the deflating background: solution of phi_bos'' - gsum phi_bos' + w2 phi_bos = exp(gsum x4) from phi_bos = phi_bos' = 0 at x4 = 0, divided by the exact particular solution exp(gsum x4)/w2; w2 = k^2 + m_bos^2	w2; gsum; x4; tau_d = gsum x4; ratio (tends to 1: the field of the source is the Yukawa field of the mass m_bos, with a strength that grows as exp(gsum x4))
superfluid_checks.csv	superfluid	PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of superfluid dark matter on the deflating spacetime	
symmetry_checks.csv	symmetry	PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham: checks of the symmetries on the deflating spacetimes (isometries, internal symmetry of the fields that do not depend on the extra coordinates, zero modes on a vortex line, reduction of the couplings)	
thermal_history.csv	history	temperature at which the expectation value of the Vorton-Scalar melts, along the deflation clock (g_VS grows as exp(gsum x4/2))	m_s (units of m); g; alpha; lambda = g^2 m_s^2/(2 m^2); C; T_high, T_largeN: closed estimates (units of m); T_half, T_zero: temperatures at which the expectation value has fallen to one half and to 0.02 of its vacuum value in the Hartree approximation (units of m); phi_before: expectation value over its vacuum value just below T_zero (a value of order one: a discontinuous transition); valid: 1 = the three preceding numbers describe a melting; 0 = the Hartree potential ceases to exist at 1.05 v_VS before the expectation value has fallen (the one-loop potential turns over too close to the vacuum value, see vacuum_history.csv), no melting temperature can be given (NaN)
unified_phase_mode.csv	kohnsham	a bare Dirac mass and the phase mode of the Vorton-Scalar (ordinary space, D = 3; fermions of the Fermi sea and of the filled states of negative energy below the cutoff Lambda; units m = 1). The mass term of the fermions is m_D + m_Y exp(i theta chi) for the phase theta of the scalar, of magnitude m(theta); the second derivative of the energy density with respect to theta at theta = 0 is E'' = (|<psibar psi>| - rho_S) m_D m_Y/m, and the mass of the phase mode is m_ph^2 = E''/v_VS^2 = C m_s^2 E"/(1 - f_D)^2. The Fermi momenta are 0, 1/4, 1/2 and 3/4 of the cutoff (the effective theory ends at the cutoff)	Lambda: cutoff; f_D = m_D/m; kF: Fermi momentum of the matter; rho_S: its scalar density; condensate = |<psibar psi>| of the vacuum; d2E: second derivative of the energy density, numerical; formula: (condensate - rho_S) f_D (1 - f_D); mph2_over_C_ms2: m_ph^2/(C m_s^2); ratio: m_ph^2 in matter divided by m_ph^2 in the vacuum = 1 - rho_S/condensate

data file	command	content	columns
unified_states.csv	kohnsham	states of the functional with bare mass, contact couplings and Vorton-Scalar (D = 3, no-sea approximation, units m = 1): eps = (1 - f_D)^2 (x^2 - 1)^2/(8 C) + e_kin(k_F, M) + (1/2) G_V rho^2 - (M - f_D - (1 - f_D) x)^2/(2 G_S), minimized over x = phi/v_VS and M	f_D; C; G_S; G_V; kF; x: expectation value of the scalar in units of its vacuum value; M: mass of the particles in matter; rho; rho_S; eps; e_bind: energy per particle minus m; slope = (M - 1)/rho_S, which tends to G_S - C at low density
unified_thresholds.csv	kohnsham	contact interaction, bare Dirac mass and Vorton-Scalar together, ordinary space (D = 3, no-sea approximation, branch connected to the vacuum): the smallest coupling C = g_VS^2 m^2/m_s^2 of the scalar at which uniform matter is self-bound, when a fraction f_D of the vacuum mass of the fermions is a bare Dirac mass and the contact couplings are G_S = -kappa_S C, G_V = kappa_V C (units m = 1). Without bare mass and contact terms the value is that of the main project, 3.701102. All three couplings grow as exp(tau_d), so that the last column is the shift of the clock reading at which matter becomes self-bound	f_D = m_D/m; kappa_S = -G_S/C; kappa_V = G_V/C; C_bind; dtau = ln(C_bind/C_bind(0, 0, 0)): negative = earlier
vacuum_history.csv	history	the one-loop potential of the Vorton-Scalar in the vacuum along the deflation clock: where it turns over	C; x_max: field (units of the vacuum value) at the maximum beyond the vacuum; x_zero: field beyond which the potential lies below its vacuum value (inf = beyond 1e30)

7. The figures

Every figure is drawn by code/scripts/make_figures.py from data files; none contains a number typed by hand. python3 code/scripts/make_figures.py draws all of them; python3 code/scripts/make_figures.py ks_flat unified draws the named ones.

figure (files figures/<name>.pdf, .png)	commands	data files
fig_binding	history, matter	dimer_history.csv, dm_resonance.csv
fig_bounds	cosmology, forecast	deflation_bounds.csv, deflation_fisher.csv
fig_cjt_history	paired	cjt_gl_history.csv, cjt_history.csv
fig_cjt_sea	kohnsham	cjt_sea_saturation.csv, cjt_sea_vacuum.csv
fig_clock	history	events.csv
fig_closure	matter, spectrum	cmb_deflation_cl.csv, cmb_deflation_closure.csv, cmb_deflation_gaps.csv, cmb_deflation_growth.csv, cmb_deflation_summary.csv, growth_deflation.csv
fig_creation	channels, modes	boson_creation.csv, creation_anisotropic.csv, creation_history.csv, creation_single.csv
fig_extinction	channels, modes	boson_bounded.csv, extinction_profile.csv
fig_families	families	dr_deflation_joint.csv, dr_deflation_profiles.csv
fig_forecast	cosmology, forecast	deflation_fisher.csv, deflation_profile.csv, deflation_residuals.csv, hubble_deflation.csv
fig_gap_history	history	gap_history.csv
fig_geodesics	matrices	geodesic_1644d.csv, geodesic_1671d.csv
fig_gravity_deflating	gravity	gravity_deflating.csv, gravity_vacuum_perturbation.csv

figure (files figures/<name>.pdf, .png)	commands	data files
fig_gravity_deflating_sector	gravity	gravity_deflating.csv, gravity_vacuum_perturbation.csv
fig_gravity_reference	gravity	gravity_reference_path.csv
fig_hubble	cosmology, matter	growth_deflation.csv, hubble_deflation.csv, hubble_mass_loss.csv
fig_isometries	symmetry	isometries.csv
fig_kernels	channels	pair_channel_kernel.csv
fig_ks_binding	kohnsham	ks_reduced_saturation.csv
fig_ks_deflating	kohnsham	ks_deflating_71.csv
fig_ks_flat	kohnsham	ks_flat_eos.csv
fig_ks_gas	kohnsham	ks_reduced_fixed_density.csv
fig_ks_halves	kohnsham	ks_44c_polarization.csv
fig_matter_history	history, paired	matter_saturation.csv, medium_history.csv, thermal_history.csv, vacuum_history.csv
fig_pair_contact	pairing	pair_gap_history.csv
fig_quench	quench	quench_history.csv, quench_summary.csv
fig_scalar	channels	scalar_zero_modes.csv, static_source.csv
fig_sfdm	superfluid	sfdm_deflation_summary.csv, sfdm_deflation_transfer.csv
fig_spectra	spectrum	cmb_deflation_cl.csv, cmb_deflation_summary.csv
fig_unified	kohnsham	unified_phase_mode.csv, unified_thresholds.csv

8. The tables and figures of the two manuscripts

The manuscripts are assembled from one source, docs_src/manuscript/*.tex, in which every table of computed numbers is a placeholder that code/scripts/build_manuscripts.py replaces by rows that code/scripts/write_doc_tables.py wrote from the data files. The two tables give, for each table and figure of each manuscript, its label in the source, the beginning of its caption (formulas replaced by dots), the commands and the data files.

Manuscript on the 4+4 deflating spacetime (53 items):

label	kind	caption begins	commands	data files
fig:isometries	figure fig_isometries	Symmetries of the deflating spacetime	symmetry	isometries.csv
fig:geodesics	figure fig_geodesics	A classical particle of the mass ... with the momenta ... along ... and ... along the deflating coordinate ..., for ... and ..., started at ...	matrices	geodesic_1644d.csv, geodesic_1671d.csv
tab:fierz	table	The Fierz matrix	selfcoupling	selfcoupling_fierz.csv
fig:extinction	figure fig_extinction	Extinction on the deflating ... spacetime	channels, modes	boson_bounded.csv, extinction_profile.csv
tab:exponents	table	How the quantities of the four-dimensional description follow the deflation clock	symmetry	coupling_exponents.csv
fig:clock	figure fig_clock	The deflation clock	history	events.csv
fig:scalar	figure fig_scalar	Modes of the Vorton-Scalar that do not depend on the deflating coordinates	channels	scalar_zero_modes.csv, static_source.csv

label	kind	caption begins	commands	data files
tab:ksthr	table	The smallest scalar attraction at which uniform matter is self-bound, for … and … space dimensions with momentum and three ratios of the vector repulsion to the scalar attraction, with the Fermi momentum of the bound state at the threshold (units …)	kohnsham	ks_reduced_thresholds.csv
fig:ksbind	figure fig_ks_binding	Self-bound matter in the sector of fields that do not depend on …, along the deflation clock, counted from the moment at which matter first becomes self-bound	kohnsham	ks_reduced_saturation.csv
tab:ksgas	table	A gas of fixed Fermi momentum in the sector of fields that do not depend on …	kohnsham	ks_reduced_gas.csv
fig:ksgas	figure fig_ks_gas	A gas of fixed density in the sector of fields that do not depend on … (…, …), along the deflation clock counted from the binding threshold	kohnsham	ks_reduced_fixed_density.csv
tab:kshalves	table	Covariant form on …	kohnsham	ks_44c_polarization.csv
fig:kshalves	figure fig_ks_halves	Covariant form of the self-interaction on …	kohnsham	ks_44c_polarization.csv
fig:kernels	figure fig_kernels	Exchange amplitudes in three pair channels, divided by the square of the coupling of the moment, against the scattering angle, for momenta of the magnitude … in the plane of … and …	channels	pair_channel_kernel.csv
tab:gap	table	The gap along the deflation clock for two of the four sets computed	history	gap_history.csv
fig:gap	figure fig_gap_history	The gap along the deflation clock, for four sets of the mass of the radial mode and of the Fermi momentum (in units of …)	history	gap_history.csv
tab:gapcontact	table	The gap at the Fermi surface in units of the Fermi kinetic energy, with contact couplings, at the clock reading {{GEN:gapc_tau}} […], for the Fermi momentum … and three masses of the radial mode	pairing	pair_gap_history.csv
fig:paircontact	figure fig_pair_contact	The gap of the hidden-axis pair along the deflation clock with contact couplings, for the Fermi momentum … and three masses … of the radial mode of the Vorton-Scalar	pairing	pair_gap_history.csv
tab:dimer	table	Two fermions as the clock passes the moment of binding	history	dimer_history.csv
fig:binding	figure fig_binding	The bound pair along the clock	history, matter	dimer_history.csv, dm_resonance.csv
tab:events	table	The order of events along the deflation	history	events.csv

label	kind	caption begins	commands	data files
tab:quench	table	The paired state while the pairing strength grows from 0.15 to 0.5, for seven rates of deflation	quench	quench_summary.csv
fig:quench	figure fig_quench	The paired state while the coupling grows	quench	quench_history.csv, quench_summary.csv
tab:cjt	table	The paired state along the clock from the CJT potential, for ...	paired	cjt_history.csv
fig:cjt	figure fig_cjt_history	The paired state along the clock, from the CJT potential, for the two sets of Table tab:cjt	paired	cjt_gl_history.csv, cjt_history.csv
tab:saturation	table	Self-bound matter along the clock, in the mean-field approximation without the negative-energy states, for ...	paired	matter_saturation.csv
fig:matter	figure fig_matter_history	Matter, vacuum and temperature along the clock	history, paired	matter_saturation.csv, medium_history.csv, thermal_history.csv, vacuum_history.csv
tab:cjtks	table	The Kohn-Sham method (KS) and the CJT effective potential compared, without the filled states of negative energy (units ...)	kohnsham	cjt_vs_ks.csv, kohnsham_checks.txt
tab:cjtvac	table		kohnsham	cjt_sea_vacuum.csv
fig:cjtsea	figure fig_cjt_sea	The filled states of negative energy	kohnsham	cjt_sea_saturation.csv, cjt_sea_vacuum.csv
tab:unified	table	Bare Dirac mass, contact couplings and Vorton-Scalar together (..., without the filled states)	kohnsham	unified_thresholds.csv
fig:unified	figure fig_unified	Bare Dirac mass, contact couplings and Vorton-Scalar together	kohnsham	unified_phase_mode.csv, unified_thresholds.csv
tab:phasemode	table	The phase mode of the Vorton-Scalar when a tenth of the fermion mass is a bare Dirac mass (...; ...; units ...)	kohnsham	unified_phase_mode.csv
tab:states	table	The potential ... of the paired state for six states of the order parameter, in units of ..., where ... is the gap of a unitary state	paired	order_parameter_states.csv
tab:gravref	table	The reference calculation	gravity	gravity_reference.csv
fig:gravref	figure fig_gravity_reference	The reference calculation of Table tab:gravref	gravity	gravity_reference_path.csv
tab:gravdefl	table	The deflating spacetime with matter as a source of gravity	gravity	gravity_deflating_summary.csv
fig:gravdefl	figure fig_gravity_deflating_sector	The deflating spacetime with matter as a source of gravity (matter of the quantized sector; the runs of Table tab:gravdefl)	gravity	gravity_deflating.csv, gravity_vacuum_perturbation.csv
tab:gravvac	table	The tuned Einstein-Lovelock gravity without matter, started at ... with the two values of ... that the constraint allows	gravity	gravity_vacuum_perturbation.csv

label	kind	caption begins	commands	data files
tab:hubble	table	The Hubble constant inferred from the angular scale of the sound horizon if Newton's constant grows with the deflation (assumptions A1 and A2)	cosmology	hubble_deflation.csv
fig:hubble	figure fig_hubble	The Hubble constant and the growth of structure on a deflating spacetime	cosmology, matter	growth_deflation.csv, hubble_deflation.csv, hubble_mass_loss.csv
tab:bounds	table	Upper bounds on the present total rate of deflation, each under the assumption named	cosmology	deflation_bounds.csv
fig:bounds	figure fig_bounds	Upper bounds on the present total rate of deflation in units of the Hubble constant, each under the assumption of Table tab:bounds	cosmology, forecast	deflation_bounds.csv, deflation_fisher.csv
tab:spectra	table	Spectra of the microwave background on the deflating spacetime at a fixed angle of the sound horizon (assumptions A1, A2, A3; three equal rates)	spectrum	cmb_deflation_summary.csv
fig:spectra	figure fig_spectra	The spectra of the microwave background at a fixed angle of the sound horizon	spectrum	cmb_deflation_cl.csv, cmb_deflation_summary.csv
fig:closure	figure fig_closure	The closure of the perturbation equations, at the largest rate that nucleosynthesis allows	matter, spectrum	cmb_deflation_cl.csv, cmb_deflation_closure.csv, cmb_deflation_gaps.csv, cmb_deflation_growth.csv, cmb_deflation_summary.csv, growth_deflation.csv
tab:fisher	table	Forecast for the rate of deflation from mock data of the microwave background (Fisher matrix; three equal rates; assumptions A1, A2, A3)	forecast	deflation_fisher.csv
fig:forecast	figure fig_forecast	Mock data of the microwave background against the rate of deflation	cosmology, forecast	deflation_fisher.csv, deflation_profile.csv, deflation_residuals.csv, hubble_deflation.csv
tab:families	table	The seven families of dark radiation on the deflating spacetime (Planck-like mock data; assumptions A1, A2, A3)	families	dr_deflation_joint.csv
fig:families	figure fig_families	The seven families of dark radiation on the deflating spacetime (Planck-like mock data)	families	dr_deflation_joint.csv, dr_deflation_profiles.csv
tab:halos	table	Could the fermions of the model be degenerate, or their bound pairs condensed, in observed systems of dark matter? …	matter	dm_halo_estimates.csv
fig:sfdm	figure fig_sfdm	Superfluid dark matter on the deflating spacetime	superfluid	sfdm_deflation_summary.csv, sfdm_deflation_transfer.csv
tab:higgs	table		cosmology	higgs_scale_exponent.csv

Manuscript on the 7+1 deflating spacetime (55 items):

label	kind	caption begins	commands	data files
fig:isometries	figure fig_isometries	Symmetries of the deflating spacetime	symmetry	isometries.csv
fig:geodesics	figure fig_geodesics	A classical particle of the mass … with the momenta … along … and … along the deflating coordinate …, for … and …, started at …	matrices	geodesic_1644d.csv, geodesic_1671d.csv
tab:fierz	table	The Fierz matrix	selfcoupling	selfcoupling_fierz.csv
fig:creation	figure fig_creation	Creation of particles on the deflating … spacetime	channels, modes	boson_creation.csv, creation_anisotropic.csv, creation_history.csv, creation_single.csv
tab:exponents	table	How the quantities of the four-dimensional description follow the deflation clock	symmetry	coupling_exponents.csv
fig:clock	figure fig_clock	The deflation clock	history	events.csv
fig:scalar	figure fig_scalar	Modes of the Vorton-Scalar that do not depend on the deflating coordinates	channels	scalar_zero_modes.csv, static_source.csv
tab:ksflat	table	Uniform matter on the flat … spacetime at saturation, for …, … (units …, seven space dimensions)	kohnsham	ks_flat_reference.csv
fig:ksflat	figure fig_ks_flat	Uniform matter on the flat … spacetime (seven space dimensions; units …) for two pairs of mean-field couplings	kohnsham	ks_flat_eos.csv
fig:ksdefl	figure fig_ks_deflating	Matter of the … model with momentum along all seven space coordinates while … deflate, without collisions	kohnsham	ks_deflating_71.csv
tab:ksthr	table	The smallest scalar attraction at which uniform matter is self-bound, for … and … space dimensions with momentum and three ratios of the vector repulsion to the scalar attraction, with the Fermi momentum of the bound state at the threshold (units …)	kohnsham	ks_reduced_thresholds.csv
fig:ksbind	figure fig_ks_binding	Self-bound matter in the sector of fields that do not depend on …, along the deflation clock, counted from the moment at which matter first becomes self-bound	kohnsham	ks_reduced_saturation.csv
tab:ksgas	table	A gas of fixed Fermi momentum in the sector of fields that do not depend on …	kohnsham	ks_reduced_gas.csv
fig:ksgas	figure fig_ks_gas	A gas of fixed density in the sector of fields that do not depend on … (…, …), along the deflation clock counted from the binding threshold	kohnsham	ks_reduced_fixed_density.csv
fig:kernels	figure fig_kernels	Exchange amplitudes in three pair channels, divided by the square of the coupling of the moment, against the scattering angle, for momenta of the magnitude … in the plane of … and …	channels	pair_channel_kernel.csv

label	kind	caption begins	commands	data files
tab:gap	table	The gap along the deflation clock for two of the four sets computed	history	gap_history.csv
fig:gap	figure fig_gap_history	The gap along the deflation clock, for four sets of the mass of the radial mode and of the Fermi momentum (in units of …)	history	gap_history.csv
tab:gapcontact	table	The gap at the Fermi surface in units of the Fermi kinetic energy, with contact couplings, at the clock reading {{GEN:gapc_tau}} […], for the Fermi momentum … and three masses of the radial mode	pairing	pair_gap_history.csv
fig:paircontact	figure fig_pair_contact	The gap of the hidden-axis pair along the deflation clock with contact couplings, for the Fermi momentum … and three masses … of the radial mode of the Vorton-Scalar	pairing	pair_gap_history.csv
tab:dimer	table	Two fermions as the clock passes the moment of binding	history	dimer_history.csv
fig:binding	figure fig_binding	The bound pair along the clock	history, matter	dimer_history.csv, dm_resonance.csv
tab:events	table	The order of events along the deflation	history	events.csv
tab:quench	table	The paired state while the pairing strength grows from 0.15 to 0.5, for seven rates of deflation	quench	quench_summary.csv
fig:quench	figure fig_quench	The paired state while the coupling grows	quench	quench_history.csv, quench_summary.csv
tab:cjt	table	The paired state along the clock from the CJT potential, for …	paired	cjt_history.csv
fig:cjt	figure fig_cjt_history	The paired state along the clock, from the CJT potential, for the two sets of Table tab:cjt	paired	cjt_gl_history.csv, cjt_history.csv
tab:saturation	table	Self-bound matter along the clock, in the mean-field approximation without the negative-energy states, for …	paired	matter_saturation.csv
fig:matter	figure fig_matter_history	Matter, vacuum and temperature along the clock	history, paired	matter_saturation.csv, medium_history.csv, thermal_history.csv, vacuum_history.csv
tab:cjtks	table	The Kohn-Sham method (KS) and the CJT effective potential compared, without the filled states of negative energy (units …)	kohnsham	cjt_vs_ks.csv, kohnsham_checks.txt
tab:cjtvac	table		kohnsham	cjt_sea_vacuum.csv
fig:cjtsea	figure fig_cjt_sea	The filled states of negative energy	kohnsham	cjt_sea_saturation.csv, cjt_sea_vacuum.csv
tab:cjtsea	table	The reference example (…, …, …, units …) with the filled states of negative energy below the cutoff …	kohnsham	kohnsham_checks.txt

label	kind	caption begins	commands	data files
tab:unified	table	Bare Dirac mass, contact couplings and Vorton-Scalar together (…, without the filled states)	kohnsham	unified_thresholds.csv
fig:unified	figure fig_unified	Bare Dirac mass, contact couplings and Vorton-Scalar together	kohnsham	unified_phase_mode.csv, unified_thresholds.csv
tab:phasemode	table	The phase mode of the Vorton-Scalar when a tenth of the fermion mass is a bare Dirac mass (…; …; units …)	kohnsham	unified_phase_mode.csv
tab:states	table	The potential … of the paired state for six states of the order parameter, in units of …, where … is the gap of a unitary state	paired	order_parameter_states.csv
tab:gravref	table	The reference calculation	gravity	gravity_reference.csv
fig:gravref	figure fig_gravity_reference	The reference calculation of Table tab:gravref	gravity	gravity_reference_path.csv
tab:gravdefl	table	The deflating spacetime with matter as a source of gravity	gravity	gravity_deflating_summary.csv
fig:gravdefl	figure fig_gravity_deflating	The deflating spacetime with matter as a source of gravity (matter with momenta along all seven space coordinates; the runs of Table tab:gravdefl)	gravity	gravity_deflating.csv, gravity_vacuum_perturbation.csv
tab:gravvac	table	The tuned Einstein-Lovelock gravity without matter, started at … with the two values of … that the constraint allows	gravity	gravity_vacuum_perturbation.csv
tab:hubble	table	The Hubble constant inferred from the angular scale of the sound horizon if Newton's constant grows with the deflation (assumptions A1 and A2)	cosmology	hubble_deflation.csv
fig:hubble	figure fig_hubble	The Hubble constant and the growth of structure on a deflating spacetime	cosmology, matter	growth_deflation.csv, hubble_deflation.csv, hubble_mass_loss.csv
tab:bounds	table	Upper bounds on the present total rate of deflation, each under the assumption named	cosmology	deflation_bounds.csv
fig:bounds	figure fig_bounds	Upper bounds on the present total rate of deflation in units of the Hubble constant, each under the assumption of Table tab:bounds	cosmology, forecast	deflation_bounds.csv, deflation_fisher.csv
tab:spectra	table	Spectra of the microwave background on the deflating spacetime at a fixed angle of the sound horizon (assumptions A1, A2, A3; three equal rates)	spectrum	cmb_deflation_summary.csv
fig:spectra	figure fig_spectra	The spectra of the microwave background at a fixed angle of the sound horizon	spectrum	cmb_deflation_cl.csv, cmb_deflation_summary.csv

label	kind	caption begins	commands	data files
fig:closure	figure fig_closure	The closure of the perturbation equations, at the largest rate that nucleosynthesis allows	matter, spectrum	cmb_deflation_cl.csv, cmb_deflation_closure.csv, cmb_deflation_gaps.csv, cmb_deflation_growth.csv, cmb_deflation_summary.csv, growth_deflation.csv
tab:fisher	table	Forecast for the rate of deflation from mock data of the microwave background (Fisher matrix; three equal rates; assumptions A1, A2, A3)	forecast	deflation_fisher.csv
fig:forecast	figure fig_forecast	Mock data of the microwave background against the rate of deflation	cosmology, forecast	deflation_fisher.csv, deflation_profile.csv, deflation_residuals.csv, hubble_deflation.csv
tab:families	table	The seven families of dark radiation on the deflating spacetime (Planck-like mock data; assumptions A1, A2, A3)	families	dr_deflation_joint.csv
fig:families	figure fig_families	The seven families of dark radiation on the deflating spacetime (Planck-like mock data)	families	dr_deflation_joint.csv, dr_deflation_profiles.csv
tab:halos	table	Could the fermions of the model be degenerate, or their bound pairs condensed, in observed systems of dark matter? …	matter	dm_halo_estimates.csv
fig:sfdm	figure fig_sfdm	Superfluid dark matter on the deflating spacetime	superfluid	sfdm_deflation_summary.csv, sfdm_deflation_transfer.csv
tab:higgs	table		cosmology	higgs_scale_exponent.csv

9. The scripts

script (in code/scripts)	what it does
make_figures.py	draws every figure from the data files
write_symbol_registry.py	writes the registry of symbols provenance/symbols_master.csv and its tables
check_symbols.py	the seven checks of the symbols (Section 5.14)
write_doc_tables.py	writes every table and every computed number of the documents from the data files, and the tables of provenance of this document, into provenance/generated/tables_cjt_ks.md
assemble_docs.py	builds each Markdown document from its source in docs_src by inserting the generated tables and the reports
build_docs.sh	typesets every Markdown document (pandoc, XeLaTeX) and checks its layout with check_layout.py
build_manuscripts.py	assembles and typesets the two manuscripts and writes their Markdown versions with tex2md.py
check_numbers.py	verifies that every number of the running text of the documents is a number that a program stated (Section 5.15)
check_self_contained.py	verifies that no document sends its reader to another document (Section 5.16)
verify_deflating_independent.py	independent verification of geometry, Dirac field, modes, cosmology (Section 5.11)
verify_cjt_ks_independent.py	independent verification of self-interaction, Kohn–Sham and CJT calculations, pairing, gravity (Section 5.12)
mutation_test_independent.py	the test of the two independent scripts with deliberate errors (Section 5.13)

script (in code/scripts)	what it does
vendor_eb.py	writes the four modules of the Einstein–Boltzmann program of this folder from those of the library crate, applying a listed set of changes; with --check it verifies that they are up to date

10. The source files of the program

The program consists of 33 source files with 15007 lines, comments included, and 68 unit tests. Each file begins with a comment that states the equations it implements.

file	lines	content
selfint.rs	1444	the self-interaction: the 256 matrices and nine invariants of the three models (7+1; 4+4 in the quantized-sector and in the covariant form) in exact integer arithmetic; Fierz matrix; reduction to four couplings; Hartree-Fock couplings; pair couplings; symmetries
kohn_sham.rs	704	the Kohn-Sham method for uniform matter with scalar and vector coupling: Fermi integrals over a ball and over an ellipsoid of momenta, gap equation, energy density, pressures, saturation, thresholds; the two halves of the 4+4 field; the functional with bare mass, contact couplings and Vorton-Scalar
cjt_contact.rs	365	the CJT effective potential of uniform matter with contact interaction, without and with the filled states of negative energy; vacuum mass; critical coupling
pair_contact.rs	291	pairing amplitudes of the contact channels in the hidden-axis channel (closed forms and direct evaluation); the pairing kernel with contact couplings and exchange of the Vorton-Scalar
gravity.rs	404	the Einstein-Lovelock equations for two scale factors from the pair curvatures; constraint; evolution with Kohn-Sham matter as the source
cmd_selfcoupling.rs, cmd_kohnsham.rs, cmd_pairing.rs, cmd_gravity.rs	1357	the commands selfcoupling, kohnsham, pairing, gravity
rat.rs	223	exact rational numbers with overflow checks; the trait that lets the geometry be computed in rational or in floating-point arithmetic
qmat.rs	201	exact matrices with complex-rational entries
metric.rs	749	the deflating metric in the sign convention of the hypothesis: Christoffel symbols, Riemann, Ricci and Einstein tensors from their definitions and in closed form; Killing vectors; Lie derivative
geometry.rs	447	the exact geometry checks G1 to G7, G10 (Lovelock tensors) and the geodesics G8, G9
curved.rs	548	coordinate matrices, spin connection, Dirac operators; checks C1 to C8, D1 to D7
modes.rs	766	integration of the mode equations (commutator-free Magnus method), creation of particles, bounded solutions, modified Bessel function of complex order
scalar.rs	295	modes of a scalar field: exponents, static source, creation, bounded solutions
history.rs	274	the couplings along the clock; thresholds and events
quench.rs	280	time-dependent mean-field dynamics of the shell model
cosmo_defl.rs	512	expansion with deflation, sound horizon, age, bounds on the rate, growth of pressureless matter for the closures
spectra_defl.rs	611	the assumption about the perturbations and its closures; the Hubble constant at a fixed angle with the full thermal history; potentials and gaps of single modes; derivatives of the spectra

file	lines	content
`eb_cosmo.rs`, `eb_boltzmann.rs`, `eb_spectra.rs`, `eb_fit.rs`	1891	the Einstein-Boltzmann program on the deflating spacetime: copies of four modules of the library crate with a listed set of changes, written by `code/scripts/vendor_eb.py`
`symmetry_defl.rs`	217	test of the 28 rotations and boosts; algebra of the vortex zero modes
`cmd_symmetry.rs`, `cmd_channels.rs`, `cmd_matter.rs`, `cmd_paired.rs`	1036	the commands `symmetry`, `channels`, `matter`, `paired`
`cmd_spectra.rs`	1425	the commands `background`, `spectrum`, `forecast`, `families`, `superfluid`
`main.rs`	832	the commands `matrices`, `modes`, `history`, `quench`, `cosmology`; the list of commands
`io.rs`, `lib.rs`	135	writing of data files and reports; list of modules

Appendix A. The symbols, and the names used in programs and data files

Columns: the symbol as it is written in the documents; the names that programs and data files use for it; its meaning; its origin.

1 Coordinates and indices

symbol	names in programs and data files	meaning	origin
$X = \{x0, \dots, x7\}$	`x0` `x1` `x2` `x3` `x4` `x5` `x6` `x7`	the eight coordinates of both spacetimes, never renamed: $x0$ hidden space-like; $x1, x2, x3$ ordinary space; $x4$ ordinary time; $x5, x6, x7$ additional coordinates (time-like in 4+4, space-like in 7+1)	N01 (text at its head) and N04 `X` (section DEFCHART): the same eight names with the same meaning. N02 uses the same eight names with `x0` as the time: related by swap04
swap_{04}	`swap04`	the exchange of the index values 0 and 4; it carries the labels of `7+1.nb` (time at index 0) into those of the project (time at index 4)	new
A, B, C, D	—	index with the values $0, \dots, 7$; always labels the coordinate xA	new
a, b	—	indices whose range is stated at each place of use: the four extra axes, values 0, 5, 6, 7 (in everything that concerns pairs and the order parameter), or the four flat coordinates other than the time, values 0, 1, 2, 3 (in the statements of the machine checks). The letters `a`, `b` in sans-serif type are the scale factors of the Einstein-Lovelock evolution; a_{sp}, a_{sc}, $a(T)$, b_t and b have their own entries	new
i, j	—	index of space-like coordinates; values 1, 2, 3 (ordinary space) unless the range is stated: $j = 0, 1, 2, 3$ for the four flat coordinates other than the time where the deflating coordinates occur as well, and all seven space-like directions in the Hamiltonian of 7+1. In the Einstein-Lovelock equations i, j label the seven space directions	new
h (index)	—	index with the values 1, 2, 3 that labels the 4×4 matrices s_h, t_h	N01, N02, N05 `h` in `s4by4[h]`, `t4by4[h]`

2 Matrices of the 4+4 spacetime (tag 1644)

symbol	names in programs and data files	meaning	origin
η_{1644}	`eta1644`	metric of the flat 4+4 spacetime and matrix of the squares of its Dirac matrices, diag$(+1,+1,+1,+1,-1,-1,-1,-1)$: space-like $=+1$. It equals the flat metric $\eta^{(4+4)}$ of the author's sign convention for the deflating spacetimes	N01, N02, N05, N10–N12 `η4488`; in N01 also `η8` and `g_AB`; D08 G_{AB} with the index value 8 in the place of the value 0
s_h, t_h	—	the three self-dual and the three anti-self-dual real antisymmetric 4×4 matrices	N01, N02, N05, N10–N12 `s4by4[h]`, `t4by4[h]`; the letters are those of D06 (s^h self-dual, t^h anti-self-dual)
τ^A_{1644}, $\bar\tau^A_{1644}$	`tau1644` `taubar1644`	the real 8×8 blocks of the 4+4 Dirac matrices (reduced generators of the split octonion algebra)	N01, N10–N12 `τ[A]` and `τ` with an overbar; N02 `τ[A]`, `τbar[A]`; matrices of this kind are the τ^A, $\bar\tau^A$ of D07 and D08, whose index runs from 1 to 8 (the value 8 there is the value 0 here)
σ_{88}	`sigma88`	the 8×8 matrix with the 4×4 blocks $(0, 1_4;\ 1_4, 0)$	N01 `σ`, `σ88`; N02 `σ`; N05 `σ88`
γ^A_{1644}	`gamma1644`	the eight real 16×16 Dirac matrices of the 4+4 spacetime	N01, N10–N12 `T16^A[A]`; N02 `T16[A]`; the t^A of D08 are built in the same way
Σ^{AB}_{1644}	`Sigma1644`	$\frac{1}{4}[\gamma^A_{1644}, \gamma^B_{1644}]$, the scaled commutators; they span $\mathfrak{so}(4,4)$	new name; the same matrices are `SAB` and `sAB[A][B]` of N10–N12; NOT `sAB` of N02, which is a sign
ω_{1644}	`omega1644`	the product $\gamma^0_{1644}\cdots\gamma^7_{1644}$; real chirality matrix of 4+4, diag$(-1_8, +1_8)$	N01 `T16^A[8]`; N02 `T16[8]`
P^L_{1644}, P^R_{1644}	`PL1644` `PR1644`	the real chiral projectors $\frac{1}{2}(1\mp\omega_{1644})$	N01 `P_L`, `P_R`; N02 `PL44`, `PR44`
C^-_{1644}, C^+_{1644}	`Cm1644` `Cp1644`	the products $\gamma^0\gamma^1\gamma^2\gamma^3$ and $\gamma^4\gamma^5\gamma^6\gamma^7$ of the 4+4 matrices; $C^\mp\gamma^A(C^\mp)^{-1} = \mp(\gamma^A)^T$	N01, N02, N10–N12 `σ16` (C^-); N02 `S44` (C^+)
W_{1644}	`W1644`	$i\,C^-_{1644}\gamma^4_{1644}$, the grading: Hermitian, $W^2=1$; density of the covariant current	new
K_{1644}	`K1644`	generator of the evolution of dirac1644 in $x4$: $\partial_4\psi = K\psi$	new
β_{1644}, α^j_{1644}	`beta1644` `alpha1644`	the Hermitian matrices $-i\gamma^4_{1644}$ and $-\gamma^4_{1644}\gamma^j_{1644}$ ($j=0,1,2,3$) of the quantized 4+4 theory	new
h_{1644}	`h1644`	Hamiltonian matrix of the quantized 4+4 theory, iK_{1644} for fields independent of $x5, x6, x7$	new
Π_{1644}	`Pi1644`	the projector $(i\gamma^A_{1644}k_A + m_{1644})/2m_{1644}$ on the plane-wave solutions of the worked example	new

3 Matrices of the 7+1 spacetime (tag 1671)

symbol	names in programs and data files	meaning	origin
η_{1671}	`eta1671`	the matrix of the squares of the Dirac matrices of 7+1, $\gamma^A_{1671}\gamma^B_{1671} + \gamma^B_{1671}\gamma^A_{1671} = 2\,\eta^{AB}_{1671}$: diag$(-1,-1,-1,-1,+1,-1,-1,-1)$, time at index 4. The flat-space part of the project uses it as the metric (time-like $= +1$). It is minus the flat metric $\eta^{(7+1)}$ of the author's sign convention for the deflating spacetimes (space-like $= +1$)	N02 `η17` with the index values 0 and 4 exchanged; NOT `η71`
τ^A_{1671}	`tau1671`	the real 8×8 matrices of the 7+1 construction: $\tau^4_{1671} = 1_8$, the other seven represent the imaginary octonion units	N02 `τ0[swap04(A)]`
γ^A_{1671}	`gamma1671`	the eight real 16×16 Dirac matrices of the 7+1 spacetime (contravariant index)	N02 `Γ17up[swap04(A)]`; NOT `Γ71[A]`
Σ^{AB}_{1671}	`Sigma1671`	$\frac{1}{4}[\gamma^A_{1671}, \gamma^B_{1671}]$, the scaled commutators; they span $\mathfrak{so}(7,1)$	N02 `Σ17[A,B]`, built from the `Γ17` with lower index: $\Sigma^{AB}_{1671} = \eta^{AA}_{1671}\eta^{BB}_{1671}$ `Σ17[A', B']` with $A' = \mathrm{swap}_{04}(A)$, $B' = \mathrm{swap}_{04}(B)$
ω_{1671}	`omega1671`	the product $\gamma^0_{1671}\cdots\gamma^7_{1671}$; real, $\omega^2_{1671} = -1$	N02 `Γ17[8]`, `ω17`
χ_{1671}	`chi1671`	$i\,\omega_{1671}$, the (complex) chirality matrix of 7+1	N02 `χ17`; NOT `χ71`
P^L_{1671}, P^R_{1671}	`PL1671` `PR1671`	the complex chiral projectors $\frac{1}{2}(1 \mp \chi_{1671})$	N02 `PL17`, `PR17`; NOT `PL71`, `PR71`
U^c_{1671}	`Uc1671`	the unitary matrix $(1 - i\gamma^4_{1671})/\sqrt{2}$ that carries the complex projectors of 7+1 into the real projectors of 4+4	new
C^+_{1671}, C^-_{1671}	`Cp1671` `Cm1671`	γ^4_{1671} and $\gamma^4_{1671}\omega_{1671}$; $C^\pm\gamma^A(C^\pm)^{-1} = \pm(\gamma^A)^T$	new
β_{1671}, α^j_{1671}	`beta1671` `alpha1671`	the real symmetric matrices γ^4_{1671} and $\gamma^4_{1671}\gamma^j_{1671}$ (seven space indices j) of the Hamiltonian	N02 `βD`, `αD[swap04(j)]`
h_{1671}	`h1671`	Hamiltonian matrix of the 7+1 theory, $\alpha^j_{1671}p^j + \beta_{1671}m_{1671}$	N02 `h`
Π_{1671}	`Pi1671`	the projector $(\gamma^A_{1671}k_A + m_{1671})/2m_{1671}$ of the worked example	new
E^L_{1671}, E^R_{1671}	`EL1671` `ER1671`	the chiral frames: 16×8 matrices whose columns span the two eigenspaces of χ_{1671}	N02 `EL`, `ER`
M_{1671}, U^+_{1671}, V^-_{1671}	`M1671` `Upos1671` `Vneg1671`	the 8×8 matrix $m_{1671} + p^j\tau^j_{1671}$ and the matrices of positive- and negative-energy spinors of the worked example (check `71-D8`)	N02 `M` of its Sec. 7.4 (in N02 the letter `M` also denotes other matrices and a mass); the two spinor matrices are new names

4 Fields and their equations

symbol	names in programs and data files	meaning	origin
ψ_{1644}	`dirac1644`	the field dirac1644: 16 complex Grassmann components on the 4+4 spacetime; $(\gamma^A_{1644}\partial_A - m_{1644})\psi_{1644} = 0$	new; `Ψ16` of N01 and N10–N12 is a column of 16 real quantities (in N10–N12 functions of `x0`, `x4` called the wave function of the universe), not this field
ψ^L_{1644}, ψ^R_{1644}	`psiL1644` `psiR1644`	the two 8-component chiral parts of dirac1644 (upper and lower eight components)	new
m_{1644}	`m1644`	mass of the field dirac1644	new
$\tilde{\psi}$	—	the covariant adjoint $\psi^\dagger C^-_{1644}$ of the 4+4 theory (invariant under $\mathrm{Spin}_0(4,4)$)	new
j^A_{1644}	`j1644`	the conserved current $i\,\tilde{\psi}\gamma^A_{1644}\psi$ of the covariant 4+4 theory (fermion number)	new
ψ_{1671}	`dirac1671`	the field dirac1671: 16 complex Grassmann components on the 7+1 spacetime; $(i\gamma^A_{1671}\partial_A - m_{1671})\psi_{1671} = 0$	N02 `dirac16complex`
ϕ^L_{1671}, ϕ^R_{1671}	`phiL1671` `phiR1671`	the two 8-component chiral parts of dirac1671	N02 `φL`, `φR`
m_{1671}	`m1671`	mass of the field dirac1671	N02 `m`, `mD`
$\bar{\psi}$	—	the adjoint $\psi^\dagger\beta$ of the quantized theories ($\beta = \beta_{1671}$ or β_{1644}); in 4+4 it differs from $\tilde{\psi}$	new
β, α^j, χ, h, m	—	in the sections common to both spacetimes: the matrices and the mass of the spacetime at hand, i.e. the tagged symbols with the tag dropped; $\chi \equiv \chi_{1671}$ in 7+1 and $\chi \equiv \omega_{1644}$ in 4+4	new
$U(\vec{p})$	—	the 16×8 matrix of orthonormal positive-energy spinors at momentum $\vec{p}$	new

5 Symmetries, pairs and order parameter

symbol	names in programs and data files	meaning	origin
$U(1)_F$	—	the phase symmetry $\psi \to e^{i\epsilon}\psi$: fermion number	new
$U(1)_{VS}$	`UVS`	the chiral symmetry $\psi \to e^{i\epsilon\chi}\psi$, $\Phi_{VS} \to e^{-2i\epsilon}\Phi_{VS}$, broken by the vacuum value of the Vorton-Scalar	new
$SU(2)_A$, $SU(2)_B$	—	the two factors of the internal symmetry of the fields that do not depend on the four extra coordinates (zero-mode sector), in both spacetimes; $SO(4) = [SU(2)_A \times SU(2)_B]/Z_2$	new
$SU(2)_X$, $SU(2)_Y$	—	the two factors of the rotation group of the four space coordinates $x0, \dots, x3$ of the 4+4 spacetime (NOT the internal $SU(2)_A \times SU(2)_B$)	new
$SO(3)_T$, $SU(2)_T$	—	the rotations of the three additional times $x5, x6, x7$ of 4+4, and its covering group; the diagonal subgroup of $SU(2)_A \times SU(2)_B$	new
$U(1)_E$, P (generator)	—	the second central phase symmetry of the zero-mode sector and its traceless generator, $P^2 = 1$: $P = +1$ on the doublets of $SU(2)_A$, -1 on those of $SU(2)_B$	new
Δ^A_{1644}, Δ^A_{1671}	`DeltaV1644` `DeltaV1671`	the vector pair operators $\psi^T C^- \gamma^A \psi$ of each spacetime	new
Δ^{AB}_{1671}	`DeltaT1671`	the two-form pair operators $\psi^T C^+ \gamma^A \gamma^B \psi$ of the 7+1 spacetime	new

symbol	names in programs and data files	meaning	origin
Δ^a	—	order parameter: expectation value of the vector pair operator along the extra axis $a \in \{0,5,6,7\}$; $\lvert\Delta\rvert^2 \equiv \sum_a \lvert\Delta^a\rvert^2$	new
$\Phi_X(\vec{p})$	—	two-particle content $U(\vec{p})^T X U(-\vec{p})$ of the pair operator $\psi^T X \psi$	new
M	M_gap	the 8×8 gap matrix on the internal states of a pair (sans-serif; M_gap in the statements of the programs)	new
$d_\pm$	—	the two fourfold gaps of a general state, $d_\pm^2 = \lvert\Delta\rvert^2 \pm \sqrt{\lvert\Delta\rvert^4 - \lvert\eta_{ab}\Delta^a\Delta^b\rvert^2}$	new
n^a	—	real unit vector: direction of the order parameter in the space of the four extra axes	new

6 The Vorton-Scalar (suffix VS)

symbol	names in programs and data files	meaning	origin
Φ_{VS}	PhiVS	the Vorton-Scalar: a complex scalar field	new (the name is that of the author's project NCSS, where the scalar is real)
v_{VS}	vVS	vacuum value: $\langle\Phi_{VS}\rangle = v_{VS}/\sqrt{2}$	new
λ_{VS}	lambdaVS lamVS	self-coupling of the Vorton-Scalar	new
y_{VS}	yVS	Yukawa coupling of the Vorton-Scalar to the fermion field	new
s_{VS}	sVS	the radial mode of the Vorton-Scalar (real field)	new
m_s	msVS	mass of the radial mode, $\sqrt{2\lambda_{VS}}\, v_{VS}$	new
ϑ_{VS}	thetaVS thVS	the phase mode of the Vorton-Scalar (real field, massless)	new
g_{VS}	gVS	coupling of both modes of the Vorton-Scalar to the fermions, $g_{VS} = y_{VS}/\sqrt{2} = m_Y/v_{VS}$, where m_Y is the part of the fermion mass that the Vorton-Scalar gives (equal to m/v_{VS} when there is no bare Dirac mass)	new
α_{VS}	alphaVS aVS	$g_{VS}^2/4\pi$	new
D_{VS}	DVS	$\alpha_{VS}\, m/m_s$: two fermions bind if and only if $D_{VS} > D_c$	new
J^A_{VS}	JVS	the neutral current of the Vorton-Scalar (bilinear in Φ_{VS})	new
J^A_5	J5	the axial current $\bar{\psi}\gamma^A\chi\psi$ of the fermion field	new
G_{VS}	GVS	constant of the Fermi type of the neutral-current interaction, $G_{VS}/\sqrt{2} = 1/(2v_{VS}^2)$	new
$\mathcal{L}^{VS}_{1644}$, $\mathcal{L}^{VS}_{1671}$	LVS	the Lagrangians that define the Vorton-Scalar in the two spacetimes	new
C (coupling)	—	$g_{VS}^2 m^2/m_s^2 = g_{VS}^4/2\lambda_{VS}$, the one number on which the fermion mass in matter depends	new

7 Pairing, gap equation and CJT effective action

symbol	names in programs and data files	meaning	origin
ν	—	the number of internal states of a particle of given momentum, $\nu = 8$	new

symbol	names in programs and data files	meaning	origin
F_S, F_P	`F_scalar` `F_pseudoscalar`	exchange amplitudes of the radial and of the phase mode in a pair channel	new
V_{NC}, $V_s(r)$	—	contact interaction of the neutral current in the hidden-axis channel; Yukawa potential of the radial mode	new
σ_X	—	the sign $+1$ (vector pairs along the extra axes) or -1 (their two-form pairs) of the phase-mode term of the kernel	new
$W(k,k')$	—	s-wave average of the exchange attraction: kernel of the gap equation	new
$\Delta(k)$, Δ_F	—	gap function; its value at the Fermi momentum	new
ξ_k, E_k, $\kappa(k)$	—	$\sqrt{m^2+k^2}-\mu$; quasi-particle energy $\sqrt{\xi_k^2+\Delta(k)^2}$; pair correlation $\langle b(-\vec{k})b(\vec{k})\rangle$	new
k_F, E_F, μ	`kF` `kF_total` `mu`	Fermi momentum; Fermi kinetic energy $\sqrt{m^2+k_F^2}-m$; chemical potential. The number density is $n=\nu V_D k_F^D/(2\pi)^D$ in D space dimensions ($n=\nu k_F^3/6\pi^2$ for $D=3$)	new
$N(0)$	—	density of states at the Fermi surface per internal state, $k_F\sqrt{m^2+k_F^2}/2\pi^2$	new
λ, λ_S, λ_P	—	dimensionless pairing coupling $N(0)W(k_F,k_F)$ and its parts from the radial and the phase mode	new
T_c	—	critical temperature of the paired state	new
D_c	—	critical value of D_{VS} for a bound pair in the Yukawa potential	new
a_{sc}, a_{BB}	—	scattering length of two fermions; scattering length of two bound pairs	new
$\Gamma[\phi,G]$, Γ_2	—	the effective action for composite operators of Cornwall, Jackiw and Tomboulis; its two-particle-irreducible part	new (method of the cited paper)
$\mathcal{S}$, D_s, D_ϑ	—	propagator of the doubled fermion field; propagators of the radial and of the phase mode	new
$V[\Delta]$	—	the CJT effective potential of the paired state, measured from the unpaired state	new
$\eta(T)$	—	largest eigenvalue of the linearized gap equation, $\eta(T_c)=1$	new
$A_2(T)$, A_4	—	coefficients of the expansion of the CJT effective potential V along a gap function of fixed shape	new
$a(T)$, b, K_{GL}	—	Ginzburg-Landau coefficients of the weak-coupling free energy (quadratic, quartic and gradient term)	new
c_s	—	speed of first sound of the superfluid, $c_s^2=dP/d\rho$	new

8 Matter, temperature and vacuum

symbol	names in programs and data files	meaning	origin
ϕ	—	expectation value of the Vorton-Scalar, $\langle\Phi_{VS}\rangle=\phi/\sqrt{2}$; the fermion mass is $g_{VS}\phi$	new

symbol	names in programs and data files	meaning	origin
m^*, x	`M` `M_ks` `M_cjt` `M_cjt_global` `M_plus` `M_minus` `M_relaxed` `x`	m^*: the mass of the fermions in matter; $x = \phi/v_{VS}$: the expectation value of the Vorton-Scalar in units of its vacuum value. With a contact interaction $m^* = m + G_S\, n_S$ (gap equation of the Kohn-Sham and CJT calculations); with the Vorton-Scalar and a bare Dirac mass $m^* = m_D + m_Y\, x + G_S\, n_S$, so that $x = m^*/m$ if the whole mass comes from the Vorton-Scalar and no contact coupling acts. $m^*_\pm$: the masses of the particles of the two halves $W_{1644} = \pm 1$ of the 4+4 field in the covariant form of the self-interaction. The data files and the statements of the programs write m^* as `M`; the author's document KOHN_SHAM_GRAVITY writes M	new
$\varepsilon(k_F, x)$, $V_{\text{sea}}(x)$	—	energy density of matter as a function of the Fermi momentum and of $x = m^*/m$ (in units of m^4 in ordinary space); renormalized energy of the filled negative-energy states. ε without arguments is the energy density of matter in the units of the calculation at hand	new
B	—	energy density of the state without expectation value of the Vorton-Scalar (bag constant)	new
E_x	—	exchange (Fock) energy of the two modes	new
M_s, M_ϑ, a_s, a_ϑ	—	thermal masses of the two boson modes in the Hartree approximation, and the curvatures U''_{eff}, U'_{eff}/ϕ	new
$P(M)$, $Q(M)$	—	thermal part of the boson loop and of $\frac{1}{2}\mathrm{Tr}\ln$	new
T_{high}, T_N, $T_{1/2}$, T_0	—	two closed estimates of the temperature at which the fermion mass vanishes; temperatures at which the expectation value has fallen to half and to zero	new
$\sigma(p)$	—	self-energy that a pair condensate in the vacuum gives to the fermions: mass splitting $m \pm \sigma$	new
α_c, Λ (cutoff)	`Lambda`	critical value of α_{VS} for a pair condensate in the vacuum; Λ: a momentum cutoff (of the ladder equation in the vacuum, and of the filled states of negative energy in the calculations with contact interaction). Λ_8 and $\bar{\Lambda}$ are cosmological terms and have their own entries	new

9 Cosmology

symbol	names in programs and data files	meaning	origin
N	—	density of the dark radiation today in units of the density of one massless neutrino species	new
z_t, T_d	—	transition redshift of the stepped dark fluid, $m = T_{d0}(1 + z_t)$; temperature of the dark sector	new
r_{step}	—	factor by which the density of the dark radiation rises when the fermions annihilate, $15^{1/3}$ for this model	new
ω_b, ω_c, h, H_0, A_s, n_s, τ	`tau_reio`	the six parameters of the standard cosmological model: densities of baryons and of cold dark matter (Ωh^2), Hubble constant $H_0 = 100\,h$ km s^{-1} Mpc^{-1}, amplitude and spectral index of the primordial fluctuations, optical depth of reionization (`tau_reio` in the data files; the letter τ with the index d is the deflation clock)	standard notation
$\Delta\chi^2$	—	worsening of the best fit to the mock data relative to the standard model	new
θ_s, r_s, D_A	—	angular size of the sound horizon; sound horizon; distance to last scattering	standard notation

symbol	names in programs and data files	meaning	origin
c_{s0}	—	sound speed of superfluid dark matter today at the mean cosmic density	new
ρ_0, R_{TF}	—	constant of the equation of state of a condensate of bound pairs; Thomas-Fermi radius of a core supported by its pressure	new
m_{deg}, $m_{B,\max}$	—	largest fermion mass for degeneracy, $m_{\mathrm{deg}} = [6\pi^2\hbar^3\rho/(\nu\sigma_v^3)]^{1/4}$, and largest mass m_B of a boson for condensation, $m_{B,\max} = [(2\pi)^{3/2}\hbar^3\rho/(2.612\,\sigma_v^3)]^{1/4}$, in a halo of the mass density ρ and the velocity dispersion σ_v	new
σ_v; κ_{SI}	—	the velocity dispersion of the dark matter of a halo (with the index v; σ_8 is the amplitude of clustering and $\sigma(p)$ a self-energy); the bound on the cross-section per mass of dark matter from colliding clusters, about 1 cm^2/g (with the index SI; κ_8 is the gravitational coupling)	new

D1 Deflating spacetimes: hypotheses and geometry

symbol	names in programs and data files	meaning	origin
g55, g66, g77	`g55` `g66` `g77`	the three rates of deflation of $x5$, $x6$, $x7$ (positive constants, dimension 1/length). These are the author's names; they are written upright and never with a subscript. They are *not* components of the metric: the components are g_{55}, g_{66}, g_{77}	author
rate_t	`rate_t` `rate`	the rate of the coordinate xt in a statement that holds for $t = 5, 6, 7$ alike: $\mathrm{rate}_5 = \mathrm{g55}$, $\mathrm{rate}_6 = \mathrm{g66}$, $\mathrm{rate}_7 = \mathrm{g77}$	new
gsum	`gsum` `gam`	$\mathrm{g55} + \mathrm{g66} + \mathrm{g77}$, the total rate: the proper volume of the three deflating coordinates falls as $e^{-\mathrm{gsum}\,x4}$ (`gam`: gsum in units of 100 km/s/Mpc)	new
gsq, gprod	`gsq` `gprod`	$\mathrm{g55}^2 + \mathrm{g66}^2 + \mathrm{g77}^2$ and $\mathrm{g55\,g66} + \mathrm{g55\,g77} + \mathrm{g66\,g77}$; $\mathrm{gsum}^2 = \mathrm{gsq} + 2\,\mathrm{gprod}$	new
const5, const6, const7; const_t	`const5` `const6` `const7` `const_t` `cst`	the three positive constants of the author's metric (his word: const): the values of the scale factors at $x4 = 0$	author
b_5, b_6, b_7; b_t	`b_5` `b_6` `b_7` `b_t`	the scale factors of the deflating coordinates, $b_t(x4) = \mathrm{const}_t\, e^{-\mathrm{rate}_t\, x4}$: a coordinate distance dxt is the proper length $b_t\,\lvert dxt\rvert$. Always written with the index 5, 6, 7 or t (the letter b without index is a Ginzburg-Landau coefficient, and b in sans-serif type is the common scale factor of the three deflating coordinates in the Einstein-Lovelock evolution)	new
V_d	`Vd`	$b_5 b_6 b_7 = \sqrt{\lvert\det g\rvert}$, the proper volume of a unit coordinate cell of the deflating coordinates; $V_d \propto e^{-\mathrm{gsum}\,x4}$	new
μ, ν, ρ, σ (indices)	—	coordinate indices with the values $0, \ldots, 7$ on the deflating spacetimes (only as sub- or superscripts; as stand-alone letters μ is the chemical potential, ν the number of internal states and λ the pairing coupling). The capital indices A, B, C, D, also with a prime, label the orthonormal frame: the frame vector A points along the coordinate xA (as a stand-alone letter C is the coupling $g_{VS}^2 m^2/m_s^2$, and D the number of space dimensions with momentum). In the statements of the programs the indices are spelled out inside tensor expressions, as in `g^(mu nu)`	new
t, s (indices)	—	indices of the three additional coordinates $x5, x6, x7$, values 5, 6, 7; s is a second such index where two are needed (the matrices s_h carry the index h)	new

symbol	names in programs and data files	meaning	origin
${e_A}^{\mu}$, ${e^A}_{\mu}$	`e_A^mu`	the orthonormal frame and its inverse: ${e^A}_{\mu} = \mathrm{diag}(1,1,1,1,1,b_5,b_6,b_7)$	new
${\Gamma^{\rho}}_{\mu\nu}$	`Gamma christoffel`	the Christoffel symbols of the deflating metric (three indices; $\Gamma[\phi, G]$ with square brackets is the CJT effective action). The author's document KOHN_SHAM_GRAVITY writes the Christoffel symbols with braces and uses Γ^A for the Dirac matrices, which are γ^A here	new
${R^{\rho}}_{\sigma\mu\nu}$, ${R^{\mu}}_{\nu}$, R	`riemann ricci Ric44 Ric55 Ric66 Ric77 Rscalar`	Riemann tensor, Ricci tensor and curvature scalar of the deflating metric	new
${\mathcal{G}^{\mu}}_{\nu}$	`Ein00 Ein44 Ein55 Ein66 Ein77 einstein_mixed`	the Einstein tensor ${R^{\mu}}_{\nu} - \frac{1}{2} R\, \delta^{\mu}_{\nu}$ (script letter, so that it cannot be taken for Newton's constant or for a constant of the Fermi type)	new
$\mathcal{X}$; $\mathcal{X}_{\mathrm{scale}}$, $\mathcal{X}_{(s)}$	`killing_vectors killing_count`	a Killing vector field of the deflating metric (script letter; the plain letter X is the set of the eight coordinates); the scaling vector $\partial_4 + \sum_t \mathrm{rate}_t\, xt\, \partial_t$; the three special Killing vectors of the case of equal rates, $s = 5, 6, 7$	new
$\bar{x}4$; $\bar{x}5$, $\bar{x}6$, $\bar{x}7$	—	coordinates of the case of equal rates in which the line element of $x4, \ldots, x7$ is a multiple of a flat one: $\bar{x}4 = e^{\mathrm{g55}\, x4}/\mathrm{g55}$, $\bar{x}t = \mathrm{const}_t\, xt$	new
${\mathcal{H}^{\mu}}_{\nu}$; α_{GB}, Λ_8	`Lanczos alpha_GB Lambda_8`	the Lanczos tensor of the Gauss-Bonnet term, ${\mathcal{H}^{\mu}}_{\nu} = 2(R\, {R^{\mu}}_{\nu} - 2{R^{\mu}}_{\alpha}{R^{\alpha}}_{\nu} - 2{R^{\mu\alpha}}_{\nu\beta}{R^{\beta}}_{\alpha} + {R^{\mu\alpha}}_{\beta\gamma}{R^{\beta\gamma}}_{\nu\alpha}) - \frac{1}{2}\delta^{\mu}_{\nu}(R^2 - 4{R^{\alpha}}_{\beta}{R^{\beta}}_{\alpha} + {R^{\alpha\beta}}_{\gamma\delta}{R^{\gamma\delta}}_{\alpha\beta})$ (script letter; H is the Hubble rate); the coupling of the Gauss-Bonnet term and the cosmological constant of Einstein-Gauss-Bonnet gravity in eight dimensions, whose equations without matter are ${\mathcal{G}^{\mu}}_{\nu} + \Lambda_8\, \delta^{\mu}_{\nu} + \alpha_{GB}\, {\mathcal{H}^{\mu}}_{\nu} = 0$ (metric with positive space-like components). In the author's Einstein-Lovelock notation $E^{(1)} = -\mathcal{G}$, $E^{(2)} = \mathcal{H}$, $\alpha_2 = -\alpha_{GB}$, $\bar{\Lambda} = \Lambda_8$	new
K_{AB}	`ksec`	the sectional curvature of the plane of the frame directions A and B (metric with positive space-like components): on the deflating spacetimes $K_{4t} = \mathrm{rate}_t^2$, $K_{ts} = \mathrm{rate}_t\, \mathrm{rate}_s$ for $t \neq s$, zero if one direction is flat; for scale factors that depend on $x4$ in any way, $K_{4i} = H_i' + H_i^2$ and $K_{ij} = H_i H_j$ with the rate H_i of the direction i (the pair curvatures of the author's Einstein-Lovelock equations). Always with two subscripts: the letter K without subscript is the generator of the evolution of the 4+4 field in $x4$, K as an index is an index set of the self-interaction, and K with an order as subscript and with an argument is a modified Bessel function	new
ς	`affine`	affine parameter along a geodesic; for a massive particle on the 7+1 spacetime, the proper time divided by the mass	new
P_{μ}	`P_1 P_5`	the conserved covariant momenta of a geodesic (with an index; P without index is the generator of ${U(1)}_E$)	new
$x4_{\mathrm{turn}}$	`x4_turn`	the value of the ordinary time at which a classical particle with momentum along a deflating time turns back, and beyond which a mode that depends on a deflating time is extinguished (4+4): there $k_d = E_{\perp}$	new

D2 Deflating spacetimes: matrices, fields and their equations

symbol	names in programs and data files	meaning	origin
γ^{μ}_{1644d}, γ^{μ}_{1671d}	`gamma_1644d` `gamma_1671d`	the Dirac matrices of the deflating spacetimes in coordinate components, $\gamma^{\mu} = e_A{}^{\mu}\gamma^A$: $\gamma^t = \gamma^{A=t}/b_t$ for $t = 5, 6, 7$, the other five are the constant matrices γ^A_{1644}, γ^A_{1671} of the frame	new
$\omega_{\mu AB}$	`omega_(mu A B)`	the coefficients of the spin connection, $\omega_{\mu AB} = \eta_{AC}\, e^C{}_{\nu}(\partial_\mu e_B{}^{\nu} + \Gamma^{\nu}{}_{\mu\lambda} e_B{}^{\lambda})$ with the flat metric $\eta = \eta^{(4+4)}$ or $\eta^{(7+1)}$ (three indices; ω_{1644}, ω_{1671} with a tag are the products of all eight Dirac matrices)	new
Ω_μ	`Omega_mu` `omega`	the spin connection as a matrix, $\Omega_\mu = \frac{1}{4}\, c_\gamma\, \omega_{\mu AB}\gamma^A\gamma^B$; the covariant derivative of a spinor is $\nabla_\mu\psi = (\partial_\mu + \Omega_\mu)\psi$. On both deflating spacetimes $\Omega_t = \frac{1}{2}\,\mathrm{rate}_t\, b_t\, \gamma^t\gamma^4$ (frame matrices) for $t = 5, 6, 7$ and $\Omega_\mu = 0$ otherwise. (Ω_{CJT} is an effective potential)	new
ψ_{1644d}, ψ_{1671d}	`dirac1644d` `dirac1671d`	the fields dirac1644d and dirac1671d: the 16-component fields on the deflating spacetimes. With mass term and self-interaction they obey $(\gamma^\mu\nabla_\mu - m)\psi_{1644d} = \sum_r g_r \sum_{\lvert K\rvert=r}(\tilde{\psi}\Theta_K\psi)\Theta^K\psi$ (covariant form of 4+4) and $(i\gamma^\mu\nabla_\mu - m)\psi_{1671d} = \sum_r g_r \sum_{\lvert K\rvert=r}(\bar{\psi}\Theta_K\psi)\Theta^K\psi$ (7+1)	new
$\hat{\psi}$	`psihat`	the weighted field $\sqrt{V_d}\,\psi$, which obeys the equation without spin connection	new
$K(x4)$, $\mathsf{h}(x4)$	`K h`	the generator K_{1644} and the Hamiltonian matrix h_{1671} with the wavenumbers along the deflating coordinates divided by the scale factors: without self-interaction $\partial_4\hat{\psi} = K\hat{\psi}$ (4+4), $i\partial_4\hat{\psi} = \mathsf{h}\hat{\psi}$ (7+1)	new

D3 Deflating spacetimes: solutions mode by mode

symbol	names in programs and data files	meaning	origin
k_5, k_6, k_7; k_t	`k_5` `k_6` `k_7` `k_t`	the conserved wavenumbers of a mode along the deflating coordinates (covariant components of the wave vector k_A)	new
$E_\perp$	`E_perp`	the energy of a mode, or of a classical particle, without its motion along the deflating coordinates: $\sqrt{m^2 + k_0^2 + k_1^2 + k_2^2 + k_3^2}$ (with the wavenumbers or momenta along $x0, \ldots, x3$)	new
k_d	`k_d`	the physical wavenumber along the deflating coordinates, $k_d(x4) = \left[\sum_t k_t^2/b_t(x4)^2\right]^{1/2}$; it grows as the coordinates deflate	new
$\mathsf{h}_\perp$, $K_\perp$	`h_perp` `K_perp`	the parts of h and of K that do not contain the scale factors: the operators of the modes that do not depend on the deflating coordinates, $\mathsf{h}_\perp = m\beta + \sum_a p_a\alpha^a$ with $\mathsf{h}_\perp^2 = E_\perp^2$, and $K_\perp = -m\gamma^4 + i\sum_a k_a\gamma^4\gamma^a$ with $K_\perp^2 = -E_\perp^2$	new
γ^d, α^d	—	the frame matrix along the direction of the wave vector in the deflating coordinates, $\gamma^d = k_d^{-1}\sum_t (k_t/b_t)\gamma^t$, and $\alpha^d = \gamma^4\gamma^d$; constant in time when one rate governs	new
w; f_+, f_-; Z_+, Z_-	—	auxiliary objects of the exact solutions of the equations without self-interaction: a constant eigenvector of $\mathsf{h}_\perp$ or of $K_\perp$; the two coefficient functions of the ansatz; two cylinder functions whose orders differ by one	new

symbol	names in programs and data files	meaning	origin
u_d	`u_d`	k_d/rate_t for a mode that depends on one deflating coordinate: the argument of the Bessel functions of the exact solutions	new
n_{cr}	`n_cr` `n_min` `n_mean` `n_max` `n_exact` `fd5` `fd6`	the number of particles per internal state that the deflation creates in a mode that starts empty (7+1)	new
T_{defl}	—	the temperature of the thermal spectrum of the created quanta, $\text{rate}_t/2\pi$ in units with $\hbar = k_B = 1$ (7+1)	new
S_{WKB}	`S_wkb`	$\int \sqrt{k_d^2 - E_\perp^2}\, dx4$ from the turning value on: exponent of the extinction of a bounded solution (4+4)	new
φ_{bos}, $\hat{\varphi}_{\text{bos}}$, m_{bos}	`phi_bos` `Shat` `Shat2` `m_bos`	a mode of the Vorton-Scalar (the radial mode, $m_{\text{bos}} = m_s$, or the phase mode, $m_{\text{bos}} = 0$), its weighted form $\sqrt{V_d}\,\varphi_{\text{bos}}$, and its mass	new
ϖ_b	`varpi_b`	$\sqrt{k^2 + m_{\text{bos}}^2 - \text{gsum}^2/4}$: the frequency of the weighted boson mode that does not depend on the deflating coordinates	new

D4 The deflation clock and the couplings

symbol	names in programs and data files	meaning	origin
τ_d	`tau_d`	the deflation clock, $\tau_d = \text{gsum}\,(x4 - x4_{\text{ref}})$: the number of e-foldings by which the volume V_d has shrunk since the reference time; in an evolution in which the rates change, $\tau_d = -\ln(V_d/V_d^{\text{ref}})$ (the letter τ without the index d is the optical depth of reionization)	new
τ_{pair}	`tau_pair`	the clock measured from the moment at which two fermions become bound: $\ln(D_{VS}/D_c)$	new
$\hat{\Phi}_{VS}$	`Phihat`	the weighted Vorton-Scalar $\sqrt{V_d}\,\Phi_{VS}$	new

D5 Dynamics of the paired state

symbol	names in programs and data files	meaning	origin
Ξ_d	`Xi_d`	$\text{gsum}/(\lambda\,\Delta_{\text{eq}})$: decides whether the paired state can follow the deflation ($\Xi_d \ll 1$: yes)	new
n_k	—	occupation of the pair of states of the single-particle energy ξ_k in the shell model of the paired state (with the index k; the letter n without index is a density)	new
Δ_{eq}	`gap_eq`	the equilibrium gap for the pairing strength of the moment	new
E_{sh}	`E_sh`	half-width in energy of the shell of states around the Fermi surface in the model of the pair dynamics; unit of energy and of inverse time there	new
n_{ex}	`n_ex` `n_ex_final`	density of broken pairs (in units of E_{sh} times the density of states)	new

D6 Cosmology and bounds on the deflating spacetimes

symbol	names in programs and data files	meaning	origin
G_N, G_8	`G_N` `GN_early` `GN_star` `G_8`	Newton's constant of ordinary space, and the gravitational coupling of the eight-dimensional Einstein theory (assumption A2): $G_N = G_8/(\text{proper volume of the extra coordinates}) \propto e^{\text{gsum}\,x4}$	new
a_{sp}, H	—	scale factor of ordinary space (assumption A1) and its rate $H = (da_{\text{sp}}/dx4)/a_{\text{sp}}$; H_0 is its value today. (The rates of the Einstein-Lovelock evolution are H_{a}, H_{b}; the author's Einstein-Lovelock notebooks use the letter H for the rate in the exponent of their metric, which corresponds here to g55, g66, g77)	new
t_0	`age` `age_Gyr`	the age of the universe in the model at hand	new
θ_*, z_*	`theta_star` `z_star`	the angle under which the sound horizon at last scattering is seen, $r_s(z_*)/D_A(z_*)$ with the comoving distance D_A, and the redshift z_* of last scattering at which the optical depth without reionization is 1 (the $100\,\theta_*$ of the Planck papers). The angle θ_s is the same ratio at the maximum of the visibility function; for the model without deflation $100\,\theta_* = 1.03965$ and $100\,\theta_s = 1.040358$	new
δ_m, f_g, σ_8	`growth` `f_growth` `growth_ratio` `f_growth_ratio` `f_sigma8_ratio`	density contrast of matter in ordinary space; its growth rate $d\ln\delta_m/d\ln a_{\text{sp}}$ today; the amplitude of clustering (only ratios of σ_8 are used)	new
v_{EW}, n_{EW}	`v_EW` `n_EW`	the electroweak scale (the vacuum value of the Higgs field of the Standard Model, 246 GeV), and the exponent with which it would follow the coupling under the hypothesis H2: $n_{\text{EW}} = d\ln v_{\text{EW}}/d\ln\alpha_{VS}$	new
ϱ_c	`rho_c`	abbreviation in the formula for n_{EW} of the scale-invariant ladder equation: $\varrho_c = \frac{1}{2}\sqrt{\alpha_{VS}/\alpha_c - 1}$ (the letter ρ without index is a density)	new
m_p/m_e	`mu_pe`	the ratio of the proton mass to the electron mass (always written as this ratio in the documents; `mu_pe` in the statements of the programs)	new
G_{Fermi}, α_{em}	`G_Fermi`	the Fermi constant of the weak interaction and the fine-structure constant (they enter only in the discussion of bounds)	new

D7 Symmetries and vortex lines on the deflating spacetimes

symbol	names in programs and data files	meaning	origin
P_c	`Pc`	$i\alpha^1\alpha^2\alpha^3\chi$, the product of the five matrices of the sector that does not depend on the extra coordinates; equal, up to sign, to the generator P of $U(1)_E$	new
θ_{12}	—	the angle around a vortex line that lies along $x3$, in the plane of $x1$ and $x2$ (with the index 12; θ_* is the angle of the sound horizon and $\theta(\cdot)$ the step function)	new
w_z	`w_z`	a constant spinor of a fermion zero mode on a vortex line	new

D8 Tags and identifiers of the deflating spacetimes

symbol	names in programs and data files	meaning	origin
1644d, 1671d	`1644d` `1671d`	tags of the deflating 4+4 and 7+1 spacetimes and of every object defined on them (the tags 1644, 1671 of the flat spacetimes with the letter d)	new
A1, A2, A3, H2, HV	—	A1, A2: the two assumptions of the cosmological part (ordinary space expands; gravity is the eight-dimensional Einstein theory). A3: the assumption about the perturbations of ordinary space (the scale factors of the extra coordinates are not perturbed, and the agent that holds them has no energy density), with its closures. H2: the author's hypothesis that the superfluid state is the Higgs field. HV: the different assumption that the vacuum value of the Vorton-Scalar sets the electroweak scale	new
"Momentum", "Pressure", "Shear" (closures)	`Momentum` `Pressure` `Shear` `closure`	the three closures of the perturbation equations under assumption A3, named after the Einstein equation of ordinary space that holds with matter and radiation alone besides the energy constraint: the momentum constraint, the trace (pressure) equation, or the equation of the shear. In data files the suffixes `_M` and `_P` mark the first two; `_N4` marks the four-dimensional growth equation, which is none of them	new

D9 Perturbations of ordinary space and the microwave background

symbol	names in programs and data files	meaning	origin
$\tau_{\rm cf}$	`tau_cf` `tau0_Mpc`	the conformal time of ordinary space, $d\tau_{\rm cf} = dx4/a_{\rm sp}$, in Mpc (`tau0_Mpc`: its value today). In the equations of the perturbations a prime is the derivative with respect to $\tau_{\rm cf}$	new
$H_{\rm cf}$, H_e	`Hc` `He` `hub` `hub_e` `a_gsum_over_Hc`	the conformal Hubble rate $H_{\rm cf} = a_{\rm sp}H$, and $H_e = H_{\rm cf} - a_{\rm sp}\,\mathrm{gsum}/2$, which takes its place in three of the Einstein equations of ordinary space when the extra coordinates deflate	new
k (perturbations), ℓ, D_ℓ^{TT}, D_ℓ^{TE}, D_ℓ^{EE}	—	the comoving wavenumber of a perturbation of ordinary space (in Mpc^{-1}; the physical wavenumber is $k/a_{\rm sp}$); the multipole; and the spectra of the temperature, of the polarization and their cross spectrum, $D_\ell = \ell(\ell+1)C_\ell/2\pi$ in $\mu\mathrm{K}^2$	new
$h_{\rm sy}$, $\eta_{\rm sy}$, $\alpha_{\rm sy}$	—	the two functions that describe a scalar perturbation of the metric of ordinary space in the synchronous gauge (the h and η of Ma and Bertschinger; here with the index sy, because h is the Hubble parameter and η the flat metric), and the combination $\alpha_{\rm sy} = (h'_{\rm sy} + 6\eta'_{\rm sy})/2k^2$	new
$\mathcal{U}$	`potential`	$\eta_{\rm sy} + \alpha'_{\rm sy}$, the sum of the two gravitational potentials of the conformal Newtonian gauge (script letter): the potential that light feels	new
$\mathcal{D}$, $\mathcal{T}$, $\mathcal{P}$, $\mathcal{A}$	`d_sum` `three_dpres`	$8\pi G_N a_{\rm sp}^2$ times the perturbation of the density, of the divergence of the momentum density, of the pressure, and of the anisotropic stress of matter and radiation together (script letters; in the notation of Ma and Bertschinger: $\delta\rho$, $(\rho+P)\theta$, δP, $(\rho+P)\sigma$)	new
$\mathcal{E}$, $\mathcal{E}_P$, $\mathcal{E}_m$, $\mathcal{E}_r$, $\mathcal{B}$	`grho_sum` `half_rho_plus_p` `b_trace` `n_bg`	$8\pi G_N a_{\rm sp}^2$ times the density and the pressure of the unperturbed universe, summed over the species ($\mathcal{E}_m$, $\mathcal{E}_r$: the parts of $\mathcal{E}$ from pressureless matter and from radiation); and $\mathcal{B} = H'_e - H_e^2 + \frac{1}{2}(\mathcal{E} + \mathcal{E}_P)$, which vanishes without deflation (script letters)	new

symbol	names in programs and data files	meaning	origin
p_{agent}	—	the pressure in ordinary space of the agent, that is, of whatever keeps the three rates of deflation constant, in the unperturbed universe (under assumption A2 the agent has no energy density)	new
$\mathcal{Y}_M$, $\mathcal{Y}_T$, $\mathcal{Y}_S$; $\mathcal{Z}$	`momentum_gap` `trace_gap` `shear_gap` `x_gap` `shear_check` `trace_check`	the gaps: what the momentum constraint, the trace equation and the equation of the shear lack when only matter and radiation stand on their right-hand sides; the agent that holds the extra scale factors supplies them. $\mathcal{Z} = h'_{\text{sy}}\,\mathcal{B} - \frac{1}{2}a_{\text{sp}}\,\text{gsum}\,(\mathcal{D} - 3\mathcal{P})$ (script letters)	new
Y_{He}	—	the mass fraction of helium after nucleosynthesis (with the index He; `Y_He` in the data files)	new

10 The metric of the deflating spacetimes in the author's sign convention

symbol	names in programs and data files	meaning	origin
$g_{\mu\nu}^{(4+4)}$, $g_{\mu\nu}^{(7+1)}$	`gs`	the metrics of the deflating spacetimes in the author's sign convention of November 13, 2026 (space-like components positive, time-like components negative): $g^{(4+4)} = \text{diag}(+1,+1,+1,+1,-1,-b_5^2,-b_6^2,-b_7^2)$ and $g^{(7+1)} = \text{diag}(+1,+1,+1,+1,-1,+b_5^2,+b_6^2,+b_7^2)$, that is $g_{tt}^{(4+4)} = -\text{const}_t^2\, e^{-2\,\text{rate}_t\, x4}$ and $g_{tt}^{(7+1)} = +\text{const}_t^2\, e^{-2\,\text{rate}_t\, x4}$ for $t = 5, 6, 7$. A component of the metric carries the superscript $(4+4)$ or $(7+1)$ and two coordinate indices; it is not a rate (g55) and not a coupling (g_r, g_{VS}). The first revision on the deflating spacetimes wrote $g_{1644d} = g^{(4+4)}$ and $g_{1671d} = -g^{(7+1)}$ (time-like component positive on 7+1); these two symbols are retired	author
$\eta^{(4+4)}$, $\eta^{(7+1)}$	—	the flat metrics of the author's sign convention, with which frame indices are raised and lowered on the deflating spacetimes: $\eta^{(4+4)} = \eta_{1644} = \text{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$ and $\eta^{(7+1)} = -\eta_{1671} = \text{diag}(+1,+1,+1,+1,-1,+1,+1,+1)$; $g_{\mu\nu}\, e_A{}^{\mu} e_B{}^{\nu} = \eta_{AB}$	author
c_γ	`clifford_sign`	the sign that relates the Dirac matrices to the metric of the author's sign convention, $\gamma^A\gamma^B + \gamma^B\gamma^A = 2\,c_\gamma\,\eta^{AB}$ with $\eta = \eta^{(4+4)}$ or $\eta^{(7+1)}$: $c_\gamma = +1$ on 4+4 and $c_\gamma = -1$ on 7+1. The Dirac matrices themselves are those of the flat-space part of the project, unchanged	new
b_t^{ref}	`s5` `s6` `s7`	the value of the scale factor b_t at the reference time $x4_{\text{ref}}$. A particle that keeps its wavenumber k_t has the proper momentum k_t/b_t, larger than at the reference time by the stretch factor $b_t^{\text{ref}}/b_t = e^{\text{rate}_t (x4 - x4_{\text{ref}})}$ (columns `s5`, `s6`, `s7`)	new
τ_d^0	`tau_ref` `tau_ref_end`	the reading that the deflation clock would have if the rates had kept their initial value, $\tau_d^0 = 3\,\text{g55}\,x4$ in the evolutions with gravity (equal rates), where the clock actually reached is $\tau_d = -3\ln \text{b}$	new

11 Mass term and self-interaction of the fermion field

symbol	names in programs and data files	meaning	origin
$\mathcal{L}_{\text{self}}$	—	the self-interaction of the fermion field (the author's definition, document KOHN_SHAM_GRAVITY): $\mathcal{L}_{\text{self}} = -\frac{1}{2}\sum_{r=0}^{8} g_r\, I_r$, a sum of nine quartic invariants	K1
g_r; $g_0, \dots, g_8$; g_S, g_V, g_A, g_P	`g_0` `g_1` `g_2` `g_3` `g_4` `g_5` `g_6` `g_7` `g_8`	the nine couplings of the self-interaction, one for each rank r (the author's symbols; dimension length^{D-1} in D space dimensions); $g_0 = g_S$ (scalar), $g_1 = g_V$ (vector), $g_7 = -g_A$ (axial vector), $g_8 = -g_P$ (pseudoscalar). Such a coupling carries one digit as subscript; it is not a rate of deflation (g55: two digits, upright, no subscript), not a component of the metric (superscript $(4+4)$ or $(7+1)$, two indices) and not the coupling g_{VS} of the Vorton-Scalar	K1
K, J (index sets); r, r' (ranks)	`mask` `rank` `J`	K, J: a strictly increasing list $A_1 < \cdots < A_r$ of indices out of $0, \dots, 7$ (there are 256); $r = \lvert K\rvert$ is its rank, $0 \le r \le 8$; r' is a second rank and l the summation index in the closed form of the Fierz matrix. In the data files an index set is given by its bit mask (`mask`: the sum of 2^A over its indices)	K1
Θ^K, Θ_K; ζ_r; sgn_K	`eps_K` `phase` `eta_K`	the 256 matrices of the self-interaction (the author's symbols Θ^K, Θ_K): $\Theta^K = \zeta_r\, \gamma^{A_1} \cdots \gamma^{A_r}$, with the phase $\zeta_r = 1$ or i chosen so that the bilinear formed with Θ^K is real, and $\Theta_K = (\Theta^K)^{-1} = \text{sgn}_K\, \Theta^K$ with a sign $\text{sgn}_K = \pm 1$. On 7+1 and in the quantized-sector form of 4+4: $\zeta_r = i$ for $r = 2, 3, 6, 7$ (the author's i^{σ_r} with $\sigma_r = r(r-1)/2 \bmod 2$) and sgn_K is the product of the squares of the matrices in K (the author's η_K). In the covariant form of 4+4: $\zeta_r = i$ for $r = 1, 2, 5, 6$, and sgn_K is $(-1)^r$ times that product	K1; the covariant form of 4+4 is new
γ^A_{44q}	`gamma44q`	the matrices with which the quantized-sector form of the self-interaction on 4+4 is built: $\gamma^A_{44q} = -i\,\gamma^A_{1644}$ for $A = 0, \dots, 4$ and $\gamma^A_{44q} = \gamma^A_{1644}$ for $A = 5, 6, 7$. They have the squares of the 7+1 matrices, and $\gamma^4_{44q} = \beta_{1644}$	new
I_r	—	the invariant of rank r (the author's symbol): $I_r = \sum_{\lvert K\rvert = r} (\bar\psi \Theta_K \psi)(\bar\psi \Theta^K \psi)$ with $\bar\psi = \psi^\dagger \beta$ on 7+1 and in the quantized-sector form of 4+4, and with $\tilde\psi = \psi^\dagger C^-_{1644}$ in place of $\bar\psi$ in the covariant form of 4+4. I_0 is the square of the scalar density, I_1 the square of the current	K1
F, $\mathsf{F}_{rr'}$	`fierz` `F_r0` `F_r1` `F_r2` `F_r3` `F_r4` `F_r5` `F_r6` `F_r7` `F_r8`	the Fierz matrix (9×9): $\mathsf{F}_{rr'} = \frac{1}{16}\sum_l (-1)^{rr'-l} \binom{r'}{l} \binom{8-r'}{r-l}$, with $\mathsf{F}^2 = 1$. For anticommuting fields the nine invariants obey $(1 + \mathsf{F})\, I = 0$. (Sans-serif type; F_S, F_P, F_0, F_1, F_7, F_8 in italic type are pairing amplitudes. The author's document writes F_{rs}.)	K1
g_0^{eff}, g_1^{eff}, g_7^{eff}, g_8^{eff}	`g0eff` `g1eff` `g7eff` `g8eff` `g_0eff` `g_1eff` `g_7eff` `g_8eff`	the four independent couplings that remain for anticommuting fields, when $I_2, \dots, I_6$ are expressed by I_0, I_1, I_7, I_8: $g_0^{\text{eff}} = g_0 - 3g_2 - 4g_3 - 5g_4 - 4g_5 - g_6$, $g_1^{\text{eff}} = g_1 - g_2 - g_5 + g_6$, $g_7^{\text{eff}} = g_7 - g_2 - g_3 + g_6$, $g_8^{\text{eff}} = g_8 - g_2 + 4g_3 - 5g_4 + 4g_5 - 3g_6$; then $\mathcal{L}_{\text{self}} = -\frac{1}{2}(g_0^{\text{eff}} I_0 + g_1^{\text{eff}} I_1 + g_7^{\text{eff}} I_7 + g_8^{\text{eff}} I_8)$	new
n	—	the one-body density matrix of matter (16×16), $\mathsf{n}_{ab} = \langle \psi_b^\dagger \psi_a \rangle$, with $\text{Tr}\,\mathsf{n} = n$ and $\text{Tr}(\beta\,\mathsf{n}) = n_S$; for uniform matter at rest $\mathsf{n} = (n + n_S\,\beta)/16$. (The author's document writes R for the matrix $\mathsf{n}\,\beta$.)	new

symbol	names in programs and data files	meaning	origin
G_S^{HF}, G_V^{HF}; G_S^{c}, G_V^{c}	`G_0` `G_1` `G_S0` `G_V0` `hf_rho` `hf_rhoS` `hf_rhoW` `hf_rhoSW`	the values that the Hartree-Fock approximation gives to the mean-field couplings of uniform matter: $G_S^{\mathrm{HF}} = (15g_0^{\mathrm{eff}} - 8g_1^{\mathrm{eff}} - 8g_7^{\mathrm{eff}} - g_8^{\mathrm{eff}})/16$ and $G_V^{\mathrm{HF}} = (-g_0^{\mathrm{eff}} + 22g_1^{\mathrm{eff}} - 6g_7^{\mathrm{eff}} + g_8^{\mathrm{eff}})/16$ on 7+1 and in the quantized-sector form of 4+4, where the interaction energy density is $\frac{1}{2}(G_V^{\mathrm{HF}} n^2 + G_S^{\mathrm{HF}} n_S^2)$. In the covariant form of 4+4 it is $\frac{1}{2}(G_V^{\mathrm{c}} n^2 + G_S^{\mathrm{c}} n_S^2 + G_V^{\mathrm{HF}} n_W^2 + G_S^{\mathrm{HF}} n_{SW}^2)$ with $G_V^{\mathrm{c}} = -(g_0^{\mathrm{eff}} + g_8^{\mathrm{eff}})/16$ and $G_S^{\mathrm{c}} = (-g_0^{\mathrm{eff}} - 2g_1^{\mathrm{eff}} + 2g_7^{\mathrm{eff}} + g_8^{\mathrm{eff}})/16$. (The author's document and the data files write G_0, G_1 for G_S^{HF}, G_V^{HF}; the data files write `G_S0`, `G_V0` for G_S^{c}, G_V^{c}.)	K1; the covariant form of 4+4 is new
G_{pair}	`G_pair`	the contact coupling of a pair of two particles at rest: the interaction energy of the pair times the volume; attractive if negative. For the pair along the hidden axis $G_{\mathrm{pair}} = g_0^{\mathrm{eff}} + g_1^{\mathrm{eff}} + 5\, g_7^{\mathrm{eff}}$ (7+1 and quantized-sector form of 4+4)	new
n_+, n_-, n_{S+}, n_{S-}; n_W, n_{SW}; pol_W	`rho_W` `rho_SW` `pol_W`	number densities and scalar densities of the particles in the two halves $W_{1644} = \pm 1$ of the 4+4 field; $n_W = n_+ - n_-$, $n_{SW} = n_{S+} - n_{S-}$; the polarization $\mathrm{pol}_W = (n_+ - n_-)/n$	new
$\mathcal{L}_{\mathrm{kin}}$, $\mathcal{L}_{\mathrm{mass}}$	—	the kinetic term and the mass term of the Lagrangian of the fermion field, $\mathcal{L} = V_d\,[\mathcal{L}_{\mathrm{kin}} + \mathcal{L}_{\mathrm{mass}} + \mathcal{L}_{\mathrm{self}}]$	new
I_r^{cov}	—	the invariant of rank r of the covariant form of the self-interaction on 4+4, written with the matrices of the quantized-sector form: $I_r^{\mathrm{cov}} = \sum_{\lvert K\rvert=r} \mathrm{sgn}_K (\psi^\dagger W_{1644} \beta \Theta^K \psi)^2$	new

12 Kohn-Sham and CJT calculations with contact interaction

symbol	names in programs and data files	meaning	origin
G_S, G_V	`G_S` `G_V` `GS_abs`	the scalar and the vector coupling of the Kohn-Sham (mean-field) description of uniform matter (the author's symbols): interaction energy density $\frac{1}{2}G_V n^2 + \frac{1}{2}G_S n_S^2$; attraction for $G_S < 0$, repulsion for $G_V > 0$; dimension length^{D-1}. The author's example: $G_S = -10^5$, $G_V = +10^4$ in units $m = 1$, $D = 7$	K1
n, n_S	`rho` `rho_S` `rho_ks` `rho_cjt`	number density $\langle\psi^\dagger\psi\rangle$ and scalar density $\langle\bar\psi\psi\rangle$ of the fermions of matter (the filled states of negative energy not counted). The data files and the author's document write ρ (`rho`) and ρ_S (`rho_S`)	K1 (there ρ, ρ_S)
D	`D`	the number of space dimensions in which the particles of matter have momentum: 7 (all space coordinates of 7+1), 4 (fields that do not depend on $x5, x6, x7$) or 3 (fields that depend on ordinary space only). $V_D = \pi^{D/2}/\Gamma_{\mathrm{E}}(D/2+1)$ is the volume and $A_D = D\,V_D$ the area of the unit ball in D dimensions (Γ_{E}: Euler's gamma function)	new
$E^*(k)$; $\langle f\rangle$	—	$E^*(k) = \sqrt{k^2 + m^{*2}}$, the energy of a particle of matter without the vector term; $\langle f\rangle = \nu \int d^D k\, f/(2\pi)^D$ over the occupied states: $n = \langle 1\rangle$, $n_S = \langle m^*/E^*\rangle$	K1 (there ω for E^*)

symbol	names in programs and data files	meaning	origin
$\varepsilon_{\rm kin}$; p_j; $p_{\rm int}$	`e_kin` `eps` `eps_ks` `eps_cjt` `eps_rel` `p` `p_ks` `p_flat` `p5` `p6` `p7` `p_a` `p_b` `pa_rel` `pb_rel` `Pi`	$\varepsilon_{\rm kin} = \langle E^*\rangle$: the kinetic part of the energy density $\varepsilon = \varepsilon_{\rm kin} + \frac{1}{2}G_V n^2 - (m^*-m)^2/(2G_S)$; p_j: the pressure of matter along the coordinate xj, $\langle k_j^2/E^*\rangle + p_{\rm int}$; $p_{\rm int} = \frac{1}{2}G_V n^2 + (m^*-m)^2/(2G_S)$: the interaction part of the pressure, which is the whole pressure along a coordinate on which the fields do not depend (`Pi` in the data files, Π in the author's document)	K1
V_0	`V_0`	the vector mean field $V_0 = G_V\, n$ (the author's symbol); it shifts the chemical potential, $\mu \to \mu - V_0$	K1
$\Omega_{\rm CJT}$	`minus_Omega`	the CJT effective potential of uniform matter with contact interaction per unit volume, a function of the mass m^* and of the vector field V_0 at the chemical potential μ: $\Omega_{\rm CJT} = -\nu\int \frac{d^Dk}{(2\pi)^D}\,(\mu - V_0 - E^*)\,\theta(\mu - V_0 - E^*) - \frac{(m^*-m)^2}{2G_S} - \frac{V_0^2}{2G_V}$; at its stationary point $-\Omega_{\rm CJT}$ is the pressure	new
G_c	`Gc`	the critical coupling of the vacuum: with the filled states of negative energy below the cutoff Λ a massless field acquires a mass when $G_S < G_c = -(2\pi)^D(D-1)/(\nu A_D \Lambda^{D-1})$; $G_c\Lambda^6 = -90\pi^4$ for $D = 7$	K2 ($D = 7$)
$m_{\rm vac}$, $m_{\rm bare}$; $\langle\bar\psi\psi\rangle_{\rm vac}$	`M_vac` `m_bare` `condensate`	with the filled states of negative energy: the mass of the fermions in vacuum, the mass parameter of the Lagrangian (bare mass) that produces it, and the scalar density of the filled states, $\langle\bar\psi\psi\rangle_{\rm vac} = -\nu\int^\Lambda \frac{d^Dk}{(2\pi)^D}\frac{m_{\rm vac}}{\sqrt{k^2+m_{\rm vac}^2}}$ (negative); $m_{\rm vac} = m_{\rm bare} + G_S\langle\bar\psi\psi\rangle_{\rm vac}$	new
m_D, m_Y, f_D	`f_D`	when a bare Dirac mass and the Vorton-Scalar are both present: the bare Dirac mass m_D, the part $m_Y = g_{VS}v_{VS}$ of the mass that the Vorton-Scalar gives, and the fraction $f_D = m_D/m$ of the vacuum mass $m = m_D + m_Y$	new
m_ϑ	`mph2_over_C_ms2`	the mass of the phase mode ϑ_{VS} of the Vorton-Scalar: zero while the chiral symmetry $U(1)_{VS}$ is exact; a bare Dirac mass breaks the symmetry and gives $m_\vartheta^2 v_{VS}^2 = (\lvert\langle\bar\psi\psi\rangle_{\rm vac}\rvert - n_S)m_D m_Y/m$	new
$-G_S/C$, G_V/C	`kappa_S` `kappa_V`	the contact couplings in units of the coupling $C = g_{VS}^2 m^2/m_s^2$ of the Vorton-Scalar (units $m = 1$); they are written as these ratios, without a letter of their own (columns `kappa_S`, `kappa_V`)	new
$\mathsf{h}_{\rm KS}$	—	the Hamiltonian matrix of a particle of matter in the Kohn-Sham description, $\mathsf{h}_{\rm KS} = \sum_j \alpha^j k_j + \beta\, m^* + V_0$	K1
$\Gamma[S]$; S, S_0	—	the CJT effective action as a functional of the propagator of the fermions, $\Gamma[S] = -i\,\mathrm{Tr}\ln S^{-1} - i\,\mathrm{Tr}(S_0^{-1}S - 1) + \Gamma_2[S]$, with the full propagator S and the propagator S_0 of free particles of the mass m (the argument in brackets tells it from the action $\Gamma[\phi, G]$ of the model with the Vorton-Scalar)	new
$n_{\rm ref}$	—	the number density of matter at the reference time $x4_{\rm ref}$; matter that keeps its particles has $n = n_{\rm ref}\, e^{\tau_d}$	new

13 Pairing with contact couplings

symbol	names in programs and data files	meaning	origin
F_0, F_1, F_7, F_8	`F0` `F1` `F7` `F8` `F0c` `F1c` `F7c` `F8c`	the pairing amplitudes of the four independent contact channels in the pair channel along the hidden axis, for particles of mass m^* that scatter from the momenta $\vec{p}, -\vec{p}$ to $\vec{p}\,', -\vec{p}\,'$ in ordinary space ($E = \sqrt{m^{*2} + p^2}$, E' likewise): $F_0 = (EE' + m^{*2} - \vec{p}\cdot\vec{p}\,')/(2EE')$, $F_1 = (3EE' - 2m^{*2} - \vec{p}\cdot\vec{p}\,')/(EE')$, $F_7 = (3EE' + 2m^{*2} - \vec{p}\cdot\vec{p}\,')/(EE')$, $F_8 = -(EE' - m^{*2} - \vec{p}\cdot\vec{p}\,')/(2EE')$. $F_0 = F_S$ and $F_8 = -F_P$	new
m_{ct}	`m_ct`	the range parameter of the contact couplings in the gap equation: each contact coupling g is replaced by $g\, m_{\mathrm{ct}}^2/(q^2 + m_{\mathrm{ct}}^2)$, the form that the exchange of a boson of mass m_{ct} gives (q: momentum transfer)	new
V_{ct}, V_{VS}	—	the pairing potentials in the hidden-axis channel: of the contact interaction, $V_{\mathrm{ct}} = \sum_{r=0,1,7,8} g_r^{\mathrm{eff}} F_r\, m_{\mathrm{ct}}^2/(q^2 + m_{\mathrm{ct}}^2)$, and of the exchange of the two modes of the Vorton-Scalar, $V_{VS} = -g_{VS}^2[F_S/(q^2 + m_s^2) + F_P/(q^2 + m_\vartheta^2)]$; attractive where negative	new
Φ, $\Phi(\vec{p})$	—	a state of a pair of two particles, an antisymmetric 16×16 matrix; $\Phi(\vec{p})$: the state of the pair along the hidden axis with the momenta $\vec{p}$, $-\vec{p}$, the matrix $X^* = (C^- \gamma^0)^*$ between the projectors on the states of positive energy. (Without suffix; the Vorton-Scalar always carries the suffix VS.)	new

14 Einstein-Lovelock gravity with Kohn-Sham matter

symbol	names in programs and data files	meaning	origin
${E^{(k)\mu}}_{\nu}$	—	the Lovelock tensor of order $k = 1, 2, 3$ in the author's notation and sign: $E^{(1)} = -\mathcal{G}$ (minus the Einstein tensor), $E^{(2)} = \mathcal{H}$ (the Lanczos tensor). For a metric whose scale factors depend only on $x4$: ${E^{(k)m}}_{m} = (-1)^{k+1} k!\, 2^{k-1}$ times the sum, over the sets of $2k$ frame directions other than m and over their divisions into pairs, of the products of the sectional curvatures K_{AB} of the pairs. (With the order in parentheses as a superscript; E_F, E_k, $E_\perp$, E^* are energies.)	K1
α_1, α_2, α_3; $\bar{\Lambda}$; κ_8	`LambdaBar` `kappa` `kappa_eps`	the couplings of the Lovelock terms ($\alpha_1 = 1$), the cosmological term and the gravitational coupling of the author's Einstein-Lovelock equations: $\sum_k \alpha_k {E^{(k)4}}_4 - \bar{\Lambda} = \kappa_8\, \varepsilon$ for the time component and $\sum_k \alpha_k {E^{(k)m}}_m - \bar{\Lambda} = -\kappa_8\, p_m$ for the space direction m. $\alpha_2 = -\alpha_{GB}$ and $\bar{\Lambda} = \Lambda_8$. The author's document writes κ for $\kappa_8 = 8\pi G_8$ (the letter κ with an argument, $\kappa(k)$, is a pair correlation). A Lovelock coupling carries a digit as subscript; α^j with a superscript are Dirac matrices	K1
a, b; H_{a}, H_{b}; n_{a}, n_{b}	`a` `b` `H_a` `H_b` `Ha_over_g` `Hb_over_g` `Hb0_over_g` `a_end` `b_end` `Ha_end_over_g` `Hb_end_over_g`	the two scale factors of a spacetime whose space coordinates form two groups of n_{a} and n_{b} coordinates, and their rates $H_{\mathsf{a}} = \mathsf{a}'/\mathsf{a}$, $H_{\mathsf{b}} = \mathsf{b}'/\mathsf{b}$ (a prime: derivative with respect to the time $x4$). In the author's document: a for ordinary space ($n_{\mathsf{a}} = 3$) and b for the four auxiliary coordinates. On the deflating spacetimes with equal rates: a for $x0, \ldots, x3$ ($n_{\mathsf{a}} = 4$) and $\mathsf{b} = b_5 = b_6 = b_7$ ($n_{\mathsf{b}} = 3$); the undisturbed deflation is $H_{\mathsf{a}} = 0$, $\dot{H}_{\mathsf{b}} = -\mathrm{g}55$	K1 (there a, b, H_a, H_b)

symbol	names in programs and data files	meaning	origin
loading	`loading`	$\kappa_8\,\varepsilon(0)/(3\,\mathrm{g55}^2)$: the energy density of the matter at the start of an evolution, in units of the energy density that would make three coordinates deflate at the rate g55 in Einstein gravity	new
$T^{\mu}{}_{\nu}$; p_{a}, p_{b}; δH_{b}	—	the energy-momentum tensor of uniform matter in mixed components, $T^{4}{}_{4} = \varepsilon$, $T^{m}{}_{m} = -p_m$; its pressures along the coordinates that carry the scale factors a and b; the deviation $H_{\mathsf{b}} + \mathrm{g55}$ of the rate of the deflating coordinates from the undisturbed deflation	new

15 Tags and identifiers of the calculations with mass term and self-interaction

symbol	names in programs and data files	meaning	origin
71s, 44q, 44c	`71s` `44q` `44c`	tags of the three models of the self-interaction: 71s on 7+1; 44q the quantized-sector form on 4+4 (bilinears with $\bar\psi = \psi^\dagger\beta_{1644}$ and the matrices γ^A_{44q}); 44c the covariant form on 4+4 (bilinears with $\tilde\psi = \psi^\dagger C^-_{1644}$)	new
identifiers of the machine checks	—	`44d-..`, `71d-..`, `def-..`: checks on the deflating 4+4 spacetime, on the deflating 7+1 spacetime, and of results that hold on both; `44-..`, `71-..`: checks of the flat spacetimes, executed again; `71s-S..`, `44q-S..`, `44c-S..`, `self-X..`: the algebra of the self-interaction; `ks-K..`: Kohn-Sham calculations; `cjt-J..`: CJT effective potential with contact interaction; `uni-U..`: contact interaction, bare mass and Vorton-Scalar together; `pc-P..`: pairing with contact couplings; `gr-G..`: gravity with Kohn-Sham matter. A leading capital I (`Icks-..`, `Idef-..`, `I44d-..`, `I71d-..`) marks a check of the independent verification in Python	new

Key to the column “origin”:

code	source
N01	the author’s notebook `4+4.nb` (Dirac matrices of 4+4; split octonions)
N02	the author’s notebook `7+1.nb` (octonions; Dirac matrices and Dirac equation of 7+1)
N04	the author’s notebook `Dirac_Einstein-Lovelock_4+4_Nash.nb` (meaning of the coordinates)
N05	the author’s notebook `Dirac_Nash_Identity.nb`
N10	the author’s notebook `Pair_Creation_of_Universes_WaveFunctionOfUniverse-4+4-Einstein-Lovelock-Nash.nb`
N12	the author’s notebook `WaveFunctionUniverse-4+4-Einstein-Lovelock.nb`
D06	P. L. Nash, “Identities satisfied by the generators of the Dirac algebra”, J. Math. Phys. 25, 204 (1984)
D07	P. L. Nash, “On the structure of the split octonion algebra”, Nuovo Cimento B 105, 31 (1990)
D08	P. L. Nash, “Second gravity”, J. Math. Phys. 51, 042501 (2010)

code	source
K1	the author’s document KOHN_SHAM_GRAVITY (folder 7+1; sections 15 and 16 of the notebook 7+1.nb): nine-channel self-coupling, Fierz matrix, Kohn-Sham theory, Einstein-Lovelock evolution
K2	the author’s document DIRAC16COMPLEX (folder 7+1): the critical coupling of the vacuum in seven space dimensions
author	a name that an instruction of the author fixes (the rates g55, g66, g77; the constants; the signs of the metric)
new	introduced by this project